SEQUENCE LISTING

<110> Tel HaShomer Medical Research Infrastructure and Services
       Ltd.
       Achiron, Anat
       Gurevich, Michael

<120> METHODS OF TREATING AUTOIMMUNE DISEASES

<130> 53050

<150> US 61/457,396
<151> 2011-03-17

<160> 126

<170> PatentIn version 3.5

<210> 1
<211> 296
<212> DNA
<213> Homo sapiens

<400> 1
gatgaccaga tcatcaactg gctgctagaa ttccgttctt ctgtcatgta cttgacaaaa   60

gactttgagc aacttatcag tattatattg agattgcctt ggttgaatag aagtcaaaca   120

gtagtggaag agtatttggc ttttcttggt aatcttgtat cagcatagac tgttttcctc   180

agaccgtgtc tcagcatgat tgcttcccat tttgtgcctc cccgagtgat cattaaggaa   240

ggcgatgtag atgtttcaga ttctgatgat gaagatgata atcttcctgc aaattt   296

<210> 2
<211> 476
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (58)..(58)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (68)..(68)
<223> n is a, c, g, or t

<400> 2
tttccttcaa cttgtggatg caggcaaggt ggatgatgcc agagctctcc tacagagnat   60

gtggtgcnaa ttgctgaaca aaccccgatt ttgttgttgt tcctccttag gaattctagg   120

aaacaaggaa aggcatcaac tgtgaaatct gtgttagaat tgattcctga attaaatgaa   180

aaggaagaag catacaattc cctcatgaaa agctatgtct cagagaaaga tgtcacatct   240

gctaaagcac tgtatgaaca tttgactgca aagaatacaa aattggatga tctgtttcta   300

aagcgttacg catctttgct gaagtatgct ggagagcctg tccctttcat tgaacccccct   360 gaaagctttg aattttatgc acagcagcta agaaaattga gggaaaactc ttcttgaaat 420

aaccaggcga tactttgttt tgtatatatt tgtgattctg tgtctacatg ttattt 476

<210> 3
<211> 533
<212> DNA
<213> Homo sapiens

<400> 3
aggcttagca ggatgggcgc agtggctcac gccactaatc ccaacatttt aggaggccta 60

ggcaggagca atcacttgtg cctgggagtt ctagactagc ctgggcgaga cttcatctct 120

acaaaaaaag caacaacgac aaaaaaaatt agccaagcat agtggcacac ccctgtagtc 180

ccagctactt gggaggctga ggtgggagga ttgcttgaac ccaagaggtc gaggctgcag 240

tgagccaaga ttgtgccact gcactccagc ctgggtgaca gagcaggacc ctgtctctat 300

tttataaatt aaaaaaggct gggtgtggtg gctcacaccc ataatcccaa cactttggct 360

cagcagattg cttgaaccca ggaattcaag tccaatctgg gcaacatggg gaaaccccag 420

ctctacaaaa aaaattagcc tggtgtggtg gcacatgcct gtagttccag ctactcagga 480

ggctgaggtg ggagaatctc ctgagcctgg aaggtccagg cagtgagcca aga 533

<210> 4
<211> 411
<212> DNA
<213> Homo sapiens

<400> 4
ccaggcatat gacatggatc cccctgggga accaggctga tctctttcca gagggcacta 60

tccgaccagt gcatgatgat atcctcatcg ctcagctgcg gcctggccaa gaaattgacc 120

tgctcatgca ctgtgtcaag ggcattggca aagatcatgc caagttttca ccagtggcaa 180

cagccagtta caggctcctg ccagacatca ccctgcttga gcccgtggaa ggggaggcag 240

ctgaggagtt gagcaggtgc ttctcacctg gtgttattga ggtgcaggaa gtccaaggta 300

aaaaggtggc cagagttgcc aaccccccggc tggatacctt cagcagagaa atcttccgga 360

atgagaagct aaagaaggtt gtgaggcttg cccgggttcg agatcattat a 411

<210> 5
<211> 291
<212> DNA
<213> Homo sapiens

<400> 5
gttgagtcaa cggggggtgtt gccaccagat gtgctggtga gtgaagccat caaagtactg 60

atggggaagt gccggcgctt cttggatgaa ctagatgcgg ttcagatgga ctgagcttgg 120

atgcttctga ggcaagctga agctttgggt tctgactgac ccaccctaca ggactgctga 180 acagagagcc cagtgtgact agggatcctg agttttctgg gacaattcca gctttaatca 240

atacattttg ttaaatgtgc cataaaatga gactttttac gcctttataa g 291

<210> 6
<211> 441
<212> DNA
<213> Homo sapiens

<400> 6
gcacgaggaa gaccataccc taggaaattc tctacgttac atgatcatga agaacccgga 60

agtggaattt tgtggttaca ctacgaccca tccttcagag agcaaaatta atttacgcat 120

tcagactcga ggtacccttc cagctgttga gccatttcag agaggcctga atgagctcat 180

gaatgtctgc caacatgtgc ttgacaagtt tgaggccagc ataaaggact ataaggatca 240

aaaagcaagc agaaatgaat ccacattcta gtcctttatg cagtatacaa ggagaactgt 300

cctgtaggat attctcttcc tgatggtgca gaacccagaa ttagaagttt gtggttacag 360

catactctgt ccttcagaaa ggcgtgattc tagctgttga ccccttgcag ctgttggaat 420

ctctgcaaga acctctgtat t 441

<210> 7
<211> 393
<212> DNA
<213> Homo sapiens

<400> 7
gctatcagcc cggatgacag tgacgaggag aactgagggc acgtggggtg cggcagcggg 60

ctagggccca gggcagcttg cccgtgctgc cgtgcagttc ttgcctccct cacggggcgt 120

cacccccagc ccagctccgt tgtacataaa taccttgtga cagagctccc ggtgaacttc 180

tggatcccgt ttctgatgca gattcttgtc ttgttctcca cttgtgctgt tagaactcac 240

tggcccagtg gtgttctacc tcctacccca cccacccct gcctgtcccc aaattgaaga 300

tccttccttg cctgtggctt gatgcggggc gggtaaaggg tattttaact taggggtagt 360

tcctgctgtg agtggttaca gctgatcctc ggg 393

<210> 8
<211> 238
<212> DNA
<213> Homo sapiens

<400> 8
gttctggcct agagctgtat gcggaatgga agcacgttaa tgaggactct caggagaaga 60

agatcctgct gagtccagag cgagtgcatg agatcttcaa acgcatctca gatgaggagt 120

gttttgtgct gggcatcgag ccccgctatg cacggccaga gtcgatgatt gtcacagtgc 180

tgcccgtgcc cccgctgtcc gtgcggcctg ctgttgtgat gcagggctct gcccgtaa 238

<210> 9
<211> 545
<212> DNA
<213> Homo sapiens

<400> 9
tttattggcc ccacttatta ccagcgtttg aagcatatgg tggatgataa gattcactct 60

cgtgctaggg gacctattca gatcctcaat agacagccca tggagggtag atctcgtgat 120

ggtggcctgc gttttggaga aatggaacga gattgtcaga ttgcccatgg agcagcccag 180

tttttaaggg aaagattgtt tgaggcatca gatccatatc aggttcatgt ttgcaatctt 240

tgtggaataa tggcgattgc caacaccagg acccatacat atgaatgcag gggctgccgc 300

aataaaaccc agatttcttt ggtgcgaatg ccttacgcat gcaaactatt gtttcaggaa 360

cttatgtcta tgagtattgc accgcgaatg atgagtgttt agctatttta caggagtcaa 420

caagataatt aaatatcttg gtgtcttgtt tctattgtgt ggctttataa aaatgacaaa 480

tatgtactgt gttgtgataa aaagtatttt atttgtttaa tgatatgcat gcttttcttc 540

tgtaa 545

<210> 10
<211> 439
<212> DNA
<213> Homo sapiens

<400> 10
ttaactgcag cttgcctgct tcagcaaaaa cggagattca ggccagcagc tggatatggg 60

ggtctctctt cagactcttc tcgtttctga gaatctagtc tactgttggt tgagcttctt 120

ggcaggacat cagtaccaac tagaagtggg tcatagatag attaccaggg atgcagtggt 180

gtttaggcag gataggtctt tactggccct gactgctgtt aataattggc agcagtgctc 240

cccagatccc agaaggtccc tgctggagtg tttccagtgc acctgtaggg aaccaactag 300

acttctctcc tggttagtcc agctctttac tctaaaccct ttctgtccaa aatgagtcat 360

tttcagttgt accttagatg tctggtgttg aggatcaagt gccatagcct ttattcaggg 420

ggcctataaa cccttccag 439

<210> 11
<211> 350
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (55)..(55)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (68)..(68)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (74)..(74)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (76)..(76)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (138)..(138)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (174)..(175)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (180)..(180)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (185)..(187)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (189)..(190)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (193)..(193)
<223> n is a, c, g, or t

<400> 11
caattcccag ggtgtcccta atttgagaag taacctttg gaatagcatt agacnctggc   60

tgtccccntc cccncncaaa taaacatgat atttcattct ctgtccagca gtcatgaacc   120

ccttcacctc caatgacntg atcatttagt ttggtggggg tgggggtggg ggtnngggtn   180

ggggnnnann canccgcagg agcaagggcc cctcccacat acacaggagg agtatttcat   240

ttctccttaa tgaaggctct ggccctaacc cctcagcact gtctccagat aggaacatgc   300

acaaagcagt taattaggca gcctggagaa aaccagagat ccagtacaga   350

<210> 12
<211> 554
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (270)..(270)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (361)..(369)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (376)..(376)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (507)..(507)
<223> n is a, c, g, or t

<400> 12
cctgctagcg cttcaaattc ccaggtgtcc ctaatttgag aagtaacctt ttggaatagc 60

aggagcaagg gcccctccca catacacagg aggagtattt catttctcct taatgaaggc 120

tctggcccta acccctcagc actgtctcca gataggaaca tgcacaaagc agttaattag 180

gcagcctgga gaaaaccaga gatccagtac agaaaggaaa ggatatttat tgattaacag 240

aagttgtctt tttaaaagtg tttatttttn gcaataaaga gcacaacata atctccttca 300

tgcgtcatcg tgccacttag ctgagccagc cagctctgcc tccctcccca aagacctttg 360

nnnnnnnnng ctttanagga tgaatctgtt gaggaggtct gtttccgagc tggcctgggc 420

tgccccatac ccctcaccag catagctgac ggcttcggcg atgttctggg aaccagtgat 480

gtggtgccag taggatcgcc gcggcanact gcgtcgtggg tgtgcaggga tacgagtaat 540

gttccgtgcc cttg 554

<210> 13
<211> 338
<212> DNA
<213> Homo sapiens

<400> 13
ggtcgtgcca aagaggtcca caaccaggtg tgcactgttc actgcagccc atttgctgta 60

tgaactgtgg ttgttgtgtg cccaatgaca aggctactaa gaaattcatc atttgaaacg 120

tagaggccgc agcagtcagc gatgtttctg aaatgagcat ccttgacgcc tgtgtacttc 180

ccaggctgga tgtgaagcta cattaccatg tgagttgtgc cattcacagc acagtggtga 240

ggaattgagc tcatgaagca ggcaaggacc gaacacctcc accccaacgt agacctgcag 300

gtgctgcccc atgacctcca ccaaagccca tataagga 338

<210> 14
<211> 349
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (73)..(74)
<223> n is a, c, g, or t

<400> 14
tttttactt cataggccag gcttgaagtt ctgaacactt tgaagtctcc aattatgaga 60

gatccagtct aannctctgg cctgctaatt agcaataagt gctttatttg gaaggaggga 120

gtcatccact cttgagccac tgcagtgaag tcacttgatc tcagtctggg ggaaaacact 180

tcaatagcta aacattctag ctttgatttt tctgaaggga atacacttgt tttcaatttt 240

ggggtttttc tttggggcac ttgcttgact ctgtatgaac ttgtgatcca aggaaaaagg 300

agaaagaaca gtgttggctt ttaaaatcag gatggtttta tgtttgcta 349

<210> 15
<211> 416
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (45)..(45)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (179)..(179)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (202)..(203)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (286)..(288)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (334)..(334)
<223> n is a, c, g, or t

<400> 15
accgaccaga tgtttgtgtt ctttccagag gagcccaagg tgggnatcaa gaccatcaag 60

gtgtactgcc agcgcatgca ggaggagaac atcacacggg ctctcatcgt ggtgcagcag 120

ggcatgacac cctccgccaa gcagtccctg gtcgacatgg cccccaagta catcctggna 180 gcagtttctg cagcaggagc tnntcatcaa catcacggag cacgagctag tccctgagca 240

cgtcgtcatg accaaggagg aggtgacaga gctgctggcc cgatannnta agctccgaga 300

gaaccagctg cccaggatcc aggcggggga cccntgtggc gcgctacttt gggataaagc 360

gtgggcaggt ggtgaagatc atccggccca gtgagacggc tggcaggtac atcacc 416

<210> 16
<211> 339
<212> DNA
<213> Homo sapiens

<400> 16
gccccattca cctgtggatt tgtgcgagat gcagcctcag aaggaacaag gcccccagag 60

ggaggtcacc tgggggcaac tggtgccggg tcttcaccca gaccacgctg ggtcccctct 120

gttgggggtt tggggtccgg gtctcccacc agccactgct tcctcctggg ccctcaattc 180

cacccctcgt cttccctccc tcggggccct gatgcgtggc ggcccccacc cggcctcggc 240

tctttactcc attcacagcc atgcacgcgc tcaagtacac cagggtgcga gatgccagct 300

ctggagttct cggttgttgt aggaggttgg gtgttttca 339

<210> 17
<211> 485
<212> DNA
<213> Homo sapiens

<400> 17
agataagcta ggagccgcgc gagtcgtagt gtcgctgttt gcgggtctcc gcgcgggacc 60

ggggcgcagc ggggtcgctg aggcgagggt gtcatgtcag acaacgagga caattttgat 120

ggcgacgact ttgatgatgt ggaggaggat gaagggctag atgacttgga gaatgccgaa 180

gaggaaggcc aggagaatgt cgagatcctc ccctctgggg agcgaccgca ggccaaccag 240

aagcgaatca ccacaccata catgaccaag tacgagcgag cccgcgtgct gggcacccga 300

gcgctccaga ttgcgatgtg tgcccctgtg atggtggagc tggaggggga gacagatcct 360

ctgctcattg ccatgaagga actcaaggcc cgaaagatcc ccatcatcat tcgccgttac 420

ctgccagatg ggagctatga agactggggg gtggacgagc tcatcatcac cgactgagct 480

ggagt 485

<210> 18
<211> 543
<212> DNA
<213> Homo sapiens

<400> 18
gtatggcttt gtaattgctg tcaccaccat tgacaatatt ggtgctggtg tgatccagcc 60

aggccgaggc tttgtccttt atccagttaa gtacaaggcc attgttttcc ggccatttaa 120 aggggaggtc gtggatgctg ttgtcactca ggtcaacaag gttggactct tcacagaaat 180

tgggcccatg tcttgcttca tctctcgaca ttccatccct tcagagatgg agtttgatcc 240

taactccaac ccaccatgtt acaagacaat ggatgaggat attgtgattc agcaggacga 300

tgagatccgc ttaaagattg tggggacccg tgtggacaag aatgacattt ttgctattgg 360

ctccctgatg gacgattact tggggcttgt aagctgagcc tggtggcctc ctacccttgg 420

tcctactcta ggaagtgtga ttgtcacact tatcatgttg tccagaggtc cagtctggct 480

gctgttgtgg aggcaaggaa ggcaactcat cccagaaggc atctggtgct tcttgtagct 540

taa 543

<210> 19
<211> 377
<212> DNA
<213> Homo sapiens

<400> 19
ccttccaggg ctgaccagtt tgagtatgta atgtatggaa aagtgtacag gattgaggga 60

gatgaaactt ctactgaagc agcaacacgc ctctctgcgt acgtgtccta tgggggcctg 120

ctcatgaggc tgcaggggga tgccaacaac ctgcatggat tcgaggtgga ctccagagtt 180

tatctcctga tgaagaagct agccttctga acctcgcctg aagccagcct ctctgccaag 240

tcactcaggt catgggcatt gttcaagcct gagtggcagc cgctcttgct cacctgttga 300

ggaagggctg gctcactgtc caccgtggcg gcatctttaa ctggcctcca ctcaatggga 360

aactgactcg cctgtga 377

<210> 20
<211> 305
<212> DNA
<213> Homo sapiens

<400> 20
tggagcccga cgggacttac gagccgggct tcgtgggtat tcgcttctgc caggaatgta 60

acaacatgct gtaccccaag gaagacaagg agaaccgcat tctgctctac gcgtgccgga 120

actgtgatta ccagcaggag gccgacaaca gctgcatcta tgtcaacaag atcacgcacg 180

aagtggacga actgacccag attatcgccg acgtgtccca ggaccccacg ttgccgcgga 240

ccgaggacca cccgtgccaa aagtgcggcc acaaggaggc tgtgttcttc cagtcacaca 300

gtgcg 305

<210> 21
<211> 515
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (173)..(173)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (175)..(175)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (177)..(178)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (180)..(180)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (182)..(182)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (298)..(299)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (301)..(301)
<223> n is a, c, g, or t

<400> 21
gttcttgctc ttcgagggcg agaagaagat caccattaac aaggacacca aggtacccaa 60

tgcctgttta ttcaccatca acaaagaaga ccacacactg ggaaacatca ttaaatcaca 120

actcctaaaa gacccgcaag tgctatttgc tggctacaaa gtcccccacc ccntngnngn 180

anaagatcat catccgagtg cagaccacgc cggactacag cccccaggaa gcctttacca 240

acgccatcac cgacctcatc agtgagctgt ccctgctgga ggagcgcttt cgggtggnnc 300

nataaaagac aagcaggaag gaattgagta ggggccagag ggggctctgc tcggcctgtg 360

agccccgttc ctacctgtgc ctgaccctcc gctccaggta ccacaccgag gagagcggcc 420

ggtcccagcc atggcccgcc ttgtggccac ccctcaccct gacaccgacg tgttggccac 480

ccctcaccct gacaccgacg tgtcctgtac ataga 515

<210> 22
<211> 354
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature

<222> (252)..(253)
<223> n is a, c, g, or t

<400> 22
gttcttgctc ttcgagggcg agaagatcac cattaacaag gacaccaagg tacccaatgc 60

ctgtttattc accatcaaca aagaagacca cacactggga aacatcatta aatcacgtgc 120

ctgcttcccc ttcgccttct gccgtgattg tcagtttcct gaggcctccc cagccacgct 180

tcctgtacag cctgcagaac tctgccccag agcacatcag ctatgtgccc cagctctcaa 240

acgacacctt gnncggggag gctcaccctg tccaccttca cgctggagca gcctctaggc 300

cagttcagca gccacaacat ctctgacttg gataccatct ggctggtggt ggcc 354

<210> 23
<211> 299
<212> DNA
<213> Homo sapiens

<400> 23
cagccggtcc cctggataaa gaggtgaggg ggtgtgaagt ctgggagccc ccctcccagg 60

gccaggctac aggggacag cagggctata ccaggagtgc cacagagctt gcatttgggt 120

gtgatctaat ccccagtcct gccctgacct cctgggtgct cttagccaag ccctcttccc 180

tgtgtgtgct agaaagtgag ggtcactccc ttccacacaa gggagtgaat tgagggtggg 240

gatctgagct ctctgagcag caaagctgga cacaggccaa attgccaatc ctgagtccc 299

<210> 24
<211> 537
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (83)..(83)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (92)..(92)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (141)..(141)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (154)..(154)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (173)..(173)

<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (177)..(177)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (191)..(191)
<223> n is a, c, g, or t

<400> 24
ccagtgaact cctgaaacct caaacagtac caaaccttta tacactgttt tttccatata 60

tatataccta tgataaagta tanatgtata anattaagca tagcaagaga ttaataataa 120

tgtaatagaa caatgataac natactataa taanaagtta tgtgaatgtg gtntggntct 180

ctcttgcttt ncaaaatatc ttcttgtaca gtactcacct attttagaat gtggttgact 240

acaggtaacc aaaaccacag aaagggaaac tttggatgag gggggcacta ctgtacttag 300

gaatacaact atatacatat gattttattt ttaagaccat attatatttg ggtatctact 360

aatattttgt ataaagcaat tttttgttcc attacgtgac tttttgtttt attgtatatg 420

taatttaaca cacaataaag ggtaaagttg cttccccaaa ccacactttt aatcaaaacc 480

tagaatcatc tgcagtcctt gttaaaaatg caggtttcta gaaccctctg aagttct 537

<210> 25
<211> 253
<212> DNA
<213> Homo sapiens

<400> 25
ttccgtgctg tgtaggggct aacaatggac acccagaagg acgttcaacc tccaaagcag 60

caaccaatga tatatatctg tggagagtgt cacacagaaa atgaaataaa atctagggat 120

ccaatcagat gcagagaatg tggatacaga ataatgtaca agaaaaggac taaaagattg 180

gtcgtttttg atgctcgatg aatgctggga attcagagga atgtcttcac ttatacttgg 240

atttgctctc ttc 253

<210> 26
<211> 231
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (63)..(63)
<223> n is a, c, g, or t

<400> 26
ctttcacact gtggcagccc agtgaagcag actgggccat gaactctcct agccctgggg 60

ccnagcctgt tccacaggca cccctgcagg aggcgctgcc aggagagcct tccatctcgg 120

ggctctttga ggttccctcc ttctgggtgt tcttcaggct gagcagagag gctcctgtac 180

cctctctctc ggaatctgaa gagccagatt taggccgggc aaaggggctc a 231

<210> 27
<211> 349
<212> DNA
<213> Homo sapiens

<400> 27
ctgcttcact tgtggcaaga tcgtcggcaa caagtgggag gcttacctgg ggctgctgca 60

ggccgagtac accgaggggg atgcgctgga tgccctgggc ctgaagcgct actgctgccg 120

ccggatgctg ctggcccacg tggacctgat cgagaagctg ctcaattatg cacccctgga 180

gaagtgacca cgctggaacc cacccacccg ctgtgctgac catgggccct gagcgtcctg 240

ccccgaattc acgaggctga ggcatccggg agctggcgta atgcctggcc gcagtgtgtg 300

tgtatcccat accccactct ggaaggaacc atccagtaaa ggtctttca 349

<210> 28
<211> 503
<212> DNA
<213> Homo sapiens

<400> 28
aagaattccc ttgcgtattc tctctctaaa acaaccaaaa aaaaatggag aggcttttta 60

tatactctaa gactggctaa acaaccttga tcattgagcc tcgagccatg ggagagatgc 120

tgaccatgtg gactgcaagg ctgcttgatt cacagatgga tgtgacctaa aggataaata 180

agctattact tatgtgctga tctcttgaca ttcactcatt agaagacctt actccttcaa 240

gcaaatgttt ggggtcaaat ttaccatatc ttctggctaa ccatattcaa gattcttctg 300

aaacttggag gatgtaaaga atccatttga tttggtcagc ctggcttttg tcgtggtggc 360

tggctcggat aaattttccc aacaattaaa tcttgccttt acacacccaa actttgtaat 420

tttagtcttg gtgaaatata atgaatttgt tcctaccttg tcaagcaaga atgtcgtctt 480

ctcctatgga ctcaattgct att 503

<210> 29
<211> 276
<212> DNA
<213> Homo sapiens

<400> 29
ccccatccag gaccttctat ttatatactg tgaacatcct gtcagctgcc cgaatgttgt 60

tcgacaggtg ctacaagagc atagccaact tgatagaaag gaggcaattt gaaaccaaag 120

agaataagcg tctactagaa aaatctcaga gggtagaagc catcattgca tctatgcagg 180 ctactggtgc agaggaagca cagttacaag aaatagagga gatgatcaca gctcctgaac 240

gtcagcagct agagaccctg aaacgtaatg tcaaca 276

<210> 30
<211> 517
<212> DNA
<213> Homo sapiens

<400> 30
gagtgagatt accacttcct ctagtgctcc cttcacccag ccattgtctt ccaatgagat 60

cttcagatcc ctacctgttg gctataacat ctctaagcaa gttcttgatc agtatctcac 120

tctgctggca gatgatccac tagagtttgt tggaaagtct ggcgacagtg gtggaggaat 180

gtatgtcatc aacctccata aggcattagc atccctagcc acagccactc tgaagtccgt 240

cgtacaggag agatttgggt ctcgctgtgc tagaatattc cgtctagttt tgcagaagaa 300

acacatagag cagaagcaag tggaagactt tgcaatgatt cctgcaaagg aggcaaagga 360

tatgctatat aagatgctct cagaaaattt catgtcactc caggttggat gccagtgaga 420

tccaggtgga cgaaaccatc ttcctgctgg agtcttacat tgagtgcacc atgaagagac 480

agtgatccag aagaagcatc ttcctcagaa gatctgg 517

<210> 31
<211> 530
<212> DNA
<213> Homo sapiens

<400> 31
cctgctccag acatttgtt cttgaatctg tgagacccag aagggcccac tgagcccact 60

cactccacct ttggcaacca ttgttccagg tcccccaggg cttcctccca cagcagctgt 120

gaatggcaca gtgaccttcc tgcagcgtgg agatggcaca tccttgctgc tggggacttg 180

gccctgctat ttatttttgt atttatgtct taatctcttc cactgatgca tcctccaagg 240

gtagatgggg agggtctgtg tgaaggggcc ggcttctctt ggtgcctgct gggttgcagg 300

ggcaggaagc gtgtggactg cagcttctgc tggtgctccc cccgtcctcc tggaggcagt 360

ataggagaga gagcaaggat tgagtctgag acttaagcac tcggtcccag cttgccagtt 420

cctggttctg tgtccttgga aaactaccta acctttctga gcctcctata ctatccgaca 480

caaatgggga tgatacctac ctccagggtt ggcgtgagga ttcatgggct 530

<210> 32
<211> 382
<212> DNA
<213> Homo sapiens

<400> 32
ggtacgccga aacatgatcc agtctcggtt gactcaagag tgtggagaag atctcagtaa 60

acaggaggtg gataaagtac taaaggactg ctgtgtaagc tatggtggca tgtggtacct 120

taaagggaca gtacagtctt gacaatagta gcaaactact aacccagcaa atctaagccc 180

aaggaagaag ggcggaacca gaagtagggc ctcgacttgc ttcagacgac acagagcaag 240

aggaactgac catctcatga cctgtggcat tgcacggtgc agtggacaga agggattatc 300

ctcagccagt cgcagggtca gcttaagtta gttagatcac tcccagaaga gaccagctgg 360

gaccttcttt gcagtacaat tt 382

<210> 33
<211> 216
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (55)..(56)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (59)..(59)
<223> n is a, c, g, or t

<400> 33
agagtggaac atgccttttc cgcacaatat taattccttt ttgtatcaga aaggnnctnt 60

taggagttat gctaccatac ttacttcaaa cccaatgact actgtcaagg tcatattttc 120

agtacataaa tactatcatt ttcattctaa agaatatttt cactgttcct tctttcttaa 180

agtcttatgt ttcactcttt aactcaaatg tattct 216

<210> 34
<211> 391
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (295)..(295)
<223> n is a, c, g, or t

<400> 34
gataaccact tgaaccaggc aaatacactg aagataaagg ttatttcttt ttttagcttt 60

gaatttgtca tgaccatttt agtcttgcag atagcagggc agccccttgg ggcaaggatt 120

tactctggga ggtaccgtta agagccttct ttcccccttt gaaagatcct tttacaatgt 180

taaagtatac tagttgcaag aacaagcagg atttgcaggt tgctttacca gcatgagtct 240

catttttctg gcttaaaatc tgggactgtg aaattattcc ataggaaagt gaatnttatt 300

ttgcagaatt agcctcttac ataaaagtat ttgttgaagt gtctttaaaa ttgctatcat 360 gagcaaaact ggttgctgta atgcttgttt t 391

<210> 35
<211> 442
<212> DNA
<213> Homo sapiens

<400> 35
ggataccaga ttggagaaga cttccaagag agatgatgcc aagaaataaa tgtaaaaaag 60

caggcccaaa acccaaaaag gcaaaagacg caggcaaagg cacaccactc actaatactg 120

aagatgtgtt gaaaaaaatg gtggaattgg aaaaaagagg tgatggtgaa aaatcagatg 180

aggaaaatga agagaaagaa ggaagcaaag agaaaagtaa agaaggtgat gatgacgatg 240

acgatgatgc cgcagaacag gaggaatatg atgaagaaga gcaagaagag gaaaatgact 300

acattaattc atactttgaa gatggagatg attttggcgc agacgtgatg acaacatgga 360

tgaggcaacc tattaggcat gaaatttttc aaaaaatatt tttatgatgc agcttctgaa 420

catttggaca gacttgattt gt 442

<210> 36
<211> 299
<212> DNA
<213> Homo sapiens

<400> 36
ctgcgggaac gggctgatcg tggaggaggg acaacgctgc caccgcttcg cctgcaacac 60

gtgcccctac gtgcacaaca tcacccgcaa ggtaacaaat cggaagtacc caaaactgaa 120

agaagtggat gatgtgcttg gtggagcagc tgcctgggag aatgttgact ctactgcaga 180

gtcgtgtccc aaatgcgaac atcctcgtgc ttacttcatg cagcttcaga cccgctctgc 240

agatgagccg atgaccacct tctacaagtg ctgcaatgct cagtgtggac accgctgga 299

<210> 37
<211> 441
<212> DNA
<213> Homo sapiens

<400> 37
tatcgcctgg actccaaggt caagcaaata ggaggtggaa ttcagagcat cacctacacc 60

cacaacggag acatcagccg aaagcccaac acacgtaagc agaagaacgg cttcccgccc 120

aacttcatcc actcgctgga ctcctcccac atgatgctca ccgccctgca ctgctacagg 180

aagggcctga ccttcgtctc tgtgcacgac tgttactgga ctcacgcagc tgatgtctcc 240

gtcatgaacc aggtgtgccg ggagcagttt gtccgcttgc acagcgagcc catcctgcag 300

gacctgtcca gattcctggt caagcggttc tgctctgagc cccagaagat cttggaggcc 360

agccagctga aggagacact gcaggcggtg cccaagccag gggccttcga cctggagcag 420 gtgaagcgtt ccacctactt c 441

<210> 38
<211> 406
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (73)..(73)
<223> n is a, c, g, or t

<400> 38
gctgaccttc gaggcggtgg atgtacccat gctttgcccc ccgctgccct ggacatcgcc 60

gcactctggt gcnttcctgc tcagccccac caagctgatg cgcacggtgg aaggcgccac 120

gcagcaccag gagctgctgg aaacctgccc gcccaccgcg ctgcatggcg cactggacgc 180

cctcacccaa ctgggcaact gcgcctggcg cgtcaacggg cgcgtgctgg acctggtgct 240

gcagctcttc caggccaagg gctgccccca gctaggcgtg ccggccccgc cctccgaggc 300

gccccagccg cccgaggccc acctgccgca cagcgccgcg cccgcccgca aggccgagct 360

gcgccgtgag ctggcgcact gccagaaggt ggcccgggag atgcac 406

<210> 39
<211> 449
<212> DNA
<213> Homo sapiens

<400> 39
ctctgaatgt gatcgacgcc cagcaaggac aagctttaaa atgtctgcgg tctgcccttt 60

tgaagcagga ctggctcact ctgtcattgg gagctgtcag ctgcgactgc aggttctcta 120

ggaggcattc cagaatagag tagcacactg tgtctgcagt tctcgatgac cgaaagttat 180

caaaaatatt taaaatattt aaattgtgaa cctattgata aagaatattt ataaaaactg 240

atctgtaggc ctgtactaat ctctacgcat tagcaatatt gactgtaaac ccacattaag 300

gaaaccacta cgggtctggc agtgcgtgtc ccgtggggtg tgcattttaa aactcgattc 360

atagacacag gtaccatgtt ccatttccgt catggtgaag caaatgaatt ggcctggcta 420

ccactgtggt cgcgtgctac aggtttgac 449

<210> 40
<211> 201998
<212> DNA
<213> Homo sapiens

<400> 40
aagcttgcag taagccgaga tcgcgccact gcactccagc ctgggtgaca gagggagact 60

ccatctcaaa aaaaaaagag agaaagtaaa ggaataaaag aatggctacc ccatagacgg 120

agcagccgtg agggctgctg gttgcccatt tttatggtta tttgttgatg atatgctaaa 180
caaggagtgg atttttcatg cctcctcttt ttagaccata tagggtaact tcttgatgtt 240
gccgtggcat ttgtaaactg tcatggtgct ggtaggagtg tagcagggag gatgatggga 300
ggtcagtctt gtctctattt tggttttggt gggttttggc cagctccttc actgcaacct 360
gttttatcag caaggtcttt atgactggta ttttgtgctg accttctatg tcatcctgtg 420
acttagaatg ccttaaccat cagggaatgc agcccagtag tttcagcctc atttttcccg 480
gctcctattt aagatggagt tgctctggtt cacacacctc tgacatgatc attgcccact 540
gcggcttcca cctcccgggt tcaagagatc ctcctgcctc accctcccaa ggtgctggga 600
ctacaggtgt gtgccaccag ctcagctaat ttttgtattt tttgtagaga tggtgttttt 660
ccatgttgcc caggctggtc tcaaactcct gggctcaagc aatccttctg tctcagcctc 720
ccaaagtact gggattacag gcatgtccca ccatgcccag actaatattt acttttaatc 780
agactaagat agggttacta cttgagttgc tatggctcca gctgaaagaa agcccgtgca 840
gtcatatcac gcgtaaacat ttgctttatg ctaaaaatat ggtggacctg gcattacagc 900
tattacaaat ctcctaagat gtctcgggta gtgtattagt tacttttcat actgctatga 960
agaaatactg gaaactgggt aatttataaa gaaaaagagg tttaatgtac tcacagttcc 1020
acaaggctgg agaggcctca gaatcatggt ggaaggcaaa gaaggagcaa aaaggtatgt 1080
cttccatggc agcaggcaag agagcacgtg cagggaaact gccctttata aaaccatcag 1140
atttagtgag atgtattcac tatcacgaga acagtatggg aaaaacctgc ccccatgatt 1200
cgattacctc ctaccgggtc cctcccacga cacatgggga ttatgggaac tacaattcaa 1260
gatgaaattt gggtggggac gcagccaaac catatcgggt agcaacaacc tagggtcagt 1320
tttgcaggtg gtaaagccat ttaccaagat agttgtgggt aaagaagggc agatttatta 1380
gagaaattgt gaaaatatgt tgcagtgggc agctcagcag agaaggggct acctgcaaag 1440
aggcaagggc tggaggaaag ttttataggg tcctgctgaa gggtgctacg tgtggaatga 1500
ggtcattgtg cccgcaggtt gtttgtgatt agctgtctct aacaattgtt catacaataa 1560
ttgttcatta ttgttctcaa cttggggctc tccccaacct ggggaccctt ccttattgtt 1620
gcttacttat caggtctcca cataaaggtg tggaaacttc attcattcat atcttcaaca 1680
caaattgtag gtagcctgtt ttttaaaaca tttattcaac aaatatttag tccaagccac 1740
tattacttac taccttctct actattgtat ggacttttaa ctatctctga cactattcac 1800
tattcttcca cattctctat tatttatacc tatggtaaaa tttgccagtt tgaccataca 1860
actaatactc acagggaata tatagagtct agaagaaaat atacaggtcc ttaaaggctg 1920
ccctgccaac aaaaccataa cgcaggaaca aacatcacaa ctatgccaaa taatcaatcc 1980
tacaatgtcc aaaattttac tttaaaactg gaattaccag acttcctttc tgcattaacc 2040

agtttaacta gacagtaacg aaatattcct actttatgct gtgatagttt gtttgtttgt 2100
ttgtttgttt atttatttat ttatttattt aagacagagt ttcgctcttg ttgcccaggc 2160
tggagtgcag tggcacgatc tcagctcacc acaacctccg cctcccaggt tcaagcgatt 2220
ctcctgcctc agcctcccga gtagctggga ttacaggcat gtaccaccac gcccgggtaa 2280
ttttgtattt ttagtagaga cggggggttt ctccatgttg gtcaggctgg tctagaactc 2340
cagacctcag gtgatacccc tgcctcagcc tcccaatgtg ctgggattac agctgtgaag 2400
ccaccgcgcc cggctgctgt gatagttgag atgtaaacca aaaataaaat tctaagccac 2460
ccaatccgac tgaatggacc cttcctgttg agcaaggaca ttccaaagta aactgaaaag 2520
accagcttag gccatgatgg gaaggggagg tgtcaacatg cctcattcta ccttcctccc 2580
tctggaatcc agacacaact gaccagcatt aacattaaaa cagagatctt aagctgggca 2640
cagtggctca tgcctgtaat cccagcactt tgggaggcca aggtgggatc acctgaggtc 2700
agaagttcaa gaccagcctg gccagtatgg tgaagccatg tctctactaa aaatacaaaa 2760
ttagccggac attgtggtgc acgtctgtca tcccagcaag gcaggcgaat cacttgaacc 2820
caggaagcag aggttgcagt gagccaggat catgccattg cactccagcc tggtcaacag 2880
agcgagactc cgcctcatta aaaaaaaaaa aaaaaaaaaa ttagccgggc gtggtggcgg 2940
gcacatgtag tcccagctac tagggaggct gaggcaggag aatggtgtga accagggagg 3000
cggagcttgc agtgagccga gattgtgcca ctgcactcca gcctggacag aagtgcattt 3060
cataatgcat tttaattgca ttagcagtga tttaattttt ttagatgcta aaacttatgg 3120
gtgaaagtgg attaaatgta gccaaatgca acatcaaaat cttcaggcac aaaaacccat 3180
taactttttc atactctcag aaggtgaacc taatttcaaa tgaaagctgc ctccagaata 3240
tattgttaag cgtattctag atataattca ttttggcaaa catactgtag aaattcacat 3300
aacattttac tgtactaaaa gtaaattgcc catgtaacaa aaaatatctt ttcagagctt 3360
gaaatgaatt ttaaaggatg actgatggtc cctggaagag aaacagtaaa caaataaggt 3420
ttgtagcaat gatgtatgag ttagaaattg cagttccaga tgatctcttt attaaagaga 3480
cgatctacac ttaatttgat caagtgttat gaacatagtt catgttaagt ctccatttaa 3540
atacaacctg aaataccaaa gttaattttc ttttctttct ttcttttttt tttttttag 3600
aaggagtctt gctctgttgc ccttcctgga gtgcagtgac gtgatcttgg ctcactgcaa 3660
cctccacctc ctgggcttga gcgatcctac tgcctcagcc ccccaagtag ctgggaggac 3720
aggcgcaagc cacggcactc agctaatttt tgtatttttc gtagagatag ggtttcacca 3780
tgttgcccaa tttggtctcg aactcctgag ctcaagtgat ccgcccgcct tggcctccca 3840
aagtgctggg attacaggca tgagccaccg tgcctggcca gaaaattgta aacacacaca 3900

aactctcaag tggcctaatt ccctctcacc aaaccaatca caatacagat aaaagagaat 3960
aacttgtgtt catttttgta caaacaaaaa agatataaat tgtgaatgat gcatgatttt 4020
taattacaag taaactgggc aaatgcttct gcattattta aagctaaaag gtgatcagtg 4080
gaaactttcc tctgttagta ctctaatact ttttatattt atcggctcac tacaacctgt 4140
gcctaccagg ttcaagcgat tctcctgtct cagccacctg agtagccgag accacaggca 4200
cgcactacca tgtccggcta attttgtatt tttaatagag acagggtttc accgtgttgg 4260
ccatgctggt cttgaactcc tgacctcaac cgatccgcct gccttggcct cccaaagttc 4320
tgggattaca agcgtgagcc acagcgccca gccttattat aattgttact atttaaatct 4380
cttttgctct ctccttcaag agagacctca tcccattcag ttgcttccat ttatttattc 4440
atcttctgcc tcctgggctc gagagatcct ccagcgtgag tctcccaagt agctgggact 4500
acaggctcac accaccaagc ttggctaaat tttgtaggtt ttggagagac aggctcttgc 4560
cacgttgcct aggctggtct caaactcctg ggctcagatg atccacctgc cttcgcctcc 4620
caaagcactg ggacatgagc caccacgccc agccgcaagt acttttacac aaaatgcaaa 4680
caccattctt ccatcataaa agtgatacca cagcttccgt gaagttttgc caggtagtac 4740
tcataattac cttgggtaaa ctttttgatg ttaaactgta tcttcttatt acgagttttt 4800
ccattgtatt aactgctttt acaacaacac aaataacaag ttattttaca aaccatttag 4860
aaatttctgt actatggtcc cagtaatgta aaatatatta atgcctatta cattcagata 4920
aattatacac ttggaaacca catacttatg acttacagaa acttacataa acaaattata 4980
gaaattacat gctcaatttt taggtatata gtcttaaatt aagcttaaat atacattctc 5040
aagataaatt aacagttcag ggcttcacaa cttgaaatct gtggaagatg acattggaga 5100
caacagaact ctggtggaat tcttagatgg aatttgccga aacttttttt tttttttttt 5160
tttgagatgg agtgtcgctc tgtcgcccag gctggagtgc agtggcgcaa tctcagctca 5220
ctgcaagctc tgcctcccgg gttcacgcca ttcttctgcc tcagcctccc gagtagctgg 5280
gactacaggc gcccaccgcc acgcccggct aatttttttat atttttagta gagatggggt 5340
tttactatgt tagccaggat ggtctcgatc tcttgacctt gtgatctacc cgccttggcc 5400
tcccaaagtg aaacttttct ttaaaataga gatgggatct tgctgtattg cccaggctgg 5460
tctcagactc cttgccttaa gcagtcctcc cacctcagcc tcctaaagtg ctgggattac 5520
aagcgtgaag cattacatcc aagtgaaact tcttgagatg gttacataat gtctaaatct 5580
gctggtgtag aagttaataa agtgtagaac tgaataacta ttaaatatta gatcaagttt 5640
ctcatgttta tcttaacgta taacgattta tcttaaagca ctgattttca caaaataaca 5700
tcagtgtgaa attggaaaag aagccaaata ttttatttca cgtatctggg aaatgaggtg 5760
ctttagtcaa ctgaatctgc ccaaaactaa aaagcattaa ttaaaaagta cttaactcag 5820

aaattataaa aataggagac atcaataaaa tacattctac acagaatacg ccaaccatac 5880
actactcttt tttgataata aaaaatgtat ttactgagcc agttgtggtg gctcacgcct 5940
ataatcccag caccttggaa ggccaatgag agtggatcag ttgaggccag gagtttgaga 6000
ccagcctggc caacatggtg aaatgccgtc tctactaaga atacaaaaat gagccgggca 6060
cggtggcacg cacctgtaat cccaggtact ccgaaggatg aggcaggata attgtttgaa 6120
ctcaggaggt ggaggttgca gtgagccaaa atcatgccac tgcactccag cctgggtgac 6180
agagtgagtc tctgtctcaa aaaaaaaaaa aaaaaaaaag aaaaaaagtc agttgcagtg 6240
gctcacgcct gtaatcccag cactttggga ggctgaggca ggcggattac aaggtcagga 6300
gatcgagacc accctggcca acatggtgaa acctcctctc tactaaaaat gcaaaaatta 6360
ggctgggcac ggtggctcac acctgtaatc ccagcacttt gggaggccga ggcgcggaga 6420
tcacgaggtc aggagattga gaccatcctg gctaacacag tgaaaccctg tctctactaa 6480
aaatacaaaa aattagctgg atgtggtggc agcacttgta gtcccagcta cttgggtggc 6540
tgaggcagga gaatggcgtg aacccgggag gcagagtttg cagtgagccg agatcccacc 6600
actgcactcc agcttaggcg acagagccag actgtgtctc aaaaacagga aagaaaacaa 6660
aagaaaattt ggactattgc caattacaaa tatttttaga gaagaattca aaacagtaac 6720
tgtggatgat ggaaacaata gttatgataa aagtctgatg aaacttccca gttcacaagg 6780
aaatttaatt acttatgtgc agcattttaa gacagtaatc agaatcatga ctgacagcat 6840
catatcaggg ccagcagact tttataaatt tcatacaatc ttcagaaata ataacttttt 6900
tttttttttt tggatagatt ctacctttgt cacccaggcg ggagtgcagt ggcatgatct 6960
cggctcacta caacctccgc atcctgggtt caagcagttc tcctgtctca gcctcccgag 7020
tagctgagat tacaggcatg tgccaccagg catggctaat ttttgtattt ttagtggaga 7080
cagggtttca ctctattagg ctggtctgga actcccgacc tcaggtgatc cacgtgcctt 7140
tgtctcccaa agtgctggga ttacaggcat gagtgacagt gcccagccat tcgtgacatg 7200
tttatacaaa tataacttta gcaaatattt agcataacta tcaaaattac aaatcatatt 7260
aaatttgtat aaatgtatgc aattttcgga acacgcatat caacaacata cccataaata 7320
taactgagat gagatctaat gtcacctcac ttgacagtgc cctcccatgc agtatcgcca 7380
catttgacaa tgcctgccca tttaatctac caaataaatc gaatcactta atacctctac 7440
aagatgagag atacattctt tagactcccc aagggatgca gctgaaaaaa atcccaaagt 7500
tagttttaag ccaaaaagac ttgatttagg attttgacac tggagaaacc catcaaagat 7560
gtcaagtttg aaaacacttg atcaaaacag aatcacaggt cactattaaa agagtattaa 7620
tttaaccaga gacttccaaa gcaatacaga aacttacatg gatataaaaa ccctaaccct 7680

tttaaaggtc agatttgcta agtgatcaaa aggggtactt gaattgaatc gacacaggaa 7740

gagtgtgtac agggttatga gtgtaggcag gtggttactt tggtcatatc tccatttgcc 7800

acctgattac acatgagaat ggcatcttta ctcaccagaa agccagtatt ataggaggtg 7860

taggaggcat tcttggactt gagacaagaa cattgttgtg tagaaatttc attgactgtg 7920

ttaaaattat tctccatggg ctggagaaca cataacatgg cctttagaat gagacgggca 7980

ttgattggat gcaaggtctc cacacttact agctgtgtga cattggacag agtgcttcat 8040

cattccgaga ctcagttttt aaaggaaaaa caactaacta ccttgcaagc ttgctagcag 8100

gtttaagtgt aataatgtgt gggaatgact gcaccgtgac taacatgcag tgacagctta 8160

attaatgtta acccttatca ttatcatata agaatgtgag ttacataaga gaggagtcct 8220

gtcagttcgt tctctgctgt gtccccaaga ccatgaatca tggctggcat gtagtaggca 8280

tttaataata tatgttcaac aagtatttgg cagtcttgga gggcagaaaa ggaggtgggg 8340

aagattttta aataacattc tttaaaaagt cacattgtcc tacaataccg atttttcttg 8400

catatttagg aaattgaggg tttttttcta aaacatgcgg acatatggga aataggatgc 8460

aacatttgca ctaatgtttc agacacagtt agaggtttcc aagagatttt gcgctgggga 8520

ggctgcttgc tacaagctcc caaagctctg ggaggacata gtattcattc ctccctcagc 8580

agaagcggtg aggcaagaag ctctggggag cacccagcgt tggactttta gcatagtgtg 8640

tcaggtcttc atagtttggg cccagggcac agagaagtca cagctctccg gcatcctgtg 8700

acctttaccc tctttgccaa gggaaaatgt ggcccaccaa agcaagaaac ttgagggcat 8760

gggtcacccc agccctggca tctgcccaga gcccgagaag gaaggaacaa tgatcctcca 8820

gctacctcac ggggctggca caggtgacca ctgccctggc atcacccagc tgtgtccggc 8880

agcctgaacc ccatctgtgg ggatgcgagg aggaaaatac aaaagtcctt aggtgaacac 8940

tgagaaggca gatgcagcag aaacctccag gccagaacta cccagtcttg gacctatggt 9000

ggagatagag catagctggc gatcatgtgt acttacactc taaggtcacc tggttgcact 9060

atggcctcat ctgtggctct gaaaatgaag atttggaagg agatcatcac agctaatgtt 9120

taacaagccc ctcctgtgtg ccaaatcatt cacccctcac cacaaccgaa tgagctaagg 9180

attctcatta tatatagttt atggagaggg aagtgcagac ataaagaggt gaattatctt 9240

acccagatca cacagctgat aagtggtgga ggcagaatag aatctaaaca gtgtggctcc 9300

ggagcccaca tgcattgatt cgacaagtgt ttattgagca cctgccgcgg acaaggcctt 9360

gtgtgattaa atagggttat aattagtaat ataaaaatga gaaatcacta atgcttttta 9420

gacttaacat tttgtttttt tgtaggtttc aggcacagaa ctgtatatcc aataatagtg 9480

aaatggatcc cactaattat gacagaaatg atgatacatt taaatgactt ggatgtttta 9540

taggtatgat ctcgtgaaat cttgagagaa actgaatgac gaatgaaact attgttcctg 9600

tttcacacag aagaaaactg aggttaaaag gggtaaagta attttgcatg gcatgaagta 9660
gaaattcaaa gtacaggaat ttgaacttgg ttctgtcctt ttctgaagcc cttgaccact 9720
atagactcaa acatcacctt gtttttccac tcattcaaca cttttttttt taaattatct 9780
aataggttgg cactcatcat gagcccctgt tctcattctg caaatggtga agctctctat 9840
tgtcctgacc ccacagttcc tgtcccatga ccagggccag ctcaccaagg agctgcagca 9900
gcatgtaaag tcagtgacat gcccatgcga gtacctgagg aaggtgagtg agtgcagaca 9960
gatggggcct ggtgcccttg agcagttccc gggtctcagc tgccacacat ctcatagccg 10020
gtgatgctgg gggaagctta cgcagtcaca gtactggctt cttcctcttt ttctttccat 10080
acaagtggct tagggatggg gtagagtagt tgacttattt ggatgaaaac cactatcttc 10140
tgtcagaaac tcaaaaggaa tcattgctgg catggtaacc taaagaaaaa caaccagaca 10200
agtgcccaac gacacttaaa aaggtgattt attatcttgc caagtttagg ctgggcatgg 10260
tgactcatgc ctctaatccc agcattttgg gaggctgagg ctggtggatc accggaggcc 10320
aggactttga gaccagcctg accaatatgg caaaacctcg tccctactaa aaatacaaaa 10380
attagccggg catggtggtg tgagcctgta gtcccagcta ctcaggaggc tgagacagga 10440
gaattgctta gattcaggag gtgggggttt tagtgggccg agatcacgcc attgcactcc 10500
agactgtgcg acagagcgag actctgtcaa aaaaaaaaaa aaaaaattat cctgcaaaat 10560
ttgaaaagga aattcaaatc aacagcttct aaactacttt ttaacatgac tcataataat 10620
acattctata gtacatatgt atgttctata actttgaata aaagagttaa ccacatcaca 10680
tttattttat aacatgtaat acatattttt tattctcctt catttgtttt gaatgctctg 10740
tgcagtctac aaaaagtcca atagtaataa ttaaattagt cattaagttg aacattatct 10800
tgtcttttaa aatgataatc tcaaaaatga tcttttattt ttgagattta tatagataca 10860
cacacacaca cacacacaca cacacacaca cacacacaca cacacacaca catatttttt 10920
gagacagagt ttcactctgt cccccaggct ggagtgcaat ggcacaatct tggctcactg 10980
caacctctgt ctcccgggtt caagcaattc ctctgcctca gcctctgagt agctgggact 11040
acaggtgtgt gccaccatgc ccagctaatt tttgtattct tagtagagat ggggtttcac 11100
catattggcc aggctcgtgt caactcctga cctcgtgatc tgcccaccgc ggcctcccaa 11160
agtgctggga ctataggtgt gagccgctgc acccggtcca agtaaaatta ttttaacaat 11220
atactatgaa gagaaaaaca ctggctatga aagaatatgc atagttttac cctgtttaaa 11280
aataaagatt gaaagaatac atatgcaaat aagtttactt ttatttttgg taacacttta 11340
ctgcattgtc tgaatattga caatcagtat gcattatgaa gctacctggc taacattgtg 11400
tactcactgt gtgtgccagg ccctgggttc aatgctctac atgcacttat atttcattta 11460

attctctctg caacctgaga tggtatagcc acctcatttt acagagttga aactgaggct 11520

cagagactga aagttaagcc tgaggttgca gtcaataaga ggcagagctg gaactgaaac 11580

ctacctgtgt ctgaccacca gttcgtgttc tgacggcagg ctagtctgca tcacagagtg 11640

tggagtagat ggtgcatgcc tgctaggatg ggctaggtat cactgtaggt aagaaacagc 11700

cccaaactat ggaaatgtac accactgaag gctcttttcc tgcccatgct gcacatcctc 11760

catggctctc ctgtgccctg tgccccacat gccctcatcc tgccacgaga ataaaggagc 11820

agcctccata tgggagctgt cagctgctct aagagatgaa ggagagagtg gcccgtctca 11880

atggctccca actcttttgc ctcgaggtga cacgcttcac ttccacgcac atctcctggg 11940

tcaaagcaaa tcccatgggt acatccactt tcaagtggcc caggagagaa cctgaaatac 12000

tcggtggact ccattaaggc cgtcatatgg tgtcagcctg catgggagac tgtggagggg 12060

cagaggagga gagtggggaa ctgatgggaa atgacaggag gactaagtca ccgcagattt 12120

gctttatctt cagccaggtg gagtttgtcc cagagccgca caaaatcatc accagcatga 12180

ttaaacggag tagacttcag aaaaagcagt ttggtcggat gtaatcagca gtgaactcag 12240

aatcaattga gtgacattga gtcagtaaat ctctgactgc ctcagttacc ccatatgata 12300

gttttgagga tgggaacatt gagagagttg atttggaagg atatcaagag taaaaattcc 12360

aacattttta gttcctttaa gttaaatcca ggcactgtct ttcctgcaag tctcctgttc 12420

ctttcagatt gcacaggtga gagtgctcag attagggctg gaggttgtaa accattgctc 12480

ccacactgac agtgcccccg tgtcgtgcgt gtattctgcg cattttcctg tgctaaacac 12540

tctcccaaaa catcgtgggg cctgattctt cctctttgtt ccaatggccc tgggtgactc 12600

aagtgcccat tcaatgacca ggacacagag gtcttagaga gatgctcctt gaggccccag 12660

gtgcgagcct gtaccctgcc ggagcatgag gcaagggaca gggcatcgtc tgtggggata 12720

gtgggggtag tgggggtagt ggtcagccag atttggtgac tctacttgct caccagacga 12780

tcctacacct gccacctccg atggatccac tgcctctgtg cctgcctgta ctgctgatgc 12840

tccagtggat aactcagcat cccagcctag gcccaatgcc actgaagatg gacctgcccc 12900

ctggggaccc aggagtccta ccactcagct gtccccagga gtgcccagac cctcattctt 12960

atccaggacc taggagccct acccctggcc ttccctcatc agccgtaaat gatgatttac 13020

tgctgttacc atcatcactg ccttcagtga ccaagggcct tccaaggtgc cagctctgga 13080

acgaaggatg cccttgggag gtgatgacac tcaggtacac gggtgctcaa cagattgctt 13140

cctcctatcc tcagacggtc tttgcatgca tgcagccatt ggcactccca ttgtgtggaa 13200

ggaaaccagc ccagggtcac acagctggtc agcagcaaca tagctggtct caaatctaag 13260

gtgcctgacc atgcctccat gagggaccgc ctccaaggga ggttgatcct ggctttgggg 13320

agcctttcct gggctgcacg aataacctcc attgttcgag accccaaact ctgctcacat 13380 cttcctttcc ctatctctgc ttgggctatg atcacggtga ctctagcagc ccttcatgga 13440

cattatagta ctctctgcca ttcacttttg ctctaatctg acttcaaccc ccacttactt 13500

ggtctctcct tttacaacca ccacaaccga aatctagggc tgcttttttt tttttttttt 13560

tttgagacag agtctcattc cattctgtca cccaggctgg agtgcaatgg tacgatctcg 13620

gctcactgca acctccgcct cccgggtcca agggattgtc ctgcctcagc ctcctgagta 13680

gctgggatta caggcgtgtg ccaccatgcc tggctaattt ttgtattttt agtagagacg 13740

gggtttcacc atgttggtca ggctggtctc gaactcctaa cctcgtgatc cgcctgcctc 13800

agcctcccaa agtgctggga ttacaggcgt gagccaccat gcccagccaa atctagggca 13860

ggaacatggc tgcagcatat aaaaagaatt gaattccata cttttgttaa ccctgttttt 13920

tgtttgtttg tagttgttgc tgtttttgag acagagtctc gctctgtcgc ctaggctgga 13980

gtgcagtggt gcaatctcgg ctcactgcag actctgcctc ccgggttcaa actattctcc 14040

tgcctcagcc tcccaagtag gtgggactac aggcgcccac caccacaccc ggctaatttt 14100

tgtattttat tagagacagg gtttcaccat attggccagg ctggtctgga actcctgacc 14160

ttgtgatccg cccacctcgg cctcccaaag tgctgggatt acaggcgtga gccaccacac 14220

ccagcccctg ttttgttttt gttttgcttg cttcttaggg ttgtttttct atttatggta 14280

aaggcattgg ctttccattt gtagcatcaa tagaatattt cctgtttaca ataaccttat 14340

gtcatagtaa atggtaaagg gatttaaagc agtggttttc agctgccaga ggcctgagag 14400

agtttgggca tactctgtgt gatcgggcag aaggcctgtg ggaagtttag cagaggacag 14460

ggccaggaaa ggtgatggac agtgggggtc tgtcctggtc accaggcccc tgggtcctgc 14520

ccacctgctt ggagctcccc acccatcaca catgatgctg ccaagccctc tgggtattgt 14580

gggcaaatac cttaggagag aagctgatga actttgtttc ttgaaatgca cagattcctt 14640

ggacgtccct gagagctcag tcatgaaagt caacttggtt ttctccccct catttgggtt 14700

cagaatttaa agtccacaca cacaggcagt aagatgatat agataaggac gtcatcactc 14760

ggtttcggat gttaaaatgt ctaggtgggt tagcggtgat ttgagatcac acaaccttgt 14820

gccacaaaga ggaattccca ggccagaggg agacatttta ttgccatgtt atgatctcat 14880

cattgagttg aaaggcaatc ttgtttcatt ttggattctt tcttatgttt atgtcttata 14940

agggcacttt gaatttccaa gcaaataata attttgaatt agcttttaat cattgacttc 15000

tagcacagtt atatgatcag aaacatgctg tgtgatttga ttgctctcaa atatattgag 15060

atttgctgga acaaaataag tcaggttaat ttttgtaaat gtaccaggca tgcttaaaat 15120

gaatgtatct acatttgttc ccgagataca ggttgatgga cggatggcta catggatgtg 15180

atggagatgg tttactatcg ggaccttccg caccctgctg atgttttgtt gcttaggata 15240 tgaatggctg agcggaggct gtaaaacctg gcactctgct tgggtatgag gttcttcctg 15300

ccatcctgcc atcatttgtt ttttatgttt tgtcgccata agtgaccttg aggaaccctg 15360

ggagctcagg aaggaaggag cgcccagaag cagggacagg gagctggttg gggaggacca 15420

gaaatcaggt ttgtgaaggt tccagagagg acctgtcttt gggaggagtg tgggagactg 15480

agatggggga ggggtcattg gaatgatgcg ggcgctactt ggcattgtcc attgtgaggc 15540

actgtccatt gtgaggcacc accggggtca tcagggattg gtggagaggg agtataaagc 15600

cccagggttg gtaagggagg gcccagaccg aagaaggttt ggtggatagc agaacctttt 15660

tgtctccctc tgattgctcc taagcctcac gctcccttgc cccgcgtgtc ctgttgcttc 15720

cctgatcttc tccgtgacct gtagctaaac cttccaccag cgcttgagaa cttaatttga 15780

accggatcct ttcccagacc cctttcttct tctcctcctc ctcctccacc tcctccaggt 15840

gcccaacagc ccccttctcc tcctttccct tcccttactt ccccccttcc cctccccttc 15900

ccctccccct cccctccccc tcccctcccc ctcccctccc caactcagat ccggccccgg 15960

tccccgtccc cttccctccc ccctgcccta agccacctcc acctctgtcc tggccgcctc 16020

agggcgccct gaaaggacca ggacatgcgg gtgcggtgga tgctcttttg gctcctcttt 16080

gggctcctac tggaatttat cagccatcag tgcatctctg tgagtagacg ctggacccgt 16140

ggggtttctt cctttttact gggctgtatc acgtggcatg aaattacaca gctcaggcct 16200

gtaatcccag cactttaggg ggccgaggtg ggcagatcac ttgagtccag gagttgaaga 16260

ctagccaggg catcatagcg aaaccccatc tctacaaaaa attccaataa agattagtcg 16320

ggcctggtgg tgcgtacctg ttatcccagt tactggagag gctgaggtgg gaggatcgct 16380

tgggcccagg agctggacgt tgcagtgagc cgagatggcc ccgctgcact cttgttttta 16440

acaaagaaaa tggaccaaaa caaagtgaaa tgtcatttga tttgtgtcat ctggtttgat 16500

gacttttttt tttttttttt ttttttaga cagagtctca ctctgtcgcc caggctggag 16560

tgcagtggca agatctcggc tcactgcaac ctccgcttct ggggttcaag caattgtcct 16620

gcctcagcct cctgagtagc tcagattaca acgcctggct aatttttgta tttttagtag 16680

accaccacgc ctggctaatt tttgtgtttt tagtagagat ggggtttcac catgttcgcc 16740

aggatagtct ccatctcttg acctcgtgat ctgcctgcct cagcctccca gtgctgggat 16800

tacaggcgtg agccaccgcg cctggccaaa atatataacc ttaagtgtaa gtttactaac 16860

tttggaaagt acatacacca gcataaacca accccctttc aagatctaca ttattttatt 16920

tatttattta tttattttga gacagtttct cccttgttgc ccaggctgga gtgcaatggg 16980

gcaatatcag ctcaccgcaa cctctgcttc ccaggttcga gcgattctcc tgcctcagcc 17040

tcccgagtgg ctgggattac agacatgtgg caccactccc agctaatttt gtatttttag 17100

tagagatagg gtttctccat gttggtcagg ctggttttga actcccgacc tcaggtgatc 17160 cgcccgcctc ggcctcccaa agcgttggga ttacaggcgt gaaccaccat gcccagccaa 17220

gatctacact attatgtcac cccagaaagt gaactctcag tcttcccagc cagtctcttt 17280

cttatcatag gttagcttgc ttattctgga atttcgcgta tacagatgca tgccatgcca 17340

taggtactct tttgtgtctg ctttgttctg ctcaacacca tgtttctgaa atcattacca 17400

ttgttgtatg gttctctaac ttcatcattt ccatttcaga ctcagcatat gctgagttca 17460

acctgttgaa gggctatctc tgtttaattc accatcttga aagaaacatt taaaattgag 17520

atgttttcaa gaatatatag ttaaatcctg aggaatcgac gtagaaatgt tatcacaagc 17580

tgtctgaact tactcagggg aagtcttcgt cttcactcac ataagagtct aatggaatta 17640

atatcaacaa tcttagagaa atcccacact attcatgcca ttttcatgat ctccaccttg 17700

ataatttttt tttttttttt tttttttttt tttttttttt tttgagacag agtctcgctc 17760

tgtcacccag gctgaagtgc agtggtgcga tctcggctca ctgcaacctc tgcctcccgg 17820

gttcaagtga ttcttctgcc tcagcctccc aagtagctgg aactataggc atgtgccacc 17880

atgccctgct aatttttgt atttttagta gagacgggtt tcaccgtgtt agctaggatg 17940

gtctcaatct cctgatctcg tggtccaccc acctcggctt cccaaagtgc tgggattgca 18000

ggcgtgagcc accacgccca gcccaccttg ttaattttta agcactaaaa tttgatactt 18060

atttgtgaat gaagtaatct cttcattgta tttttttttt tttttactta tgctgagatt 18120

taaatgacaa agattcatat aatccaagag agaagtatta tttagaggga ttcttttacc 18180

atgtgatata taataaatgc atccaatgtt atacatcaat ttaaaaaaca agtaaataac 18240

tttaaagaaa agataactac tggccaggtg cagtggctca cacctgtatt cccagcactt 18300

tgggaggcca aggcaggtgg atcatgaggt caggagttgg agaccagcct ggccaagatg 18360

gtgaaaccct gtttctacta aaaatacaaa aattagccga gcgtggtggc aggcgcctgt 18420

aatcccagtt actcagtagc tgaggcagga gaatcgcttg aacccgggag gcggaggttg 18480

cagtgagttg agatcatgcc actgcaatct agcctgggtg acagagcaaa actttgtctc 18540

aaaacaaaaa gaaaagaaaa gataagataa ttactttata cttagcttgt cttacccatg 18600

agtgacgggc tgcatgtggc ccaggacagt tttgaatgca gttcaacaca aatttgtaaa 18660

ctttcttaaa acattaggag attttggcca ggtacagtgg ctcatgcgtg taatcccagc 18720

actttgggag gctgaggcgg gcagattacc tgaggtcagg agttcgagac caccctgacc 18780

aacatggcaa aaccccatct ccacaaaaaa tacaaaaatt tgctgagtgc actgtcaggc 18840

acctgtactc ccagctactc aggaggctga ggcaggagaa tcacttgaac ctgagaggca 18900

gaggttgcag tgagccggga gcacaccact gcactccagc ctgggtgaca gagtgagacc 18960

ccatctcaaa aacaacaaac aaaaacaaaa acaaaaaaat ggctgggcac ggtggctcac 19020 acctgtaatc ccagcacttt gggaggccga ggcaggcaga tcgcctgtca ggagttcaag 19080

gccagactgg ccaacatggt gaaacctcat ctctactaaa aatacaaaaa tgagtcgggc 19140

atggtggcag agacctgtaa tctcagctac tcgggaggct gaggcaggag aatggcttga 19200

gcccaggagc tggaggttgc agtgagccga gattgcacca ctgcactcca gcctgggcga 19260

ctgagtggag cggaactctg tctcaaaaaa aaaaaagagg ttttttttag atcatcagct 19320

attgttagtg ttagtgtatg ttatgtgtgg ctcaagacaa ctttgcttct tttaatatag 19380

gcagggaagt caaaagattg gatatccctg ctttatacca agaaagacaa cacccccacat 19440

ttgcaatgcc tgaaaacact accagccatc tgaaaaacat gtgacttcta acttctgttc 19500

ttttttgtag cagtggaatc ccacggtgat atctgaggga tgtggttacc ttttggagga 19560

ggttgacggt ttctaaggat gattctttct gagtgaaata ttgtcagtgt cattgacctt 19620

ttcattattt caactattat tattccaggt tatcaatact ctggctgacc atcatcatcg 19680

tgggactgac tttggtggaa gtccttggtt acatgtcatt attgcatttc cgacaagtta 19740

taaagttgtc attaccctct ggatagttta cctttgggtg agtatactaa ctttctgtag 19800

aggtatactt gtaatcacaa ataagaataa attatataaa acaattcaca tttctggact 19860

tcattatgaa tatgtggttt tacccaaaaa atcagggaaa tgatttatta gtataagaat 19920

tatgaaaaca tctgccattt gcattatgaa aattaaatag gtcggtgttt gtttaataga 19980

atgtcaacag agcttttggt caaaaataag tttttttaac ctttgtgcta tttatcacaa 20040

atggagtatg aggtttcgtc acttaaatag gaaattcttt ctaaactctt ctgctttata 20100

gttctatcgt atgggtggaa ggaaagcttc caatctcctc tctgaagatt cactgcagaa 20160

atgagctgac aacagacagc ttaacaggaa aagaaaaaca tagaacaggc ataaacatgg 20220

gaaccagctg aaaaatgaga ctgctagaag ggccggatgg ctgatgctta aagagcaccc 20280

tcttctgagg ggagagggag atagatggag atgtaggcca tttagagggg cagcaaatga 20340

tttttagggg aaatgaaaga ggccaaggaa caaacaattg gcctgagaca aagttcctgt 20400

gaggtcatag ggacgaggtg acaaactgcc ggaaggtgaa gggcagaact gcactgcgtc 20460

tcatgatgca gagaaagccc cagagactct tagaactgcc ctccaagaga atcaatgaaa 20520

agtgtgtctg ggcagggtaa ttttgaatga catcattcaa agtgcatgtt ccgacttgga 20580

actggagaga gatcagtatg tcaaaagtct gtacttggta agaatttggc tgctaagttg 20640

tgccataatt tgtcttttga gcctttttc ctttgggtaa gttgagctct acattttgtc 20700

ttgccattca tgacagtaaa aatgtggttg tctgggggct gaacctcctt ctgaacaatg 20760

atccaagata aaagtactaa taccacaatg ctttttata ttcaagggaa gaggaagtat 20820

gtttcagttt taccacctag ataattacac gtcatttggc actgcctttc aagatatgta 20880

gaaaacagaa aatatatgag ttatgaagat atctaggcac atttaacatt ctctatgcca 20940 cttagtcctg aacagagaat tttcggtata aattggagga agcttttttc tttttttttt 21000

ttcttttctc accccgaaga cgagtctcct tctgttgccc aggctggagt ataatggtgt 21060

gacctcggct cactgcaacc tccacctcct ggcttcaagt gattcccctg cctcagcctc 21120

tcaagtagct gggattgcag gtgcccacca ccatgcccag ctaatttttg tatttttagt 21180

agagtcgggg ttttaccatg ttggccaggc tagtctcaaa acccgacctc aaatgatcca 21240

ccaacctcag cctcccaaag tgctgggatt acaagcgtga gccaccacgt gagccagggg 21300

aagtttttaa atttaccact ttttaacaat tccacttagg aaagttcagt tgagctgttg 21360

gacttggaca acttcgcacc tctcatcttt gtccttgtca tctagtcatc tataccatta 21420

cctccttagc agggacatca tgggtgccat gaagcattca tgcgtgatgg catttcttgg 21480

cttctcattt cttcatgtgt ttgacatttc ccctagctcc aaactgggcc agctaccttt 21540

cctatgaaat ctagcagtag ctgtgggatt gacgtggttg ctcttttcat ctttttagat 21600

tacccattgc ttctctcgaa atcctagtac atgatttttt ttttatccta tgtgcagaaa 21660

tcaggaaaaa acaaattcta caaagaattt gaaagatatt atttcaggcc aggtgtggtg 21720

gctcatgcct gtaatcccag cactttggga ggctgaagca gatggatcat ttgaggtcag 21780

gagttcaaga ccagatgggc caacatggtg acaccccatc tctactaaaa agacaaaaat 21840

tagccaggca tggtagcagg cacctgtaat cccagctact tgggaggctg aggcacaaga 21900

atcgcttgaa tctgggaggt ggaggttgcc gtgagccaag gtagcgccac tgcacttcag 21960

cacggttgag agtgacactc tgtctcaaga aaaaagtcat ttcaatgacc acctcaggag 22020

attcataggt atctgaccca catctgagat gggatttgca ttgcatttta gctatgatga 22080

gaagaaatat ttaatatctt agaagattaa aagcatactg tgataatatg gaaatcttgg 22140

tgggaattca gtcattagtg agaatgtttt gcgttaagtt caaaccagcc tcaatgaagc 22200

tgatgtgagg gaagggaaag tgaactctga gtagagcagg gacagaagga agatgctcca 22260

gtgcagatca ggaaggagca ggggatgaaa tgttacaaat tctagaactc agagagctga 22320

aggtaattac ttccttttca agttgtgaaa catgttaacc tgtggtaaaa tacttataag 22380

atgataatta ccatctaacc gtgttgaagt gtacagttca gttgtgtgaa gtatattcat 22440

gtcatttttt tttttttttt tttttttgag acgaagtctc actctgtcac caggctggag 22500

tgcagtggtg ggatcttggc tcactgcaac ctctgcctcc tgggttcaag cagttctcct 22560

gcctcagcct cccgagtagc tgggactaca ggcgtgcatc accatgctca gctaattttt 22620

gtatttttag tagagacggg gtttcaccat gttgcccagg atggtctcca tctcttgacc 22680

ttgtgattca cccgcctcgg cctcccaaag tgctgggatt acaggcgtga gctaccgcat 22740

ctggcctatt tttttttttt tttttttttt ttttttgag acagagtttc aattttgttg 22800 cccaggttgg agtgcaatgg cacaatctca gctcaccaca agcttttcct gctgggttca 22860

agtgattctc ctgcctcagc ctcccgacta gctgggatta caggcatgca ccaccatgcc 22920

tggctaattt tgtattttta gcagagacag cgtttctcca tgttggtgag gctggtctca 22980

aactcccgac ctcaggtgat ccgcctgcct cggcctccca aagtgctggg attacaggag 23040

tgagccaccg tgccagcctc atgtcattct tgtgtgtgtg tgtgtgtatg tgacagagtc 23100

tcattctgtc gctcaggctg gagtgcagtg gtgtgatctc ggctcactgc aacctccgcc 23160

tcccagcttc aaacggttct ctgcctcagc ctcccgagta gcttggatta caggcgcccg 23220

ctgccatgcc cggctaattt ttgtattttt agtagagacg gggtttcacc atcttggcca 23280

ggctggtctt gaactcctga ccccgtgatc cacctgcctc ggcctcccga agtactggga 23340

ttatacgcat gagccaccgt gcccagccgt cattcttata ttattatttc ctaggtgtct 23400

ctcctgaaga ctatcttctg gtctcgaaat ggacatgatg gatccacgga tgtacagcag 23460

agagcctgga ggtccaaccg ccgtagacag gaaggtatgg ctctgttgga atccgcatag 23520

tgtggaaatg agtttgccct ggaaagggaa agaacagctt cttgccctca ggtttctcac 23580

cttctcctct cctcactctc accaagggct gaggtccatt tgtatgcaca caaagaaaag 23640

agtttcttcc tttcgaggaa ataaaattgt cctgaaagac gtcattactc tacggagaca 23700

tgtggaaaca aaagttagag ctaaaatccg taagaggaag gtgacaacga aaatcaacca 23760

tcatgacaaa atcaatggaa agaggaagac cgccagaaaa cagtaagatg tgccttgaca 23820

caaatactgt tgtatgaacc atgtgccaat caaagtagac aactgtaaag tccttgagaa 23880

tattttctac aatatttgtg gcaaattcag tgggttcaaa attgagtttg tcctttctgc 23940

ttcattagtt taagctgtat aattcctttc ccttcctaca ttcttgtttt catttttttcg 24000

gaggaagagg agttgctagt actggcattg gttttccttt ctcttttttt tttttttttt 24060

ttcctgagat ggagctttgc tgttgttgcc caggctgtag tgcaatggca caatctcagc 24120

tcactgcctt ttgggttcaa gcaattctcc tgcctcagcc tcccaagtag ctgggattac 24180

aggtgcccac caccacgccc agctaatttt tgtattttta ctagagatgg ggtttcacca 24240

tgttgtccag gctggtctcg aacttctgac ctcaggtaat ccacctgcct cagcctccca 24300

aagtgctggg attagaggcg tgagccacca cagccagcct tttttttttt tttttttttt 24360

tttaattttg cgatagagtc tcgctctgtc gcccaggctg gagtgctatg gtgcaatctt 24420

ggctcactgc aacctctgcc tcccagtttg aagcaattct gcctcagctt cccgagtagc 24480

ttggattaca ggtgtgtgcc accacatttg accaattttt tttttttttt ttttttttga 24540

gacagagtct cactctgtca cccaggctag agtgcagtgg catgatcttg gctcactgca 24600

acctccacct cccaggttca agcgattctt atccctcagc ctcttgagta gctgggacta 24660

caggcatatg ccaccatgcc cggataattt ttgtattttt agtagaggcg gggtttcacc 24720 atattggcca agctggtcta gaactcctga catgatccgc acacctcggc ctcccaatgt 24780

gctgggatta caggcgtgag ccaccgtgcc cggcccaatt tttgtatttt tagtagagac 24840

aggggttcac catgttggcc aggctagtct tgaactcctg acctcaggtg atctgcctac 24900

ctcagcctcc cagtgtgagc caccgcaccc agcctggatt gttgaattca atgcttgggt 24960

cacctccaga ttcattttca cagtctttca tgttttggtc atatgacatt gtattttgct 25020

gccatatgac tgatcttttt ttgttaaatg tgagatactt gttaaaaaat gtttagcaat 25080

gaattgaggc ctagtagcat gttatcttgc tgcagaagag atgggagtct acttctgggg 25140

gatggtcagg ggtcctccat acaagctgca attgaagtcg tcggtgcagg ctcagtccct 25200

acaaaggcca gggtatttcc tgtccacctt tattctgatg catgactctt ctgggtctca 25260

accagagcca gcggacttca gtatgggtcg ctttcattgg cagaccctca atccacttgt 25320

tttccatcta atcccacgca tgtgtgcaaa agctgctgtg cttctttgca tctcagtagt 25380

tccttctgga attcagcaat gaaactcagg gaaatgggtt ccaaatgcga ggctgacttt 25440

cgtcctgggt ttccttcttc tccatcttca cctcatgtct gtttactgcc atgttagcaa 25500

tttgatgtat tcaatcatgg gttttatatt ctgtttggtg tcccccattg ttctcattgg 25560

agatcagaag cttcagatgc acttatgtca actcaagagt agaatgcttc cttagcttcc 25620

ctccagagtc aggttttgtg tttctagttc ccaagtgcac agcaggagta gtgatgtcct 25680

cactggcttc tcatttgcat taagctgtga gcttctttag cgtggggaca ggaccctgct 25740

cccattgcat tctcagcacc acaccacaca ctccttgttt gaggccactc cagacagcat 25800

gtgctgaagg atgccttgtg gtcagaaaca agttcattaa ctttctcttt gaagtgtttt 25860

cgcccctgtt tcctagcgtt ctgggaattt tacacatcct tcctataaag ccaagtatca 25920

ggtgagatcc ttaggatcag gaccatgaat caagtggtgt gagggcaaca cagcaaactt 25980

acccttttta ggccgtttcc tttttctgcc ctcaatctct gtgaactgaa ccttgttaaa 26040

gtcagtcaac accagggtgg atggtttgcc gttgtcacct attttcagga cataacaccc 26100

tgacttagga gccattccga tcatttctaa ttcaatagat gcgcccagca ttcagattgc 26160

cttttctctc aaccaggatc tttaaagttg atgacaagag ttccagtcct gaatcatggc 26220

aaagtgcagt agtgaactgc ggggttattc tggaaggatc tctctatggc tgatggtctc 26280

agttccggca tcagcctctg actgagaatc aggtctcaca caggaggagt cagatgagga 26340

gcaatcctct gcttccgatg gagttagttg tgatgaattg gtgaggtctg gtttttcaca 26400

ctgaactaaa atgagctttc gctgtgtcaa gcacaagact gaccccagag acacacatag 26460

tgcacctcat agaagctttt aatagtcttt atatttacta aagaatagga ctaactatgg 26520

aactatgaag atgagctgga aatgacaggt gacttgccag caggccagag tgtgactttt 26580 ttttgtccct caatgggagg tgtcaattct cccttcggtt gtgagaatca gttggttcat 26640

ttgtgggaag gttgcagggg ggaatctttg aatcacagcc ttcagatgcc agaagggcag 26700

agggaatccc acacgggctg gtggatcatg tgtgtgcatt tctctccctt ctaatctgag 26760

gaaactaagc gtgaaagaat gtgagcatgc agaaaaggag aggcaggtat cagaggcaga 26820

ggaaaatggg aaattggata tgaaagaaat acacacctac aagtgagttc agaaactgta 26880

ccccaccctc ttgggaaacg cccattggag tgttgttttt aacctttgta cagtatttag 26940

acccagtaaa tgcagaaata gaaacaaacg gtcagaagac atatcgtgag agagagcgag 27000

agagagttca caaaacagaa aacaaagtac cttaatattt accagtgacc aaaagatgtg 27060

aagtagcaaa acgtctcctg accccattgc cagctagact gtgtggaaac tcggttcata 27120

ccagccattc taggggtggg gtgagttgtt gtcatcctta ggaaagtgtg ttgttgtagg 27180

atcaaccaca tccttcaaaa ggactatgcc tgtttataag cccagctgtt tctgccctgt 27240

gaaacacggt aaggatatta atacaaagag aatacagctt tatgataaaa gatgctcaat 27300

gaaggatgaa ttagggatgt actgagaatg gggaaggaaa ctatcatctc agaagtcagc 27360

aggcagtaag caagaggagg aatcaataca gcaacagttt ggatcagact gtacagtttt 27420

tttgtttttg tttttgtttt tctgagatgg agtctcgctg tgtcacccag gctggagtgc 27480

aatgacgtga tcttggctca ctgcaacctc cgcctcccag gttcaagtga ttcccctgcc 27540

tcagcctccc gagtagctgg gattacaggt gcctgccacc acgcctggct aatttttgt 27600

atttttagta gagaaggggt ttcaccatat tagccacaat ggtctcaatc tcctgacctc 27660

gtgatccatc cgccccgccc tcccagagtg ctgggattac aggcgtcagc caccgtgacc 27720

ggctcagact gtactcttct agccatctga aatacgtttt ctaggtagag atagattgtg 27780

taagggtaca gttgtgagga taacagaaac atggcagatt atttaaaatc atcctgaaag 27840

tggtgcttta tctgatgaaa gtgattgtaa tccataggaa aatgtttcaa cgtgcgcaag 27900

agttgcggcg gcgagcagag gactaccaca aatgcaaagt aaggagcttc ctccctgcag 27960

ttgcaggata gttcagtgct gatgcagatg atgccacggc ccttagactc tctcaacatt 28020

caatttctca tgtgttggct ttttcagatc ccccttctg caagaaaggc tctttgcaac 28080

tgggtaagtt tgcttgtttt ccttgctttt ggacatagtc tgccaggtca ggacatggat 28140

acatttttct ccctacagct ctgtgctcaa gccctgcaga gggagatggc agagagaaag 28200

gctgcctaca agcatcacag tcccatccct gttggtaacc gtgttgcgca aaaacacctt 28260

catccccacc cagtggggcc cctgatctaa tattctaagt gtcagaggtt ccgtatttgt 28320

aatagcagat gggccctgac tgtaaactag tgaagagtga atgtaactta ttacccacag 28380

ggacaattcc aaatgaaggc cttaaatgat gctcagctaa gctggttctt gtgtggcctc 28440

tgtaccttca aaagctgccg agtcctatga ttacacgtga tgggacttgt acacttgaag 28500

tgaaacacag ttttaaaact tgctttgttt agaattccca cctcattttt ccatggacaa 28560
aagtattctt tatgtcctag tgcacttaca atttggtatt acctgggagt gaaaagaaat 28620
attacagcca tgcctaagtg acttcttgag gtgagattgt tctgtcagaa aaccctctcc 28680
cagttcccct gcagctcttc aggaatccac atctctccag agctctttgt tctcatgggt 28740
ggcacctcca gagtgaagaa gatcctttgt caagaaggga aacagagggg aaatgagagg 28800
gtcctgcagg cagagctgga atcaacttcc actctgcctc ttgcaagctg tgtgaccctg 28860
ggcacaattt ctccttcctc tggaaacctc tgttttctta gatttggagc agggtggtca 28920
cactgacctt gcagagttct gagaatcaga gacagaacat aaaaggcctg gaaaacattc 28980
tccaaaaaga agctgcaaca tgtgtggaca gtgggctttt catgcctctc ttactgtctc 29040
ttactgtctg ttgacctggt gcaagaaaca tgctctggtg atggctgtga gggaggaatg 29100
aggatagaca tagacactcc tgtgtctcaa acatgcttct ttattactct gttatgactc 29160
tgtcttccct ggggcaggac cccagcctgc ctacatttgc agacagacac agtggcatgt 29220
ggagacaaca gtgtgtccca atgactttcc tttaccctcc agctgtcggc agtactcagt 29280
ggaagggtga tattatgaca ctgatacttc tattttgaaa cctggaggat ggaaaggtgc 29340
aaaaatctat caccagcaac agaaggtgca gactgtgttg gtggcggtaa ttttgtccat 29400
caaatgaata tgtgtgaaaa cattccctcc tttggcccta caggtcagaa tggcggcagc 29460
ggagcatcgt cattcttcag gattgcccta ctggccctac ctcacagctg aaactttaaa 29520
aaacaggatg ggccaccagc cacctcctcc aactcaacaa cattctataa ctgataactc 29580
cctgagcctc aagacacctc ccgagtgtct gctcactccc cttccaccct cagcggatga 29640
taatctcaag acacctcccg agtgtgtgct cactcccctt ccaccctcag cggatgataa 29700
tctcaagaca cctcccgagt gtgtgctcac tccccttcca ccctcagcgg atgataatct 29760
caagacacct cctgagtgtc tcctcactcc ccttccaccc tcagcggatg ataaactcaa 29820
gacacctccc gagtgtctgc tcactcccct tccaccctca gctctaccct cagctccacc 29880
ctcagcggat gataatctca agacacgtgc cgagtgtctg ctccatcccc ttccaccctc 29940
agcggatgat aatctcaaga caccttccga gcgtcagctc actccccttc caccctcagc 30000
tccaccctca gcagatgata atatcaagac acctgccgag cgtctgcggg ggccgcttcc 30060
accctcagcg gatgataatc tcaagacacc ttccgagcgt cagctcactc cccttccacc 30120
ctcagctcca ccctcagcag atgataatat caagacacct gccgagcgtc tgcgggggcc 30180
gcttccaccc tcagcggatg ataatctcaa gacaccttcc gagcgtcagc tcactcccct 30240
tccaccctca gctccaccct cagcagatga taacatcaag acacctgcct tccaccctca 30300
gcggatgata atctcaagac accttccgag cgtcagctca ctccccttcc accctcagct 30360

ccaccctcag cagatgataa tatcaagata cctgctgagc gtctgcggat tccgcttcca 30420

ccatcagccg atgataatct caagacacct tccgagcgtc agctcactcc ccttccaccc 30480

tcagctccac cctcagcaga tgataatatc aagacacctg ccgagcgtct gcgggggccg 30540

cttccaccct cagcggatga taatctcaag acaccttccg agcgtcagct cactcccctt 30600

ccaccctcag ctccaccctc agcagatgat aatatcaaga cacctgccga gcgtctgcgg 30660

gggccgcttc caccctcagc ggatgataat ctcaagacac cttccgagcg tcagctcact 30720

ccccttccac cctcagctcc accctcagca gatgataata tcaagacacc tgccgagcgt 30780

ctgcgggggc cgcttccacc ctcagcggat gataatctca agacaccttc cgagcgtcag 30840

ctcactgccc ttccaccctc agctccaccc tcagcagatg ataatatcaa gacacctgcc 30900

gagcgtctgc gggggccgct tccaccctca gccgatgata atctcaagac acctccctta 30960

gctactcagg aggctgaggc agaaaaacca cgcaaaccca agaggcagag ggcggctgag 31020

atggaaccac ctcccgaacc caagaggcgg agggtcggtg acgtggaacc gtcacgcaaa 31080

cccaagaggc ggagggccgc tgacgtggaa ccatcatcac ccgaacccaa gaggcggagg 31140

gtcggtgatg tggaaccgtc acgcaaaccc aagaggcgga gggccgctga cgtggaacca 31200

tcatcacccg aacccaagag gcggagggtc ggtgacgtgg aaccgtcacg caaacccaag 31260

aggcggaggg ccgctgacgt ggaaccatca ttacccgaac ccaagaggcg gaggttgagc 31320

tgagaagagg ccagtgcact caagcctgag caataagaat aaaaccgagt agaacaaaat 31380

aaaaaattca aaaaacaaaa caaaacccac actccaaaaa ctaacaaaga ataaataaat 31440

aatataaaaa taaaataaat actgcagtcc ttatgttatt gctttgtttc gatatctggt 31500

atgattgcct gagggacctg aggtttttaa tcataggggt ttttttttaa tctttagaag 31560

tggttggtta tgtaaaatat tattattttt ttttttgaga ctggattttg ctgtgtcacc 31620

caggctggag tgcagtggct cgatcacagc tcactgcagc ctcaacctcc tgggcttcaa 31680

gcaatcctcc tgcctcagcc tcccaagtag ctgggatcac agatatgtgc caccacgcct 31740

ggccaatgtt aaaaaatcct ttaacttttt tgtagagatg cactcctgga ctcaagcgat 31800

cctcctactt gtcccgacca ccagcccctt tctgataaac aaacatttac actgtttatt 31860

atctgatgcc atttctatct tcttccttgt cgtccagaca tcgaataatt aggtttcttc 31920

agggttttct ttttcaagtg ctcagtgtta aagatcactc acattagggc cagacaccat 31980

ggctcatgcc tgtaatccca gcactttggg aggccgaggc gggcagagca cttgaggtgg 32040

ggagtttgag accagcccgg ccaacttggg gaaaccccac ctctactgaa aaaaatacaa 32100

aaattagctg ggcgtgatgg tgcatgtctg tagtcctagc cacttgggag gctgaggcat 32160

gagaatcgct tgaacccagg aggcagaggt tgtagtgagc caagatcacg tcagcacact 32220

ctagcctggg tgacagagtg agactctgac tcaaaaaata aataaaataa atatcactta 32280 catgagatat acccaagggg tggtctacag agacttggaa gcagtggtta ttgcaacagg 32340

ggcacggaag tcatctggct atgccagggt gcccagggga tactcggggt gggtggcatg 32400

gtgctgctgg ggactcatcg cacaggacgc tctgattgac gcactgccag gagtagcgct 32460

ctgtcttggg gctgcagccg gcctcctcag ctcgagtgta acatcagtcg tggccatggc 32520

agcacctgcg gatgtcacat gggcaggaca gcaggtgggt gaagctctct cctggccctc 32580

ctctcttgcc aggaccatgg gtgactgaag acccccaggg aggcacagca tcctcttatc 32640

taagattttt ttttttttaa gagacagggt ctttaaaaaa gtcctgcagt ctgcagtcgc 32700

ccaggctgga ctgcagaggc acaatcatag ctcacggcag ccttgaactc ctgggctcag 32760

cgatcctccc acttcagtgt cccaagtagc tgagactaca ggcacacgcc agcatgtccg 32820

gctggttttt tagtttgtat ttcctttgag acagcatatc tctctgtcgc tcaggctggg 32880

gtgcaatggc tcaatcagct cactttagcc ttgaactccc gggctcaagt gatactgcca 32940

cctcaacttc ccaagtatgc tactacagga acacaaactc ctttttttaaa ttttttgtgg 33000

atatggggtc tcactatgtt gcctaggctg gtcttgaact cccaggctca agcagtccta 33060

cctcagcctc cccaaatgct gggattacag gtgggagcta ctgtacgcct ggccttatct 33120

aagctgtttc cctgaaaatg cccgtcttgg gtaatgattc cattggcccc accatgccct 33180

gtcctgcctt cctggctgtg cccaagcttg gtccctgcct gcctgcctca ctctctgggt 33240

ctcgagctcc tgtgacacat gactcctctc tcttcctgga gtgatccaag ccctgccact 33300

tcctgacttt gcccacactg taccctctgc ctggggcaac ttcatgtctg cccattgtcc 33360

cttaggcctc agcccaggca caagcccctg cctccggagg tcatccaggc ctcaccaggc 33420

tacaccctct cgtaaaattg gattccctcc cttcagggca ggtttataat gaaatcctcc 33480

tcagaggcca ggtgcggtga cacccatctg taatcccagc actttgggag gctgaggtgg 33540

gaggatcact tgaggccagg gggtcgagac cagcctgggc aacataagag agactcttgt 33600

ctctcttgtc tctataacaa atttaaaaat tagctcacca ggccaggctc agtggctcat 33660

gcctgtaatc ccaacacttt gagaggccga ggcaggtgga tcacgaggtc aggagttcga 33720

gagcagcctg accaacacgg cgaaaccctg tctctactaa acatacaaaa ttagccaggc 33780

atggtggcac gcacctgtaa tcccagctac tcgggaggct gaggtaggag aattgcttga 33840

accccggagg tggaggttgc ggtgagccaa gatcacgcca ttgcagtcca gcctgagcaa 33900

cagagcaaga ctctgtctcg agagaataaa aacacacaaa aaattaactc gccaggatgg 33960

cacatgccta tagtcctaac tacttgggag gctgaggtgg gaggattccc ttcagcccag 34020

gagtttgagg ctgcagtgag ccactgtgat tgtgccactg cactctaacc tgggcaaaag 34080

cgagacccca ggctagagtg catgattttg ggtcactgca acctccacct cccaggttca 34140 agtgattctc ctgcctcagc ctcttgagta gctgggacta caggcatgtg ccaccacgtc 34200

tgggtaattt ttgtattttt agtagagaca gggtttagta gagaccatgg tgaaacccca 34260

tctctattaa acaaatctct actaacccca tctctacaaa aaacagctgg gcgtggtagt 34320

gcacacctgt aattccagct acttgggagg ctgaggcacg agaatcattt gcatcttgga 34380

ggcagagttt gcagtgagct gagatcgcac cactgcactc cagccgggat gacagagcaa 34440

gaccctgtct caaaaaaaaa aaaagggccg ggcgcggtgg ctcacgcctg taatcccagc 34500

actttgggag gccgaggcgg gcggatcacg aggtcaggag atcgagacca tcctggctaa 34560

cacggtgaaa ccccgtctct actaaaaata caaaaaatta gccgggcgtg gtggcgggcg 34620

cctgtagtcc cagctactcg ggaggctgag gcaggagaat ggcgtgaacc tgggaggcgg 34680

agcttgcagt gagccgagat cgcgccactg cactccagcc tgggcgacag agcgagactc 34740

cgtctcaaaa aaaaaaaaaa aaaaaaaaaa agaaaaaaga acaaacaaca gcaacaacaa 34800

caaaaaaacc tctgtgtcaa tcacagcctt cgagctaggg gagaggcggc cgaattctgc 34860

cctccgctaa cgagctatag ctttgtggaa atgggcgagt ggcgtgccct tgtgagcctc 34920

agggccgcat ctgtaaaatg ggcataactg tcatgcctgt ctttaagaac agccttgggg 34980

gtaaatgagt ggaactaatg gaaagatctc agcccacaac cttccacaga acaggcgctt 35040

ctcacacagt aagtagcagg agtgcagagg ctgcaggcat gaatccagtc agactgcaga 35100

ctgcctgggt tcaagtccca gctcccacgt cttggtaact aagtggcctc agacaagtta 35160

cttagtattt cttcttcttc tttttttttt tttttcagac ggagttttgc tctgtcaccc 35220

aggctggagt gcagtggtgt gatctcggct cactgcaacc tccgcctccc gggttcaagc 35280

aattctcctg cctcagcttc ctgagtagct ggaattacag gcacctgcca ccacatccag 35340

ctaatttttg tatttttagt agagacaggg tttcaccata ttggccagga tggtctcgaa 35400

ctcctgacct cgtgatctgc ctgcctcagc ctcccaaagt actgggatta taggcgtgag 35460

ccaccgcacc tggacacatt acttaatatt tctgtgcctt ggtttcttca tctgtgaaat 35520

gggattgttg tgagaatgca aagggattcc cagggcagtt cctagtgcat agtctggctg 35580

cctttgtgtg tgtgtgtgtg tgtgtgcatg tgtgtgtgtg tttaatatag agacagggtc 35640

tcactatgtt gcctaggctg gtttcaaact cctgggctcc agtgatcctc ctgcttccac 35700

ccaaagtggt gggataacag gtgtgagtca ccacacctgg tcactttata ttattttttt 35760

cttttgagac agggtctcgc actgttgccg aggttggaat gcagtggtgc aatctcaact 35820

cactgcaaac tccgcctccc gggttcaagt gtttctcctg catcagcctc ttgagtagct 35880

ggtactatag tcaccgggct ccttgcctgg ctaagttttg tatttttagt agagatgcgg 35940

tttagtgatt ctcctgcatc agcctcttga gtagctggta ctataatcac ttagctcctt 36000

gcccagctaa tttttgtatt gttagtagag atgcggtttc cttttttttt tttttttttg 36060 agatggagtt tcgctcttgt tgcctaggct ggagtgcagt ggtgctatct cggttcacca 36120

cagcctccgc ctcctgggtt caagcgattc tcctcctcag cctcccgagt agctgggatt 36180

acaggcatgt gccaccgcac ctggctaatt ttgtattttt agtagagacg gggtttcccc 36240

atgttggtca ggctagtctc gaactcctga tgtcaggtga tctgccttcc ttggcctccc 36300

aaagtgctgg gattacaggc atgagccagc atgccaggct ggcccttttt tttttttttt 36360

taatcactta acatatatct tggagaactt tccattttgg gagttaaaga gatttgtttg 36420

tttgtttgtt ttgagacagg gtctccctct gtggcccagg ctggagctgg accttggctc 36480

agtccaactt ccaccccccg ggctcaagca atcctcccac ttcagccttc caagtagctg 36540

ggtctatggg cacatgccac cacatcccgc tactttttat agtttttgta gagataggat 36600

tttaccatgt tgcccaggct ggtcttgaac tcctgagctc aagtgatcca cctgcttcag 36660

cctcccaaat tactgggatt acaggcatga gccatcttgc ccagcctgtt ttttatttaa 36720

tatctactaa gtgccaacta ccatagagga cataaagatg attcagtctc tgcagaagtc 36780

attttctttc tctttcctgt tgtacagcac aaaattaatg gactaaatag tctgtcacta 36840

gataaagaag ccctaagtaa tcaggcactt gctgcagttt ttacaaagtt taaaaagcca 36900

tatgaaacac agtatactcc aagtaataag aggcaaaata tgtgaagtgt tactgctggg 36960

gaatttacgg actattcttt tctacaatat atctgctgtg gtctgaatgt gttccccaag 37020

attcatatgt taaaacttaa ccaccagtct ggtagtatta agaggtgggg catcattaag 37080

tcatgagggc atgggattag taacctcata aaaaggttgg cgggaacgag ctaggccctt 37140

tcattgccct tacatccctc tgtcacttga agacacagca ctggtccccc cgggaggaca 37200

tagcagcaag gtgccgtatt ggaaatggcc accatgccct caccagatac caacctgctg 37260

gtaacttaat cttggccttc ctagcctcca gaactgtgag aaagaaattt ctgggttttg 37320

tttgtttgta tgagacaggg tctctgtcac ctaggctgga gtgcagtggc acgatctcgg 37380

ctcactgcaa cctctgcctc ctgggttcaa gtgattctcc cacctcagcc tcccgagtag 37440

ctgggattac aggtatgcac caccacaccc agcttttttt ttttttcata gcgttgtaca 37500

gatagggttt cgtcatgttg cccaggctga tctcgaactc ctaaggtcac acgatccacc 37560

tgccctggcc tcccagcatg ctgggattaa aggcgtgagc cactgtgcct ggccaaaatt 37620

tccgtttttt ataaataacc cagtctctgg tactttctta tagccgcacg aacagataaa 37680

gactgtacct atctgttgac tgggcgcagt ggctcacgtc tgtaatccca gcacttcggg 37740

aggcttagac aggtggatca cgaggtgagg agatcgagac catcctggct aatatggtga 37800

aaccccgtct ctactaaaaa tacaaaaaat ttagctgggc tcggtggcgg gcgcctgtag 37860

tcccagctac tcaggaggct gaggcacgag aatggcatga acccgggagg cggagcttgc 37920 agtgagccga gatcatgcca ctgcagtccg tcctgggcga aagagcgaga ctccgtctca 37980

aaaaacaaac aaacaaacaa acaaacaaaa aagaccgtgc ctttctgtct gtctccctca 38040

taggtcagtt tccacctgat tgtaaccaca tcaagtatcc tagtatattt catatttaca 38100

gaaaaataaa tgggcaaata ctgtcattta cagagaacct gccctgtcct gtacactgtg 38160

acatattttg tggtttgtga ttatgtgctc tgatccttac gatagctcta aaatagctca 38220

aaaagttatt cccattttgt acataagaaa attgaagttc tggaaacata aggcaattgc 38280

ccaaagtaat agagtaaatg acaaagctag gatttctttc tttcttccat taattaatta 38340

attatttgag atagggtccc tgttgtggga tgcagtgggg agatcatagc tcactgcagt 38400

ctcgacctcc tgggctcaac tgatcctcac gtctcagcct cctgagtagc cgggactaca 38460

ggtgcacacc aacactatgg ctaattcttg tatttttctg tagaggtctg cctaggctag 38520

gcagagattc tggctaggct ggtctcaaac tcctgggctc aagcaattct cctgcctcac 38580

cctcccatat agctgggacc gcaggtgtgt gccaccactc ccagctgact tgtttatacc 38640

agtggttctt aattggaaga gttttgtacc ccggggacat ttggcaacat ctggagatgt 38700

ttttgaatgt tacagtgggg aggaggggaa tgctactgtc acctactgcg tagaggcaag 38760

ggatgctgct gaacatccca caatgcacag aacagcctcc tccccaccca acagagaagt 38820

atctggttgg cccaaaatct caatgatgcc aagtctaaga aactgctttg tatttctctg 38880

agatactggg atgaggaacg ctctaaatta gttgtctttg aggataatgt atgcatacat 38940

acacatatat gtgtatacat gtatataact aaagatatat agttgagatc ttgcatttgt 39000

gttaatgaat gtagttttta taaagataat tgactcatac attaattgtt gactctttgg 39060

aaaggaaaaa ggacaaatga gatacatgct attggttttg ttatttagaa attgttcgtg 39120

tggatcctcc ctaccattca gccatttgca aaccattagt tgagcataac ttaatccctt 39180

ctagttcaca gcaaagattc atcaaaagcc atttggtata ctgtcacctc atctctctac 39240

cagtgttgat tagattacag gagggttaca ggagggcacc ttatccaaat tggccaatga 39300

taaattcttt tttggtacct tactatatat ttgtatcctc agaacaaatt ctcctttctg 39360

cttaagctgg cttgacttca tttctttcac tagcaagcta agaggctttg gccaacagaa 39420

ctactctagg ggtagctgta aaatttatct ctaaggaaag gctgctcttt tgaaaacaca 39480

gtttaatggg tcttctgggt atatgacgat agtaaagcaa aattttgttc tggaatgaag 39540

aagctgatta cattgttgta ttacaaacaa tatattactc agtgaagtga aaggccacac 39600

tgtgggcagg gtggtgactt tgcagatagc atgttaatcc ttgcatgctt tttgttgttt 39660

ctctttttgg agggggaggg tcttaatttt ctttggccct cccccacccc ctttattttc 39720

tctggggaag gcaagactgg ataaggaggc ttttatcctg ccaagatgag ttggcccaca 39780

ggacaatttg actgaataga ggctccacaa agaacggaca tggccaaaga attacaacag 39840 gaaattatgc atttagtttt aaggcttcct ttttctcctt tttttttttt tttttttgag 39900

accgagtctt gctctgttgc ccaggctgga gtgcagtgtt gcgatctcag ctcactgcaa 39960

cctctgcctc ccgggtccaa gcgattctcc agcctcagcc tcccgagtag ctgggattac 40020

aggtacctgc caccacacct ggctaattcg tatttttggt agagatgggg tttcaccatg 40080

ttggcccggc tggtcttgaa ctcctggtct caagtgatcc acccgcctca gcctcccaaa 40140

gtacagggat tacaggcatg agccacaggg cctggccagt atttttcttg tataatgctc 40200

agagtttcct ttaaatcttt ttaaaaactt ggcagcctta gtcttagctg tgctttggga 40260

aaataaggag ctgcactcaa gcttaagaag atattgctgg aagtgtacat tggcacagcc 40320

ttcgaagctt cttggtgccc tcctgtaacc caatcgacac tggctagaga gatgaggtca 40380

tattgcatca aatggcttcc agtgattctt tgctgtggac aagaaagttg ctctttttct 40440

ttgagaaagt tgctcttttt ctttgagagg gagttttgat gcctaggctg gagtgcaatg 40500

gcagtgatgt cggctcactg caacctctgc ctcccaggtt caagtgattc tcctgcctca 40560

gcctcccgag tagctgggat tacaggaatg tgccaccatg ccaggctaat tttgtatttt 40620

tagtagagac agggtttttc catgttggtc aggctggtct cgaactcccg acctcaggtg 40680

atccacccat gttggcctcc aaaagtgctg ggattacagg cgtgagccac cacgcccagc 40740

caagaaggtc actctttatg ctcaacaaat gtattgcaaa tggctgaaag gtagggagga 40800

tccagatgaa caatttctaa ataggaaaac taatagccag acacgatggc tcacacctgt 40860

aatcccagca ttttaggagg tcgaggcagg cagatcacct gatgtcaggc atttgagacc 40920

agcctgggca acacggagaa accccatgtc tactaaaaat acaaaaatta gccaggtgtg 40980

gtggcgggca tgtgtagtcc cagctactca ggaggctgag gcaggagaat cacttaaacc 41040

cgggaggcag aggttgcagt gagctgagct tgtgccactg tactccagcc tgggcaacag 41100

agagagactc tgtctcaaaa aaaaaaaaaa aaaatcgtag atatccaatt tgttcctttc 41160

tcctttccta aaagcttttt gtgccccctc caacccaaca gttccaccct agaaaaacac 41220

ttgtacacat acatcaggaa acgatacaag aatattcagg ctggtcgcgg tggcttgcct 41280

ataatctcag cattttggga ggccgaggtg ggaggacggc gagagatcat gccactgcgc 41340

tccagcctgg gcaacagagg gagactccgt ctcaaaaaaa aaaaaaaaaa aaagcataat 41400

gttgaaaaca aacagcagta taatacagca tggagataac tttataaaat gcaaagacaa 41460

ataaaagtaa atatagtgca ttgtttagga gtataaatat ggtaattatg caaagaaaac 41520

taaaatgact tttaaaaatc atgatggtga ctatttttgg gaggaaagca ggtagaaggt 41580

gggatctgag agcagcacac agtaagtgtc aattgtccta gttaatgttc ggcttcttaa 41640

attgggtgat atgttcacag gtgttcatta tgttcttttt tttttttttt tttttgagac 41700 ggagtctgac tctgttgccc aggctggggt gcagtggtgc gatctcggct cactgcaagc 41760

tccacctctt gggttcatgc cattctcctg cctcagcttc ccaagtagct gggactacag 41820

gtgcccgcca ccacacccgg ctaatttttt tttttttgt atttttagta gagacggggt 41880

ttcaccgtgt tagccaggat ggtctcgatc tcctgacctc gtgatccacc tgcctcggcc 41940

tcccaaagtg ctgggattac aggcgggagc cgctgcgccc agcccattat gttcttataa 42000

cttacattaa catattattt tatatatgtc aagctttttt tcccttttttt ttttttaaga 42060

gatgagattt tgctttgtca cccaggccag agagtgtcgt ggtgcagtca tagctcacag 42120

cagcctccaa ctcctgggtt caagtgattc tcccacctct gtctgccaaa taattggaac 42180

tacaggcatg tgccaccatg cctgggtaag tttttttttt taatttttta tttattatta 42240

tactttaaga tttagggtac atgtgcacaa tgtgcaggtt agttacatat gtatacatgt 42300

gccatgctgg tgcgctgcac ccactaactc gtcatctagc attaggtata tctccgaatg 42360

ctatccctcc cccctccccc caccccacaa cagtccccag agtgtgatgt tccccttcct 42420

gtgtccatgt gttctcattg ttcaattccc acctatgagt gagaatatgc agtgtttggt 42480

tttttgttct tgcgatagtt tactgagaat gatgatttcc aatttcatcc atgtccctac 42540

aaaggatgtg aactcatcat tttttatggc tgcatagtat tccatggcat atatgtgcca 42600

cattttctta atccagtcta tcattgttgg acatttgggt tggttccaag tctttgctat 42660

tgtgaataat gccgcaataa acatatgtgt gcatgtgtct ttaaaaacagc atgatttata 42720

gtcctttggg tatcctgggt aagtttttta aaagtgttat ttgtagagac ggagtctggc 42780

tgtgttgcct gggctggtct caaactcctg gcttcaaggt atcctcctgc ctcagcctcc 42840

caaagctctg gcattacagg tgtgagccac tgcacccaga ctcagacttt ttttaaggga 42900

aagaatggga gtgtaggtgg ggagacacac tttgggaggc caaggtggga ggatcacttg 42960

ggccgggggt tcaagaccag tctgggcaac aaagtgagac cctcgtattta ccaaaaatac 43020

aaaaaattag ctgggcttgg tggtgtgggt ttgggaggct gaggtgagag gattgcataa 43080

gctgtaggag cccgaggctg caacgagctg tgatcgcgcc attgcgctct aacttgggct 43140

agacaatgag atcctgtctc aaaccaaaac aaaacaaaac agataattgt caaattgctg 43200

ttttgctatt gttgcttttt gtttttgctt cgctttgcct tggaagtgaa gaagagattc 43260

tcatttaaac agttatcttg aagtatcttt gtgaactagg gtgcaattat ttcctctgtc 43320

cttgagacac agatgattcc tgtccaacat tcccaaggaa ctcagtaagg accaaataga 43380

gactcaggaa agacagttac tgattttaca ctgttgcaaa acagagctat ggtttatgtt 43440

taacaaactg ctggtggggc gtggtggctc atgtctgtaa tcccagcact ttgggatgcc 43500

aaggtgggtg gatcacttga ggtcaagagt tcgagaccag cctggccaac atggtgaaac 43560

cctgtctcta ctaaaaatac aaaaattagc agggcatggt ggtgcatgcc tgtaatccta 43620 gctactgggg agggtgaggc acaagaatcg cttgaacctg ggaatcggag gatgcagtga 43680

gccgagatca cgacactgta ctccatcctg ggtgacggag cgagactgtc tcaaaaaaaa 43740

cacaaaaaac aaaaaaacca aattgctgta ttttattttg tgaaataggg tctagctctg 43800

ttgtccaggc tggagtgcag gggtgcaatc acagctcact gcagccttga cctccagggc 43860

tcaatcgatc ctccctcctc agttttcaag tagctgagac tacaggtatg caccaccata 43920

tgctgcccag gctggtcttg aactcctgga gagagataca tacacacaca cacacacaca 43980

cacacacaca cacacacaca cttttttttt tttttgaga cacagtttcg ctcgtcaccc 44040

aggctggagt gcaatggcac aatcttggct cattgcaacc tctgcctcct gggttcaagc 44100

tattatcctg cctcggcctc ccaagtagct gggattagta aggcactgcc accatgcctg 44160

gctaattttg tatttttagt agagacaggg ttttgtcatg ttggccaggc tggtctcaaa 44220

cttctggcct caggtgatcc acttgcctcg gcctcccaaa gtgttgggat aacaggcatg 44280

agccactgcg ccgggcccat acatatgcat tttaaaaaat ttatttattt atttcgagac 44340

agggtctcac tctgttgccc aagcaggagt gcagtggtgc tatctcccag gctcaagcaa 44400

tcctcagcct cccgagtagc tgggactaca ggtgtgtgcc atcacaccca gataattttt 44460

attattttta ttttttaaat tttttgtaga gatggagttt caccgtgtca cccaggctgg 44520

atatttttgt atttttgata ggcctgtaca gtttccaaag ttgcaacctt tccccctccc 44580

tgagagtagg ggcagccccg gctctccctc tacatcctcc acagtcccga ggttttggcc 44640

tctgtttcct ctgtttccta tgcttggaac accagtcgct cttttgttgg tctggctgac 44700

ttctgttcct cttttaaaaa tttaagtttg gccgggtgcg gtggctcacg cttgtaatcc 44760

cagcactttg ggagtccgag gcgggtggat gacctgaggt catgagttca agaccagcct 44820

ggccaacaca gtgcaacccc gtctccacta aaaatacgaa aattagccgg gtgtggtggc 44880

atgcgcctgt aatcccagct acttgggagg ctgaggcaag agaattgctt gaactgggga 44940

ggcggaggtt gcagtgagct gagatcacgc cactgcactc cacctgggca acagggcaag 45000

actcggtctc aaaagaaaat aaataaataa ataaataaat aaataaataa ataaataaag 45060

tcaagggggt aacacctctt ggtaactctc ctgttgtttc tcatgccagc atcatcacag 45120

ccttgaggct ctggggtagg tcacttcgtc gagctcgttt ccatgaggat aacgttatct 45180

tgggtgtctg tgagaatgct gcactgagta tagagcccag gctcctgggt cagccgggtt 45240

cgaatcccct ttcctccgtg aagatctggg tcagtcacaa gtgcttcagt ttcttcgatc 45300

tgactgaggg aggctttgac tccaaaaaat taacacttga gtgtacctgg ccacagctta 45360

gcacatccag ggtgtttcca ccctttcttt gggatcctca gggctggatg gagccgatcc 45420

ttcccgtctc tccttacact cgcgcactca cgctggctgg aacaagtcct ccaagtagaa 45480 cgaagagcgc gttttagcgg cgctctagcc cgccgagagc atacgccctc cccacacggg 45540

gcccctgatt gtctgaaggt tgcgctggca cgcgcaactt ccgggacaga ggctgtggct 45600

ggaaggagct gggcatccgg cctgaggcgc agcggtcgcg ttagttcggc ccaatggcgg 45660

caccgctgct tcacacgttg tttgtcggga gatgcggccg cttcgtcctc tgcagtcaag 45720

acgctgggcg cgtcgaggac tgggtaagat tcaggccgct tccttctgcg cgtctgggac 45780

caaagctcag gaccgcgctt agaggagcgg attgaaagga tgtgggacaa agctaatggc 45840

gtgtgatagg agcacggggt cgagggtcat ctcacgttca cagaaatgag ctcattcctc 45900

ctaactgggt aatagacatg ggtggggcct ggaaaagtga gtatgttctc tgttctggag 45960

gcccactttc ccgactgtgt ctcttcgtga tttcccaggc ctgggtactg ccttctgcgc 46020

cttgacccct cttccttccc tcttcttcgt ccaaatttgg aagggatttc cctgggctat 46080

gtgggttatc agccgaacgt tgtcactcat ggcaaattga atattacatc tttttgttt 46140

gaaatttgtt tcgacacacg tatttgtttc gcagtcttta ttttgctcca cttttaaaat 46200

ccctaacccc catagcactc ttggcgttta actttcagag tcattaggat gctatgtttt 46260

ttcattaatt tactacgtgt aagtgaagca aaccttgtaa aacaattagc gtaatatgat 46320

tcctaatatt tatcgagctc ctgcttactg tgttaaacac tggggacagt ggtttatcca 46380

aagacactaa tgtccctgct ttctacagag cttacagcat agggtgggaa ggcagtacac 46440

aggccaataa ataaacgaac acgatgattt cagttataca acaaggtaat gggggaggga 46500

gaagggagag aaggagtgtt cgagatttct cattgggaag acatctgtca tttcagcttc 46560

tacttgaatg aagacaaaaa tctaggccgg gcgcggttgc tcacgcctgt aatcccagca 46620

ctttgggagg ccgaggtggg cggatcacct gaagtcagga gttcgagacc agcctggcca 46680

acatgcgaaa cctcgtctct actgaaaata caaaaattag ccaggcgtgg tggcagggtg 46740

cctgtaatcc cagctactca ggaggctgag gcaggagaat tgcttgaacc cggagatgga 46800

ggttgcagtt aagctgagat cacaccactg cactccagcc tgggcagcag agcaagactc 46860

catctcaaaa aaaaaaaaaa atccagccac gtagagattt gggaaaagag tatttccaat 46920

agagtgaaca gtaagtgaaa tgagaaacag cttggcttgt ttgaagagca gaaaagacat 46980

gatggctgta gtaaaacaag ttgttggaga tgaggtgaga gaggtaggcg ggggccagat 47040

taatgtagga ttttaaagac cacactgaga gatttggatt gttactgtaa gtgcagtggg 47100

aagacagtta ttggttgctg agcaaaggag ggattgttgg ttaagagtgg aagcagggag 47160

accagtaaag aggccttaac aatagtccca ctgaattatg ttggttcaat aaaggttggc 47220

tattataatt tttattattt tcatgaactt aatagcttgt taatcttggt tccacaggat 47280

ttcaaatatg cgtgcattag agaatgattt tttcaattct cccccaagaa aaactgttcg 47340

gtttggtgga actgtgacag aagtcttgct gaagtacaaa aaggtaagag gagataatgt 47400 gtgaggtttg cttttggtca ggtcagaata caactattgc tgttatacta aagaccaata 47460

gaaatagcaa gattaattaa gataccagtt gaaatcaaat atttaataat agcatgatgc 47520

cgtcagtgca aaattagagt aatagtgtcc ttttttccc ccaccttggc ccatttcaca 47580

ggtaataatg agagagtaat aatgtcttta ctgaggtttc acctcttcaa atgctttatt 47640

tacaaagcat cttttaactt tagcaagtgc tagaattaaa aacaattaca gcattttatt 47700

tatttattta tttgtgctgg agtcttcctc tgtcacccag gctggagtgc agtggcgtga 47760

cctcagctca ctgcatcctc cacctcccag gttcaagcaa ttctcctgcc tcagtcttct 47820

gaatagctgg ggttacaggc acgcaccacc acacctggct aattttttaa tttttttaat 47880

agagacgggg tttcaccatg ttggcctggc tggtctcaaa ctcctgagct tgtgatccac 47940

ccgccttggc ctcccaaagt gctgggatta cagatgtgag ccaccatgcc tggcctacag 48000

cattttattt tttgaggaac ttacctaagc attactttgg gacagtaaac cggttctctg 48060

aatagggatt tttgtttttg tggtagttta gaagcatttc tactatatct cagcagtaga 48120

gggaaaatgt taagtaaccg tatgtttata tgaaatatcc atttgtatcc atatttgagt 48180

gaatactttt ttagatcctc ctgaattaga tcattagagc tggctgtttt ttcccctcat 48240

gctttttgag aattcgcagg agtatcaact attatattca aatgtcaata cagaagtata 48300

gctaaatgta gtttatcatt ttcctttttc caagccctct ggctgcacta acatgagtgt 48360

ttaaattttt gtagtcatga ttttataatc cgcaattgac atgtgaaagt tagtgttcct 48420

tttataattt catctgatgt taaagtacgg ttaaaagtct tgctgttgat actaaacagg 48480

aaacaaaagc ataacttaat tctttcccct tcttgttaag ggtgaaacaa atgactttga 48540

gttgttgaag aaccagctgt tagatccaga cataaaggta attaattttg tgtttgatca 48600

ttagcaaaat tattgccact ttatacagac atagtttgct ctttgggtcc cattctgttc 48660

tgcagaactt gctctctcca tggtcctccc ttactttaat ctggtggttc tcaaccaggg 48720

acagttttac cccctagaag acatttggtg atggctgcag atatttttgt cacaactggg 48780

aggaaagggt gctactggca tctagtgggt gaatgacaga gatgctgcta aacatctcac 48840

agtgcacagg gcagcctccc ataaccaaga gtgatccagc cccaaatgac aacagtgttg 48900

aggctgggaa accctgctct aatgcttcct ttctattaga ttactacctc tttcctccat 48960

gctgcatgca actcttttgt ctctttaaag ctaaaacaaa ccaaaaaaaa aaaacaaacc 49020

actgtttcag catttccagg ttcgagatac acctatcatg tagtaaaacc ttaatacatt 49080

ttgtttcacc attcttcctt tactgcccag ttttgaagag aatggtttat taccatggca 49140

gtggtagtta gattgcctgg aatgaaattc caattttatt atccagtgtg tgatcttgag 49200

caaattgttt taacctctct gcctctattt tccactgtgt gaaaccaaga aaacaataga 49260 gatttaaaaa atatggagtg ttttgttttt taagagatgt ggtcttgctg tgttgctcta 49320

gctattcaca ggtgtgatca tagtgcacta cagccttgaa ctcctggcct caaatgatcc 49380

tttctcttca gccttctaaa aagctgggac tataggtgca tgccactgtg cctggcttta 49440

aacatggaaa tacttaacaa ggattcaatg agctaatatg caagaagcac ttagaacagt 49500

ctctgactca aagtaagggc agtaattgtc atctgttgtt tttgttccag ctgactgtgc 49560

tgtatcattt ctcactcaca tttaagtcca ctgttcttat cactgtagta attaccctga 49620

cagattaccc atgttttttt tttacatgct gatttcagtg gacttttttt gagacaaagt 49680

ctccttcttg tcacccaggc tggagtgcag tggtgtgata tgggctccct gcaacctttg 49740

cctcctgggt tcaagcaatt ctcctgcctc agcctcccaa atagctgaga ttacaggcac 49800

ccgccaccat gcctggttaa tttttttatt tttagtagaa acggggtttc accatgttgg 49860

ccaggctggt cttgaactcc tgacctcagg tgacctgcct gcctcggcct cccaaagtgc 49920

tgggattaca agtgtgagcc actgagccca gcctcagtgg acttactttt ttaagccttg 49980

tattccttgt atcagccgac actgttggcc acccacttct taaaacttca gtgtttctga 50040

tcctcctgtc ttctgatcct ttaatctctc tctttttttt tttttttttt tttttgctct 50100

gtcgcccagg ctggagtgca atgacgcaat cttggctcac tgcaagctcc acctcccgag 50160

ttcaagtgat tctcctacct cagcctcccg agtagctggg actacacgcg cccgccacca 50220

cccccagcta atttttgtta tttttagtag agatggggtt tcaccatgtt agccaggatg 50280

gtatcgatct tctgacctcg tgatccaccc gccttggact cccaaagtgc taggattaga 50340

ggtgtgagcc accacacccg gccagtgatc ctttaagctc tagtatctct cgataggttc 50400

ttgatcttaa atttggtgtt gattgggctt caaaacttga ctcttttctc actctgttga 50460

ttcttctgtg tgatctcctc atctcccttc atggctttga aatctacctg tgtcctaata 50520

tatttgtgtc tgtagccaag attgctcttg tgggctccag acttatttca ttttcatttt 50580

tggggatggg cagaacagag tcttgctctg tcacctaggc tgtagtgtag tgggatgatc 50640

ttggctcact gcaacctctg cctcctgggt tcaagccatc ctcccacctc agcctcctga 50700

gtagctgtgc caccacgccc agctaatttt tttgtatttt cagtagattt ggggtttcac 50760

catgctggcc aggctggtct cgaactcctg acctcaagtg atccacccgc ctcagcctcc 50820

caaagtgctg ggattataga cgtgagccac tgcacccggc ctagacttgt ttcttaactg 50880

tctgttagat gcatttaccc agaatcatca tagatgctcc aaacttatca tgtccactct 50940

tggctgggct ccatctttca tggagctttc cctggttctc tctaagcaca tggttgttcc 51000

ttcattgagt ctatttccca cacttccaga tctctctagt tacagatctg gtttacaagg 51060

ccctccatgg tctatttggt gcttctttgt tccccagatt tattatctgt tggcttggtc 51120

actatacatg ccagccatac tgaacatttt tcagttttct gaaaacatac ttttccttct 51180 gtaagaagca gaacttccag aaaagactca actgtgttac tgtttaaaga cagctgaagc 51240

atcactttct ctttaaagct tttcctgacc cctgcctcct ttcccagata caaagggaca 51300

ttttctttgt gttccactgt attttgtatc agcagttctc attcttggta ttttgacata 51360

ccaagaattg cactagttgt gtggagtgtt gcaagtagaa ctttttcgt cttgagacag 51420

ggtctcgctg tgtcacccag gctggagtgt aatgggcccg atcatggctc actgcagcct 51480

caacctccca ggctcaagca atcttcccac ctcaggctcc cgagcacctg ggaccacaag 51540

catgtgctac catgcctggc taatttttct agagacgagg tctccctatg ttgcccaggc 51600

tggtctcaaa ttcctgggct caagcagttt cctcctgcct tggcctccga aaagtgctgg 51660

gattacaggc atgagccact gttcctgcct gctagtagaa ataataatag ttcagtacta 51720

aagcatcaaa gtctgcaact gatttacttt tttttttttc tttttgagac atagttttgc 51780

tcttgttgcc caggctggag tgcagtggca tgatcttggc tcactgcaac ctctgcctcc 51840

gagtttttta agtaattctc ctgcctcagc ctcccgagta gctgggatta tgggcatgca 51900

ccaccacgcc cagctaattt tgtattttta gtagagacgg ggtttctcca tgttggttag 51960

gctcgtctcg aactcccaac ctcaggtgat ccacccacct cggcctccca aagtgttggg 52020

attacaggtg tgagccacca cccccagcct gatttacttt taaaaatggt acagtttaaa 52080

tgttatcctt atagttttgt tgcagtcttt ttagtggaaa agagatagga tagattattt 52140

tatttacaca ctaacttagc ttgttttcta catgcctttg gccttagtga gctaccgtta 52200

atgttatcct taacagtttt ggacatatga aattaccata gtacaaatga gttgtggttt 52260

tactttattt tactgccagg ctacttggga tttcatcaga aaatggttga tctgtgggag 52320

tttgacacat ggatatggca tagtaagcac tcagtagctg aattaaggtg gggaaaaggg 52380

gacagcttct tctccgcata tagggaggca tgtgggatgg tggacagagg atagccttga 52440

ccgagacaga tgggtttgga cctgcttctt tactggcctc ttggttgggc agattgctta 52500

ttaatcgttc ttagcctcag cttcctgaac agcaaaatgg gaataactaa acatcttgca 52560

gagttcttag gattagaaga agatatatat gcggagtgtc aggcaccatg cctggcatat 52620

ggtgtattct cactaaatga taactccata tgaatatccc tgtaggtatg atgaccttgt 52680

gttgctttta tttatatgtc taagccttcc acaaattagg ggctttttct taatggtttt 52740

tgtcctgtgc agtatatatg catgaatata attaatatag taatatttta cataattgac 52800

actgtatttt ataccttgcg tttcaaattt agcagttctt ctcatgtcac taacaattac 52860

tatgaacagt aatttgattg cctgaaaaat atttcatgga ggaatggggc tatcatttat 52920

acagaacaaa cacatcataa tatatttaaa ctcagccaca gatttggttt agaaaagtta 52980

tgtttattca tgaccccaat tgatcagcct agactgaatt ttatcagcat gcttcctggt 53040

cagcttgaat atagaggaaa tagaggtagc tattgttcct ttgtgatctt ctaatatttc 53100
aatctgctag aattctgcag tttttaaaag tcccaggtgt caacatttga ggtgatttca 53160
ctttttcagg gcaaacaaaa gtgatcaggc tgaagtattg catttaagtc tttctcctgt 53220
gtattagagt tactagatta ctttcttaaa acgattaagt ttattgtgac atctttttct 53280
gttttaatac ccagttttga tttccttcca agtttgtgac cttttccccc caacctattc 53340
ttgataaatg attgatataa gatagctgta aatttctgtt atcttagagg atttgtgatt 53400
ttgaaagtac tctttgttta acttaaggat gaccagatca tcaactggct gctagaattc 53460
cgttcttctg tcatgtactt gacaaaagac tttgagcaac ttatcagtat tatattggta 53520
agttcaccat ttattttact gtcaagtatg taattcagaa ctttggtaat agtatatgtt 53580
atattaataa catgctgctt ttatctttct tcccccactc tagagattgc cttggttgaa 53640
tagaagtcaa acagtagtgg aagagtattt ggcttttctt ggtaatcttg tatcagcata 53700
gactgttttc ctcagaccgt gtctcagcat gattgcttcc cattttgtgc ctcgtaagtc 53760
attactcttt gcttgcttgg aattttcttt tcttttcttt ttaatacttc tttgttaaaa 53820
taccaccttc cccttatata tgagagactg ctaccatgga agattccaga tgcatattgg 53880
caccaggtct ggtagacata tattccccgt aatgacccct atggaggtgt ctagattcat 53940
ttgttgctgt gagtttgatg aattataact tgctttattg aactcctggt gaaatctagg 54000
aatttttagc catttaaaaa ctataaagtt gcattacttt ttttcagatt gtgcatttaa 54060
ttaatcattg ggctaacttt ggattatgga aaaataactt tttttatagc tgttcattgt 54120
ctaggtcaat aacttttttt gtatagccat tcattgtcta gatcaatgac agaacaacat 54180
attttctttt tccctcaaaa gcccgagtga tcattaagga aggcgatgta gatgtttcag 54240
attctgatga tgaagatgat agtaagtata aaaaggttta aagcctgggc acagtagctt 54300
acacccataa tcccagcatt ttgggaggcc aagatgggag gatcacttga ggccaagagt 54360
ttgagaccag cctgggcaac atagtgagac cttgtctctg caaaaaaaca ttttttttca 54420
aatattttct taaaaaaggc ttaaagtaga actaggcagg gtagtgtgtg tctttagtca 54480
cagctacctg ggaggcttaa gtgggtggat tgcttgagcc caggagttca agctctgcct 54540
ggtggcaaga ctctgtcttc tttaaaaaaa aaaaagtaaa gcacagaata cctggcatct 54600
attctaataa gtagactgca acaaatgaca acttttgatg taatcttttt gttatattta 54660
ccattgatat gcagtcagtt gtcctgaatg cattatttat ataattagtc catttaattt 54720
tcattgatgc tggtggagaa aagtcttgaa attattattt ctctgataaa ttattccgtt 54780
ttggttagca tgtgttttta gcttcaagta tgtcactttt tgtttgtttg tttgtttttt 54840
gagacagagt ctcgctctgt tgcccaggct ggagtacagt ggtgtgatct tggttcgctg 54900
cagccttcac ctcctaggtt caagtgattc tcctgcttca gcctcctgag tagatgggac 54960

tacaggcgtt tgccaccatg cccggctaat ttttgtattt ttagtagaga tggggtttca 55020

ccatattggt caggctggtc tcgaactcct gacctcaggt gatccaccca cctcggcctc 55080

ccagagtgct gggattacag gcgtgagcca ctgtgcccgg ccagcattta tttttagttt 55140

caagcatgtc gcccttcagt tttgttttga tgctcatact ctgaactttt cttctttcag 55200

atcttcctgc aaattttgac acatatcaca gagccttgca aataatagca agatatgtac 55260

catcatgagt atacttttcc ttattttgaa tgtttaattc tcaagaaaat tgtaatcaat 55320

tagtaaaaat tataaaatgt taatagtatt aaagcttgag tcttacattg catttttttt 55380

ttttgtatcc acttgaggaa acattacatt ctacaaaaag tggcatttcc attttctatt 55440

tattctcttt aattgttttt caaagttcgt atgcagattc tcccccaatt ttgtatggtg 55500

gttggaattt tgtttttatc ttcaacagat atgctatcca aaatttttca gtgagaaacc 55560

cctgggtgtg tttgtgtcat gccatatgaa taaaaattgc acttctaaga aaagcttttc 55620

aggtttgtgg gtttcttttg gaggggtgga tttctagttc cctctgtctg ttgattattt 55680

gttaacttaa aaaaatccaa cttgattatt ttttcttctt ttaaaaataa tatacatgtg 55740

tagtgggaaa tgtcagcaaa agtgctgtta tgtttctgtg ggagagaagc tccctctttg 55800

atttgctgtt gatatcagag ttaacagaag cttattttct ctaagttgtt atagactttc 55860

tcagaagcta tacattgtaa gttccagttc tggccgggcg cggtggctca cacctgtaat 55920

cccagcactt tgggaggctg aggcgggcgg atcacctgag gtcgggagtt cgagactagc 55980

ctgaccaaca tggagaaacc ccgtctctac taaaaataca aaattaggtg ggcgtggtgg 56040

cgcatgcctg taatcccagc tgtttggggg gctgaggcag gagaatcgct tgaacccggg 56100

aggcagaggt tacagtgagc cgagattgta ccactgcact ccagcctggg caacaagagc 56160

gaaactccgt ctcaaaaaac aaaaaaagtt ccagttcttt gaggtaaggg ttcctgtttg 56220

cctcctatgt ctatcgatat ttgcttttag aatggtagtt ttccttttta ttccttttct 56280

agaaagtaaa gtcaacatgg attgatttaa ttttttaaaa atagggcacc gtggtttctc 56340

atgccaatac tggtggaaaa atttccattt gttcgaaaat cagagagaac actggtaaga 56400

aatcttttca ttgagaacat catggaaaag ttgtttgtat gatttcattt tagatgatat 56460

taggtctttt tctttctttt tctgtcttta tttttatttt tcttttttga gaccgagtct 56520

cactctatcg cccaagctgg agtgcaatgg cgtcatcttg gctcactgca acctctgcct 56580

ctcgggttca agcaattctc ctgcctcagc ctccccatta gctgggactg caggcaccta 56640

ccaccatgcc cagctaattt ttgtattttt agtagagaca aggtttcacc atattggcca 56700

cactgttatc gaacgcctga ccttgtgatc tgcctgcctc ggcctcccaa agtgctggga 56760

ttggtgaacc accgtgcccg gctgatgtta ggtctttttc ttaaaggtta ctttgtcttc 56820 tagactttaa actgatgtct aagaatttga ctcagattcc tttcttataa agcggctatt 56880

ggggattcgc agtgcctttt tctgttatta ctatgtgcaa gtcaaggtct gagttcattt 56940

ccggaatatc tgtagtggct ttatgctcat acgggcaaga attactagaa gataatagtt 57000

catgtattac taattgtgaa catgccttat tttaacctga aaacaaagcc ttccatagaa 57060

gaattctgct taagtttttg tacaatgttc agatcatctg tgcagttttt aataattaat 57120

agtggttgcc ttagtagaaa accgaatcta gtagcataca aaaagaatta cgtaccatga 57180

ccaagtgcga ctgatgttag gaatgcaaga ttgatttttt ttttcggggg tgggggggaca 57240

gtctctgtct gtcacccagg ctggagtgca gtggcaccat ctcagctcac tgcagcctct 57300

gcctccaggg ttcaagtgac tctcccacct cagcttccca agtaggtggg actatagaca 57360

tggggcacca caccccacta atttttgtgt ttttggtaga gatggggttt tgccatgttg 57420

gccagactgg tcttgaactc ctgacctcaa gcgatctacc cgtctccacc tcgcaaagtg 57480

ttgggattag aggcgtgaac caccgtgacc ggctgagatt gagttagtac ctgaaaatga 57540

attaataaaa tattttgtag caatagaaca aaggacaaaa accacataat catctcagta 57600

gatgcagaag tgtgtgacaa acaccaatat ccctttatga gaaaaacaga aggaaatttt 57660

ctcaacctga taaagggcat ctgaaaaacc cacagctaac atcatattca gtggtgaaag 57720

accaaaagtt ttttcctaag acaaagaaca aaacaaggat ttccgctctt gctgcttgtc 57780

tagccaaggc agttaggcaa gaaaaagaat taaaagcatc cagatggaaa ggaaggcgta 57840

aactctcttt tgcatggtga ttttatatgt cattctaaga agtttacaca cacacaagaa 57900

attttagaga taataaatga gttcagcatg gttacgggac agaagactaa catacactaa 57960

ccagttgttc aagacaattg aataggggag aatagtcatt tcaacaaatg ctgctggcag 58020

aagtggatat gaacatgcaa aagagtgaag catatggata tccatataca aaaatgaact 58080

caataaaagc cctacatgaa gtgtaaaaac tgtaaaactc tgagaagaaa acgagtacat 58140

tttcataatg ttggattagg cagtaatttc cagatttgat gcctaagcac aagcaaccaa 58200

agaaaaaatg catcaattgt acttcaaaat taaacgttgt tatgcttcat aggacatctt 58260

caagaagatg aaaagaatcc ccaaataatg ggaggaaata tttctaaatt ttatgtctgg 58320

taatggactt gtatatgtaa agaactctta taattgaata ataaaagggc aaatagccca 58380

actgaagagg gcaaaggatc tgaataggca tttctgcaaa acacatgaaa agaagctcaa 58440

catcattagc catcagggaa atgatttcac ttaatgccca caaggatggc tataatcaga 58500

acgagaagac agtaacaagt gttcacaagg atatggagaa atgggaacgt tggaactgtc 58560

atatgttgct gtgagaatgt aaaatggtgc agccgttttg gaaaatagcc tggcatttct 58620

tcaaggttaa atgtagaatt aacacgtgac tcagcagttc catttctggg tttataccca 58680

agagaaatga aaatatatgt ccacagaaaa acttgtacat ggatgttcat agcagcagca 58740 tccataatag cctcaagtag aagcaactca aatgtctgtc aactgatgaa cagatgacaa 58800

aacatggtac aatggaatat tactcagcaa tgaaaaggaa tgctttatat gttacaacat 58860

gattggaccc taaaaacatg ccaaaaggct gtgtattata tgactccatt gataggaaag 58920

gaatggttta catgttacaa catgattgaa ccttaaaaac atgccacaaa ctgtgtatga 58980

ctccattgat atgagaggaa tggtttacat gttacaacat gattgaaccc taaaaacatg 59040

tattatatga ctccatttat atgaaatgtc tcaaagaggc agattcatag aaagactagt 59100

ggttgccaag gtcttcattt tttaggggtg cactaatgga tgtaggattt ctttttagag 59160

tgattaaaat gttacaaaat tgctggctgg gtgcagtggc ttatgcccat aatcacagca 59220

cttcgggagg ctgaagtggg aagatccagg agttgaagac cagcctgggc aacatagtga 59280

gaaaatgtct ccctaaaagg aagaattaac ctcatgtggt ggtgtgcacc tgtagttcta 59340

gctactaggg aggctgagga ggaaggattg cttatcccgg gaattcaagg ttgcagtgag 59400

ctatgattgc acccactgta cctcatcctg agagagagag caagaccctg tctctaaaag 59460

aaaaataaat gttctgaaat tgattatgtt gacggtcaca taactgaata tattaaaaac 59520

ttaaattgta tactttaagt tggtgattgt atgatatatg agttttatca atacagctac 59580

ttaaaaacct atagttatgc aaattaaaaa tttcatttac tggggataat tgaaatgatt 59640

ataccgaaca taatacatgt agaaacagta tagtttttgt attgctggat agtctgtttt 59700

tttctttttc aatatttgaa actaaaggtc atgtaattga tgtttttctt acataactgt 59760

gaaatattta ttctctgttg aaatgtttta tcttacgttt tctcctttag gaatgttacg 59820

ttcataactt actaaggatt agtgtatatt ttccaacctt gaggcatgaa attctggagc 59880

ttattattga aaaactactc aagctggatg taagtattga gtaatctatt tttattttct 59940

tttttttatt ttttattttt ttattttcat ttactgactt gaatttgtta taatcacagt 60000

atgtggaaac aatagtcagt gatagaaaag aatccacttg gccaggcatg gtggctcatg 60060

cctttattcc cagcactttg ggaggccgag gcaggcagat cacctgaggt caggagttca 60120

agaccagcct ggccaacatg gcgaaacccc gtctctataa aaaaaaaaaa aaaaaattag 60180

ccaggcatga tggtgggtgc ctgtaatccc agctactcag gaggctgagg tgggagaatt 60240

gcttgaatcc gggaggcaga tcttgcagtg agctgagatc gtgccactgc actccagcct 60300

gggcgacaga gcgagactcc atctcaaaaa aaaaaaaaga aaagaaaaga aaagaaacca 60360

ctagcaccat tctttgcttc ctttctttga atgtgtcttg aactccatct gtgcatgtgc 60420

tgggagttgt agacagttcc ttctcatgat tggagaacaa ggcgttaaat acatagttat 60480

ccaaatgtaa aagtatggtt gtggaaaatg ccatgaatga aacatacatt atgagttaga 60540

gaacctgata gaatcacagt ggggtcagga agggattcct acggaagtga tttttcctgt 60600

ttggcctttc ttaagggcag attataatta taaacagtta aaactttgtt taaggaggcc 60660
cgcactaagg tgcagtggga atgaaaggaa gtggtagatt ctagtgacat tgtgaggaaa 60720
ggtgaactgg tccttgagac tggtttggag gaggggaggc agacagtaag ggaaaggaat 60780
ccttcaatag ttgctccctg tggaatcgaa tcttggtgtt gccattaatg gtagttagaa 60840
atatgaagag gaggctgggt gtggtggctc acgcatgtaa tcccagcact ttgggaggcc 60900
gaggcgggcg gatcacgagg tcaggagatc gagaccatcc tggctaacat ggtgaaactc 60960
tgtctcacta aaaatataaa aaattggccg ggtatggtgg tgggcaccta tagtcccagc 61020
tactcgggag gctgaggcag gagaatggtg tgaacctggg aggtggagct tgcagtgagc 61080
ccagattgtg ccactctgct ccagcctggg tgacagagca agactctgtc tcaaaaaaaa 61140
agaaaaaaaa aatatgaaga ggaggcagtt ggaagagtag ttccatcttg gccaggttca 61200
gttgctggtg ggcagcctac cagagaatac tcacaggcag tcgtggctgc agatggggac 61260
ctgagcataa acctttggaa agatgcagtt taggacaggg gaggagaagg gtgatcagaa 61320
gtatggggaa aaccaagagt ctggatgctc aggaagaatc cgctggaagg aggagtttgg 61380
tcagcagcat cagatactgc tgtcattttt tagaaagatg aaaagagcaa cagtccttgg 61440
atttagtggt tagaaggtag tctttgttgc tttctggagg accatgtcag tgaagacgca 61500
gaaactgcat ttcgggagag gatgtggatg gtggggaagc agaattgggg ctgttagaga 61560
ccttggtgca gggttgtggt ggaaggaggg gatggagcag ggctaagagg cgtggtttag 61620
gagtggggag atgtgagcag gtttgtggac tgaggggaga ggagctttgg tggaggaaaa 61680
cattgatgct ataggaaagc aggaagatgg aacaaggtct tagaagagct ggagcttggg 61740
ctcactggtg cagtgctccc ttggagttgc acctctctgg ccaactgtat atgtactctt 61800
tatagtcttt ctctggtata tacttaagga acattttaga atgtttacaa agaaggtcag 61860
gcatagttaa taaaaaatgg catggtttga gtggtatgtt aagatatttg aatggtgata 61920
taccaaaata aatattgcat catgcacatt tggcttgcag ttcatcattt ttctgctcaa 61980
ttgattgacg atatgtttat tacacaatgt gtctgtgagt gtcttgtgca tagagattgt 62040
attagtccat tttcacactg ctgataaaga catagctgag cctgggaaga aaaagagatg 62100
tttttgtttg tttgtttgag atggtgtctc gctctgttgc ccaggctgga gtgcagtggt 62160
gcgatctcgg ctcactgcaa cctccacctc ccgggttcaa gcagttctcc tgcctcagcc 62220
tcctgattag ctgggattac aggcacgtgc caccatgccc ggctaatttt ttgtattttt 62280
agtagagatg gggtttcact gtgttagcca ggatggtctc aatctcctga cctcatgatc 62340
catccacctc ggcctcccaa agtgctggga ttacaggcgt gagccactgc acctggccaa 62400
aaaagaggtt taattggact tacagttcca catggctggg gagccctcag aatcatggcg 62460
ggaggtgaaa ggcacttctt acatggtggc ggcaagagaa aatgaggaag atgtaaaagt 62520

ggaaacccct gataaaacca tcagatctcg tgagacttat tcactatcat gagaacagta 62580
tgggggaaac ctaccctatg attcaaatta tctcccacca gtcccccccc caacaacatg 62640
tgggacttac aggagtacaa ttcaagatga gatttggggc caggcgtggt ggctcatgcc 62700
tgtaattcca gcactttggg aagctgaggc cggtggatca cctgaggtca ggagttcgag 62760
accagcctga ctaacatgga gtaaccccat ctctactaaa aatacaaaat tagctgggca 62820
cagtggcaca tgcctgtaat cccagctact caggaggctg aggcaggaga atctcttgaa 62880
cctgggggc ggagtttgcg gtgagccaag atcttgccat tgtattccag cctgggcaac 62940
aagagcaaaa ctctgcctca aaaaaaaaaa aaaaaaaaag tgagatttgg gtggggacac 63000
agagccagac catatcagag ctagaataaa tgttgaattt gttgaggcta cctgacatag 63060
agcatcatgt gatagttgtc gattttatat aagtatgtag taaaaggggc ttggtttatt 63120
atatttaaat tccttcatga cctaggtcag tttacaggct tgcaccataa ttgtgtattg 63180
tgttggggtg tgatataagg cactaatctg gacaccttga acgtgtgtat atcagatgaa 63240
tttccatccc aaaataacat agttgtattt tttaaatcct tttattcttt tttccccccc 63300
ctttgttata ggtgaatgca tcccggcagg gtattgaaga tgctgaagaa acagcaaatc 63360
aaacttgtgg tgggacagat tccacggaag gattgtttaa tatggttagc agtttattaa 63420
tgaaagtgga gatgaagttt atcataatca aagggtggaa acagctagtg ctgctcatct 63480
ttgttaaggc tttagattga aagaattaaa atagtttagc aaacttgaaa acgattcctt 63540
atatgagtaa tttgctgcca tgtcatttag cacttagcat aattggtcta tttccaaggc 63600
tttgaatttg ggtttggtga agtatgtttc acttttgttg ttgtaacttt cagtgtttgt 63660
ttttgtaagc cagatgctgt ctgtgagggc gtggctaatg gaaaagcata actgtttaat 63720
ttctgcattt taccacttgt acactttata gcattacttc ttacgagtag ctgggaccac 63780
agattaccat gcctggctaa tttttgtgtt tttagaagag gcggggtttc accatgttgg 63840
ccagactggt tgtgaactcc tgaccgcaag tgatctgccc gcctcctgcc ccgcaaagcg 63900
ttgggattac aggtgtgaac aactgtgcct ggcccacgtt cccttctcag tacacttgga 63960
gagaaaacag attgctgcct gccagcccag ctaggtgctc tgaaaatgtc atcctgcctt 64020
tggtcactag gtggtggtct tcccttaagc ctttctctat taaaatctca tatggggtaa 64080
ttaacagtat ttcctttatt ctttccaagg gttgagttgt aactagccca aaccaactta 64140
ttaatctaga attttaaaaa ctttaggctt tgacttttct tcttcttctt cttttttttt 64200
ttggtggggg aaagaatata gaaggctttt ccttctctgc aacgattttg tggcttccta 64260
gaggtcagga gagtgttggt catgggaaag aaggttgaat tcagtctgcc cacatgggcg 64320
tgcctagctt tagaacagcg ctatttagga gaagttggaa gttacaccct ttggtgagaa 64380

gctgtgtctg ttttttttcca tgattggcat aattaactca aataccagct gtacattagt 64440
ccgtatttct gttcatggtt gagttcagtg tgtccagaga ccggaaggtg ctttgcactc 64500
acaggagtgc ccatgtggag ctccatggga tgtgaattat tgttggtcac tagttctggc 64560
tgacattgga atcacttgaa gagtttttat aatatgtgga ttccaaagcc ctgtcacaaa 64620
cctattgaat ttgtacctcc caggttgaat tttttgttgt tttttgtttg tttgtttttt 64680
gagatggagt ctcactctgt cacccaggct ggagtgtagt ggcatgatct cagctcactg 64740
caacctctgc ctcctgggtt caagtgattc tcctgcctca acctcccaag tagctgggat 64800
tacaggcacc tgccaccacg cctggctaat ttttgtattt ttagtagaga gagggtttct 64860
ccatgttggc cagtctggtc tcgagctcct gaccttgggt gatccacccg cctcggcctc 64920
ccaaagtgct gggattacat ttacaggcgt gagccactgt gcctggccct gggttgattt 64980
tgatgcttag ctaggtttgg gatccattgg attatttaac acccgaggtg ccttttgttt 65040
ctaatgatat tctctcaacg tgttttaaaa atgaagccca tgagatagtt atgagatagt 65100
agaacttttc cctacattgg tgaagtaaaa atcttgggat tttgatagtc agattatctt 65160
aggcattaaa aaatatcaca ccgatgccct ctctttttat agggattcgc agaggcattt 65220
ttggaacatc tttggaaaaa cttgcaggat ccaagtaatc ctgccatcat caggcaggct 65280
gctggaaatt atattggaag ctttttggca agagctaaat ttatttctct tatgtaagta 65340
gcctaatttg ccgagtactt tttaatatca tgctttaaaa agagtatagc attgtctcaa 65400
gtcagaaata tctcccatat gtttttggca tgtttttaaa gtgaataaaa ttcctactct 65460
gtgcaagatg tttatatttc taagtggtga ttttagaata aagtgtctcc ttttttatta 65520
taaaaccctg tatgtaaggc ttttgtcatc tcttttgtgt ggttgcactt aaagatccat 65580
ttgttttgtg gatagaggac agtgttgtat actgttttga ttctttttgt aggtttgtca 65640
tttttttcat ttgcattcca aatctattgt atctgttaaa gctgaagaaa aaccctttta 65700
aaggtaatag acctatctag gaggccagtt tctttcagtg gcccatgaag atatctttgg 65760
acaaggatgc tgttgaaacc cttccccaag aacaaaatta ttcacccata ggacttgact 65820
ggatgcatca gggaatactg aagtccacca gactgtcttt ctcttgagac gtgttggtga 65880
acatgtccta tttggccaat caccctaaga ggggtgcctt tgagatggtt aggagaacct 65940
gctttccatc ccttgggacg ttcttagggg ctcacctgtt cctagaaggt cagagctact 66000
ctgccttgta attggaaggt tgtcttccta cgcacccatc cttatccttc ctttctttgc 66060
ttttcctctg tacccatggg tattatttaa agaaacctat gaacttactt agcatggttt 66120
gtaatgaaag gcagttgtgt gtttttatgt tattctggtt tttttatgaa gtgtaaagtt 66180
gacttgaatt tttcttttct ctagtactgt aaaaccatgc ctagatcttt tggttaactg 66240
gctgcacata taccttaata accaggattc gggaacaaag gcattctgcg atgttgctct 66300

ccatggacca ttttactcag cctgccaagc tgtgttctac acctttgttt ttagacacaa 66360
gcagcttttg agcggaaacc tgaaagaagg tcagtgttgt gggagtgctg gactggattt 66420
tccttgtgtt cttgtcaccc ttcagaatgg tgattcatta ctttttgag atttttataa 66480
aaactggatt cagaaaactg catgtacact caaacttcta ataataattt caagcagctc 66540
ataggcccct acaaacccct taagatagat ttgagcttga gaaccctaca aaccccttaa 66600
gatggatttg aggttaagaa agaggtttct gcctttgaag gtttgaaatg tgaagatgtc 66660
tccagaggtg aggctgagcc ctgggctgtg ccagcgccct gtacaaagct tcagttggat 66720
gcaccttctc tttgttgtcc ttgtaacagc ccagtaaatg gcaggtattc tccctttaca 66780
gacagcacca aagcacgggg aagtcatttt cccaagatca catggttact ggcaggatta 66840
gaaaactgaa gccaggttca gctgacccta aagtttgagt ttatatagat tacactctgc 66900
ctgaagcctt gagacttaat tgaccagtat tgttttgcta atttctaaga gttacttata 66960
attcaaatct atcagttgaa acttactaga ttagcgtatt ttagttgaag agagtctcca 67020
agaacagtgt ttataagtca ttgtaaattg ttctgtttat gtttatgaat aattcatatg 67080
gttttgtggg tcacttcctc taaaccaggg tctgtcaacc ctgcatgatt gccatttggg 67140
ctgcatcatc cttcatcgtc gggggctgtc ctctgcactg taggatgttt aacagcctcc 67200
acctactagc tgccaacagc agtccctgac caccccagc tgtgacaact aaaagtgtct 67260
ccagacactg ccaggtgtcc tctgtggggg ttgcagtctc cttaggttaa cagccacagc 67320
tctaaactga cagttgtacg tgttgcatta tatatgttta cctacatcct acatgcttct 67380
aaaagatgtt gtatgaacta gtaggatgag gttttatcac aaggtaagta aatacaagct 67440
ctgcttttct ttgtataaat taatgccagg aatctgaatt aaatatcttg tttttgtaag 67500
cagtgacatc ccatttaggt aatttttatt gaaatatgca tcaaagaaac tcctaagaaa 67560
atatacttaa gtacaagttg gtcagcttgc ctcttaaaat aaatgtgatg tctttatttt 67620
actcatgtag gaaagaattg tattcactaa gtctaagaaa gtggcttctg tctaaatttg 67680
ccgtccgttg aggtagaagg caaatttgga gttttcttgt ttagaaaaaa aactacagat 67740
gactactgtg cacctgaaaa cagcactcag cttcactaac gagacatgca agctagaatc 67800
aaattgctgt tttgttttgt tgcctgtcgt gattgttagc tgaaaccaaa tcacaaggtc 67860
tttttctccc tctgtattag ctcagcatac actgagctta caaacgtatg aacttcacgt 67920
tgtcgtggaa tcttacagcc tgctacttcc taagtttcct ttagagaagc tgccttggtg 67980
accaatgaat gtggttagcc tagtgatact cttctgggcc atatactgtg tgactatctg 68040
catggacctt tattgaaagc atttctgcaa ataatttttt taagtttttt ttaaatgtgt 68100
gataatttgt gcttttaaag atatcttaca cttttcactt atttgtacct ttaaaaatct 68160

tttttttttt taaaccaaag gtttgcagta tcctcagagt ctgaattttg agcggatagt 68220
gatgagccag ctaaatcccc tgaagatttg cctgccctca gtggttaact tttttgctgc 68280
aatcacaaag taagttattt atgctttctt gatgggagtt atttaaaata tttttattta 68340
tgtttctcta gtattgtaag agtctgttaa atttctatga aattagtaac attataaaag 68400
gccaggcgtg gtggctgacg cctgtaatct caacattttg ggaggctgag gtgggaggat 68460
tgcttgaggc caggagttaa aagaccagct tgagcaacat agtgagaccc tatctctata 68520
aaaaaatttt aaaaattagc tgggtgtggt tgcctgtgcc tgtagtccca actactcagg 68580
aggctgaggt aggagaatca cttgaaccca ggaggcagag gttgcagtga gctgagatcg 68640
tgccattgca ctccagcctg ggtgacaagg gcaagactcc atcaaaaaaa aaaaaaaaaa 68700
aaggaatttc tgcaatacgc tacaacatga atgattttgt aggacattac gctaagtaaa 68760
ataagccagt cacagaaaaa caaatactgt atggttctac ttaaaggaag cccatagagt 68820
tgtcaaaatt agagagacag aaagtagaat ggtggtcccc agcggctgca gaaagacaga 68880
gtggggaaat tattgtttaa tgggtacaga gttttcattt tacaggatga agacttgtgg 68940
atatggatgg tggtgatggt tgcacaacaa tatcaattta ttttatacca ctgaaccgtg 69000
cacttcaaaa tggttaagat ggtaagtttt atgttgtgta ttttaccata ataaaaaaaa 69060
ttgtagaggg aaaaacagtc tgcctccact tttgatatgg gactgctaac atcttccacc 69120
ctccctctcc ccctctgccc cacatctggg caagctaaga aagcctgctg ctctctcctc 69180
tggcaccagc tggaaattca tacccaacaa gccctagccc tcccaccaga cccacatttc 69240
atccccatcc ccatcgcatc cccatcccca tccccatccc taaccaccat aaatgctaag 69300
ggagtttcct tgcctggttt tctgaaacca tttttggacc tgcttgggaa tctgccctgc 69360
tctctcagaa agcttcatta tatgagcaat aaaccttttc ctaccctctt ggtgcatgtg 69420
gtgtatcatc agtcttgaca tctaaaacaa attttgggtg gtggggtcca tgtctttgca 69480
gggtgaccac aatagtacct ggcacattat gtgtttaata aacagagatt actgtcatat 69540
ttattttatt ttattttttg agatgaaatt tcactctttt ttcctaggct ggggtgtagt 69600
ggtgcgatct tggctcactg caacctccac ctcccgggtt caagtgattc tcctgcctca 69660
gcctcccaag gagctgagat tacaggcatg cgccaccaca cctggctaat tttgtatttt 69720
tagtagggat ggggtttcac cacgttagcc agactggtct cgaactcctg acctcagatg 69780
atccacccac cttgacctca cttacaggcg tgagccaccg cgccttgtct ctgttatatt 69840
tatttctcta tttaaattga tggatatatg caaacctgat cattatcata cttatgcctt 69900
gacacaagag aggcaataaa ctaatctaag tgatgcttgt gatgccaaag atgtcagaac 69960
actttctggg ccaatggcag atacctcatg tcaccagatg ctaagggtcc acaataaaaa 70020
gcgttgaatg aaaattttga ggataaatat ctccaggttg aggaagaagg ttgcacatat 70080

cgggtgctca ataaatattt gtcgaatgaa tgaatgagtg aatggcccca gtgtgtgggg 70140
cttgggaagt gattggatat aggcagagaa aaggaacaag tcaaaaataa ttcagaaatc 70200
aagaacaagc aagttgcctt gatatacttc attcctacac ttggcaaact ttagtgatta 70260
aggaaacaat gttttaaaaa aagttttggt gatgagacat tcaggaagat ctatcaataa 70320
atagcaaacc tggtcctttt taagacactg tgtataaaaa aattccaaaa agattaaaat 70380
cagtgcagaa accaaagaac cattttttc tatatcatat tgatcatttc aagtggaact 70440
gttagctatc ttagaaaaat tgtggttctc aattggtttt gcctctatct ctgaaccacc 70500
attcctaaag gaaaacattc aaccagaaaa tttcagcaca tcacaatctc tctgaagatt 70560
aagaagtctc tgtgaaggac tgaatgtata aattgaaaaa tttttgctgt cacatttagg 70620
taaaagagaa atcgttcttc atatcccctt cctttccatc tgtagctcac ctatgtcaac 70680
tttcctctca gaagtgaaat aaaattaaag ctatgacact gagtgtcagt catggaggga 70740
cacgttcccc acttagcctt tgctgaagtg tttccagaga caactgtcta atgccaggtc 70800
cccacttagt gggctgcact tttctactca tttgcacata atccagaagt cacattttgg 70860
gttcacagtt tccactgggg taacctcatt aggcctggcc acctctgtgc ttcttgtgaa 70920
gtctctgatt tgaaggatta gtatcctttc cagactgtgt gggttgactt tcacccatct 70980
ggagttgctt ggaacaaaga taactacctc aactccttgt catgaaagca aaaaaaaaat 71040
agcattagag ttctgagaac aggatgtcat tacgggttgt gccttgtcag ctacaggtaa 71100
gatatttgag tggctcctca agcctcccca actccctctc caacctcgct acacagtctt 71160
cccaggtctg gtatttgacc tccagactgc cgcccagaaa gcaactcaga gctcagcaac 71220
accatgatga tgaaacacaa atatgtgggc ccccaacttt ggagactcca ctggttgctt 71280
ttgtgcagct ctttctatat atgactttca aaatggcagc caaccctctc aactgttggt 71340
tgatctgaat atgtaaagtt cagccttcaa acccagcaaa cagtcagcta taggatagag 71400
ttcaggtgct ggggaatgac cacggttggc tgctacgttg ggaaccctgg tgacatcata 71460
tatggacctg ggaatgtaaa gaaacgtagg aaatctgaat ttatgacttt tctaatctct 71520
ctatcgggat ccttttttgg aatcaagatg attttccttc taaaaggtca ttttattaca 71580
gtaatgggca gggtagacat acctcactag cgtactctca aaatttcttg catgcatatg 71640
ctttctccgg catgcatatg ctttctacta caatgtgaac aagcctacga aagctagcct 71700
aatggaggat gagagaccac gtggaggaga gctgaagtgc cccagccagc agctagccta 71760
cacccacagc tattttgctt gtgttccact cgtcctattt gcataatatt gttctttaag 71820
ttaacccatt cttaatttat tgttttaatc ttaagcaata atattcaaga actaatgagt 71880
tttgtgtatt cattatattt ttccataata tatattatac taataaatgc ccattcaaat 71940

ttttgtttga gtgctcaagt gggtatcatt taaaataatc ttatatacta tatataacag 72000

ccaaggtttg ggcaacacaa cagaaactgc atgagtttat tttatcagaa ttttttaaac 72060

ggtatgggag aactaaaaaa gtaaaaaagg gaacccttga gttaacaagg agataagaac 72120

tacgtaaagc agttactgtc ctgaaggaat aaagggaaga gcatggggtt attagaacct 72180

agaagtttgg aaccaccagg agctgggacc ctatgaggag agggttggcc cctgacaatg 72240

ctggtgtctc taagggagct cctgaggctg attctagcag tgtagggaaa gaaactggaa 72300

actggaacaa tttcctctgt aatcaatgac ccttgccagg gtaaagaatc actgctgaag 72360

agatgctact ggaaaagcaa gcaaacaaaa aggagggtgt cccttcccct tcttccttcc 72420

ttccagtctc cctcatgaca gagcatctgg ctggtgacgg gaaaaggtgc tccacagagc 72480

accaccccaa catcacacgg gcatctgctg acccacagct gactgcagat gaggaagctc 72540

agcctaggtc acacaaacgc ctcccagctg agtctagcct aaattgccaa cctgccaaat 72600

catgaactaa taagtgacta ttgttttaag ccactacctt taagggaatt cagtatgcag 72660

caatagctat ctgatccata taggcctaca aggctataga aaaactatca catgtaatcc 72720

cagcactttg ggaggccagg gcgggtggat caactgaggt caggagttcg agaccagcct 72780

gaccaacatg gagaaacctc gtccctacta aaaacacaaa attagccagg catggtggtg 72840

catgcctgta atcccagcta cttgggaggc tgaggcagga gaatcacttg aacctgggag 72900

gcagaggttt cagtgagcca agatcgcacc attgcactcc agcctgggca aaaagagcga 72960

aactctgtca aaaaaaagaa aaagaaagaa agagagagag agagagagaa agcagagagg 73020

ctactgcaga gaaaagtcta gaaggatggg ttcatgggtt catcgagaga caatagctta 73080

acaaccagca caccatagtt ggcaaaacac tatcattgaa aaaaaaacat gctcaaaagg 73140

ggaaatgcca gtttgggtaa atatgctttt gtgttggaga gaaagaattt ggaacaggct 73200

tttcagaccc ccttaaggcc caacaaacaa attataattt agacaagtct gggattcttc 73260

acagctcagc ttgtggtgat ggtattagct tcacaactcc aaacaagtta agctgtctgt 73320

gtgaaatctc ctcaacaaca cctcactggc aaacctggag gtgctgaaaa cagagctttc 73380

aattcttgtt tgcaaccaag ggagttgagt tggcagatgg gcactgtgtc cagccttggg 73440

aaaggacatc gcagactttg catcctaaga actcataacc acaacggcaa ggtaagacac 73500

aagctcttga aagtttccat cacagtgcag cacaaatgac cttggctatg tgccctgtta 73560

ttgctggtcc ctgcttaaaa atctcctgtg acttccaacc acacaaattt cctacctggt 73620

tgcaaaaatg cccttgataa ttcacccctc cctctatctt gccccccttta caatgtggct 73680

tggcagctcc tcccatcaag agttaaaatc tatttcctca ccccttgaat ctaggctggc 73740

catgggactt gctttggcca atagatgtgg cagaaattat ggcgtgacag ttctaagcat 73800

gagtctcaag aggctttgca tgcagcaact ttctcttaga accctgccac catgtgaaca 73860

atcctgcctg ggctagccta atggaggatg agagaccatg tggaagagag ctgaggtgcc 73920
ccagccaaca accagcctac cccagaagca gaggcatctg ctgacccaca gctgactcca 73980
gatgcataag ggagctcagc ctagatccag aacgcctccc agctgagtct agcctagact 74040
gccagcctgc caaatcatga agtaataagg gactattgtt ttaagtcact accttttggg 74100
ggaatttgat atgcaacaat agctctctga tacatatagg cctacaagtc tatagaaaaa 74160
ctatgctgcc tctctctcca gccacacaat ttctttctct tctcatttac tattctaatt 74220
cctctgtgtt atagtctgtg tcccaaaatt catgtcaaaa tcctaatctc caaggtaatg 74280
gtattagaag gtaaatcttt ggtaggtgat caggtcatga gggtggagcc ctcatgaatg 74340
ggattagtaa ccttataaaa gagaacccag agagctcatt tgctgcttct gccatgtgaa 74400
gatacagtga aaaaagaagc aggcccttgc cagatacgag tttgccaatg ccttgatctt 74460
ggaattccca gcctccagaa ctgtgagcag taagtttcta ttgtttataa gctacccagc 74520
ctatggcatt ttgttacggc agcctgaatg gactaagaca gtctacctag accattattt 74580
ccctttcatc atccaccagc caattccagc acatctttta gatctcagct taaatactcc 74640
ctccaagacc tccctctatc tctaatatga atgaaatcca tatctcaagt tcttcacaga 74700
atcctctact ctttccttca tggcatttgt cataatttgt aattatatat ctagcaaagt 74760
tctttgttgt taaacatcta cctcctccac tctcctagaa actccacaag gacatccctg 74820
cacccagtgc ctaggcaatg ccagacacat agcagatgct ccattaatta tctgtcgaat 74880
gactgaatgg cttccaagtt agttaactgg gcacccttga taacagattc tggcctattt 74940
gaaggatcaa agaagaaagt ggtgctacct tctcccctgc cactatcttg cccacttgtg 75000
gtgccagttc aggaggtttg gaatggatgt ggctaatgat agacgtagac ctattgcctt 75060
tcttggatca taattctgcc aggctctgag tccatgtggc atcgatggct aattgtcctc 75120
caaaatttat cctctcttct tccatttata ccctcccatg gagttttaac agggcatgtg 75180
gtcaccctac tgggatctca cttctcagct tcccttgcaa ctggatgtgg ccttgtgact 75240
aaattctcat gaacagaatg tgagtgcaag tgatgtgtca gtatcttcat cactttccta 75300
aaaagggaac tgctggtcct ccacttcctc tctttcaccc ttccaatgag ccagaacatg 75360
catgtgatgc tggtgagtca gcttcagtca catgaataaa aacaaactcc aggagatgac 75420
taagcaataa gacagaagga acccaagtcc ctagacgagt tcacagaacc aagctaccta 75480
tccaaccctg ggcccacctg gattataaca tgagaaaaac ataagtccta atcatatttt 75540
tgaagcactg cattttaggg cttctttgtg acagcagcct accctctagt ctaatcaata 75600
tacctcacca agtctcctgc tcctaaggga gacaaagaag caaaatgagt ctcaaaacat 75660
catccaaatg gaatagatac agacctgtaa tcccaacact gtgggtgccc aaggcgggtg 75720

gatcacttga ggtcaggagt ttgagaccaa cctggccaac atggcaaaac cctgtctcca 75780

ctaaaaatac aaaaattagc cggacgtggt gttgtgcacc tgtaatccca cctacccacg 75840

aggctaagcc gggagaattg cttgaaccca ggaggggag gttgcagtga gccgagatca 75900

tgccactgca ctccagcctg ggtaacagag tgagactctg tctcaaaaaa ataaataaaa 75960

ataaaaataa atagaccatt aattaataga tatagccttg gtctgtgacc aaagctcaga 76020

atgttatgat attcctttcc tatgtcacct caacttgccc ctgtcatcag acaggacaaa 76080

ttccccactg gtcctttgca ctcacagctg ttacatttga aatgggagct tagccttccc 76140

tgccctggtt cctccttaga ctcatttggg aaaacaggaa acgtaattat ttctgccatt 76200

acctttatct catggagcct gacagagtgt aaccaatggt aggaattaaa acattctaat 76260

tgccaactca caacaactcc cgaaaaaaat cattttaact cattatacat attaaattat 76320

gacatgctta atgtccaaac ctaatagatt cagtactcag gaaatccctt atacaggtag 76380

acacctttcc tcctgtactt taagaaaatc ttacatcaat atgcgggact tctcaaattt 76440

ttctatcaca gttttcttaa taggaaggag aatttgtgcc aaaagatgta tggaaattta 76500

gcacaaagta gccctctaca agcggaggat ttctttaaag cattgtgttt tatctcaaga 76560

ttccatggca aatgtttatc ttctctgctt gttttagtat gaaatagttt tcatttgctt 76620

gtcatcattt ttaaggagtt gaaaatacaa tcaacctcac tcatcataaa ataaaagcaa 76680

attaaaacta ctatgacata ttttcaccta caaaattgag aacattttaa aatgtgataa 76740

tagttctatt aacaagggtg tgagaagaca attctcaggc agggcatggt ggctcacacc 76800

tgtcatccca gagctttggg aggctaaagt gggaggatcg cttgagctca ggaattcaag 76860

atccacctag gcaacatagt gagactctgt ctctacaaaa agtaaaatat aaattagctg 76920

ggtgtggtgg catgcggcta tcatcccagc tactggggag gctgaggcgg gaggattgct 76980

tgaactgggg aagaccaggt tgccatgagc catgatcacg ccactgtatt ctagcctggg 77040

tggcaggacc ggaccctgtc ccaaaaaaaa aaaaaaaaaa aaaaaaatcc ctcttccctg 77100

attggtggga ctgtaagctg gtgcaacctc ttgggaggag caggtggtta gaaaacatgt 77160

atcaaatttt tatgcaaaat ttaaatagac ccacaattag accctaggaa tttattctcc 77220

agatattctc atgcatgcgt gcaaggtata tttgcaaaga tttgtatcca gcactatctg 77280

taattgcaaa aatccagaag caacctcagt gtcaatcctt agaagactgt gtacatgaga 77340

tactgtacat gttggttggg ttctccgaga agcgcatgcc cggatagaat gcaaaagatt 77400

cattagggag taacacctct gccagaaaag cagaagagat aggattgggc aggaggggcc 77460

attagagcac aatttagagc taccaccatt tcagaaggca gcaaagattg cctgttagag 77520

aaatggtcag gcccttgtac cctcagtcac tggatgggag ccactccaag aagagcatga 77580

ccatgactta acagctaagg ggaccctgaa agagctgcca ggttaaggct atcagctcct 77640

cactccccac agctggacag agagcctttg tttaaggagg atctgagcag ctcatctcca 77700
ggtctggcac aagggttacc cataggatga agatggaaga catgaatcta tatgtattag 77760
tatggaacaa tctccatgat agatttttaa gtgaaagagc aatagtaaat tgcaaaagag 77820
aacataacgt tttgggacta ggcagaagtg ataaccgaac aacagtgtga atggagtaaa 77880
atgtcaccga attgtacact ttaaagtggc taattttgct atgtgaattt cacctcaatt 77940
tttttttttt tttttgagat ggtgtttcac tcttgttgtc caagctggag tgcaatggca 78000
aagtctcggc tcaccacaac ctccgcctcc tgggttcaag caattctcct gcctcagcct 78060
cttgagtagc tgggattaca ggtgcgtgcc accacgcccg gctaattttg tatttttagt 78120
agagacgggg tttctccatg ttagtcaggc tggtcttgaa ctcccgatct caggagatct 78180
gcctgcctcg gcctcccaaa gtgtgggatt acaggcgtga gccaccgcgc ctgaccccat 78240
ctcaatgttt taaaagagag agaatacagc atgctgtcat ttgtgttaat tttaaaaagg 78300
aaatgaattt atgtgcatgt ataaatgcta gacatggaat ctctctgaaa ggagccatga 78360
aacactcata ctatgatctc cagtcaggaa agagacttaa ttttcactgt acgcccttgg 78420
tgctgcttaa atttttatca tgtgcatgta attacaccct gtctttaaaa atcattaaag 78480
atgtttaatt gttctgatga aggaatacat tacttgccat caagaaaaat gaatgaaaaa 78540
ttttctgcga gacaattttt agcaagacac tgttgtattg atcattcaag ttcagaaaat 78600
tcagcctccg tcaaggggca caaacatcat atatcaggtt cagtttgtcc tctctctcag 78660
agtcaaagtg ctttaggaac atagacacaa taagtttctg gaaccaaatg gcaaatatca 78720
aaacttgcta gaacaggaga aaagtgtatc ttattgaaaa ttcaccagct gctataccat 78780
tcagcattgg gaaaatcagc ataccttctt agacttcatt attttaaaga tggcaaaata 78840
gccaagtcat ggatgtctcc ccctttcatc aaaatgtaaa gaactagctg cctctgggac 78900
tctccaccaa ttttcaagca cgtcttttga acccatttga tggtgtcact caataagggc 78960
accttttttca acttggctgc ctctttttga cccaaaataa tttcaaccct tttctgcagc 79020
tccgggcttc accaggcttt ctattattgc atataccttt catagtgatt ctaaaccgac 79080
ctcgaatgaa gagacaaatg atttttttatc tattggtttg attgcacttc tccttgtact 79140
gctccaagac aaggcttgtc ttttgagttg caaaaaatac tagcgctcta tttccaaagt 79200
caaacaagtg gcttttcaat gtctctgcaa gtgttttgtt tcatgcagtc agtgctgact 79260
tttctcgatg gagagacata tggtttgggc ccattttagc aactctataa taaaactgat 79320
tataaaaaata agcatctaag aatatcttag gcttttaaga ttgacaccac tgcttgctac 79380
tcaattgcta gttgtggttg gcagtgcacg cagtgtcttt gtggtcaagt tcattgtggc 79440
aagctcagag gtcatgtcag atcacaacac agggactttg aattgggtgg acatccattc 79500

attacgtggc acacgtcacg agcttcaagg ttttgcttca aaaactcttc tgtcttctag 79560
gtgaaagtat aagtttaaac ttactgctct ttaagaaagt aaatgaaaaa tgacactaaa 79620
gtcccaaaag ccagaattgt cagcaatcct aggtgcagtt cattcattca ttcattcacg 79680
tattcattca ataaatattt atggagggcc tatttacctg gcagcacttc atgaggccct 79740
ggaaatacaa tagtgagcaa taaaaacaca ctccttcacc aggtggagct atagtctact 79800
agggagatat agatgttaaa caaattatca cacaggccgg gcatggtggc tcacacctgt 79860
aatcccagca ctttgggagg ccgaggcagg tggatcacct gaggtcagga gtaggagttc 79920
aagaccaggc tgactaacat ggtgaaaccc catctttact aaaaatacaa aattagctag 79980
gcatggtggt gcatgcctgt gatccaagct acttgggagg ctgaggcagg agaattgctt 80040
gaaccgggga ggcagaggtt gcaatgagct gagatcgcac cattgcactc cagcctgggc 80100
aacaagagcg aaactccaac tcaaaaaaaa aaaaaaatta tcacacaaac aagtacataa 80160
ttctatattg tgaagtgtcc caaagaaaaa tatgccactc ttaataagta caggaggctt 80220
aatttggaaa gttagaaaag tgatgtttaa agtgagaact gtaagacaag tcactttgtc 80280
agtgcaaagt ggaaagaaag tattttaagt agcaaggaga gcatgagtaa agaccacgga 80340
gaaggaaagc gaggatgcag ttagagacat gaaaggactg cattggtggg gcaccgggat 80400
gaaggagatg atgaagaaga tgtaaccaga aaggctggca gggacaaggt catgctgggt 80460
cttgcaggcc agcatgagca cttgagattc ttaaagtaat tgcaagggag cctttgaggg 80520
ttttaatgag ggcagtatct ttatcagatg tgcacttgtt tggggtttct ctggattttg 80580
ttgaaagaac aatttaagca gaaggcagat taggaaatag gagaatcgag aggctgtata 80640
tgttgagacg cagtggtagt ctagggtgac aatgcaggaa atgggaagca gtggatgaac 80700
tggggatatg ttttgaaggt agaatagatg atggctggga agacagtgac tattcaaggg 80760
caggggggtt gggggaggta tcaagaatgt ttagaatata gctaaccaca taccctttta 80820
gaaacagtct tctctgaaca tttctcctgt ctccaagcct cagtttcctc atctatgaaa 80880
taagaacact actacttcct ttttaaggtt gttaaactgg ttaaatgaga ttatgcaggg 80940
aaagcatcac tagtcagtgc tcaaaaatgt gctttttaaa tttcctccct ttgcctctta 81000
ttctcaactt tgtcctttgt aatattattg ttctttaagt gggcttggtt ttgtcctatc 81060
tttgcccatt cactcactgc tccccatcca cccaaatccc ctctgtattc tgtttatgca 81120
agactgagtt taccccttct cagtccattg acttatctct cctcactcat tgacttgtcc 81180
caggcaattt attctgcaat cttggacaaa aatctggatt ttcagccagg tgcagtggct 81240
cacacctgta aacccagcac tttgggaggc caaggcgggt agatcatttg aggtcaggag 81300
tttgagacca gcctgaccaa catgacgaaa ccctgtctct actagaaata caaaaattag 81360
ctgggtgtgg tgacgtgtgc ctgtaattcc aggcatgcct gtaatcccag ctactcggga 81420

ggctgaggca ggagaatcgc ttgaacccag gagacggagg ttgcagtcag gcgagctcac 81480

gccactgtat tccagcctgg gcaacaaagc gagacttcat ctcaaaaaaa aataataata 81540

attcattatg taatccagct ttgaaacact ctttggctac acttttgtat gctttaagga 81600

ggaacaaaac acagatggtc tccaacttac attggttaaa tctacaattt ttcagcttta 81660

caatggtgca aaaacaatgt gcattcagta gaaactgtac ttcaagtacc catacaacca 81720

ttctggtttg ccccttcagt acaatgttca atgaattatg tgagatattc aacactttat 81780

tataaaacag gctttatttt agatgatttt gcccaacctt aggctaatgt aagtgttcta 81840

agcatgttta aggtaggcta ggctaagcta tgatgttcag taggttaggt gtattaaagc 81900

aagttttact taagatattt tcaagttaca gtgggtttat tgagatgcaa cctcattgta 81960

agtcaaggaa catctgtact tcagaagtca tcaaagctgc atgagcagga cacaagtcat 82020

atgaaaagcc aggtagacat aatgctataa aaaatccctc cattgggccg ggcacggtgg 82080

ctcatgcctg taatcccagc actttgggag gccggggagg gtggatcacg aggtcaggag 82140

attgagacca tcctgactaa cacagtgaaa ccccgtctct actaaaaata caaaaaatta 82200

gccaggcatg gtggcgggca cctgcagtcc cagctactcg ggaagctgag gcaggagaat 82260

ggcaagaacc cgggaggcgg agcttgcagt gagccgagat cacgccactg cactccagcc 82320

tgggtgacag agcaagactc catcaaaaaa aaaaaaaaat ccctccattg tcagagtgtg 82380

agcttccagc tcattatccc agaagcccga gatagcagca gttctcagat cttgtgataa 82440

aggtcatctc ctatcctggg gctctcagga ccataatgca agagtctccc tctaaacctg 82500

ccagccccag ggcttttccc gccttcctca tcctaagtcc tgaaaagttc actgggccaa 82560

atggtgaacc acgcacttat tgccccataa cccttggtac aaatgtctcc aaatatatcc 82620

atcaagccta caggtagtac tgagaataac aacagtagct aacattgatt ggacacttct 82680

aagcccctta aatccattat cttacttaat tctcacaaca ctgatcaaga gttggataaa 82740

ataatccact ctcaagccag caaatctaaa ccagccactc ttccgtatgg attcctgctc 82800

ttatggtaac aagggcttgc cttccccacc tttattctta acccttctgg aaaacctctg 82860

ctcctccttt tctgagatgg aaaaatttat aagtgaaaaa ccattccatc tttcgaggtg 82920

tggagggagg aaaacaatca ctcctgcctt caactaagag tgtgaaaaat aagcttaact 82980

aaacctgaaa tacattttca aatgcctttg aaaagactta taaatcaaat cacatttgtc 83040

catctctctg ctcttcaaaa ttatcatgca tgcacctgaa gtttaagcaa agaaatccat 83100

taaacaaaca aacctaaaat cataaaaccc agatttagag atttatccgc tcagtctaat 83160

gaatgcccaa ttcagaatac aattttgtct tcaaagagcc ctgaaggttc ttatctttct 83220

tatctttcta tagtgttaac agaaatatta catctttgaa aagaagaaaa acattattcc 83280

cagagctaaa acagaaaagg ctttgaacta ttttagggat aaatcaactc acagttacca 83340
ataaaccaaa aagaataaaa aagactgttt caaaccaagt tgactactct tacatatatt 83400
caagtgtcaa cttacaaatc agtctttaaa tatacacgta cactttctaa ctctcctgaa 83460
atgtcaccca agcccccatt caatcagcta aaaacaattt aattctttct ctagggagga 83520
aatcaggtta tcagataagt aaaccttaaa taccatttct aggcctgatg tggtggctca 83580
tgcctgtaat cccagcactt tgggaggcta aggcaggtgg atcacttgag gtcaggagtt 83640
tgagaacagc ttggccacat ggtgaaaccc tatctctact aaaaatacaa aacttagcca 83700
ggcatggtga caggcacctg taatcccaac tactcaagat aatctgcata ccaattgtgg 83760
gtagacatag gtttttgtcc agagccctcc acagacccat cccttaccta ccattgtctc 83820
tcgggcttca accttatttg aaagtcttaa tttgcagttc cacatactgc aaacacaaga 83880
cccagtcttt ctggttctta ttttacctgg agattaaaat acaggctggg cgtggtggct 83940
catgcctgta atcctagcac tttgggaggc caaggtgggt ggattgcttg aggccaggag 84000
tttaagacca gcctgggcaa catggtgaaa cccctgtctc tactaaaaat acaaaaatca 84060
tccgggtgtg gtggtgtgtg ccgacagtcc cagctactca ggtagctgag gcataagaat 84120
cgcttgaaac caggaggcag aggttgcagt gagctgaaat ctcaccactg cactccagcc 84180
tgggcaacaa ggcaagactg tctccaaaaa aaaaattaag tttctgtctt acaatatcat 84240
aagaaaatgg ctggacaggt tttcaccaaa gttggagggt acttttgtga tgggtttggt 84300
ttaaattggt ttaaaatata agacacatag tccatagaga attcacctat ggactatgct 84360
gctaagagaa tctcaaagag atgcactgtt atgctccaga gttttgtgag aggccactaa 84420
ggtcaggaga cacatgccat atatatcaag atgctgtcaa cagagaaaac cagtgaggtt 84480
tcaaacagaa gccccgctcc attcaaccag gcagccactc ctcattgcag gtgctgacct 84540
gggctttggc tgcttctcac atgggcaact ctatacactc tattcctggg agaagggcag 84600
caaagaccca cttattaaat gatgtttaac aatcctcggc cgggcgcggt ggctcacgcc 84660
tataatccca gcactttggg aggccgaggt gggcggatca tgaggtcagg agatcgagac 84720
catcctggct aacaaggtga aaccctctct ctactaaaaa tacaaaaaaa ttagccgggc 84780
gtggtgtcgg gcacctgtag tcccagctac ccaggaggct gaggcaggag aatggtgtga 84840
acctgggagg tggagtttgc agtgagccga gatcatgcca ctgcactcca gcctgggtga 84900
cggccgtctc aaaaaaaaaa aaaaaatatc ctccaggcaa ttgtgtgaca gctggaatga 84960
aaaatcaggg gcaaattgta catataaggg aacaattgtt catatttgtg taagctaccc 85020
tccggagtct acaagttaaa aggcacactt taatcaattt ggcaacttgc atggcatttt 85080
cctccactat ttgtaggatg ctggtatctc cttaacagct actgttttcc tatgcaacac 85140
acaatgactt cttgaacaca tggcagcttt tcatttgttc atttaacaaa tacttattga 85200

gttactacta tgtgccaaac accattataa aggtactgag gatacagcgg tgaacaagat 85260

ggacaaaaat ccctgccttt gtactacatt cttgagtggg tgtgaggaga caatgaacca 85320

aagaatgaac aaactgtgta ttgtgcaggt gtgtcatggg aaaaaaatga aggagggaag 85380

aaaagcggaa aagcagaaaa tgtcaggaat gcacttccgt ggcgggtggc cagacaggtg 85440

gccaagaagt gacatttgaa ctaaagaagg tataagtgag caagctatga gggaatttgg 85500

caaaacaatt tcggaggcgg aggtcacagc cagcaggtgc aaaggcctgg ggcaggagtg 85560

ggtccagggc atgggacgga tggggagaag gtcagcatgg ctgaaggaag taggggggtaa 85620

gctcaaacaa gtcgcaggtg ggaaactgag tgtattggac cttgtaggca attttaagaa 85680

ctttagtttc cactcattaa catggaaaac cactggaagg ttttgagcaa aggaataaca 85740

taatctgcct ttttttcttc aaatgctgtg aaacaaatac ttatttgacc ctatcaccat 85800

ttctaccttt ggaaaggcta tggtgtgtta ctggatgttg aggatagctt actcttcaat 85860

gtgcagtaac caaactgaat tcattctttc tcaagatgag agaaagataa gccaggtatg 85920

gtggctttct tagataagcc aggtatggtg gctcacgcct gtgagctgag gcgagaggat 85980

cacttgaggc caggagttga agaccagcct gggcaacata gtgagacccc tcatctctta 86040

aaaatttttt tttagttagc cagacattgt ggcatccgcc tgtaatccca gctctttggg 86100

aagctgaagt gagctatgat cacgccactg caccccagcc tgggtgacag agtgagaccc 86160

ccatctctaa ttttgaaaaa aagactggat agggcctggt taatacaact aactccccaa 86220

aattcaagtt tttcatatag gtcttttttta aaaaatagct ttaattgaca taaaattcac 86280

ccatttaaag tttacaattc aatggatttt tatatattca caatgtagta caaccatcat 86340

tataattttt tttttttttt tagacagagt ctagctctat tgtccagagc tggagtgcag 86400

tggcgtgatc tcagctcact gcaacctcca tctcctgagt tcaagcgatc ctcccacttt 86460

ggcctcccaa gtaactagga ttacaggcat gtgccatcac gcctggctaa gttttgtatt 86520

tttagtagag atggggtttc accacgttgg ccacgcttgt ctcaaactcc tggcttcaag 86580

tgatccacct gcctcgggtc caaagtgctg ggattacagg catgagccac agcacctgcc 86640

cgtaatctac gttagaactt tttatcatcc catcacccat ttaagtcttt acccattagc 86700

catcactccc cattttctca acccttcccg caaaacccct acccagcctt gggcaaccac 86760

taatctactt tctatacatt tgcctattct gaacatttca tatcaatgga atcatactac 86820

atgtgacatt ttgcatctgg cttcttagaa taagattttc aaggttcgtc tatgttgtag 86880

catatatcag cacttcagtc cttttgagtt tttttttaac aatctttacc attttcaagt 86940

gtatcgtttc atggcagtaa gtatattcac actgttgtgt aaccataccc accatccatc 87000

tccagaactc ttgtcacctc tccaaactga aactctgtac ccattaaaca acaattccca 87060 ttccccaccc caccccagtc cctggtatgg caaccaccat tgtactttct gtctctataa 87120

atatgactac tcatacttca ttttttaaat tgccaagtaa tgtttcattg tatggatata 87180

ctgtacaaca atttaactat ccattcagct aatggacatt tggggttttt ttaacctttt 87240

ggctatcatg aataatgttc tgatacttcg tgtgtgtgta tagatagata gataaattaa 87300

atagaagata gaagagagag agagagattg gagacagggt ctcactctgt cactcaggct 87360

ggaatgcagt ggcagggaca catctccctg cagcctcaac ctcccaggct caagtgatcc 87420

tcccacctca gcctcccaag cagctgggac tacacgcatg tgccaccaca cctgactaat 87480

gtatgtattt acttatttat ttatgtattg taaagacaga gtctagctat gttgcccagg 87540

ctggtctcaa actcctgacc tcaagtgatc cttcctacct cagcctccca aagtaccaag 87600

attacaggca tgagccccg tgcccagcct gatacttttt tttttaagta ttattccagt 87660

tgccttgttg aaaatagacc ccaagaaagc aaatctcaaa cagagaaaac tgctaggaag 87720

ttcttgctgg aatccaggtg agaacggata gaggctcaca tttaaatgaa gtagtcagaa 87780

atagccacat ttggatgtat ttttatacaa ttcctgctcc tgaagtcttc cccactcctt 87840

tttttttttt tttaaccatt actacaattg ctttgctgcc tttttgctga tttattggat 87900

cacgtgttta aaaccctgat gtgaacacct acatttatcc ttcttactgg gtatgtgtta 87960

ggtatttaac aaagtcttag ttctcctgga gtctgcctgc atgaaccaac caaatataaa 88020

tctgcaaaat gggaactcta cagtgtctct tcagttttgc tgtcaagatt tcacagcctc 88080

agcttctaaa attatttcat caagttcaat ggatacatat tcttgaactc ttttctagcc 88140

tatattttcc aacaatgttg ctaactatat ttccatacca gccttcttat ctaacatact 88200

ggttaaaatg tcaaaaagca gagggtttaa aaagcttttc tcggtggaat gtgcttctcc 88260

ttcatacatg atataacttg atttgaacaa tgtcacaaag atattttctc tgttagatta 88320

aaattttgtt tgcatgaatt tttcaatagc tttaagcagt tgaatagcaa tatatgcagg 88380

aagaagctga gagacttatg taatagatat ttcatgtatc tataacccac actgctgccc 88440

aggaaatgtg cgctgcatta atagagagga ttttttcctg ctgaatacct tgaggagttg 88500

gccaacacgt ttgggagtag aagtagaaag ggccaggtgt gatggctcat gcctgtaatc 88560

ccagcactct gggaggccaa gtggggagga ttgcttaagc ccaggacttt gaggccagcc 88620

tgggcaacag agtgagactc catctctaaa gaaaaaaaat cataaaaaac taaaattctc 88680

tgccaaaatg gacacagaaa aaactgacaa tccagagaaa gataatatgc aatgaagcta 88740

gacatggcca aattagaaaa tgatattgag agagaacaag agcaagaaag aggagccctc 88800

agcattgaga gggctgagga agcacagaaa tgactgatgg gttggttagt tagttacttt 88860

ttgtgaagtg tgcaatgtaa atttcacttt ggtctcccca ccggaatcat caactaaagt 88920

ctacactgct atatcggcta tctattgctg tgtaacaaat tattccaaaa ctcagtggct 88980 taaaacaaca catttattat ctcacagttt ctgtgggtta ggaattcgaa gatgggcccc 89040

ctgcttcagg gtctcccatg ggttgctatc taggtgtaag ctaggtctca tctcaagact 89100

caactggggc aggatccact tccaagtgca cccacatgat tattggcagg attcgtttcc 89160

catgaactgt tgtcagaagc cgctttcaga tccttgccac gtgggcctct ccgaagggca 89220

gctcacaaca caacagcttg atttatcaga gcaagcaggc aatgaggcag aacagggacc 89280

tctcttaggg acatgcagca ctcccaccct caaacataga aataaagaaa aatcttaagt 89340

tcctttaaga aaaattccag gcacttagct agcccttaaa aaataaataa ggccgggcac 89400

gggggctcat gcctgtaatc ctagcacttt gggaagccga ggcagttgga tcacttgaag 89460

tcaggagttc aagaacagct tggccaacat gacaaaaccc catctctact gaaaatacaa 89520

aattagccag gagtggtggc gcatgcctgt agtcccagct actcaggagg ctgaagcagg 89580

agaattactt gaacccagga ggtggaggtt gcagtgagcc gagatcatgc cattgcactc 89640

cagcctgggc aatggagaaa gactctgtct cagaaataaa taaatacatc aataaacaac 89700

ttaataagca agaagataat agtagcttag aataatgggc aaaaaagtta aaatcatggg 89760

atgtttggct tccctataaa aactaatgtt catagattgt ttttcaaaaa tgaggactcc 89820

ccactaaatg ggtccagcaa cacacaaggt ccagcaacac aactcagata aggggggacct 89880

gaaggctaaa ctcttaacac ttttctcagt tctaaatttc ttcctaaggg gagtagagga 89940

agtcacaccc caggccagaa ctaacattcc actgatctca aatttttaga caaggcttct 90000

cctcctaagc caattacaaa tcaaaacatc tttaaatcta cctttgaccc atgggttccc 90060

actttgagac gtcctgcctt tttaggtcaa accaatgtag agcctcccat atattgattt 90120

ataactttgc atgtaacctc tgccttcctg caattacaaa tccttaccta taagccatcc 90180

gggagcttgg gacttaagca ttaactaatt atctttgctt ggtgcccctc caataaatac 90240

cccacttcct cttgctacaa tcccaatatc aatgtttggt tttgctgtgc tgggcagggg 90300

gacccaagtt aggttcagta tcagcaagaa ggcaagacag agtgtgtgct agcaagacag 90360

aagtccgtgt gtttggtaac ctaatctcaa actgaaatgc catcaccttt gctgtgttct 90420

actgattaaa agctagtcac ccatatgttc attgcagcac tattcacaaa agcaaagaca 90480

ttgaatcaac ctaggtgccc atcaatggag aattggaaaa agaaaatgtg gtacatatat 90540

accatggaat actacacatc cataaaaagg aacaaaatca taccctttgc agctacatag 90600

atgcagctgg cagccatttt cctaggtgaa ttaacgtagg aatggaaaac caaatactgc 90660

atgttctcat ttataagcaa gagctaaacg ttgggtacac atggatataa acatatgaac 90720

aataaaaact ggggactaca gctgggcatg gtggctcaca cctataatcc cagcactttg 90780

ggaggccaag ctgggcggat cacttgaggt caggagttca agaccagcct ggccaacatg 90840 gcaaaaccct gtctctacta aaaatacaaa aattagcagg gcgtggtggc aagcacctgt 90900

aatcccagct actcaggagg ctgaggcatg aggatcactt gaatttggaa ggtggaggtt 90960

gcagtgagcc aagatcatgc cactgcactc cagcctgggc aacagagcaa gactctgtct 91020

caaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaacggg gactgctaga tgagagagaa 91080

agggagagac aaaagggctg acaaactatg ctcactattt ggatgacaga accagtcata 91140

ccccaaacat cagcaacaga caacataccc atgtaacaaa cctgcacatg tatcccctac 91200

atctaaaata aaagttgaaa ttataggctg ttcgcagtgg ctcatgccag taaccccagc 91260

actttgggag gctggggcag aaggatcact tgagcctagg aaatcaagac cagcctgggc 91320

aacataggga gacttcctct ctacaataaa attaaaaatt agccaggcat tgtgatgtgc 91380

acctgtggtc ccaactactt ggaaggctga ggtgggagga ttgcttgagc ccaggagttc 91440

aaggctgcag tgagccgtga ttgtgtcact gtactcatcc tgggccacag agcaagaccc 91500

tgtctcaaaa aaagagacag ctcactgtca gctcactgca acccccacct cctggattca 91560

agcgattctc ctgcctcagc ctcccaagta gctgggacta caagagcgca ccaccatacc 91620

cagctaattt ttgcattttt ttttttagt agagacaagg tttcactatg ttggccaggc 91680

tggtcttgaa ctcctgacct cgtgatccgc ccgcctcagc ctcccatagt gctaggatta 91740

caggcgtgag ccaccgtacc tggccgagaa attatttttt aagtgaaaat aaaaaaataa 91800

aagctagtca ctaggtccag cccacagtca gggcaagggg tcacaaaagg acgtgaatat 91860

gaggtggcag ggatcatttg gggtcatctt agaagctgtc taccacaatg ttccactatg 91920

aattatttca gaggattcac acctggggca aggaggtaca tcgatagcat gcaaccagaa 91980

ggagtcctga gacagtcatt tgcagaaacc ctggtgaagt tttggttttc cttggccatg 92040

caatcagggg ctactcactg gctgatggac tcagctgaca cccagattaa tttgagaact 92100

gtatccacag tcattaacta cggggcagtg cccaactgtc cccaagccag aagtaagagg 92160

tatgcatctt cacgacctca atgacaaact tgatcatata aaataggcag cattaattga 92220

cctggttcag gctacgccgg aggtggacta ccacctggaa acagagagac attgcagaag 92280

cttccctgag gttccccttg cagccttcag ggtgaattc tttgacactg ggtgtttgaa 92340

atggagcaat caaggcccca gcaaaaaaca gatgcacact ccaatggggt aattgaggac 92400

agtttaggaa agagattatt tacagaagtg tggacaggat taagagaaaa caaggatgtg 92460

gagcacgctg gtattatcaa cagtggggtc taaaagggaa ggagagagag agagagaatc 92520

ctggaaccca gagagagctg tcgctgtaag acagaactgc ccaacaggaa ctgtggcctt 92580

tagggagaaa ctgagctact gcaaactctc agcccggaag gaaagaagcc aaaggaataa 92640

ataccctgac ctccccttcc accctccaat ctcctactgg gtccccactg gccagtctgg 92700

ctccctgggc acaccagagt tgagaagaac ggacagcgga tttggaggcg caagaagaaa 92760

gtatccaaaa cctagacaga aagccaggca gcttcacatg agctttttaa aagtgtagaa 92820
gtcataaaaa ttctcctgga aatgagcaaa cacatgggct ttcagaagat gaacacaaaa 92880
tagcttttaa aatgaaagtc tgctctggaa gggaaaaggt ggttcctggc aatgtgcaga 92940
ggaggattaa agtccccacc ctcaactcca gcctgctgcg tcttccaaga gagaattccc 93000
cagagcctgg actagggaag cattggcgcc actgggaaaa gctgctacag cccttgttgg 93060
gcggctcttt ctcacagcta ctgaagctgc ctttgcaaag attatgacag tgagagaatc 93120
tggtgttgct gactccatct tgtttctagc ctcacaggct aactgtcctc actcattcct 93180
gggcataggc caagttaatc atgagatgaa ttttttattt ttatttattt gctttttgag 93240
acaaggcctc actctgtcgc ccaggatgga gtgcagtggc atgatcacgg ctcaccacag 93300
cctcgaactc cctggtctca gggtggtcct cccacctcag cctccagagt agctaggact 93360
gcaggcgcat gttaattttt ttgtattttt tgtggagatg gggcttcgcc atgttgccca 93420
gtctggtctt gaactcctgg gctcaagcga tctgcctgcc ttggcatccc aaagtgctga 93480
gattgcaggt gtgagctact gcacctggat aggtatttag tttatagttt aatttgaaag 93540
caaggatgat aatagtgttc cactaaaact gattccctca ttgtttcagg gctgaaacca 93600
cctttgtaaa actaaggaaa ggccacaaga ttagggaggg gcctgaattc tgctaaaatg 93660
gaggcatagt cttaggcata gttttagaac cagccattgt tctataagtc acaagatttg 93720
tgacttcccc aattgctcct atagataaca tcactattat agaacctagg attggtcttt 93780
tgagatgttt ttcagacttt gcattctggc aaataactga ccccacctgg acttgtaact 93840
catgactcaa ctggtcctgt ggcccctacc cagaggtgga ctcagagcac caggaccatt 93900
tcccacacct ctattgcatc cccaactaat cagcagcatc cattgcctag tcctctgccc 93960
accaaactat ttttgaaaaa ccacaggctg ggtgcggtgg ctcatacctg taatctcagc 94020
actttgggag gccaaggtga ggggatcacc tgaggtcagg agttcgagac cagcctgacc 94080
aacatggaga aaacctgtct ctaccgaaaa tacaaaatta gccgggcatg gtggcacatg 94140
cctgtaatcc cagctacttg ggaggctgaa gcaggagaat cacttgaacc cggggggcgg 94200
aggttgtggt gagccaagag cacaccattg cacaccagcc caggcaacaa gagcaaaact 94260
ccacctcaaa aagaaaagaa aagaaaaacc acaacctcca agtttttggg gagactgatt 94320
tgagtgataa ctccagttct tccacatggc cagcctcaag ttaattaaac tctttcttca 94380
ctgcaatacc acagtctcag caaactggtt ttgtctatgc agtgggtagg aaggttgagt 94440
aatcacacta cagttctttc tggattccag gagtctctct ccccttccct tttagacctg 94500
gttggtaagg gctctgtgct gttgacagtt ccagagtacc tcacatcctt tgtttcattt 94560
aacccagtcc atgcttctgt aaatactctc tttgctaaac tctcctcaag taattgattt 94620

taatgtgcat ccattgcctg ccaggacctg attagtccat ttcaagcatc caatatcaaa 94680
gaatgcaccc aaaacaaaga cgctgttagg aaacacagaa ctgggcactg caaaggtctt 94740
tgagcatgac acaaggatgt tgcaagagcg gcaggcccag aaggcaagtg gggattgaag 94800
aaccttaaca gcatattcac ggggctcagt cctaggattc agcatcatga catatcacct 94860
ccaagcttcc atgctttcat ttgtatagct aagacttgac aagactagtg agggtgtccg 94920
atgatgagga cagtcaagga tgttaaattc ctgactatac attaacaagg aggaaaaaaa 94980
ctaacaaacc tgcatcagtg tgaagaccag aaaggcctgt ccacagagcc agatgacaag 95040
gtacaacagg atgctgttga aaaaaatcag actagataag agcaggttgc tgtgtgagca 95100
aactcagacc atacaagaga actataggcc cctctcctcc agcccagaat tcctgccctg 95160
agtgatgcta aatcagaagg aagaaaagca ggagatgata ttggctaccc tccggggaca 95220
ggaagcatca cctggttgga ctaccagact gccattcaca cttatataca tggagtcagt 95280
cgggtatgac tcagtcccct gaggccaaat gtcagcgagg agagtgggag gagcaaatct 95340
ctctcctcca ttagagttgt ctccggaaca aacccaggtc tcaaggcaaa ggcctcacac 95400
actattttgc tattttgtca cccacgtaaa gttttcagaa acaggatccc tgtagctcat 95460
caggcactca ggtgcatcaa agctgaattc agggtaaaga ttgatgctgt ggcccagtga 95520
agctgagatg cccatactct ctctgttcag gttatagaga aaatgggcac tttgtgatca 95580
cttataccca taataaaaaa caatttgtgt gcatctcatg agcaagaaaa ataaacagga 95640
aaaaagaaag caacccaact acttgtaagt ataaggaaat ccaacccata tttgttcata 95700
taactgaaag gtccaggggc aaacctgcag gtatggtttg atgcaggtgc caacatcttt 95760
tgccaggacc cagtgtttct cacacccttt ctttttctt tttttctttc cttctttctt 95820
tctttttctt tttgtctttc cttctttctc tttttttttt ttaacaggat ctcactctgt 95880
cacccaagct ggagtgcaat ggtgcaatct cagctcacta tagcctcaac ctcccaggct 95940
caagcaatcc tcccacctca gcctcctgag tggctaggac cacaggcatg caccacaatg 96000
cccagctaat tcttttcatt atttatagaa acagggagtc tcactatgtt gccagggctg 96060
gtctcgaact cctgggctca tgtaatcctc ccacccacct aagcctccca aagttctggg 96120
attacaggtg tcagccacca tgcctgacct cacaccattt cttaactcca ttcttctcac 96180
tccatttctt aatttcattc ctttataaag cttctcttct tactatttca agatggctgc 96240
ccaattcatg tgcagaggaa agagaagttc tttctcttta ctctgacagt tgaataaaaa 96300
atccaaagcc tggctctctt tggtccatcc ctgaatcagt cattatggcc tggggaatgg 96360
agtatgctaa ttgacttaag ggaatcaggg cccagcactg gagtgaaggt ggggctaatg 96420
ccacctaatc cactggagag taccaaaagt gtgcttcccc aaaggaaatt cacaatactg 96480
tgggaaagga tgaattgatg ctgagtcact atgaatgaca aatgcaaaag ataaacatac 96540

caggccccac tccttgcagg aagcaaaaga tcctagaggg agaggctgac atggaacagg 96600
atgtctgacc aataaaactt cttccaatga ggattcacag acatagtcat accttccagg 96660
ttaagtaagg ctcaattcca ggcagctgtc tgtctcagct cctcatgcac atccgtcgct 96720
tctgtctacc cagcatttgt ttctccctta ttcagttctc attgctgtgt aacaaattga 96780
cagaagtgca tcaactaaag caacacaaat gtattatctc acagctctat aggtcaaaat 96840
ccaagcacgg ctcaaccgga ttctctgctc agggtctcat ggggctgaaa tcaaggtgtc 96900
agctggagct gtagtcttat ctaaagctca gggtcttctt ccaggatgat tggttgtttt 96960
cagacttccg ctccttctga ttatcttgag ataggaggca ggacttgact ctggaggtgg 97020
ggcttggaca ccggaccaag ttcaggacta actaaaacag agctgggagg gaagcagctt 97080
tccctaagac acacccacca gtgtgccagg tcagtttacc attgacacgg caatacctgg 97140
gagttaccac ccctttccat ggcaatgacc tgatgaccta aagttactac cccttctcta 97200
gaaagttctg cagaaaccac ccttgaatct gcatataatt aaaagcaggt ataaatatga 97260
ctgcaaaact gcccagagat gccactctct ggttacaggg tagccctgct ctgcaggagc 97320
cgtcatggag ctgtaacact gcaggagctg taacaccacc gcttcagtaa agctgttttc 97380
ttctacctcc agcttgccct tgaattcttt cctgggcaag gccaagaacc ctcacaggct 97440
aagccccagt ttggagttca tctaccctgc atcaatatga ctgaggtctt ttttcttgct 97500
ggctatcgac cagagacctc tctcacctcc taaagacaaa cctaggtcct tgccctgtgg 97560
cctctccata ggctttctca cactttgagc atctctgact tcaggaaggg cctagtccct 97620
tttaaaggtg cacctgatta ggtcaggccc acccagatgc tctccctttt gattaactca 97680
aaatcaacta actagtagcc cagtcagggc agggctattc catcacatcc tctaattgtg 97740
cagcactgga gaggaggaga ttgcacagat tgtgcacacc agggagcagg gatcttgggg 97800
gccatctcag aattctgcct accacatgtt agtaattgat ctcttctgga atcagtcctt 97860
tcctgctcag tcaatgttat ttgtggtagg atttagggta atcagaccta gccttggcca 97920
attggtgtaa ccattgtctg ggccacagtg gtcagatcag aaatgggcac ctgcctcaag 97980
caagcccaat cagacctaac ctggggactt ttgctgaagt aactaggaaa gatctctctc 98040
tctctctctc tctctctctc tctctctctc tctttcttgt tattgcggaa gtgttaggct 98100
gtaaacttag agctgatggt ggacaccatg tgctttgatt caaatgaccc cccaaaactc 98160
atgttgaaac tgaatcccaa aagtgggagc attgagaggt gggaccttta agaagcgatt 98220
gtatccctct tgagggcaga gcctttactg accacctaat gttttattaa agcttgtctg 98280
caccacccct atccctacca tcaccatctg gagctctcga tcctttcctc cagatttctt 98340
tgctccataa tatttatcac tatctgatac aggggtgtgt ggatgtgaac atgtatatgt 98400

gtgcatgtgt gtgtttatgg tccgcctctc ccagtccccg gtagactgta agcttcatga 98460
aagcagggct ttgtgtactg ttgtttctgt agcatgaggt ctgccaaaaa agtaggttcc 98520
ccataaatgt ttgttaaatg aatatattct gattctatct ccccctcctc cagaatggat 98580
ctaaagcttc ccaatctttg cctgagtgag tgtcctactg agttcaattt tgtttcagaa 98640
aggacgtgtg tatgttcacc catatgctgc acaaatatgt aatggacacc agccatgacc 98700
agcctactgg ggactgcttg gcagtagcaa catggcagcg aacaagacag atgagcctcc 98760
tgcccacaga agcttacatt cgggtggaat acaggctggt gagttgtcat gccgattcct 98820
ccaacccaaa acccttgcca gtatggcagg tcctggagct ggactggcca ccattgagaa 98880
tccacccaac tgcccatcag catctcaagt tccctgttgc aagatgccag ggagcatgag 98940
agaggaccgt gggggagagt gagctctcaa tgaacctggt gtggcaggga gaagacgaag 99000
atggggcaga tccccagggg gaagacaact taagggcaga tttgttgaac taagttactt 99060
caaagggtct tccatcttcc agcatgagca gagaagatga ccacacacct gggggagagt 99120
cagatggctt cttcaattgg aacgttcttc cctaaggatg agctaatata gcacatgacc 99180
tgaaaatcct tctgggagac tccaggtaaa agggccttga ctgcaccagg gagagaagct 99240
aagttgtaga tttgggggaa atgagtgtta gaacagtgag tgctcaaggg aagtagttaa 99300
gtgagccatg agatacaaag tctgccatgg caaagtaggt ggaacaacca tgttatgagt 99360
ataagttctg accccagctt tcctgtgatt tctacgaccc ttggtcttga agatgggaaa 99420
taaatgggtg catgacagaa gttaaagact attatcgggt atcaagtacc atagcacttt 99480
ttggaggcag aagagcttga caaatgggca tagtgtacat agtaaatgat ggtaagggta 99540
aaggataaaa tgtatcatct tcctgtcatc agaatgggac cagcactgtg taaggtgggc 99600
atgttaatga catgatatat aagacagctt ttgctacaat aacaaggaac tccctaggcc 99660
tggcacggtg gctcatgcct ataaccccag cactttggga ggctaaggcg ggtggattac 99720
ctgaggtcag gagtttgaga ccagcctggc catcatggcg aaaccctgtc tctaccaaaa 99780
aatacaaaaa ttagccaagc aaggtggcac tgcctgtaat cccagatact cgggaggctg 99840
aggcactagt ggtgcttgaa cccaggaggc agaggttgca gtgagccgag atcactccac 99900
tgcactccag cctgggtgac aaagtgagac tccatctcaa aaaataacaa taacttaaaa 99960
aaagagaaat agctctcaaa tcccaataga tccctataac aaatatttgt ttcttgcttc 100020
tgagtcttca gattggctgt gattacactg ggtttggctt ggctgagctc agctaggttc 100080
tgctgagctc agcttggctc caaggtttgg gttgagttca ggtctgctcc acatgtccct 100140
tcacatgaga agcaattgcc aagaggtcaa gtcaaatcat gcagcacatt taaaacttct 100200
gctgggggaa aatgtgtctg ctcacattcc aatgaccaaa gcccaaagtt agaaggcaa 100260
ggaattagac tctacctatt gagatgtact acaaagtcac atggcaaaaa gcttataatt 100320

ctaataaagg aactaagcag aatcactagg agcaatcacc tagtccgcca cacatggaga 100380
tgtgccacag ggactcagag actggaagga attccccagg gccagagttc tcctgggaaa 100440
ctacagcctc caccattgcc tcccagattt catcagcatc tctgtagtct ggctcatcag 100500
aggccacaac ggagataaag gcaaataaag acttcagctg ctggcaagct gcagatatct 100560
ccatggatca gccaagccca tgtctctttc tgaaacaatc agtaatcggg gaagcgacca 100620
cagaaaagcg taatacaaac tacctatggt attggaagaa tcccaggaat cgttggaggt 100680
cttgaatgaa tttgaagagg gtactcagtt caagactact ttaagacaca cactttgtag 100740
atgtcccaac tagatactgt gtggcctggg aattctgcaa tgtttacttt ttttcaatta 100800
tattatgtta taatatattt ataagaaata tatatcatat ataagattat atatagctta 100860
tatatctata tataatctag atatattata tatctaaata taatctttga tatatatctc 100920
atatattatc ttatacatat gatatattat catatacatg agttatacgt atctcatata 100980
tattattttc ttatatatat gaaatatatc tcatatatat aagatatgag agctatatat 101040
ctcatatata gatagatata gatatctatc tacatctata ttcccaggcc acacagtgtc 101100
tagttgggac atctacaaaa tgtgtgtctt aaactagtct tgaacgagta ccctcttcaa 101160
attcattcaa gacctccaac gattcctggg attcttccaa taccataggt agtttgtatt 101220
acgcttttct gtatatgaga tatatatctc atacacagat agatatagat cattccatca 101280
cccaggctgg agtgcagtgg cacaatcata gttcattgca gcctcaaaat tctgggctca 101340
accgatcctc ccacttcagc ctcctgagtg gctgggacta caggtgtgtg ccaccaggcg 101400
ggggctaatt tttctttttt tggagacaga gtctcactct gttgcccaga ctcagatgta 101460
gtggtgcaat ctcagctcac tgcaacctcc acctcccagg ttcaagcagt tctcctgctt 101520
agcctcccaa gtagctggga ctacaggcac gtgcaaccac tcccagctaa tttttttgtat 101580
tttactagag acgaggtttc accatgttgc ccaggctggt cttgaactcc tgagctcaga 101640
tacctgcccg cctcagcctc ctaaaatact gggattacag gagtgaccca cccactgccc 101700
ccagcctttt tttttttttt tttttttttt ttttttttttg gtagggacag gatttcccta 101760
tgttgcccat gctggacatg aactcctggc ctcaagtgat cctcctgcct cagcctccca 101820
aagtgctaag attataggta tgagccacca cgcctggccc aatatttcta aaggcccctc 101880
aagaggcaaa agtgggcaaa ggacttttaa ggacaaaaaa tgcctagtat tgaatattaa 101940
gttgtttact gtgttgtaac tctcttccat gacttcagta aagcagagta agcacacaca 102000
tgccccaaga cacagtaacc tctgtttggt agtacagccc atcacaagca gctgtggctg 102060
aacccctggg taccaggact agagaggata acttcattgt aagttccact ggatgttagg 102120
ctgaagttgc agccccccatc aagccaggct tggtgtctgg agcgggcctg aaggtagaga 102180

ttctgcaaaa ggatctaact ggggtgatat tcaaaaaatt tagaagccag aatggcacag 102240
gtcctgactg ataagaacca gagccaacat gctgtgatct gcatacaagt gccagtgatc 102300
cctgcctcca gctgaggagg gctgtgaagg atggaggcta gaaatccatc aggagagctg 102360
acattgaaag gtgaatggga gcagcacctg ccagcgtgac ctgctgcttt cctgcatgca 102420
aaggactgtg catttcaacc tgagaccctg gggttcctgc agggcatcag ctagatttgt 102480
cttcctgcct ggggcgatca ccaattatat gtgctgtgag gcaaactgtc ctcccaccag 102540
aggaggaggt gaagggattt acagaatctc tctctgatct gacttattag agaagttaaa 102600
ggacacctgt ccaggaggca gcacagtgca gagataaaga agtcaggatt caaatttgtc 102660
ctatctggga agcttagaca agagctcctt aacctctctc acactcagtc ccctcatctg 102720
gaaaatggga atcctaacag tgcctacctg gcaggagcat ggccacaatg aaatgagcta 102780
atgacacagt gaaagccctt agacagcctg acccagagtc agccctctgt aatgggagtc 102840
attattcaag atgggaggaa agaaacagga attagatcag aaacaaatga ccagaaagag 102900
agatgaaata caaaagctac atgaagtcgg cagtgtgaag cctgactcgc aggagagtgg 102960
attttgttac tgtttgcttt ttcttgtcca ggataatcca agactggcag aaagtgagaa 103020
taccgattga gatccaagga catcctcatg ggagtctgtg cagacacgtt ttcatcaaga 103080
acccctctcc aagcagctgg tccagcccag ctaaactggg ggctgctgtt tgtaagaaaa 103140
ttaatgctct gggccaagca cggtggctca tgcctgtaat ctcagcactt taggaggcca 103200
aggcaggcgg atcacttgag gccaggagtt caagaccagc ctggccaacg tggtgaaatc 103260
ccgtctccac caaaaatgca aaacttagct gggcatggtg gcgcatgcct gtaatcccag 103320
ctactcagga ggctgaggca ggagaatcac ttgaacccgg gaggtggatg ttgcagcgaa 103380
ccaagatcat gccactgcac tccagcctgg gtgacagagt gaggctccat ctcaaaaaca 103440
aaaacaaaaa aaattaatgt tctgtacctg aggagcaccc atttgctgct acttccctgc 103500
caggatgaag gaaaaccaag tcagacatta aataacacgt gcaggatcac aggatcacac 103560
ctttcacact gcagtcttag ttttctataa aaccacgtga cctctgaaaa tacactccag 103620
cctctgcaac acctactcat gacgctttgt aaaatccacc gccttttagg gacaccaaga 103680
ttttcggaaa catgcagttt tcttacctcc agatgagatg tctactgtgg ataggtgact 103740
aggaggagaa cccaaggtgt gctgatggca gaacagaagc acctaggatg ccaccaggga 103800
aacgccctga caagcagctg tgatgctgtc ttcgagaagg tgtttgcaaa caccactgct 103860
gcccccccttc cctgagccct gacttcctaa acctaaggct aaaagcatcc tggcaattcc 103920
cggaagtaac ttcattctag caggatcgag ccagtgggtg gagttttctc agcccgccag 103980
gatcacatca gtgactgact tacccaatgt acttttattt tttatttcaa ccaattcccc 104040
aaagcccaga gcaacttaaa accagaagag ccaccaccat ctccaccaaa aacaggaat 104100

attttgagag tttaatgata acttcacagc catccattca gctgagtccc aaggaaatgg 104160

aagacacctg aaaaatgtat ttttaaattg atttaacgtt gagccatgtt aaatgtttta 104220

acttccaacc aatgcaatgc ccaccaaaac atcaattagt cccagcaatc aggcaagaat 104280

attggcttct gctcagggat tcatttattt cagctctgct aaatatcgta gaggagaaat 104340

ccaatgtagc ctgtagcttc aagaaagtga aattaagccc aagtagaaac tagatgaggc 104400

cctggctaaa atggtgcctt ttcttgcact gcctctctct ggtatctctc taaagtgagg 104460

gcagcaactt gattgtaagg gtctcgtgct gttaacagaa aacacccttc cttccccaca 104520

ggttctctgt ctccaagctg gaaaacaaaa gaggctgcaa tgggacagag aagggcttcc 104580

tgctgaggag atgagtgggc ttaagaattt gcaagatcaa aaatcaaaac aattaaaccc 104640

atggagatag agtagaagga gagtgaacag aggctgggaa gggtaatggg agagttggtg 104700

gggaggaggc agatggttaa cgggtacaaa aaaaacagaa agatgagtaa gacctacttt 104760

tccatagcac aacaggatga ctatagtcaa cagtaatttt tttttttttt gagataacgt 104820

cactctgtca cccaggctag agagcagtgg catggtctct gcccactgca acttctggct 104880

cctcagttca agcgattctc ctgcctcagc ctcccgagta gctggaatta caggcgcaca 104940

ccaccatgtc cagctaattt ttgtattttt atagagatgg ggtttcacca tgttggccag 105000

gctggtcttg aactcctgac ctcaggtgat ccacctgctt cggcctccca cagtgctggg 105060

attacaggca tgagccactg cacccgaccc tatagtcaat aataatttaa tcgcacattt 105120

taaaataact aaaagagtac aattggattg tttgtaacac aaaggataaa ggtttgaggt 105180

aatggatacc ccattttaca tgatgtgatt attacacatc gcatgcctgt atcaaaacat 105240

ctcatgtacc ccataaatat atatacctac tatgtactaa tgaaaattaa aatttaaaaa 105300

aattaaaaat aatttgcaag atcattttta ataattattt ttcatttggg gtgcattttc 105360

accatcaagg ccccagggtt tccctgccct agatgagggt ctggatccag ggaagggtgt 105420

cccaccatct ggagcatcct gtctctacac atgttggcca gctgcttggg gcaacacagg 105480

acgatgcgct caggccactc accacaggag cttggccatt agaaccagca ggggacaacc 105540

aagccaaaga ccccacgact ggagagctcc taaaagcaaa tacaaaataa gagtacacat 105600

gaagttctgg cagcaagcag aaggtagaca catgatgtcc aactccccga gtctctctct 105660

ggggccactt gatgcagagc cttgatttgt tcgtctctta aatgggaaca atgatcccca 105720

aagaggagga ggggtggggg agtgagaatt aaatgagatt gtgtcagtca gcatgtacta 105780

gggagcttgt tagaaacacc aattccagga tcccctggag attctgattt agcaagtctg 105840

ccaggagacc caggaatttg tattttaagc atctctccca ggtgattctg ataattaagc 105900

aactttagga aagaaaagat tagggctctg gtccatagga agggttcatc caatggcagc 105960 cctcacttag gcccttgggg tgaatgaagg aggggcatcc aggatcttgc ccagccaggg 106020

acatgcagga aacatctttt taaggggccc tcatcacttt ctgacactgc tgtttcaagg 106080

gaaaactggt cacacctttc tgcaggggat ttgacaacat ccattaaaaa aaaaaattta 106140

agacagagtc tcgccctgtc acccaggatg gagtgcaatg gcgccatctc ggctcactgt 106200

agcctctgtc tcccaggttc aagcgattcc tctgcctcag cttccctagt aactgggact 106260

aaggcatgtg ccaccacacc tggctaattt ttgcattttt agtagagacg gggtttcacc 106320

atattggcca ggctcatctc aaatttctaa ccccaagtga tccacctgcc tcagcctccc 106380

aaagtgctgg gattacaggc ataagctgcc acatccgacc aacatccatc aaaattataa 106440

atgcagaccc tgtgatctag caatctcact ttgaagagtt tatcctacag acatacttgc 106500

acagatgcaa attatatcag gtcaggagct gcagcattgt tcataacagc aaaaaaaaaa 106560

aaaaaataga aacaactgca aggtccatct acagacctct ggttgaatac agttacggca 106620

tagccataaa accaacaatg gcacagctcc tgtgcacaga tacagaactc cctccaaagt 106680

atattgttag gtgaaaaata caaggtgcag aataaggtgt ataataagct tttgtgaaaa 106740

gtgcagagat catctatatt tgatttgctg tgtatgcata gaatatctct ggacaaatac 106800

acaagaagct gctaacattg gttgcctctg gggagggaac agggtggctg gggggacaggg 106860

gctggaggaa aagatttcac cacatgcctt gcagacttta tatattcgga accctgtgca 106920

ttatctattg cacatattat ctattcatgc aaaacgctat aacatttaca aataactttc 106980

aaatatatgc atatcatatt tcaatggaaa gtttttttg ttgctgtttg tttgtttgtg 107040

atggagtttt gctcttttgc ccaggctgga gtgaactggc atgatctcag ctcactgcaa 107100

cctctgcccc cagtgttcaa gcaactctcc tgccttagcc tcccgaatag ctgggattac 107160

aggtgccccc caccacaccc ggctaatttt tgtattttta gtagagacgg ggttttgcca 107220

tgttggccag gctggtctca aaactcctga cctcaggtga tccgcccacc tcggcctccc 107280

aaagtgctag gattacaggc atgagccaat gtgcccagcc tcaatgaaaa gtgttttaaa 107340

gtaataatat aatacatcaa aaggcatgat ttatgttttc aagtgccacc tctggccact 107400

tgtggctgta gaaatgggga atactccaat tcaccatccc tatgtgaagg caggacaggg 107460

agacattcaa gtcaggaaaa ggtggagtgg aggttttagt caagcaccaa gtcctctgac 107520

ttctctgggc acttgcccat gccggtacct actcctgaga cactgtctac ctcgctaagc 107580

ccatcttctt ctcaccttgt gtctcacctg agacatcact gactttagga ggtttcctga 107640

cccatgtgtg ttccccatct cagcacccag cacctgcctc atgttcacca gggtacatgt 107700

cttccctccc cgccaggctg cactgagagc tcaatgaggg cagggaacat ctctgttccc 107760

catcttgtta gaagctatct ccccagagca tggtccaaat gtggtgttaa gaacagaagt 107820

tatcacaaag aagtgagaaa gtgaagggga aatggggaga gctgtaagaa aggcatgggg 107880 gagaaaaagg aaagaaggcc atgttcccag gatccaatgc cactccctgg ggaggcttct 107940

ctatcttagc tcagccctca gtcactatca ccctccctgg tcttgtcaga tttgaaatat 108000

cttcttcatt tacaagaggg ccttatctat cttggtcacc tctgtattcc cagcccctac 108060

tgcggacttg gaacatagta ggtgcccagt aaacatgggt tgaattaatt aatattggac 108120

aggccaggct tggtggctcg cgcctgtaat cccaacactt tgggaggctg aggtgggcag 108180

atgacctgag gtcaggagtt caagaccagc ttggccaaca tggtgaagcc ccatctctac 108240

taaaaataca aaaattaacc aggtgtggtg gcacacacct gtgatcctag ctactgggag 108300

gctgaggtgg gaggatcgct tgagcctggg aggtggaggt tgcagtgagc cgagatcgcg 108360

gcactgcact ccagcctggg cgacagggcg agactctgag tcaaaataat aataataata 108420

atagttcatt attattggat agtgaggagg gagtttgcaa aggagcggct aaaatgaaat 108480

tagtcctgag tttctgatgt tatgcgttct gggtaggggc gggagaactg gaagatgttt 108540

aatgcccccc acccaagggc agccatagtt gagaatcact aattctgaag cttggtaggg 108600

aagcttgata gggactttgg gtttttcact gagggcagaa tccccagtga ctgtcagaga 108660

cctcgctctg aatcttaaga acctccccac cccaggcctg acagggagga caaggaagga 108720

gagtggggct gggacgcggg gcagagggcg gctctgctca cctgcgggag aacgcgagga 108780

tgaggttcta gtgggaccgg cgctagcggg cgcgtcctgc caggagtgag tcttggcgcc 108840

atctagcgcc tgctggaggc tctgcagctg cagtggactg agccgccggc gctgggcgtg 108900

ggtcaccgcc gcggcgttct ccggacccag ggaggcgatc tgctccgcgg acagctcctg 108960

caggaggcaa aggggcagca ccctcacacg gggcccctcc ccggatttcc cacatctcgg 109020

gcgtccgcgc aggatgcaca caactttgcc atccccttgt gtgtatcatg tgtggtcctc 109080

tttaagatat ttccttaaaa ttagatcact cttgcactac caaagtaaat gcatttaaaa 109140

atagaaaatt taaaatggaa gtttccaaaa atggaaagcc ggcagacttg ccataagcaa 109200

gcattaaata ggagccgcaa aataaaaata aaacaatatt gttgaatcct agctagattt 109260

gtccgccaac aatatctgaa cccctggcct cttggaaaaa aaagagagag agagagaata 109320

taaaatgtac tatacagggt taaattgaca cttccttctt gaagtattta gaagtactaa 109380

tggagagttg aaaagggaag catgatttcc tccctatgtg gcaatgttgt ttaatgcaat 109440

gcaggacagc ttcccagtgc ttcaagtctt ccacctcctg aaacactgat gtggaggggg 109500

aaacacaggc cttaaagatc agaggcctga attcgagccc ctgctctgcc acatacttgc 109560

tgtgtcacct tgaacaaatt acacagcctc cgtgggcttt ggggataaat gtgagacggc 109620

atagagaact catctcctct gctgactgat tctgatcctt tggtttgact gcctgagcac 109680

catgtgatga gctctgtgag ggctccatgg agggaaaatg cagtcatcta ttggtgatat 109740 ctgctatgga caacatgagt tggaaattct gccagccaga ctatgtcttc aaggactgtg 109800

aacaaggtgt cttctgaagt cacttccaga tcaaaggact tggtgactcg ttccaatggg 109860

actggaatac gaaggggact ctatatcatc atgtttattt tctaaaggcc ctgaagaatc 109920

tgggaaagat gcttatgctc cctcatccct tcctctcctg tcagtcaccc tctcctcctc 109980

ttttgtccag gtagatactt ccctggaatc atatccctct gtgtaaagcc cttccaggaa 110040

ccccactcca gaacacagtt caggctcctg gaaaaaactc tgcccacctc tccagcctca 110100

cacatggcat tcttcactcc aggagagtgg gtgccaatat ttacagttcc ctagaggcag 110160

catcttcctg tgcacctctt tctgcctgga acactgtttc ttcttcatcg ccagctgttc 110220

tccaagattc agcatctggt cggtcaagcg ccatggctca tgcctacaat cccagcactt 110280

taggaggctg aggcagaatg attaacggag ctcaggagtt cgagaccagc ctgggcaaca 110340

cagtgagacc cccсttctct acaaaaagtt tttttaatta gccaagtgtg tagtcctagt 110400

tacttaggct gacgtgagag gattgcttga gcccagggga tcaagggtgc agtgagccat 110460

cattgtgcca ctgcactccc ttgggcaaca gagtgagact ttgtctcaaa aaaattaaaa 110520

ttcagcatca cctcctctcc tctcactaca cacacacaca cacacacaca cacacacaca 110580

cacacacggt ggggtggtca gagaggtact tttttggctt ccatatacca ttaacacaga 110640

gtctgtattc taaaaattat ctcttttatc aatgttgctg acttgcgacc cacaggccag 110700

attgaatctc taggcctacg gcactgtgtt ttcaaaaata aattcagatt ggctgccaac 110760

atttaattat caagcaattc catatgaaaa tgaagatttc tgacttcttc tcgaaaaaag 110820

tgaaaaggaa aattcagcaa tcttgggccc acatggccac actgcatgaa tgtgctagga 110880

atgagaagca gcctccacat ttagacaagg ctctccagtt cacttcggtc tctacccagc 110940

ctactgcacc ttatgttacc ttttaggccc ctgaaggcac tatgtgataa ccccttcacc 111000

tgtgcatcct taggtgcacc taccctggag cctgccacaa tgtaggtgat cagtagggat 111060

ttctggaata gataaatacc tgtacaagga agcagcacaa cagcactcca aacaagacca 111120

agagaaagtg cttagcaaaa tgccagcatc caagaccaag ctgcttgtcc catgcagagt 111180

gccccagagg ggaagcaggt atttgctcaa tggctttgag gactgctatt tattgccctc 111240

aaacagctat tatgcctgca atgtgctcgg aactgtccag gatgctgaag tgattgaagg 111300

ttggctttct tttttttttt ttgagacaga gtctcgctct gtcgcccagg ctagagtgta 111360

gtggagcgat cttggctcac tgcaagctcc gcccccсggg ttcacaccat tctcctgcct 111420

cagccttctg agtagctggg actacaggca catgtgacca ccccagctaa tttttttgta 111480

tttttagtag agacagggtt ttaccatgtt agccacgatg gtctcaatct cctgacctgg 111540

tgatccaccc tcctcggcct cccaaagtgc tgggattaca ggtgtgagcc actgcacccg 111600

gcctcagcta tctttcttga aatgaacaag tgatgtggca ctgggggagt ttgtcgagcc 111660

tggtttgtgt gttttatcca gaagtgctgg acaaaatgtg ttcccacatt acgcagcaaa 111720
caggaacacc agaatgactg cctcctcatc agagcagact attcccacca aacactcccc 111780
taatcttgta tacagacaga aaaaaaggac ctgtcattag ataccctgct tcccttcctt 111840
caaatgatcc aatttcaccc tctcaatttt ttcatagcag tttttcacct gtcctcttgt 111900
gattttccta atacttttct ttaaatcaac taaaaatata tatatataga aaaaaatata 111960
tatatataga gagagagaga tggtttcgct cttgttgccc aggctggagt gcaatggtgt 112020
ggtcttggct cattgcaacc tctgtctccc gggttcaagt gattctcctg cctcagcctc 112080
ccaagtaact gggattacag gcatccacca ccacgcccag ctaattttgt atttttcata 112140
gagatggggt ttcaccacat tggccaggct ggtctcagac tcctgacctc aggtgattgg 112200
cctgtctcag cctcccaaaa tgttgggatt acaggtgtga gccaccatgc ctgacctaaa 112260
tcaactaaaa tatttttatt taattatatc ttaaaaggaa atgttacaga ggtccacaat 112320
ccctcatctg caattccaaa gtccaaaaat aatctgaaaa ctgcaagttt tcccccaaag 112380
tttaagtcaa acttatttat cagcaaaact tgacctgaag taatgtgagg ctatttattt 112440
atagtctcta tttattccaa ttagtgtggc ttgtcacaga tttcttcaca gaaatattaa 112500
tgtgtttgct tacaaggtgc ttccccagac actgctgggg gtatgaagtt acatacagta 112560
aatgtacagg gtgatctttt caaaatctga gaaattctga attctgaact atatctgtcc 112620
ccaggggttt tcgtaagggt ttatgaacct gtcataccac cataagtaga aaatcagtac 112680
cacctgctag aagaagagaa gatgaccgta aaaataaata taattaaact acatattgta 112740
attttaagaa aaatgttttt taaaaacacc taaactcaca cagatctcct ctgaccccat 112800
cagcagagcc tggtcacaag cctctaaatt ccaaggccca tcacctgttt ccctgtgtga 112860
tttgaaatgg ggtcaagctc ccatttctcc ttgaagaact gagcacctac tttgaatatc 112920
tcatcaggaa ggcattttat tgctgatggc tggaaatatg gcatcaaatc cttgtcaagc 112980
atccggagct ctgccttagt taatccagct ggggagaaaa aggaatcacg gggggtttagt 113040
tcaagccatc agaactccgc ttgttttatt aatggtgctg cataatgttc agatctgagt 113100
gttctaggca ggcatcattc cttacaaaag gccctggaaa tcacactggg gaatcaagtt 113160
ccttcatcaa ctcagaaaaa aaaaatgtgg gtcacattag ccctgattgg cctcctacag 113220
tgaaacgcat gcccagaagg aacttcaatt tacacacttt caaattttgt ataaacctac 113280
ttaggggcca attaaatcac attctaaact agcggttttc caaactttag tgtacacaag 113340
aatctccaaa agagcttgtt ttaaaagcag atttgcagac ccaccctctg caacttcaaa 113400
tcatgaaatg taggttctac tgtaacgcca ctgatgtttg ctacacatgg ccaaggataa 113460
tgttttattt tgtgtcccca catttaagtt tggaaagaga gagaaaggta tgctcagggt 113520

gagtcttacc tgcaatggtc ccaagctcct gcaagacaga actggtccac tcagtgggat 113580

ccccaaacac aacttcagcc ttcctcttaa actcggctaa gacatgtgtg ctgcagagca 113640

gggtcccaat tctggccact accaccctgg tagtggttaa agagggaggg atataatatg 113700

agcttggact cttcagccaa aaacaaacaa acacacacac acacacacac acatacacac 113760

acacactgca cagtaggctc agcagggaca gcagatccag cttatcccat tagcccagtg 113820

ggattttagc ccagaaaggt gccaagtgtc aggaggtgga atatctggat ggatggatgg 113880

atggatggat ggatggatgg atggatggat ggatgaatta acccatttgc cattttgcac 113940

attcatattt tagttacctg aattctgaga tctttataag tgggatttca gtgatgttta 114000

tagcacacag ggttgcacca agtcctacca aatgaaagct cttcaggtcc tggatactgt 114060

atcctgaatc atccaggtac ccttgcaaaa tggattcagc ctaaaaaata gtaagaataa 114120

aagataaacc atccaggaat gatcaaggtc cccaggcctg aggggataaa taagctgttg 114180

ctgtagtctc ctgactgatc tcccatccta ctcctctgga atccccactc caaaccatcc 114240

tggcctctgc atccaagttc atgtccttaa gatactactt caactgagta tgtcccctaa 114300

tctatgaaag tgtcttcagg caatgaatct cttacatcct tcaccaatat taaaggcact 114360

tgtgtcctgt gtgtctgctg cttatttcct tcagtcactc ttatgaccct cagacagttt 114420

ggacatacaa cttcttctgc atcaggatcc aaacttttcc agcatctttt tccacaacgt 114480

ttcttctccc tttttttttt ttttttcgtt ttttggagac ggagtcttgc tctgtcaccc 114540

aggctggagt gcagtggcac agctcgctgc aacctccacc tccccaggtt caagcaattc 114600

tcgtgcttta gccacccaag tagctagaat tacaggtgca catcaccaca tctggctttt 114660

tgtatttttt gtagagacgg ggtttcacca tgttgcccag gctggtatca aactcctgac 114720

ctcaagtgat ccacccacct cggcctccca aagtgctggg attataggca tgagccaccg 114780

cacccagcct ttccacaacc tccaacaaaa ccttataatt tcctgtctct ttgcctttgt 114840

tcaaaccagt cctttcatct gaaatgccct tctgcacttc caagtgcaga cattcttttt 114900

tttttttctt tggttcaact caaatgtcac cttcttcatg aggtttcagc cagaatgatt 114960

ttttcttcct ctatggtcct acagaaatat gtttacccct taatgatttt ttcttcctct 115020

gtggtcctac agaaatatgt tcacccсttc aattccaatg tttactcttc aattccaaga 115080

gtagcacaca aaatgctttg gtgttaaact attttaaact aagccttgat ttaaggcggc 115140

tgacacataa gtctccataa ttctagcaca gtggctattc atcattcata acttcctctg 115200

gagaaccacc tctcttgtat tcctgcttca tgtggttcag tcaaagctaa ctgcaccaag 115260

ttccaggagt gatgaatttc aattcatacc ttagcctagc tgcagtcact ggttctggat 115320

tggacatgtg atttaaccag agtcaaccag aactttgagt ggagcattag gggaagagct 115380

ttctttactc tggacttgaa cttagaagga tatacacaag gatctgctgg aacctaccac 115440

atgcaggcat agagcctgtc tctcaatgaa gccaacacaa agaaaagcaa agttcaaaaa 115500
catgataaga gacagattcc cacccagagt gttgaagatc ctggatccag ctgtatctga 115560
acactacccc tgaactttcc agtaatggga gtcaataaat tcctttttt tttttttttt 115620
ttgagacaag gtctcactct gtcacccagc ctggagtgca gcggcatgat ctctgctcac 115680
tacaacctcc ctctcccagg ttcaagcgat tctcatgcct cagcctcccc agtagctggg 115740
attacaggca ggcatgaaca tgcctggctg atttttgtat tttagtaga gaagtagttt 115800
cactgttggg ccaggctggt ctcaaactcc tgacctcaag tgatctgccc gtttcagcct 115860
cccaaagttc taggactaca ggtatgagcc actgcacagt gcacccagcc cctttatttg 115920
ttcaagccag tttgtgttgg gttttctgcc acttgtaact aaacatgtgc tgattcattt 115980
tatctaccta tgtaggacct gaggaggcat ccaagccaat cctatgaaga tcagctacaa 116040
aataaagtct gggctgggca cggtggctca cacctgtaat cccagcactt tgggaggccg 116100
aagcaggagg atcacttaaa gtcaggagtt tgagaccagc ctggctgaca tggtgaaagc 116160
ttgtctctac caaaaaatac aaaaattagc caggcatggt ggcacgtgcc tgtggtccca 116220
gctacttggg aggctgaggt gggaggattg cttgagcctg ggaggtggaa gttgcagtga 116280
tccaagattg tgccactgca ctccagcctg ggtgacagag ggagactctg tctcacaaaa 116340
taaagtctgg ttccttcagt gctcatggga gcaagtaaaa gagattataa gacctcacaa 116400
ggcaaagatg aggagacatc caaggagaga cccctaagtg gaagcgaaaa tcacaggcta 116460
tagtcaatct tcccaactct ctttgctttt ttttgttttt tttttttgag acagcgtctc 116520
actgtgtcac ccaggctgga gtgcagtgac atgatcttgg cttactgcaa cctccatccc 116580
ccagggttca agtaattctt atgcctctga ctcccaagta gctgggatta caggcgcccg 116640
ccaccacacc caactaattt tttgtgtttt tagtagagac agggtttcac cattttggcc 116700
aggctggtct caaactcctg gcctcaagtg atctacccac ctcagcctcc caaagtgctg 116760
ggatcacagg catgagccac catgcccgac ccccatctcc gtttaatgtt agtcatcccc 116820
atcacacaat atagatcatt aaggtgttga gagaaagtgt tgaggaagat tgtgaaatgt 116880
tgcaatgaaa ttccgtcttc atgggctggt gcttcccacc cctcaggtgt gtttaaatgc 116940
tacctactca gagaaaactg ccctctggac tctttcatct caaacagccc ttcttcccac 117000
tattcccatg agcaccttgt ccattttccg catcacttat cgtgattttt caaatttttc 117060
actttgatta ctcacttttt tgtctgtatc accaagtagg ctcccaaaag aaaaggacca 117120
tatccaggta gtttaccaat gagtccccag cacctagcat aatgcctgcc acagagtagg 117180
agttcaataa gtacttcttg agtaagtaaa tgaatgagtg aatgaatgaa tgaatgaaac 117240
aacatctgag tgagtgactt acaaaagtcg tgggcaatat atattatttg acatgcttca 117300

ttttgctgcc ttcatgtaat agactggctt caggaagatt tctcatgttt ctgaaaatcg 117360

gactggtcag tgtcctcatc aagttgtcct ccatgattac aaagctcaca gctactatgg 117420

gggcgggggа tgcagtaaca ccatgcttag acttatatgt ctttgactta atgggactgt 117480

atatccaaag tggtaatagc taacatttat tgtagctaac atttcttgag tgcatactgt 117540

gtgctggcac cactttgacc actttacaca tactatctca tctaatcctc ctaaataaca 117600

ctatgaggta agtattaata gtatccctag tttgcagatg agcaaactga ggcaaggaga 117660

ggttaagtaa ctaggccaag atcacacagc tagaaaatga tcgttctggc caggcgccgt 117720

agctcctgcc cgtaatccca gcactttggg agacaaaggc gggcagatca cctgaggtcg 117780

ggagttggag accagcctga ccaacatgga gaaaccccgt ctctactaaa aatacaaaat 117840

aagcagggtg tgatgatgca tgcctgtaat cccagttact caggaggctg aggcaggaga 117900

attgcttgaa cccgggaggc agaggttgca gtgagtggag attacactcc agcctgggca 117960

acaagagcaa aactcattct aaaaaaaaaa aagaagaaga agaaagagaa aagaaaatga 118020

tggttctagg atcaaaaccc aggcagtctg attccagggc ccatactctt agccagtgaa 118080

ggtgtttggc tatggagaaa agatggagat tcaagttagt tttcaaattt ttcttattaa 118140

gtcttcataa tcaggtttca ctagttgact cagagcaatt tgggctcctc aactatcagg 118200

cacgctcact ttaaaatgaa agtgcaaaaa tacaatttaa ataaaattac tttgaagata 118260

tggtattaaa ttgtccttgc cactgaaacc cggaaaatgc aagctcagcc tgcaaggtga 118320

taagttaaaa taaatttcct tgagtgacga gaccagtgta tatgtaatga ttccaagaca 118380

attaatacca acacttttag gcaatattaa ctgttgaaaa atgaatagct ttaagatttc 118440

aatctctctg ttcatccctc tggcttctat aatagttttt tcccttatat tggtttttga 118500

gctgaactat ctgttaatgt agttccgtca ggtctgacta ttaatctaga aacctgcatt 118560

taaggttgat tgggagctaa agttgaagaa ctgactacaa atgatacatg caaatgttag 118620

gttttcatat cctcttcaaa catattgata aatctcactg ccatccatga gaattaaaat 118680

cggtgcgaag ggaaggagat accctagttt ctgcatgtat ttttctgttt ctccgagcct 118740

gcaaatagaa tgattttaat agcacagtgt gatgtggcac ttctcaaact agtcaaaacc 118800

agttttagag tcaaagaacc atgggattat agaatatcag actggaagga acctcagatt 118860

gttggggcta accctctcat cttgcaaagg aggagactaa cccagagaaa ggggctgtaa 118920

tgatgataat gttgatgatg acaatgcaat gatgtcaata tcaacataaa aacagctaat 118980

ttttattgac tactatgtac cagacattgt tcaaaatgtt ttcaagtatt aactcatgta 119040

atcttcatgg ccaggatctg catgttgccc agctccaggg gtgcaccatt cacagtgtat 119100

tctgtttgaa tggtgccact cagagttatg aaacatggcc accgtgatta caaaaagctc 119160

aaaagagtag ggggacattc attggttttg gctgctcagc gtccacactt tttctttgga 119220 aaattaccct ccattagctt cctattgcta ctctaccaaa ttaccccaaa catagtggct 119280

ttaaaactac acgtttattc tcttatagtt ctggagactg tatgtcctaa aatggactag 119340

aaggatgggt tccttctggg agttctaaga agaattcatt ttcttgcttt ttctagcttc 119400

aagaggccac ctgcatttcc ttggctcata gccccatcct ccatctccaa aatcagcagc 119460

agagcacctt cagatctctt tctaggtccc tctgcttcca ttatcatgtc accttctcta 119520

tctatgaccc tccttgctcc ttcttaaaag gacccttgtg attacattgg gcccacccaa 119580

ataatccaag ataacctcct catctcaaga tccttaatcg catcttcaaa gtccctttttg 119640

ctccataagt taacatagtc acagcttcca ggggttcaga catggacatc tttcggggac 119700

cttaatcagc ccaccacata tccttcccca ctcggctcat gtggtcatga gatgctgata 119760

agatggactc cgctccctga tgcaggcctc gccaaccaac atattcaacc cctgaccaga 119820

gtggttgctc aggggcgggc aactacatga gtggagtcaa tgctgaggct tttccaaaaa 119880

ctaataaaga agaggcacat ttttatgggc ttgttaatta ggttgtatat aagcctagtg 119940

ctgaaaatga ccactgagag agctgcctaa gaacgaagat aacatggaag ggctaacact 120000

tccctttcca gcagtacagt ggattagaca ccctaaactc ctctcccaat caaaacaatt 120060

aaaattctgg gaagaacttc ttctttagcc catgaaggat taattaacat aggaattggc 120120

cacccaccat aaacaactag aaaatttaac aaaatacatc agacaactgc atccggatat 120180

ggaacaacag aatcataatc ccagaagaag aaaaacaaac aagatgagcc ttaaaattac 120240

cctcacttac tgccaaaaag cagtttccaa gacatggatc aggaagagga accaaaacaa 120300

gcccagtggc ccagctgagt ggaggataca gatatccaag ttcagagagg ccacggcact 120360

tatcacttgg gcaaagtatt ggagaggagg aaactgcaga gggttccaga aagctgcaga 120420

gatgtctagt actgactgct attttgcaca tgcaaagagt gaaactccac ataggcagga 120480

aaagagtcat cagtaatcag aaaggattag gctgagcaac ttccagagct catatggagc 120540

tggaaatagt tcacattctc accagccaaa gtagagagat cttgaaatac atggagcatt 120600

aggtaacatc ctcaaaggaa tcatgtcaca gtaatgatga taaattaaca acagaataaa 120660

ggtcactctg gttttaccct aacaaaactc aaaaggaagc atcaaaagga tcaagctgat 120720

ttgaaagtaa cttaagtgta tgacagaaca aagcccaata ctcttcaaag aaatacaact 120780

aaatcaaata ctcaacaatg taaaatccac aatgctcatc acccaatcaa aattgctagg 120840

cttgcaaaca aaaaaagaaa atatgactca taactaagag aaaaatcagt caacagaaac 120900

agactcaaaa atgaccatca tgaggaaatt aacagtaagg atatgaaggc agctcttata 120960

aatatggaat agttaaaaga cccagaaaca tccacccacc tgtcccgggg tccattctgt 121020

ttgccagctt agggaagcca cagtgtctat ggagctgagg tccagctgct ccagctcact 121080

ctcattaaga gccagagcaa tgcgccccag ggagacgata tggtgttctc tccagtaaga 121140
tggcatgtcc caaaccttta ggcaaaaaaa gaaaatagat tagactgaac actgtgatgg 121200
tatttacaat gatttgaatg tttgtgcccc tctaaaatgc atatgttaaa acctaatctc 121260
caatgtgata gtattggaag gttgggcctt gggagatgat taggtcataa gggtggagcc 121320
ctcatgagtg ggattagtgc ccttgtaaaa gagaccccag agagctagct acaagctagt 121380
ccctttccac catgtgagga cacagtgaga aagcaccatc tatgaaaaac aagtccttac 121440
cagatactgg atctgccagt gcttcaattt ttgatttctc accccccaga actgtgagaa 121500
ataaatttct gtttataagt taccagtcta tgatattttg ttatagcagc ctgaatggac 121560
aaaggcagta tctaaaatag acataaggag gctaggtgcg gtggctcaca cttggaatct 121620
cagcattttt ggaggccaag gcatgcagat cacatgaggc caggagttca agaccagcct 121680
ggccaacata acgaaaccct gcctctacta aaaatacaaa aaaaaaaaat tagctgtgca 121740
tggtggcgca catctgtaat cccagccact tgggtggctg aggcacaaga ctcacttgca 121800
gaagttgcag ttagccaaga tggtgccact gtactccagc ctgggtgaca gagcgagacc 121860
ctgtctcaaa gaaataaaaa taaaacagac atagggaaac ctattgtgca gcatggtgac 121920
cacagttgat aataatgtgt tgaatggttg aaaattgtga agagagtaga tgttaaatat 121980
tctcacaaca aaaactgaca acggacaagt atgaaaggtg attgacatgt tagcttgctt 122040
taaccattcc ccaatgtata catacatcaa aacatcatat tgtatatatg tatacaccat 122100
aaatatatat aatttgtatt tgtcaactat cccttaataa aaaataaata aaaattaaaa 122160
agacataggg taaatcttaa tataaagctc tcttttaaat caataataaa aacatgacta 122220
tatttgtttt aaaaaaaaga tagacaaaag acatcaacaa ggaattcaca aaagaaaaat 122280
taaatgtatg gggaaaactt tattctcagt cataattaaa gaaatataca ttaaaaacaa 122340
tcagaccatg ttgattcatc acatgtcaag aagtataaag acttatcacc atttacgata 122400
ctggaaagat acaaggtaag ggataccctc atttgttgta gttgagaata taagttggta 122460
catactcttg agagggcaat ttggctatct gtatcaaaag cttttaaagt atttatattc 122520
tttgacccag aaattccacc tctgggaatt tagcctgagt aaatgagaca agtacccaac 122580
agtatatgta taaaggcata cattgaggca tttttttaat gcaacctgaa agtctaacaa 122640
ttagctaaat aaataagaac taaccataaa taaaatgaag ttgcccctgt ggatacctcc 122700
accaaggatt ggttccagga cttcttaaag ataccaaaat ccatggatat tcaagtccct 122760
tatataaaat agtgtagtat ttgcatataa tctatgcaca tcctcatgta tactttaaat 122820
catctctaga ttagttataa tacctgatgc aatataagtg ctatgtaaat atgtacaaaa 122880
tgttatgtca aaaatgctac tgtattttta tgtgtattgt tttttattgt tatattgtta 122940
tttggatcat tttttctaaa tacattctac ctgaggttgg ttgaatctgc agatgtggaa 123000

cccactgata tggaaggcta attgtattac aaagctattt aaaatactga tataggctag 123060
atgctgtggc ttacgcctgt aatcccaaca ctttgggagg ccaggcagac agatcacttg 123120
aggtcaggag ttcgagacca gcctggccaa cattgtgaaa ccccatctct actaaaaata 123180
caaaaattag ccgggcatgg tggcggatgc ctgtaatctc agctacttgg gaggatgagg 123240
caggagaatt gcttgaacct gggaggcaga agttgcagta agccaagacc acactattgt 123300
actccagcct gggcaacaag aacaaaactc catctcaaaa aataaataaa taaataaaat 123360
actgacatag atgtacattt tggatatgaa aaaatatgca gtctgtactg ttgggtgtag 123420
aaagcaagtt attgaatagt aggtaagtat aatatcattt ttgtaaaaca tgagatatat 123480
gtatgaaata tataaaataa tattttttat atacatagtt ttggagtctg gtaagcttca 123540
agtgaatata ccaaaatatc aacagtgctt atctgctgaa cagtgcttat caaaatatca 123600
acagtgagta aaagattatc actgattttc ttctttatta ttttctgact tttctacaat 123660
aaacttgaag tactcatata atacataaat acagttatat ttataatttt aagacattga 123720
attgtttaac ccttgagggt aactagatat tccacaacca tgtaaagagc taaaacaggg 123780
ctgggtgcag tggctcatgc ctgtaatccc agcactgtgg gaggccaagg tgagcagatc 123840
tcttgaggct aggggtttga gacaagcctg gccaaacaaa ccccgtctct actaaaatac 123900
aattattagc caggcatggt ggcttgtgcc tgtagtccca gctactcagg tggctgagac 123960
acaagaatca cttgaacccg ggaggcagag tttgcagtga acaacagatc gcgctgctgc 124020
cctccagcct gtacgacaga gcaagactct gtcttaaaag aaaaagaaaa aaaaaagaaa 124080
gaaaagctaa aacaggccac aaagggacct tttcctttta tttatttatt tgagacagag 124140
tctcgctcta tcaccaggct ggagtgtagt gacgcaatct cggctcatgg cagcctccgc 124200
ctcccgggtt caagcaattc tcctgcctca gcctcccgag tagctggaac tacaggtgca 124260
tgccacctgt agagatgggg tttcaccatg tgggccaggc tggtctcgat ctcttgacct 124320
cgtgatccgc ctcccaaagt gctgggatta taggcatgag ccactgcacc cagcctattt 124380
ttatttattt ttgagacaag gtatcagctc tgacgcctag gctagagtgc actggcgcaa 124440
tcttggctta ctgcaacctc cacctcccgg gttcaagcca ttctcctgcc tcagcctcct 124500
gagtagctgg aactacaggc acatgacacc acgcctggct aatgtttgca ttttgagtag 124560
agacagggtt tcaccatgtt ggccagcctg gtctcgaact cctgacctcc cgtgatccgc 124620
ccgcctcggc ctcccaaagt gttgggatta caggcataag ccactgcacc cagccaacaa 124680
agggaccttt ttaaagatgg aaagctactt ccagtcctct ttttactcct ttctgttata 124740
ttttcagaca aatttgcaaa tgattctgag aaaacctgct gtgagcagca gctggggctc 124800
tccaggtgaa ggaataaagc ctaattcctg caaaccgcct tggtcagagg cactgggaca 124860

tccacagaac ttgattcact gagcatctgc tagatgccaa gacacatctc atcccattct 124920
ctgccacagc cctaagaggg aggaactgga aatttcctcc tttctcagat aaggagatca 124980
caatatcact gagctgatgc agtaaaattt caagatgatg tatgggaaaa ctgctcgagg 125040
agggttctat gtgcagaaat gctgaactgg ttttggtgtt tttctttttt cctctgttta 125100
atgtttcctc cctgaggtgg gcttcacctg tattgctttc tccttcaggg tcatcagctg 125160
aggccggctg aaaccctgga cagctcccag cagttccacg gtcctgatga atgtgtcttc 125220
ctccatgcac cgcaggtcct ccagggccca gcaggcgttg gcttcggcca acttgaagat 125280
gtcatcagaa gagggggcca cgactccatg gcaacctgaa gaaaggagag atgcagggaa 125340
ggagaagagg aaagagaatt tcaggaaatg ttagtgtatt gaaatgagta acattctttg 125400
ccagtcagat ttatgcatgg aagggactac agagtaaaag cagagagcag gaggcaggga 125460
ccagtttgaa tcccacttct gccatttctt gctaattact gggtaacaaa gtcatggaaa 125520
tgccagcttt ttgcacatcc acatgcacaa actaggagcg gtgcaacatt acaattaaac 125580
tacaaattcc ttgagggcag taagagctgg tggtgaacag tcctgacttt cgattcaaat 125640
ccaaggctat ttaacctctc tgtatttcag ttttcttgtc tataaaatgg aaataaaatc 125700
gcctcatagg atggttgtga aaactaaatg tgttatatgt aaagcactta gaacagtgcc 125760
acgcatacag taagcactcg atacacaatt tctgttgcct atcttacccc ctatgccttt 125820
gagtttgcag cccagcataa agattccaaa attatgcagc agccttccta taaaaatggg 125880
aagatgggct gtgcggtggc gcatgtcttt aatcccagca ctttgggagg ccaaggcaga 125940
aggatcactt gagcccagga attcaagacc agcctgggca gcatggtaag acctcacctc 126000
taaaaaaaaa aaaactacaa aatattagct aggcatggtg gcacatgcct gtagtcccag 126060
ctacccagca ggctgaggtg ggaggatcac ctgagcctgg gaggtcaaga ttgcagtgag 126120
ccgtgatcac acctattctc cagcctgggt aacggtgaga ccctgtgtca aaataaaata 126180
aaaaataaaa accaggaaga tgagttgctg tccatgaaaa tcatagtggg ggttgtggct 126240
ttcccatccc aagagaaaag agaaatgagc ctattgcatt tctcttggca acaggaaaaa 126300
catccatttc ggtctttgaa taatgtgaac cttacctact caaaccagct accaaccaac 126360
atgcaagtga gatattgtct ttgctggatc tgatgtgccc aacctgggag aaaactctag 126420
gatccttaag agcaaccctg gatttcttat cttagccgca gaccaccact gatgctgaca 126480
gatgcaccaa ccccacccaa ggccagggta tttcccgagc cccatggcct ctctcttctc 126540
attgcacctt caatggaaat tcactgcatt ccaatcacga ggcaaaagta gaccagatca 126600
aagatgccag ttgtggtcct caaacatctt tatcataagc cacttggaga ggactaaaga 126660
cccatccctc cctaggccaa cccccagccc accctctagt gaaaacaaca tcagtgatag 126720
cacagaacta gagagggcag aatggttgac tagtatatca ggaacttggt cttatcactg 126780

tgtctgtcca gggcttagaa tagagcctgg tacatcgtag gtgttcagca aatatctgtg 126840
gagtgagtga acacacacac acaaattaat ataaaaccaa gcggtaattc tgctaatcca 126900
tcaatcacta aagcacagca gcaattaata cccagaacca actaaagaac actcaaacta 126960
tgcactgata ccagagcttc cttcccctga gcatggaaag ccttattttg tgtaccacat 127020
gcacacagta agagctctgc aatatgatga catggactgc tagaccttga gggcagtggg 127080
agccagtggt ggtcaccacc aactttgggt acaaatccaa gaactgccat ttactaactc 127140
tgtgacttta gacaagtccc ttagtctcct aagccccagt ttccttcatt taaaaaacaa 127200
gagaacaccc aagcctggaa atctgcagcc ctaaatggga ataggcattc ctgttttcac 127260
gcccaaatgt taggttttgg cctgccacat cccactatcc tgtacccata taaaccccaa 127320
accccaggct ccatgagcat acaagcagat gagcagatga acagaagaga agaggagcag 127380
aagagcagca tggcagagaa aggagcatct gaaggccaag aagagtttgg ctggggacag 127440
ttggagagga gattggtcac aggacagcca aactccaggg gaagatcatc ttcccactcc 127500
atcccctttc tagctcccca tccatcccac cgagagccac ctccattacc caataaaacc 127560
ccacattcac caagaaaaaa aaaaaacggg agaacaagat aatgcatgga gggataatca 127620
cccaaggtag acacaaatcc aattgtgata cttctttgct cgatgtcaca cgatgcctcc 127680
acccagagta aaagccaaag tcattagggt ttcctttaac agtcactaca tggctggcct 127740
tgtctctcac tcccctccaa acacactggc ccttacctcc cactcatgga agtggtcctg 127800
ttgcactctc tgcctggaag actctaattg aaaatatcca catggcccac tctcccttgg 127860
gtcttcattc aaatgtcacc ttctcaataa gatgttcctt catgattctc ttttcacatg 127920
acagtcccct ctcaacactc gtggtgcatt cccttgcttt attttccttc ttggcacttt 127980
tcatacaata taccatatat tttacttatg tatggaaatt tattgtcttg ctaccctcca 128040
tttgaaagta agctctttat ttacaaaatt ggtgcttaac gaatatttgt tgggtggatg 128100
aaagcaagca ctgactgtca actactatca ctgggggtga ttaactttgt ctcctcatgc 128160
ctggccccag tctgcactta gtaggtgcat ggtaataata ataaaatatc taacacttgg 128220
acaggcatgg tagctcacat ctataatccc agcactatgg gagaccaagg caggaggatc 128280
acttgaggct cagagttcaa gatcagcctg ggcaacacag taagacccta tctctacaaa 128340
aaaataaaaa attatccaga tgtggtggtt catgcctgta gtcccaacta cttgtgaagc 128400
tgaggtggga ggatcccttg agtccaggag gtcgaggctg tagtgaacca tgattgctgc 128460
actccagcct gggtggcaga gcgaggccct gcctctataa aatcaaattt taggccgggg 128520
gcagtggctc acgcctgtaa tcccagtatt tcgggaggcc aaggcaggtg gatcacctga 128580
ggccagcgtt caagaccagc ctggccaaca ttgtgaaacc ccgtctttac taataataca 128640

aaacttagcc aggcgtggtg gcacatgcct ataatcccag ctagtcagga ggctgaggca 128700

ggagagttgc tgtaatctgg gaggtggagg ttgcagtggg ccgagatcat gccactatac 128760

tccagcctgg gtgacactcc agcaagactc catctcaaaa aataaaaaaa atcaaatttt 128820

aaaaatatat aatacttatt aaagatctgc tacatgccag gcattccggt aaacatgttt 128880

ctgggtttaa acccattaat tctcacaata acccagtgag gtagagactt tcattatccc 128940

catttgataa aggatgaaaa ctgaggcaca cagaggttaa agagcttacc caaagccaca 129000

cagccagtaa gtggcagact caggagtgaa atgtagccag cccggccctg tcactgctat 129060

gttaaaccac taatccatgt tggtcctcta agtcaatcct actgaaatgt ttgttacaca 129120

tattcacgca ttaaacctac tagcctgggt atggagtatg ggacatgatt ccaggtcact 129180

ttacaaaagt gtaactcttt ttttttgag acagagtctc gctgtcaccc aggctggagt 129240

gcagtggtgc catctcagct cactgcaacc ttcacctcct gggttcaagc aatcctcgtg 129300

tctcagcctc ctgagtagct gagactacag gtatgcacca ccacacccgg ctaatttttg 129360

tgtttctagt agagacaggg tttcgccatg ttggccaggc tggtctcgaa ctcctgacct 129420

caagtgatct gcctgcctca acctcacaaa gtgctaggat tacaggcttg agcaaccgca 129480

cctgaaccaa aagtgtaatt cttttgctta tagattttgt cattctattg ctttgcagac 129540

atttcatcca gttccctgca taaggaggcc tcttgatgtc agggaccctg cccaaaacat 129600

taattactat gtcagcaatg gctcagatac cctgaaaaca ctcaccaggc atagggtcat 129660

agctggggat cttatcactg ctgttccgaa cagactgaaa gacagcccag aggaattctt 129720

tagtgggcag ggtcctggcc atcttggaag ctgcttgcag aaggatctca ggaaactgtg 129780

tgaaagaaga tgaaagaaga ggaataataa ttaaaacccc ttaaatacag attgaaatta 129840

gagttgaaac tgtcacacta cttactgcct tcattctttg ttatagcaac ttctcatata 129900

aatctattcc ctagtcccac taatgtggtc tctataaaga atcctataaa atgatactca 129960

acttcaataa aaatccaata aagtctaaaa aacataagta aatcactttt cacccatcag 130020

attgtcaaaa ataaaatgaa gtgctaatac ttaatgctgg gcacattcat ttttccgtgg 130080

gactgtcaaa ctgtgcacta ttagcagagc aatctggtaa tatcgacaaa aaatgtaggt 130140

taggcgaggt gtggtggctc atacctgtaa tcccagcatt ttgggaggtc gaggcaggca 130200

aatcacttga ggtcaggagc tcaagactgg cctggccaac atggtgaaac cccatctcta 130260

ctaaaaatac aaaaattagt cagtcgtggt ggcatgagcc tgtcatccca gctacttagg 130320

aggctgagac agaaggatca ccagagccca ggaggctgag gttgtagtga gtcgagagca 130380

tgccactgca ctccagcctg ggtgatggga gtacaaccct gtctcagaaa aaaacaaaaa 130440

aaatgtatgt gtacctaccc cttttttggt ttggtgctat ttctaagttt ttactaaact 130500

ttatatttga catagtttgg atgtgtaccc cacccaaatc ttattgaaat gtaatcccca 130560

gtgttggagg tggggcccag tgagaggtga ttggatcatg gggcagatt tctcatgaat 130620
ggttaaatac aatcccttg gtactgtcct cacaatagtg cgtgcgttct ctcgagatct 130680
tgttgtttaa atgcatgtag cacctcccc atcacgctct tgctcctgtg tcggccaagt 130740
aagatgtgtc tgctcccct tcgccttccg ccatgattgt aagtttcctg aggccttccc 130800
agaagctggg cagatgccag catcatgctt cctgtacagt cggcagagcc atgagccaat 130860
caatcctctt ttcttgtctc tttttttttt ttttttttc agagtcttgt tctgttgctc 130920
aagctggaat acagtggtgt gatctccact cactgcaacc tccacctccc aggctcaagt 130980
aattctcgtg cctgagcctc ctgagtagct gggattacat gcatgcagca ccaagcaggc 131040
taattttttt ttattttat tgtagacaag gtttcactat cttggcctgg ctggtctcaa 131100
actcctggcc tcaagtgatt cgcccacctc ggcctcccaa agtgctggga ttacagatgt 131160
gagccactgt gcctggcagt gtttgtttgt ttgtttgttt agacggagtc tcgctctatc 131220
acccaggctg gagtgcaggg cacgatctcg gctcactgca agctctgccc cctgggttca 131280
cgccgttctc ctacctcagc ctcccgagta gctgggacta catgtgccca ccaccacgcc 131340
cggctaattt tttgtatttt ttagtagaga cggggtttca ccctcttagc caggacggtc 131400
tcaatctcct gacctcgtga tctgcccgcc tcggcctccc aaactgctgg gattgcaggc 131460
atgagccact gcacctggcc ctttttattt tatttttttt aatttaactt ttattttaag 131520
ttcaggggta catggcaggt ttgttatata ggtaagcttg tgtcatggag gtttgttgta 131580
caaattattt aatcacccag gtattaagcc tagtacccat tagttatttt tgctgatctt 131640
ctccctcctc ccacctttca ctctccaata ggccccagta tctgctgttc cattctatgt 131700
gacaacgtgt tctcatcatt tagctcccac ttataagtaa gaacatgcgg tatttggttt 131760
tctgtttctg cattagtttg ctaaggatga tggcctccag ctccatccat gtccctgcaa 131820
agaacacgat ctcattcttt tttatggcca catagtattc catgtaaacc tcctttcttt 131880
ataaattacc cagtctcagg tatttcttta tagcaatgca agaattgcct aacacaggtc 131940
aggtgcggtg gctcacacct gtaatcctag cacttgggag gcccaagcag gcagatcacc 132000
tgaggtcaag ggttcaagac cagcctggcc aacacggtga aaccccgttt ctacaaaaaa 132060
tacaaaaatt cactcggcat ggtggtgtat gcctataatc cctgctactc aggaggctga 132120
ggcaggagaa tcacttgaac ccaggaggtg gaggttgcag tgagccgaaa tcgtgccact 132180
gcattccggc ctgggcaaca cagctagact ccatctcaaa aaagagaaag aaaaaaagaa 132240
ttgcctaaca caacatttgt catgaaaagg aggtgaagaa gggttgtctc aacatcctgg 132300
gtggagacta gcaatgaggg gacttagggg actttgggtt gccttggaga aagtgccatt 132360
tctcatctag aacgaataca ctttgtcaag acttgggatt ttattagaag gcctgcccat 132420

gggcccagat aatccatcag gctctacaaa cagctatgcc tcactgggtc ccggctcacc 132480

cagagagcaa gcctctcagc ctatgtagct ccccttcta gtctcatctt cagaattaga 132540

gtcacagtct aagccagcag ttcttcaacc ctggctgtac ctaagagtca cctgggacac 132600

ccagagccag cccctaaaga ttctgactca gtcgggctgg gctggatacc atggattggt 132660

ccctcttgga atctccccag gtgattctaa tgtgcagctg aggctgaagg caaccactct 132720

aagacagtta acttttagaa agcaatatgc atagcggagt gaatggcatt ttccattttg 132780

tattttctaa agaagagtgt gtatatgctc ttgtgttttt agtagagaca gggtttcgcc 132840

atgttggcga ggcttgttat ataggtattt ttatacctat ataaaacttg ttatatagat 132900

attcaggata tttctgaaag gatgcagaag agattgggga cagcaattgc ctctgaagag 132960

ggaggctagg gaactagaag tctggggtgg gagggagact tgcttctcat cctattacct 133020

ttggtgcgat tgggatttgt taaccaggag catgtattac ttgctttatt aaacatttcc 133080

agtatttaaa aaaagagtat acagaataat acaacaaaac acccatgtcc tcacaaccca 133140

acttaagaaa taaaacatca caatataaat aataagtccc tccttcactc tttgccttcc 133200

tccttcccca gaagtaactg tcactctgaa tttggcattc acttttacac ttctcttaca 133260

tataaacttg gcttacattt gctttaaatt aattcagcta ctatttattt atttactttg 133320

tatcctacta tagatattct tttgcaatct gggttttttt ttaactctac actgtgtttt 133380

tgagatttca cagtgttgat gcaagacttt ttcaatttca gaagttgatt tttttaagat 133440

tagatagata tatggatgtt ctcactgata tgtgggagct aaactatgag gacccaaagg 133500

cataagaatg atacaatggg ctttggggac ttggatggaa gagtgggagg ggccaaggga 133560

taaaagacta caaatatgtt gcagtgtata ctgttcaggt gatgggtgca ccaaaatctc 133620

acaaatcacc actaaagaac ttactcatgt aaccaaacac cacctgtacc ccaatacctt 133680

atggaaaaat aaaaataata ataaacatct aaacataaga aaaacaagga aggaaaaaaa 133740

atagatggat atataacatt tgtgcatcag ggccgggcat ggtggctcac gcctgtaatc 133800

ccagcacttg ggaggccgag gcgggcagat cacttgaggt caggagttca agaccaggct 133860

ggccaacatg gtgaagcccc atatctacta aaaatataaa aattagctgg gcatagtggc 133920

aggggcctgt aatcccagct actcgggagg ctgaggcagg agaatcactt gaacccagga 133980

agcagaggtt gcagtgagcc gagattgcat cactacactc cagcctgggc aacagaggga 134040

gactccatcg caaaataaaa taaaataata aaataaaaat aaacatgtgt gcatcaactg 134100

cttataacta tatattcaga catccagaat atgattttac tgtgactgga cttcagaatg 134160

tgctgcgtgt gatcctaggt gaagttgtgt gtgttcaggc cctgctgagc atgtgtgacc 134220

atgtgcacct tgtgcctgca agtgcaggta tgagagtgtg tggatgtgct ttgtggggga 134280

tctgattgtc attcagcaaa cattcactcc tttcctgccc tccacctcca tggaaggaga 134340 ctccttcctg ccccattaaa gttggtcttg gtcatgtaac ttactttggc cattggagtg 134400

tggcagaagt gacagcgtgc caacttccaa cctaggactt aaggagaatt gtacttatcc 134460

cttccctttt ttggtagttt cataccttca tggtaagaaa aacaagttct agagcagggc 134520

tgccccagcg acccacaaat caggcagcaa ggaacatatg cgaattgttc tatgcccttg 134580

agattgtgtg gctttgttat gcagcacaaa tgactaattc atgcttattt tgcatgcccc 134640

ttggtctaga ctgtctaact tctggagcct tactcctaga aaagcgtcat gcccaaatgt 134700

aatgatcaat aaagacttgt tattggatta aatgtgcttg cacatgtgat tgcatacact 134760

ggacatacat gtgcatgtgc atagggcacc agccagtgtg agagccagaa cagacgtgcg 134820

tgtgagatgc acactaatag cagagtggta gctaagtaat atctgtctgc acacatctgc 134880

ctgggggggcc acacaaaagg gcctgagttc atttagctgt ggactcactc ccttttccag 134940

aaccttgcac atcctggaat ggagctggaa acatcttagc ccttggaggc agggaggaag 135000

cttccagaac catagacagc agtaaaccca atgctgtata actgaccaca ctttctctcc 135060

ccttcaaact ccttcagcct tcttaagatg gagcacaaca ttacctttgt ggctatagag 135120

cttaattcat ctcctgagaa aagtactaga aagggtccca agtccttcgt ggtctcggct 135180

gtccagtgct gagggagtct aaaaagagat aataaccagt aaagtgaaaa aacatgctgt 135240

ggtggacatc tgttgccttt tcccccagca ccccttttctt tcaggaatag attgtcttat 135300

actcatgtca atcacatggt cccacttcct tgaccaagaa aattggcatg tgatgcaggc 135360

tgaccaatca gagtcatccc tgggagtttt gatggaacta tcagagaagc tctcctttct 135420

ctggtatctc tggcagtagg ggaggatatt ggaggacatt catattacca agtggaaaaa 135480

gtaaagacca caccaaggaa cacagagctg agggatggga ggaacatatt cctgaagata 135540

tcatttgaga cacgggattc agccatgcct gaagaccacc agtggagttt cccgttacat 135600

tccattcctg aattcaatac attcccgttc tctttagttt gaattaaatt attgaatttc 135660

tgccatttac atcacagtgt gtccttcttc tccctcctca atagaagagt aattatatat 135720

ttcttccttt tactttacca taacagtctt ctcttagaat aagaaaaaac ctttctcttg 135780

aacttggcag gataaaataa aggcactgac ccagaatcca ctgttattct tgtatagatc 135840

ataaatgcct acagtgaaga gcattacact atctttggcg acatctctaa aggaggtctg 135900

cccaattagc agtgacagct ggtgggaatg caaaatcata cagccacttt ggaagacatt 135960

ttgttggttt cttacaaaag caaacatgtt tttgccatat aacccagcaa acacactctt 136020

tggtatttac acaaaggagt tgaaaactta cgtctacatg aaaacctgta tatggatgtt 136080

gatagcagct ttatccataa ttgccaaaac ttggaagcaa ccaaatgtcc ttctgtaggt 136140

gaatggctaa ataaactgtg gttcattaag acaatgaaat atgattcagc actaaaaaga 136200

aatgagctat caagccaaaa aaagacctgg agaaaactta agtgcatatt actaagtgaa 136260
agaagtctat ctgaaaaggc tatctactgt atgattccaa atatatgata ttctagaaaa 136320
ggcaaaagta tcagtggttg ccaggaatta ggagtgagag aggaatgaac aggcaaagcc 136380
cggaaggatt tttagggcag tgaaaatact ccgtatgata ctataatggt gaatacatgt 136440
tattatatac ttgtctgaac ccatagaatg taaagcacca agagtgaatc ctaatgtaaa 136500
atatggacta tggatgataa tgagaatcca atgacgataa tgtcaacata ggttcatcag 136560
ttctaacaaa tgtacaactt tggtggggga tattgatcat ggggagctta tgcgtgtacg 136620
gggtcagaga gatatgggaa atctctatct tctccatttt tctgagaacc taaaactagt 136680
attaaaaata gtctctaggg tcaggcatgg tggcccatac ctataatccc aacactgtgg 136740
gaggcttagg tgggtgaatc ccttgagccc aggagttcaa gacccaccta ggcaacatgg 136800
tgaaattcca tcccttaaaa aaaaatacaa aaattagctg ggtacagtga tgtgcacctg 136860
tggtcccagc tacttgggag gctcaggtgg gaggatcacc tgagcccagg gaggttgagg 136920
ctgcagtgag ccattactgt gccacagcac tccagcctgg gtgacagagc gagaccccat 136980
ttaaaaaaaa aatagtcttt aaactaataa taataccact accttgcatc tgtaaagggc 137040
cacctttcc aaatttcccc ttcatatgcc aagctgtgta agaaacaact ctttgagatt 137100
tttagggcag ctactattga ttccacttta cagcaaatct gaagccaagg ccaggcgcgg 137160
tgactcacgc ctgtaatccc agaactttgg gaggccgagg tgggtggatt acgaggtcag 137220
gagatcaaga ccatcctggc caacatggtg aaacccagtc tctactaaaa atgcaaaaaa 137280
tagctgggcg tggtggcaca tgcctgtaat cccagctact cgggaggctg aggcaggaga 137340
atcgcttgaa ccagggagtc agaagttgca gtgagccaag gtcgtgccac tgtactccag 137400
cctggccaca gagcgagact ccgtctaaga aaaaaaaaaa aaatctgaag ccaaaagaag 137460
aaaggtcaca tttccaaaat aagcataaga attttatctc atcctaagcc agagactctg 137520
tttgcagggc aggagagcca aggttgagtg tccctccaca gagcatacca caccaccaaa 137580
agccaaaccc agccgctcct cacccatact gtcccaggag cttgagcctc actgcagcct 137640
tttgctcagg gttgaggtct gggcagtctc tgaggccgtg tagagcagtc gcccaagccc 137700
tgggggagat ccccctgtcg atgatggctg ccggcaagtg acacagcagg ttccccatga 137760
tgtccacagt gtactcatca gcaatggagt cgtcctgaac ccaaaaagtg gagccaatgg 137820
gcagttccaa cagagcataa gatcgtgagc cattagaaac acatcctgtg gggtgggcac 137880
tgtggcttat gcctataatc ccagcactct gggaggccga ggcagacaga tcatttgagg 137940
ccaggagttc gaaatcagcc tggccaacat ggtgaaaccc catctatact aaaaatacaa 138000
aaattagcca ggcattgtgg cacatgcctg tagtcccagc tactcgagag gctgaggcac 138060
aagaattgct tgaacccggg aggcagaggt cgcagtgagc ctagatgatc atgccattgc 138120

actccagcct gggtgacaga ggaaaactct gtctcaaaaa aaaaaaaaaa aaagttccga 138180
tcccaaactt cttaaagatt ccaggatctt aactgcctac cctgccatgg attcctagag 138240
atggaacagg acctgatctt ccccagggat gcaggcctca gaaatctgga aaaccctggc 138300
ttggcacacc aggagcactc agtaattgag cgctgctgtc actgttattg ctctgagaat 138360
attccagact tcaaccacca ttatactatt ggcaccacaa caatgaaaac agcacttcaa 138420
agtttgtatg ccacttgaaa gccttcaaag ctcttttgca ctttcacaaa atgacaaaaa 138480
tagttcacat ttattgtgtg tttcctatgt gcgggcaccg tcctgagtat ctgtgggtct 138540
caatctagtc atcttcaaag taactgtaag aagtaagtac tactattatc catttttaa 138600
agaaaaagaa actgccagct agtggcaagg ctaggattca cacccagaca atctggcctt 138660
tgagccttta cattaaataa tacattgccc atgtcatgtt ttcaatactc tatgaggaga 138720
ctaggtatgg tggctcatgt ctgtaatccc aacactttgg gaggccgagg tggatggatc 138780
acctgatgtc aggagtttga gaccatcctg gccaacatgg tgaaaccctg tctctactaa 138840
aaatacaaaa aataaccggg tgtagtggta cacacctgta gtcccagcta cttgggaggc 138900
tgaggcagga gaatcagttg aacctgggag gcagaggttg cagtgagctg agatcgtgct 138960
actgcactcc agcctgggtg acagagtgag actctgtccc aaaaaaaaaa aaaaaaaaaa 139020
aactctaagg gggatgtgct actcccccca ccttacagat gagaagactg aggctcaggg 139080
aggaaaggtg attcacccaa agccacataa ccaatcactg gcacccagtc tctgggtact 139140
gtgctctttc cattcctcca gagtagcctc ccctcctttg ggccgtgggg cagaagtctc 139200
cagaaccaat ggaaaagatt gttccaacca gcaactttag agtaagcttc aagacccacc 139260
agtcagactt ctggccatca cagcaggctc aagatcctag ccagatccct cacagggctc 139320
cccaagatgg ccacagatac ccagagggag gggacaccca ggaagagttt tcttaccagg 139380
cactgctgca ctttcctgag gactgaagtc tttttgtggg aatctaaaac caaggagtcc 139440
aaccagctct tccccaggct gatcagaaag ggcacacact gggaggctgg gacagaagcc 139500
aggtagcggg ccctggagag agatggggag aacaagagga ggaacacatg gctgctcctg 139560
ttggccccta aaatcagggt tcgcctgccc aagagctcat cccaaagaca ttctttaaat 139620
gtttgttctt cccagtgatg ctttggctcc ctgcagcagg tagcaagctt cccacccatc 139680
atgctcctct tctatgggtg ggaacaaact tcccatcatg ctcctcttcc atgaatgggc 139740
aacaaacttc ccatcatgct ctcttctatg ggtgggaaca aacttcccat catgaccctc 139800
ttctgtgggt gggaacaaac ttcccatcat gctcctcttc tatgccccag agcattgatc 139860
agagaatggg ttctcgtaaa tagacatgtt gattgatgga ctggttgagg ggagatgaaa 139920
aagaaaggaa aaagtagatg aaaaaagaga caacgaggag agagaataac tagaaagtga 139980

gaaagaaaga taagaaaaag agtggaaaga gaaattctgt catctccagt ccagaactct 140040

gccttccact cttgatcctt cagctcttga caatgtattc tcaacagggt caaagtcacc 140100

ctcaagaagg agaaactggt ttggatggag ataaaaaaga ctgttatgac aacgatctgt 140160

aacccactga agctcattcc tactcaaaat atttttcctc agtgtataat ttctttcatt 140220

caggagaaat taaaattttc tcctttggaa ggcattaaca aaagttgaga aagtctgcat 140280

gaaacccatc cggcctactc aaccccacca cttcactgtt ccaaaaccag atacaacgtc 140340

ttcccccaaa catgttcttc cgcctctatg ctatcccaca ctcctttttt tctttatccc 140400

aagcctgtca ttgctatctc cttagcatta tcccaatctg ttcttacttc tggccaggtg 140460

tggtggctca cagctataat cccagcactt tgggaggcca aggtgggtca cctgagatca 140520

ggagtttgag accagcctga gcgacatggt gaaatcccat ctctaccaaa aattcaaaaa 140580

attagctggg catggtggca tgcacacctg gggtttcaac tactctagag gctgaagtgg 140640

aagaatcact tgaacctggg agacagaggt tgcagtgagc tgagatcgca ccactgcact 140700

ccaacctggg tgacagaatg ggaccccgtc tcaaaacaac aacaacaaaa aaatctattc 140760

taacttctcc attcccagta ccactgtctt atttatttct tcattcatcc agcaaatatc 140820

tacaaagcaa ctacagtaag ccaggcactg ttctaggcac caaatataac aatgaacaag 140880

acacaagtgg tccttaattc tagcagagag aagaaacaca gaaaccagtg aacaaaacaa 140940

gacatttgag ctattaatag ctggcatggt ggctcacgcc tgtaatccca gcactttggg 141000

aggccgaggc aagcagatca tctgaggtca ggagtttgag accagcctgg ccatcacggt 141060

gaaaccctgt ctctaccaaa aaatacaaaa attagccagg cgtggtggcg cacacctgta 141120

atcccagcta ctggggaggc tgaggtagga gaatctcttg aacccaggtg gcggaggttg 141180

caatgagcca agattacacc actgcactcc agcctgggtg acagagggag actctgtctc 141240

aaaaataaat aaattaatta attaattaat taatgtaagg aagaaaataa aatggctctg 141300

taaaagagaa tgacaatggg gataatcaaa cgagcccctt ctaaggcagt tccatttgag 141360

ctgagacatg aatgagtggt agaagtgcca tgccaagatc tgggttactg aaataatctt 141420

cagctgtctc ccttctttcc attcagcctc cctgcattct atcctctatg tgggtagtgg 141480

ctatgatctt tctgaaatat aaatctgatg gctgcacaca gaatatgtgt gaaacacctc 141540

actttacaaa gcccccatga tctgatgtcc cttaactctc tggccttctc tcccctatct 141600

actctgcaga tatacccccca tgtgttccca gagtgattgg cttttcttac ccactgctga 141660

accggacatc cccattccac tgcctggaat gtccatctta aactctctat ccagcattag 141720

ttcggtctcc cctgtctctc cagtctatgg gtgatattcc ctcctctatg tccagtcctc 141780

attatccctc ccctttgcct ctcctttgga aggcaataaa aaagttgaga aaggctgcat 141840

gaaacccatc tggcctagtc aaacccacca cttgactgtt caagaaccag attcaacatc 141900

ttcccccaaa catgttcttc cacctctgtc ctaacagagg tgtgtaacag cacctggctt 141960
tcttggaaag cacctagttt ttcttggaga accatctccc tcactctaag tccttgtggt 142020
ttctctccag ctccaggggt gcaatacaga ttgaagtcag tcaaagttac ccatcctccc 142080
aacctcactg attggttcaa ggaaaacaaa ttcgcccaat taaggtcaat gagagtcaaa 142140
cccaagatgt gtattgagaa tgatgggaag agaacatttt tctattttcc ctgaatatga 142200
aactgggaaa atttaacctt agatcttcca gggttcccac agaggaacaa cttgcctgat 142260
agtgaagcca acagagaaaa gctgggacaa gagacaaatg acatcatttg cacccctaga 142320
tcgagccatg cctgaagtcc ctctctgaac tttccagtta cctgaacaaa aaattccctt 142380
ttattgcata agccagtttc agtttcagtt ctgttgcttt caaccaaata tcaacctgat 142440
ataattggct tcatgtttgt ctattccctc tcccaccatg agattataag gtcttataaa 142500
ttaataggaa tttctaaatc ttcagataga aaatttagct atctgagaac tagcacacag 142560
caagtactca atgaactttt tttttttttt gaatgaacga agacaataag agcaaaaaaa 142620
ggtagaggga aataaagaag gagagaagga gagaaacaat gtccagatca tgtttgaaaa 142680
gcagggccac cctgcaggcc caaaagctca cacatgccag gagaaacgcc tactgctccc 142740
ctcaactctg attcccctgg agcctggcac agccgcaaag ccaggccaga tgggacctgc 142800
ctcactgaca ctcattcagg cttgggttgc tttggcttgg tttttagata acaggaaaag 142860
caagaaggtc tgtctcaaat gtctgtgtga tactcagaat tgaaatcctg gatctcaagg 142920
gcttaactct ctaaggcatc ctccactctg cctctggttc ctgaagaaaa cccagtgggg 142980
agagaatcat tttgacttca gtgattccca ccgatctcac tgatgagcca gaaggtgggg 143040
gctgatgttc acttacggga gtgcagccaa gaggaaaggt ggcatagaca atctggaaac 143100
ttcccagtat ttccacgcca aacaattaac ctgaaagatt aagcagttct ggggatttaa 143160
tgcagtaaga agaagaacaa acacagtgct ttccatgtgc caggtactat tctaagtgct 143220
tcatacgtgt tcatttatgc aatagagcac cctatgatgt gagtaccact actgtgccca 143280
gtttatggat gagaacgctg aggtaataac aaacatgaac acggacattt aagtgtgaga 143340
atgctcatcg caattttgct tatatctgca agaaagggag atataatttt aatgttcctc 143400
agcaggggat tggttaagta aattatggta tatacataca atggaatagt gcacctactt 143460
tgttttaata aggtagattt atatatatat taacactaaa tgatgtctgt gctattttat 143520
gcatgtgcat atatatagtt atattatata taatgtaaaa accacaaatg ggctcatatt 143580
gtacacacaa ttctgttaaa ctatatatat agcttaacta tacatatagt tatatatagc 143640
ttaactatac atatagttat atatagctta actatatata tagttttata catatatata 143700
tcttagctat atatataaac agaacagcat gtatagtatg agctcacttg tgtttttttt 143760

ttttgttttt tttttgagac agtctcactc tgtcacctag gctgtagtac agtggtgtaa 143820
tctcagctca ctgcaacctc cgccttccag gttcaagcag ttctcctgcg tcagcctccc 143880
gagtagctag gattacaggc accccccacc atgcacggct aatttttttt tttttttgta 143940
tttttagtgg gtttttttgg ggtttttttt ttgtattttt taaaaatttt tttaggttgg 144000
ccaggctggt ttcgaactcc taacctcaag taatccaccc atctcggcct cctaaagtgc 144060
taagattact ggcgtgagca ctgtgcccgg cccacttgtg gtttttatat tacatataga 144120
tacttttttta aaatttggtt tagagataga gtctcactct gtcacccagg ctgaaatgca 144180
gtgactccat cataactcac tgccaccttg aactcccagg ttcaagtgat cctcctgcct 144240
cagtttccca agtagctagg actacaggca tgtgccacca tatttggcta atttttgtat 144300
ttttttgtag agttagagtc ttgctgtatg tttcccaggt tggtttcaaa gtcctatcct 144360
caaatgaacc tcccacttca gcctcccaaa gtgctagaat tacaggcatg agtcattgtg 144420
accggcctat attttatata taaaagaaaa ctctcaaaag taatagcaaa ctgttcttta 144480
tagttacctc tgaaacacag ttgctcacct cagtattatt gacattttcg accaggtacc 144540
tctgtgtagt ggtgctgtcc tacgcattgt aggatgttga gctccacccc tggcctctac 144600
ccactagagg ccagtagctc tgctccggtt gtgacaacca aaaatgtttg cagacatggc 144660
caaaagtccc ctaagagggc aaaacagtac tctgttgaga acaactgctc tggggaaaat 144720
ttggggaaat tttactttct ctgtattggt tgtttgtgtg tgtgtgtgtg tgtgtgtgtg 144780
tgtgtgtgtg tgtgtgtgta ataattatgt gttttcagaa tgggcgtggt ggctcatgcc 144840
tgtaatccca gcactttagg aggctgaggc tggtggatca ctggaggtca ggagttcaag 144900
accagcctgg ccaacatgat gaaaccccgt ccctactaaa aatacaaaaa ttagccaggt 144960
gtggtggtgc gcacctagaa tcccagttag acaggagact gaggcaggag aatcacttga 145020
acccaggggc ggaggttgca gtgagattga accactgcat tccagcctgg gcaacagagc 145080
aaggtcctgt ctcaataata ataataatga taataataat aataataata ataatgtgtt 145140
ttagtaaaaa tataaacgag aaaggcaaat tttctaatta actggtattt gaaggctctg 145200
agagctggaa gcctaggaaa gcaccttcac tggggcaact cttcctgctc gaacatgtag 145260
gtcttcctca aagcaggtct agcttccatc catttgctca gttattggct tgcccacctg 145320
ggcaggtctt ttaatatagt tcagtggttt gtaccagcaa actgattaga aatgcaaagt 145380
attaggcctc acccccttacc tactatatgt aaaactctgg gagtggggcc cccaatttgt 145440
gtttttacag ccttccacac aatgctgatg caagctcaac tttgagaatc actaacagaa 145500
ttaacagtcc aagggaatga gagagcttca ttaaaacttt gcatattcct gtaatgatct 145560
tgaaggatta tacaccaagc actctatgct tcctggtttt ctgggagata atttactctt 145620
tggaaattct tcattctggt ctgaaacaca aggccagagt tgagaaggtg ctttttaata 145680

tccattacag gagtctgtaa gccagcggtt cacaccaaaa gttcaaatgc tgtaaggcct 145740
gtgtttacta gcttagacac tgaaaaatca gtcactggct gggtgaagtg gctcatgcct 145800
gtcatcccag cactttgaga ggctgaggca ggaggatcac ttgagcccag gaatttgaga 145860
ccagcctggg caacatatca agaccctatc tctgcaaaaa ataaataaat tagccaggca 145920
tggtggtgtg tgcctttagt tccagctact caagaagttg agacggaagg atctcttgag 145980
cccaggaggt tgaggctgca atgaaccatg attatgccac tgtactccag cctgggtgac 146040
agggcaagac tgtgtctcaa aaaaaaaaaa aaaagaaatc agttactgct tctagtctcc 146100
ccaaaggtta aatagcctcc ataacctttt agaaaatgaa gttttaagtg agaaagaaaa 146160
tgttctcaga gtactgtttg cattcacagc tactcctgga aagtgttgga atattagctg 146220
aaggagtacc ttccatccca cgggaacata attatgaaca cacaaaatac tgtcccttat 146280
gaatatattt taaggcttaa atttgagtat ggaataacca atctgttgat tctttggctg 146340
tagtgatgtg ggtatttgga gctgtttaaa aagaaactgg aaaaaacctg gctcaatctt 146400
tggccctaag tcatcacttg cgtctgaatg attctggcat aaagatcttt agatgccaca 146460
gccaatccag gcccagtgac tggggtaggg agagagaatt agaagctgga aaatgcttta 146520
acggtacctg atagggtgaa agcagggcga agttgttctg aaaatcctgg aaatgggcca 146580
gaaagaagtc agtgctcatg gcatcaatgt gtgagcaggt cacgcctttg accagctgcc 146640
cagcactaga aaacaaaaca aggcactcgg gagaggaagg aagtgaggag gagaggaaag 146700
aaagagcgag agaaggcaga tatgatcaga tacaagctac agatgtgact gcctaactct 146760
cccacctgga aaagtaggct gctctggcag aaggctctga aaggcctgtt cagcacccag 146820
gccgagttac attttcctac ctcaaaagct cctcaggcct tctggtggtc tttaacagaa 146880
gctcatacag gaacagagcc tgcaagaaga gaatgatcgt tagcactgat ggcacactca 146940
gcgtgtgccc aacactgtga ggccctatca ggaaggcatt ccacaatccc attttacaga 147000
tggggaaatg gaggcttaac aggctatgtg agttacctag gactagacca ttgctaagtg 147060
gcagtgcctg gatttgaacc agtgtcaagg taattccaga gccccactct cacactccag 147120
atgattctct acaccccatg aggcagatga gaggcaaaga cattggtggc agggctggag 147180
ttttacttcc tgtgctttag gaagctgctg accagttgag aactccctgt gtctcctatt 147240
cacatgtgca cctgccaggt tctgaacacc aggcatccca gaataaacca ggatgggcta 147300
gaaagcacca gcaataaagt acctgcagct agcagcatgc actgtcttca aaccaccccc 147360
caatgagcac tctagaacaa tgcctagtag acaggggagt catttgtttt aaggaacata 147420
attttacaga atccctaact tgcagacgca atcaagtact acatttgtct attttataga 147480
ctctgaaact ttgagtatca ttccccagac agaataaatt atgagctagc attagacatg 147540

gtagctgaga gccaaaaagc agcacaaaag cccaggcaaa aggaatggac gatggaggaa 147600

agttctggac actgtttgtg gccccagaga ttttcctggg ctcagggtca aattctggga 147660

ccctcccatg gtcttcccca tctcaggaaa cagcaatccc atcagctcaa tggccgagac 147720

aaaagtatct tgaagtcttc ggtggcagaa tgtattttcc aaaataaaac tattggctgg 147780

gtgccctggt tcacgcctgt aatcccagga ctttgggagg ccaaggcggg cagatcactt 147840

gaggtcaaga gttcgacacc agcctggcca acatggtgaa accctgtctc tataaaaaat 147900

acaaaaatta gccgggcatg atggtgtgtg cctgtaatcc cagctacttc aggaggtgga 147960

gccatgagga ttgcttgaac tcaggaggag gaggttacag tgagccaaga ttgtgccact 148020

gcaccccagc ctgggtgacc gagagagact ctgtctcaaa aaataaaaca gaataaaata 148080

aaacggttgc cactgtcact aaatttagag gtggtttgtg acacagcaat agatcagcag 148140

gacatcatcc agcctcatga gagaccctga gcaagagcgc ctggaacaga cagggatttg 148200

tgtcgtttca ttcacttttg cacactcagc actgtgagca gtgcctggcg cactgttggg 148260

ctgtgcttga taaacatttg ctgggtgcat gtaatgaaga ctgactgaat acaaaggtag 148320

tcaacgatgg tactaaggca gagcatccac ttcaacagca ccatcccccc aagcggcctc 148380

aatgctcaca gcatttaata atacacataa tagccgcgaa tacttacagt acacagccca 148440

ggtggcatgt atcagtgacc ctcattttat tatcaaaatt atccccattt cgcacatgag 148500

aaaactgagg ccatataagg aacttctcca aagctaataa ataagtggga gccagaattc 148560

atgctcaggt cttgtctaac ttttttttta agagacaggg tctcgctctg tcgcccagac 148620

tggagtggag tagcatgacc ataactcact gcattgcgga actcccatgc tcatgtgatc 148680

ctcctgtctc agcctcctga gtagctggga ccacagacat gcatcaccat gcctggctaa 148740

ttttttaact ttttgtagag acaggatctt gcttgctatg ttgcccaggc tggtctcgaa 148800

ctcctggcgt caagcgatcc tcctgcctca gcctgtccaa attcttaaca ctatactatt 148860

ctgcctccta tactaatccc acagaaataa atttctttta tcaaattaac cttaaaacag 148920

accattcatt ctcacaagac agatagtcag aaatacagga tcgatctgtg tttcatggta 148980

atacctggct ccttccaagt tccttatcct tcaggactgt agagttgaat ccaggttgcc 149040

tccttaaatc aaagagagac acttccttaa agaaagcccc ttgtatctcc acgatgcctg 149100

gggcagtgtc ttccgcttgg accatctgcc agaagcgaga agcaacaaaa caacattgta 149160

aaaaatgcat tgagctttga ggaagggcca ggcactacat cacaggcaat aaaatccatc 149220

agaaccgctc agcaacccta ggaagtggag agtagcatca tccccatttc acaggtgagg 149280

aaacagagac ttaaagtgtg atgagttgaa agtcagtaag tgacattcaa gcccaagtct 149340

gtctgattcc aaaacccgtg ctccttattt ctttatttta actgtatggg ctacttgcta 149400

tctgcaaggt attggtgttt taggccaaac cccttaggtt ttagggtttt tttcttttt 149460 tgagatggga atctccctct gtcgccaggc tggaattcag tggcatgata ttggctcgct 149520

gcagtctccg cctcctgggt tcaagcaatt tccctgcctc agcctcccga gtagctggga 149580

ctacagaagt gcactaccac acccggctaa tttttcgtat ttttagtaga gacggggttt 149640

caccatgttg gccaggctgg tctcaatctc ctaacctcat gatccgccca ccttggccta 149700

ccaaagtgct gggattacag gcgtgagcca cctcacctgg ccaagttttg gtttcttaac 149760

agattttgcc attggacaga acggacctga tagagcaaga tgtcaaaaga ctccctgaca 149820

agtaaaaaag gggccaggca tggtggctca cacctgtaat cccagcactg taggaggccg 149880

gggcaggtag atcacttgag cccaggagtt tgagaccagc ctgggcaaca tggcaagacc 149940

ccatctctag aaaaacaaaa attagtgagc agcacaggcc tgtagtccca gctacttggg 150000

aggctgaggt gggaggatcc cttgagcccc agaggtggag gctgcagtga accaagatca 150060

cgccactgca tgctggctgg ggtaatagag caagacccta tctaaacaaa caaacaaaaa 150120

aacagaatac ggacatggct gtggaccatg aaaagggctt tcagatgcac cctaggactt 150180

tgggtttatt tttagatttg aaaaacaaat tttaggccag gcacagtggc tcacacctat 150240

aatcccagca ctttgggagg ccgaagcaga aggattattt gaggccagga gttggagacc 150300

agcctgggca acatagcaat actccatctc tacaaaaaat taataaatta gccagacatg 150360

gtggcttgag gccaggagtt tgagactagc ctgcgcaaca tagcaagacc atgtctctac 150420

aaaaagttaa aaaaattagc caggcatggt gatgtatgcc tgtggtccca gctactcagg 150480

aggttgaggc aggaggattg cttgagccca ggagactgtg gtgagctatg atcataccac 150540

tgtactccag cctgggcaac acagaaagac tgtctcaaaa aaaaaaaaaa aacttgaacc 150600

tgaagtgtta taagatcagg aaacctgatt taagaaaagt cttccagatt agactgtctg 150660

gcagaacaaa ggggtctggg aagaatgaca gcatgactga agggctccgt ctggaaagga 150720

aggaaggtgt gccaccagga gaaggcagag ccaccccaga caccaacagc tgagacaatc 150780

ccagccctgg gttcatggcc caaagtcaca gcccactcac caaccccaaa aacatacccc 150840

ctgtgacatg tggctgagca ccagacatct tcctctcacc ttgctgagga taccttgctg 150900

ctgggcaggt gacaagtcgg atacatactg ggagacggca cttctcagga cctgcgagat 150960

gtccttgcgt ttcatgctgc agaaggcctg ggtgctgacc ccagccagca gtgcgcccat 151020

ctggctgaca ttgtagaaag acagcacctg gaagaggagc ggctgcgcag tcaggctctg 151080

ctcccgcctt tcacctctcc aacgtgcact cagcccatct atgtgccaag tatagggatg 151140

ggtgacacct taggggcaca gcagtgagcc agacagatgc tgcctccaca ggccttcctt 151200

ccttctatca agaaagagag ttggccaggc atggtggctc acgcctgtaa tcccagcact 151260

ttgagaggcc aggcgggtgg atcacctgag gtcaagagtt cgagaccacc tggccaacat 151320 ggtgcaaccc catctctact aaaaatacaa aaattagcca ggcatggtag caggttcctg 151380

taatcccagc tacttgggag gctgaggcag gagaattgct tgaacccagg aggcagaggt 151440

tgcagtgagc tgagattgta ccattgcact ccagactggg caacaagagc aaaactctgt 151500

cagaaagaaa agaaagaaag aaagaaagaa agaaagagag agagagggag aggaaagaaa 151560

gaaagaaaga aagaagagag agagagagaa agaaagaaag agagaaagaa aggaaagaaa 151620

gaaagagaaa gaaagagaga aagagagaaa gagagagaga aagagagaag gaaagaaaga 151680

aagaaagaaa ggaaggaagg aaggaaggaa ggaaggaagg aaggaaggaa ggaagaaagg 151740

aaaagaaaga aaggaaggag gaaagaaaga aagaaagaaa aaagaaagaa aggaaagaaa 151800

gaaagagaga gagagaaaga aagagaaaga aagaaagaaa ttatacttgg gtttttttc 151860

cttcctttag agtgaagatg ctagatagtt ttccatataa tcaaggacat gctttggttc 151920

tatgaaaaga cagctgagtg ggttccttta gttattctgt gtataatggg tgatctcatg 151980

ctgtcttcaa agaaggcaag acctttgac ctcttgcctt ctgagtaaaa gtggcctcac 152040

ccctcagtag gaagagttgg gtattaggaa agagacaaac cactttgtcc tgggctggga 152100

gggaacaaaa ccgtctccct caactcccta aaatcaaatt cagagaggac tgtcaaggtg 152160

gacccatgga gccccagtca aggtccagaa acaaggattc aaagccttca acataaagtc 152220

accacgaggc tagaagagac cagatgaatg ggctggcctg gtacctgagt cagaaagtgg 152280

gagtgcgtgg gcattggtca tggtgccata atggagacag tgagcacagg agttaaacaa 152340

gatggctctg aggccaggtg ccctgggttc aatcccagct gcgtaacttt cacgtggcct 152400

tttccagttc ccttacacac tctgtacctc acatgaatga actggaaaat gaagactaca 152460

gcactactga cttcagagga ttgttggatt aagttattaa ttcacttaga acacaacctg 152520

gcacatagta agtgttcagt aaatgtttgt tattccacac cctccctccc ttggccccgc 152580

gatggaggaa gcaggctagg accagccctc ggagctgcag ctgcccttca tccctccctc 152640

gccctctcta ccgacatcct gctccagttc ccacttggat ttactttggg gaatgtggtt 152700

ggaatatatc tggctcagaa ccatgacata ccaaacccag ctttaaaaac ttgaagaaat 152760

ttaaaaagac ttggatgcca agaagaaatt ccctactcct tctctttggg aggccaaggc 152820

gagtagattg cttgagccca ggagtttgag atctgcctgg gcaacattgc aaaatcccat 152880

ctctacaaaa aaatacaaaa atcagcccgg cgtagtggca tgtgcctgtg gtcccagcta 152940

ctcaggaggc tgaggtggga ggatcacctg agcctggggg gtcaaggctg aagtgagcca 153000

agatcagatc acttcactcc agcctgggca acatagtgag aacctgtctc aaaaaacaaa 153060

aaaagaagaa gaaacccccc agttcctgag gccaactcca gcactgcctt ctggatgcat 153120

ggattctact ctataactct taaacccctt taccgcctga atcaaaagct tttgttttca 153180

ttttaccacc tgaatcaaaa gcttttgttt tcatttccaa cctcagtgat gcgatctcgg 153240 ctcactgcaa gcaccgcctc ccaggttcac gccattctcc tgcctcagcc tcccgagtag 153300

ctgggactac aggcacccgc caccatgccc agctaatttt tttttttttt atttttagta 153360

gagacatggt ttcaccgtgt tagccaggat ggtcttgatc tcctgacctc gtgatctgcc 153420

tgtctcggcc tcccaaagtg ctgggattac aggcgtgagc catcgtgcct ggcctagtga 153480

ttttcttcct tgtgagacac tgtggtattt tgctttagaa caagcaaaat ggggcccctt 153540

acattttcaa catcttcacc tcttcccata tcctccttca agctgcatgg gaggtactaa 153600

cactagaatt acggtgctag gttagtaaac atgaccttta aggagtagtc tctcctttat 153660

tctttgggat tcctactact tttttctttt ttttttttaa gacagagtct tgctctgtag 153720

ctcagcccag tctggagtac agtggcatga tctcggctca ctgcaacctc ctctgcctcc 153780

tgggttcaag taattcttct gcttcggcct tccaagtagc tgaaattacg ggtgtgcacc 153840

atcatggcca gctaattttt ctatttttag tagagacagg gtttcaccat gttgtccagc 153900

tggtctcaaa cccctagcct caagtgatcc attcatcttg gcctcccaaa gtgctgggat 153960

tacaggcata agccaccatg cccagcctac taccttttgt caaaataaaa attgatgagt 154020

tttgtatagt tggtcagaca cagttaaaac taaattcaca gtttagcaat tataatgggt 154080

gcttgttaaa catctggtag taaattatgt tgtttcaaag taattaaaat taatatccag 154140

aagccaaaaa taaacaaatg tttgttatta ttatttgatt ggaatgggtc ctaatccaat 154200

ccttgtgagc cagtttgcat cctgggaagt ggccaggagt gtggactaaa cgagaaggac 154260

accaaaagcc aggcacggtg gctaatgcct gtaatcccag cactttggga ggctgaggca 154320

ggcggattac aaggtctgga gttcgagacc agcctggcca acatggcgaa acctcgtctc 154380

tactaaaaac acaaaaattc gctgggtgtg gtggtgggtg cctataatcc cagctactcg 154440

gaaggctgag gcaggagaat tgcttgaacc cggaggcgga ggttgcaatg agctgaggtt 154500

gtgccattgc attccagcat gggcgacaag ggcaagactc cgtatcaaaa aaaaaattag 154560

atgacaccaa aatgcttaga actcaagtct cccagatttg gagccctctg gactggaccc 154620

aacccaggga cattaattgt ccccaaaaga atgtccattt tccaccccag gagcagcaag 154680

aacacagatt aagactcctg tccccaccaa agaggactgt ctgcctgaca gttccctctc 154740

tgcagcccaa gattggaata aaatagcatc atccctaatc ccaaagtaat ggcaaaaacc 154800

gcaattactt tttgcatcaa cctaatagca gcttaacagc aggcaggaaa atcctgggat 154860

gagtcccaca cgttgatccc actcagggat ggctaatgcc ccttgagctc ctactctgtg 154920

gtgtcacctg gtggtctagc ttcttcatgt gtaccaactc acataatatt cgcagcaatg 154980

ctacaggata agtgctactg atttgttgtt gttgttgttt aattttttttt gaaatggagt 155040

cttgctctgt cgcccaggct ggagtgtggt ggcacaatct ctgctcactg caacctctgc 155100

ctcccaggtt caagccatcc tcctgcctca gcctcccaag tagctggaag tacaggcatg 155160
caccaccacg cccagctaat tttgtattt ttagtagaga cgggatttca ccatgttgat 155220
caggctggtc ccaaactcct ggcgtcaggt gatcctccca cctcagcctt ccaaagtgct 155280
gggattgcag gtgtgagcca ccttgcccgg cagtaagtgc tattgttgtc cacatattac 155340
aaatagagaa actgagtcaa agtgaaaata gcaaattgcc tagtgtcaca cagctagtaa 155400
atggcagagc aaggatttga acccaggcag tcctcctcag catcatcttc ttactgattg 155460
gttgccccct gctgtgtgtg tataactgat tgttatggat taaatatttg tgtctcccaa 155520
gaattcatat gtcaaagtcc taacccctaa tgtgatggca ttgggaggtg agacctttaa 155580
gaggccttta agggttagat gaggtcatga gataatgagg gtcaggtggg gccctgatcc 155640
aatgggatta gtttccttgt aagacgcaac accagagagc ttgctctcat gtggtcaggt 155700
gagcacacag caagatggcg gccacctaca ggccaagaga agaggcctca gaatgaaacc 155760
tacctttctg acacctcgat gttggacttc ccagtctcca gaactgtgag aaataaattt 155820
atgttgttta cgtcactcaa tctacagtat atttttatgg cagcctggca ggtctaatac 155880
acttaggtca tcttagggga ataaaaaaga gtttaaaact ccagctgacc ttctcatggg 155940
ccaagtattt ggcagacaag atgatgacct ggctcttggc ccaacctgag acctggttga 156000
gggtggagat ggctccgtgc acagcctcgg gggagaggga ttccagttgg ctgtaggtca 156060
aacctacaac tgcatccgac aaagaaccca gggtctcatt gagggtctgg ttctccatgg 156120
caatgtccag gagttctgct ttgagctgga aggagagcaa actggaatga gtgttgacag 156180
gaagcagggc ggtggctccc tctgaccata tccagggcta ttaggactcc acagcacctc 156240
ccttgaccct gttccatcaa agcttttggg ctcagaaacg atgaaggaaa gggactcctg 156300
ccatgcccag acagtgagac accacacgga ggacacctct tctggcctta gctttcatgt 156360
gccaaaaatt ccctttgtac ttcatttaaa tattaatttg ttattaaatc aaccttaatc 156420
catgtcagaa agaaatgttc cttttataac tacaccacac tcttatagca tctgttgtcc 156480
tctggtgacc ttgaactcta tatccatata tacaggctgt tttcctttca ttaaatagcc 156540
aggggccttt tcacgataat ttataatctg tgtaaaaata actttttttt ttttcttaag 156600
acagagtttt gctcttgtca cccaggctgg agtataatgg tgtggtcttg gctcactgca 156660
acactgcaac ctccaccttc cgggttcaag cgattctcct gcctcagtct tctgagtagc 156720
tgggattaca ggcacccacc accacactaa tgtttgtatt tttagtagag acagggtttc 156780
accatgttgg ccaggctggt ctcaaactcc tgacctaatg taatccacat gcctcagcct 156840
cccaaagtgc tgggattaga ggcatgagcc accacatcca gccaaaaata actcttcaca 156900
gtacaagaaa gtaaaagtct cctgaacatc ctaccacata aaaataacca tgattagcaa 156960
ttgggtgtat atgcttccaa gatttcttat acctttataa atatagagac actacatagt 157020

ttttacagag ttgtaatatg aacagaagca caatttactc ttcttaaaat tatctcttgc 157080

acctctttgc acatttctat gatcatttta ataattattt tattgcaata tgcctgccta 157140

atgatagtaa agtgtgcgtt acagctgctt tcggtaagga atgtgataaa gtcacctact 157200

atacaatgag ctctgtaaca aaaaacaaga atggttcata ttttaacacc cgaatttacg 157260

taataacgta gtcatttcag gcaggtgcac aaaacgggtt tctggcaata ttgaaatagc 157320

cactgggggg cagcagagtg aagtagaaga aacaactgtc aaagcgcctg ggttctctaa 157380

gttcggcaac tgccttacct agaaatcagt ttccacatct gtaaaacgaa ggggtggact 157440

acagtggcag ctcccaaagt gtggagcaca cccagcggca tctgcaacac ctgggaactt 157500

gttagaaacg cagattgcca ggctgctccc ggacctcctg aatcagagac tgggtggggc 157560

tccgaaatcc agggatcccc agactccggg tcacagatgg ggaccaccgg gaccctggcc 157620

tgttaggaac cagccacagc aggaggtgag cagcaggcca gtgagcatta ccgcctgagc 157680

tctgcctcct gccagatcag aagcggcatt agattctcct aagagcaaac cctattgtgc 157740

actgtgcatg caagggacct agtctgtgcg ctctttatga gaatctaatg cctgatgatc 157800

tgtcactgtc tctcatcacc cccagatggg accgtctagt tgcaggaaaa caagctcaga 157860

gctcccactg atttcacatt atggtgagtt ctatagttat ttcattatat attacaatgt 157920

aataatagaa ataaagtgca cagaccgggc gtggtggctc acgcctgtag tcccagtact 157980

ttaagaggcc atggcaggcg gatcacgagg ttaagagaat gagatcatcc tggccacatg 158040

gtgaaacccc gtctctacta caaatacaat aaatagctgg gcgtggtggc gtgcacctgt 158100

agtcccagct actcaggagg ctgaggcaga gaattgcttg aacctgggag gcagaggttg 158160

cagtgagcca agatcgtgct accgcactcc agcctggcaa cagagcgaca ctccatcaag 158220

aaaggagagg agaggggaag gggaaaggga tggggggggg aggggaggg agggagggaa 158280

ggaggaaaga aagaaagaga gagagtgaga aagagaagaa agaaaggaag gaaggaggga 158340

agaaagaaag agagagacag aaagagaaag aaagaaagaa agaaagaaag aaagaaagaa 158400

agaaagaaag aaagaaagaa agaaagaaaa ggaaggagaa ggaaggtagg gagggaggga 158460

gggagggagg aagggaggga aggaaggaag agcacaataa atgtcatgca cttgaatcat 158520

cctgaaacaa tcccccggcc ccagtccacg gaaaaaaatt gtcttccaca aaaccggtcc 158580

ctgatgccaa aaaattgtct tccacaaaac tggagcgctg ccacaagctg tatttgaaca 158640

ggccctccct gtgatttaga tgcaccctcc agtttgagaa caactgagct agatgatccc 158700

aaagggctcc ctcgtatcta attgcatgcc accttctcct tggtcagggc tgccacatgt 158760

tgagacccct gggccttaaa atacatttat cttcttattt ttgcttgagc ttgcttagga 158820

cggattcttt tccttggaaa caaaaatatg cattcacagg ctaggctcgg tggctgacgc 158880 ctgtaatccc agtactttgg gaggccaagg caggtggatc acttgaggtc aggagttcaa 158940

gaccagccta gtcaacatgg taaaacccca tctctactaa aaacacaaaa aattagccgg 159000

gcatggtggc atgcacctgt agtcccagct actccagagg ctgaggcaag gcgaaggttg 159060

caatgagctg aaattgtgcc acttgaaccc ggaaggtgga ggttgcagtg agctgagatt 159120

atgccactgc attccagcct gggtgacaga atgaaactca aaaaaaaaaa aaaaatgaat 159180

cttataaaaa aaaaaaaaaa gatgcgctga ccaaaataga tgaccacact ctcaaatgtc 159240

aaatgtgttt ggggacttta tggtgatgtg tgggaaactg ctgtgaaatt attgatggct 159300

ttatcaaaat ttcatttaat attattttca tgtcatcgtt tcgattttaa gcaaattgaa 159360

aagggccaca aaaagtggaa tggaaggggg tagaattgag acttgactgg aaggaagaag 159420

agagagatgg agaggaagga aaatgagaga gagcgtgttt gctcatctgg tcttgggaac 159480

tcataaccac cctgcaatcc agtgttctca atcctggctg cattgaaatc acctgagaag 159540

cattaaaatc aggccatacc tcaggccaca ccccagaaca ataacatcag attctctgag 159600

ggtggcacct ggcagggatg ttactaatct agaggcaggt ttgagaacca ctgctttatc 159660

tcctgccgct ccaggtgtgc tcaccagagc agcggttcca gaatcacctg gaagcttgtt 159720

ggaagtgcag cgtcagctgg gcgcaatggc tcacacctgt aatccaaaca ctttgggagg 159780

ccaagggaag aaggaggaag aagaggaaga tgggagtcct gggcctcaac caagagctcc 159840

tgaatcagaa tctgcattta agatccccag gtgatctgta tgcatattta agttcgaagt 159900

agctgtccct ttcttttctt ttctgttctt ttttttagat agggtcttgc tctgtcatcc 159960

aactagagtg caaagtcact atcatagctc actgcaggct ggcactcctg ggctcaagcg 160020

atcctcctac ctcagcctcc caagtagctg gcactatagg cacatggctt tttttttttt 160080

cttttgagac agagtctcac tctgtcaccc aggctggagt gcagtggcac gatctcagct 160140

cgccacaacc tctgcctccc aggctcaagc aattctcctg cctcagcctc ctgagtagct 160200

gggattacag gcgcatgcca ccacgcctgt ttttgtattt ttagtagaga cagggtttta 160260

ccatgttggc catgctggtc tcaaactgac ctcaggtgat ctgcctgcct tggcctccca 160320

aagtgctggg attacaggtg tgaaccaccg tgcccagcca ttttttttgta ttttttgcag 160380

agacaagggt ctcatgatgt tgaccaggct ggtatcaaac tcctggcctc aaatgatcct 160440

cccatcttgg cctcccaaag tgctgggatt acaggcatga gccaccgcgc ctggctgaga 160500

cagctttctg attaaatttt gatctcccca gggtctgttc ctttgctatt ttctgtctgg 160560

gcttcgatgt tacacttacc tcagtatttg agatgccatc tgctgtgact ggacttaaat 160620

cccgactaat gctgtgtgtt ttctcccaaa ggctaagcta ggctgttctc cccatttctc 160680

tggctagtga cactgaggca gaggcctcag ggccttgcac cccagctgta ccttctggac 160740

gccagcctgg aagcccctca ggtggctgca cttgatcatc tggtaaagca ggacttgagc 160800

cacagtggca tccagcaatt ccaggtcatt gtagaaacaa accagcagcc ccagcctgtg 160860
tgagaagagg agggatggtg gtggaggtgt aaccgcagaa ccagcccatt ctggttcaat 160920
tttgtgtaat aaaatggtga gttgtttttc agttgccacg gactcccagg ttgaaggtca 160980
cataacctga acatcctcag atgaaccaaa tgtgcaacca caggcggaac ctaactgctc 161040
agacgagacc aaggaatggg ggctgaatta tgaagtggac accacatggc atggtccacg 161100
atccaatcag aatgagtcct ggcatcacct catggcatga tccaatcaga tcacacctcc 161160
cagcatcacc ttgtagcaag acccaatcag atcaagtctc attaccctcc gcctataaaa 161220
cctgccccag tccccagctc agagacacag atttgagcac tgtcttctgt ctccttggca 161280
gttgattcac agtaaccgtt tctctctaca aaaacctagt gcttcagtgt ttggttttcc 161340
attgcgcatg ggcaaacaga cccagtttgg ttccataaca gagcgctgag agctggtgga 161400
cacgacaccc cctccccagt ccaaaccaat ggggtcttga ttatattaat agcagctacc 161460
acttatcagg tcttgcactg agccgtttac ctgcattatc tcattacagc aaccctacaa 161520
ggcaggtgct ctcattagcc ccatttcaca gatggggaaa ctgagatttg atttgtccaa 161580
ggcaagtcgg tggagctggg acttaaatgt agatccatcc acctatgcca tctctccacc 161640
tgggatgaat aaagggaata aggaaagaaa ggtgccggta gccacagaaa actccccatt 161700
ttccacatta caatcactaa attctagagc tgatctcacc ctgagggatt ctctgaggca 161760
gagatctgtt tctaggatgc cttctggtac ccacccagat gatgcaaaga tttcagaagg 161820
aaaacaaaaa agactggagg aaaacaatgg aagggcagaa gtgagattgg aaccctggta 161880
taactgagag caatgtttct aatcattaag agcattgatt caggcctact gcagtttgaa 161940
gcccagctct gtgagatgtt tgctctccac caagttactt aacttctctg tgaggccgct 162000
atacacagtt ctgcagtgca cactctgtac tcccatactg tacagagaca gggtgcctct 162060
ccctctctga gcctctaata aatgggaggt tttctttgtt catcgtagca catcataata 162120
aacatgagtg aaagaaagag gatgattgtg aaagttaaat gagaagtggt ataccaactg 162180
atttgcatac atcctggtgg aaaccaagca ctcattcatt aatgaatgct cttagagtta 162240
gaatgaagac tacctagttg accacagtca gcctggctcc accctctcag agcgatattg 162300
acatattcag acatgcccac aggagagctg ccaggtccag ggaaaagatt tgaaagtgtg 162360
tctcatgagg aaaagagatc tgcttaactt cgagtaaaga agactcaggg gaagtgaggg 162420
ctgaacgtta ataaccacgt tgagaataac tattcacaac agccaagaca tggaatcatc 162480
ctaagtgtcc aacacgtcag tggatgaatg gatttttttt ttaatttggt ggatacacaa 162540
tggaatgctc ttcagctttg aaaaagaagg aaattctatc atatgggacc ccatggatga 162600
acttggagga tcttatgcta agtgcaataa gctaggcaca gaaagacaaa tactgtatga 162660

cctcatttat attatgaatc tgaaaaacaa ataaacaaac gtcatactca tggaagcaga 162720

gccttagaat ggttggtacc aggggctgga ggtgggaagg gagttggaga gacttggaca 162780

aaggacacaa tgtttcagtt aaataggaaa ataggttctg gagatctatt gtacatgctg 162840

actacagtta ataacaatgt atcatatact caaaaattgc taagagtaga ttttaagtgt 162900

tctcaccgca aaaaaagtat gtgaggtaat gataacttaa ttagcttgat ttagccattc 162960

cacaatgaat atatatctca aaacatgttg cacaccatag atataatttt tatttatcaa 163020

tttaaaaaaa taaattataa aagaaaaata ggcttggcgt ggtagttcat gcctctaatc 163080

ccagcacttt gggaggccga ggcgggcgga tcacaaggtc aagagatcaa gaccatcctg 163140

gccaacatgg tgaaaccccg tctctactaa aaatacagaa attagctggg cgtggtggcg 163200

tgtgcctgta gtcccagcta cttgggaggc tgaggcagga gaatcacttg aacccaggag 163260

gcagaggttg cagtaagccg agattgtgct actgcactcc agcctggcga cagagtgaga 163320

ctccgtctca ataaataaat aaataaaaag aaaaaatatt attcattgag cacttactat 163380

gtgtcaagtg tgcaaatccc tacaaagacc ttaatgcagt aggcatattt ttatctccca 163440

ttttacagat ggggaagcgg aagtatagag aggtgatgta aaaatagcaa agatcacacc 163500

attagtaagc agcagggaag gattaagcca cgaagtctgg ctcgaggtcc catgttctta 163560

atcatgaggc tatttttgtg tgtgtgaaac catagaaaaa tgagacacat gctaataagt 163620

gaaaattact aaaggtaggt gaaacagatg taattgtttc ctatgagtac attccctaag 163680

agtggctcat ttgaaaaacc ttaaggtaat atgggatggg agcagtgtgg caataataag 163740

accttgagag acggacacaa gaggattgaa tgatggccca gcacagtggc tcacgctgta 163800

atctcagcac tttggaagtc caagatgggt ggatcacttg aggtcaggag tttgagacca 163860

gcctggccaa catggcgaaa cccagtctcc actaaaaata caaaaattag ctgggtgtgg 163920

tggcacacgc ctgtaatccc agctacttgg gaggctgagg caggaggatc tcttgaatct 163980

ggaggtggag gttgtagtga gctggtattg caccactgcc ctccagcctg gacaacaaag 164040

tgagactctg tctcaaaaaa aaaaaaaaaa aacaggattg aatgggccag gcgtggtggc 164100

tcatgcctgt aatcccagca gtttgggagg ccaaggcagg cagaccacct gaggtcagca 164160

gttcaagagg tcaggagttt gagaccagcc tgaccaacat ggcaaaaccc catctctact 164220

gaaaatacaa aaaatttagc tgggcatggt ggcacacatc tgtagtccca gctactctgg 164280

aggctgaggc atgaggattg cttgaaccca ggaggtggag gttgcagtga gccaaggtca 164340

catcactgca ctccagcctg ggtgacagga caagactctg tctcaaaaaa ataaataaat 164400

aaaaggaaaa ggagggagta ttgagctgta ctcaccacct ccttcccatt gcccactccc 164460

ttcctgacac ccctggaatg tagagttcct gtccctcctc ccctgctggg aatggtttta 164520

gggaggatca cattacagtc actaaattct agagctgatc tcatccaggg gctccccagc 164580

taagagtctg ataacaagtg gttctcatat atggggaccc atgagttatt tccaacactt 164640
gactcaactt aaacggaaat cacacattca ctttggaaca ggacccagtc ctgagtattt 164700
aaaatgtttc atttctgtgc tgagagacag aattagcact tgataaggtt gcataaaatg 164760
cctggcacac aggagatgct cagaaagcat ttatcctttc acccagcttc ataacctctt 164820
cataaaaaaa gttgcagaca cctctcctca catgcacaga gaaatatggg actattcaaa 164880
gagatggacc agccacctcc cttccctccc tgggtgtttt gctgctcaga gaattctgat 164940
gcttagatca catcttggga aagggctcca aggcccagag ctcatgcgct tgcctgtgga 165000
tggtggaggt attcctcatg ttaaagttgg aggagctgat cctctccaga aacgcctggg 165060
ccagctcagg tgtgatgtca tagaccatgt ccagctgctt ggtggcgttg tcatagctga 165120
taaacagccc aatctagttg gtggacaagg acgagaatat cagtgaggag ggtggaagtg 165180
gcccagtgtg gccccaccct ggtggtctgc actgtgcccc atcatggaca cttggataca 165240
cctcctggtt ctcattgtca ttgatgtctt tttttctttt cttttttttt ttttttttga 165300
gatggagtct cactctgtcg cccaggctgg agtgcagtga catgatctca gttcactgca 165360
acctccacct cctgagttca agcaattctc ctgcctcagc ctccggagta gctgggacta 165420
caggtgccca ccaccacgct tggctaatat ttgtattttt agtagatatg gggattcacc 165480
atgttgtcca gggtggtctc gaactcccag catcaagtga tccacccgcc tcggcctccc 165540
aaagtgctgg gattacaggc gtaagccacc atgcctggcc tcattgtcat tgatttctta 165600
gtggtctgta actgctactt tagtttcctc ctcaacctaa ctattcttta ggaaagaatt 165660
atttttaat atctgagaaa ctgggctttt taaaagctaa tctttgcaca tttatttcta 165720
gatttgttat atggaggtca gagaatgtgg tccacaaact ttctgctttg aagaatcaga 165780
atttttttaa tagatgaatg agtttataaa tggcccttgg gtgatggaaa agagtatgtg 165840
ttctctttgt aagacacaaa gtttggtgca tatatattca acttattaaa tatactattc 165900
agattcccaa cattcttgtt cctttttggc ccacatgtcc aagggtgcca cgtgcaagtc 165960
ttctactact acaggactca tatctatcta ttcttccttg tatttctagc agattataaa 166020
ttatatatat aattttgggg ggacactgac tcactctttt tattttattt ttgtttttat 166080
ttttattttt atttttattt ttttgagatg gagtctcgct ctgtcaccca ggctggagtg 166140
cagtggcaca atctcagctc actgcaactt cagcctcccg ggttcaagtg atactcctgc 166200
ctcagcctcc tgagtagctg ggattacagg catgtgcaaa catgcccagc taattattgt 166260
atttttagta gagacggggt ttcgccatgt tggccaggct ggtcttgaac acctgacctc 166320
acatgatcca cccgcctcag cctcccaaag tgctgggagt agaggcgtga gccactgcgc 166380
ccggccaaga cacagtctca ctcttttacc caggcttggg tgcagtggtg tgatcatggc 166440

ttattgcagc ctcgacctcc tgggctcaag tgatcctccc acctcagcct cccaagtagc 166500

tgagacctca ggcacacacc actacacctg gctaattttt aaattttttt gtagagatga 166560

ggtttcacta tgttgcccag gctggtcttg cacttctggg ctcaagtgat cctcccgcct 166620

tggcctccca aagtgctggg attactggca tgggccacgg cagccggcac attctatatt 166680

ttgagacagt attatgctgt gctcaaaggg gctcataaag ttgtctctcc tttgtggatc 166740

gtactctttg tccatataaa attccttttt aggctgggtg cggtgtctca cacctgtaat 166800

accagcactt tgggaggctg aggcaaggca gatcacctga ggtcaagagt tcaagaccag 166860

cctggccaac atggtgaaac ctcgtctctg ctaaaaatat aaaaattagc caggcatggt 166920

ggcatgtgcc tgtaatccca gctactcaag aggctgaggc aggagaatca attgaatctg 166980

ggagaaggag gttgcagtga gctgagatca cgccactgca ctttagcctg gacgacagaa 167040

cgagactcca tttcataaat aaacaaataa aataaaataa aattcccttt tagcagtgtt 167100

attcttgact tctcccttct ctacattttt agcgaataaa aagttgcaaa tgagtgcttg 167160

accttagaaa tttcctctag ggggctcagc tttctaagaa tttaagctgt tacctactca 167220

aagcaactct acattgagag agtaaatggc agtggggtgc ggggttaatt ctaggactgc 167280

gggttataaa aatcggaaga cacttgaaat actcccagcc ccctaggtaa aagagagaaa 167340

gctcgttctg cctctggttt cataataacc acacattgct gggtcaatgg gtccttgaac 167400

aggccagtca tttgatgcaa acacttctct cactggccag aagagtcagg gcacatatga 167460

cttaaaaaag aagaagaaga agaagaaaaa ttggaagcaa ggcccacagt ctcttcatga 167520

atccctttgt gttggcagca ttctttggag accgcattca agaatcatga tgcccaaggg 167580

attttagtgt cattgggtca acatcaaaag gagagggaca ggttgactag tgagttaagt 167640

aatcctagaa tgtgtgtatt tctccactgt aaactctaga attaaaatct aaccattgct 167700

catgctcaag tagctattaa cagaagtagc tataaataga agtagctatt aatccatttt 167760

gcttttgctt tgagttcatg atcccaagga tgggaaactt tatatctttt gtccttggcc 167820

ttctccagta caaaaatacc cttaaatgat ttttggttca tagccaacaa tccctttaaa 167880

tcatgtttgc tcacatcaga gatgtgatct tagttgtcca caattctttt tcttttctta 167940

gcccttccac actaaaacaa tcccatgact gcccacttga aactttcttc caatagtttt 168000

cttcagaggc ccaaggtcta ttgggctgag gagggtacac aatttccaag gcaccttgac 168060

tggctggatg aacattagag tttgagaata caccttaccc caagtgtatt agtaatggtg 168120

acatataaat gtacttttcc aacttacttc tataggacta attttcgtaa tttcttcaaa 168180

cgataggtga accatgtatc tgcccaaaac ccataagtgt tccgcactga cccatgaagc 168240

agagtcatct acaaaaaata aagaaatgaa taaatgagta catacataaa taaaagcctt 168300

tacactggta tgtgatagtc aacagcatct ttgctgtcaa gacctacttc ttgcattcat 168360 taattcgtta ttttgttttt cggggttttt tgagacagaa tctcactctg ttgcctaggt 168420

tggagtgcag tggtgtgatc tcagctcatc tcaacctccg cctgccggtt caagcgattc 168480

tcctgcctca gtctcccaag tagctgggat tacaggcacc caccaccatg cccagctaat 168540

ttttgtattt ttagtagaga cagagtttca ccatgttggt caggctggtc tcaaactcct 168600

gacctcaggt gatctgcccg cctcttgcct ccccaagtgc tgggattaca ggcatgagcc 168660

accatgccca gccacattaa tttgttatta actcattcaa caatcatata cgtagggcct 168720

acctgtgaga tcacacgtag gatgtgaaca cagattcgca ccacctcaga ccttgttact 168780

caatagtgtg gtctttggaa ttgcaacatc tgtgtcacct ggccacttgt tagaaataca 168840

gattattctt tctggccagg cacctgtaat ctcagcattt tgggaggctg aggcaggagg 168900

attgcttgag gccaggagtt tgagaccagc ctgagcaata tatcaagacc ctcatcgcta 168960

caaaaaaaaa aaaaaaaaaa aaaaaagcca gctgtagtcc cagctactca ggaaactgag 169020

aagggaggat tgcttgaaac caggaatttg aggctacagt gagctatgtg ctcactctac 169080

tgcactccag cctgggcaac agagtgagac cctgtctctt taaaaaagta ataataatta 169140

aaaataaata aataaataag aaatgcaggt tatttttaat acttaaaacc cacaaaatgt 169200

agttttattt tttcattcga ttggttacta aataatctgg aaacatgagg ccagatgtgg 169260

tggctcacac ctgtaatcac gaaatcccgt ctctactaaa aatacaaaga aaaattagcc 169320

aggtgtaatc ccagctactt gggaggctga ggcaggagaa tctcttgagc ccgggaggtg 169380

gaggtggcag tgagccggga tcacatcact gcattccagc ctaggtgaca gagcgagact 169440

ctgtctaata ataataataa tgataataat ctggcaacat gaaatatgtt tgactgtctt 169500

gcatcactac cactagtaag gatcatcctc actgattatc accatgttac tatacttgaa 169560

ccaaagcaaa ctacttgtac ctccccaaaa cattctccaa atcttggctg gccttggccc 169620

tgctcaccac ctacaatgct tttccttacc ccttctgcat ccaaatccta ctttaagatc 169680

aagctcaaat tttcctctcc cttaaagcct tctcttatct ccttcacgag aagtactttc 169740

tccctccact gaattcccag aggactctct ctgacacccc gttctggctt ttgaaattcc 169800

ccatcactga gatgtcagtg gaattttaaa aaccatcaag tcagccaggt gcggtggctc 169860

aagcctgtaa tcccagaact ttgggaggcc gaggcaggtg gatcacaagg tcaggagatc 169920

gagagcatac tggtcaacat ggcgaaaccc catctctaat aaaactacaa aaattagcca 169980

ggcgtggtag cgcacgcctg tagtcccagc tactcaggaa gctgaggcag gagaatcact 170040

tgaacccggg aggcagaagt tgcagtgagc tgagatcata ccactgcact ccagcctgag 170100

tgacagagcc agactccatc tcagaaaaaa aaaaaaaaaa aaaaaaaaat atatatatat 170160

atatatatat ctcaagtcta ggatcagact tcaagtttca ctgagctgga agtggctgcc 170220 aatgctcccc agctctttag caaaagacat ttacacacga tattgtattg gaggcatttg 170280

gggaaaatga aggaagtggg gagcatttac agggtgcagt gactctaaca tcaagagcta 170340

tttgcagaag ccgtgggcaa tgacagatgc caaaacaaga tggagaaatc aacttttata 170400

tagactgatt cacaagaaaa tatggagtgc ctctcccaaa ccaggaatca aagatgttgg 170460

ggtgacacag gcagactcct acgatcctct agatggggac cctggacatt tgccttgcct 170520

atatagaggg ctggaaactt ttgaggctag aaccacacat tcacaaagaa ccaagcttag 170580

ttgtttattt tttaacttat tactaagcat aaactgtatt tctgtagatc aatcatcccc 170640

aagcttggga tttttttttt tcttcctctg ttgtccaggc tggagtgcag ttggatgatc 170700

atagctcact gaagcctcaa actcctgcct cagcctccct agtagctagg actacaggca 170760

cacatcacca catcaggcta atcttttaat ttttttgtat gggggggggg gtctcactac 170820

attgcccagg ctggccttga actcctggcc tcaagcaatc ctccttcctc agccttccaa 170880

aatgctagga ttagaggtgt aagcgaccac acctggccag caaggttggg atatttttaa 170940

cagccaaagt atttccagtt ccctcaaggg ccttcatgaa aaaacaattt aagtccaaac 171000

agaattaatt ttaactcact gtagtttaat aatgaagcgc accgtataag aattttagaa 171060

ggaaagtctg tgcctaatta aactctggca ataaagacag agaagtctga aggtagagag 171120

gctttctcat ggttacccag tgtgagactc tgattcctgg agaccacaat tatgcaccag 171180

gcagagggaa ttctactatg catttgagac tttgattatg atgttgttta atgttcatta 171240

tgcacaaatc tcagagctga attccaggaa aagattgatt ggcattcccc atcctccagc 171300

cccatctgct ttccttatgt tttccccaca ccgagctcat tcccgtctca gggcctttgt 171360

atttcctggg atttctctct gggatgccct ttctccagag ccttacgtga ctggttccat 171420

ctcctcattt tggtcttgat tcaaatgtca cccacttgag aggtcttccc tgattcctta 171480

gtccaagagt tggcaaacta caatccatgg ggcaaactct gcccatcacc tgtttttata 171540

cagcccatga gccaaaagtg gtttttacat ttattatttt ggactttttt ttagagacaa 171600

ggtctcgctc tgtcacccag gctggagtcc agtggctcca tcacggctca ctgcagtctc 171660

aaactcctag gttcaaggga tcctcccccc ctcagcctcc agggtagctg ggactacagg 171720

catgtaccag gacaccggct attttttaaa aaatttttta agaaatgggg tcttgctatg 171780

ttgcccaggc tggtcttgaa cttttggctt caagtaatcc tcctgcctgg gcctcccaaa 171840

gtgctgggat tacaagcatg agccactgca ccctgcctgc tttttacctt ttttaatagc 171900

tgaaaggaaa atcaaaagaa gaatcctatt tggtgacaca tgaaaattat gcaaatttca 171960

gcatccatta gtaaagcttt actgggacac aggcatgctc attcatctat tgtctacagc 172020

tgtcttcaag ctgcagcgtc agagctgaat agttgaggca gagatggtag gcttacaaag 172080

cctaaaatat ttacctggtc ctttacagaa aacatttgcc aagcgctctt ctagtctaaa 172140 gtacctgtaa tatcctttct gcctgggtgc agtggtttat gcctgtaatc ccagcacttt 172200

gggaggccaa gccaggtgga tctgttgagg tcaggagttt gagaccagcc tggccaacat 172260

ggaggaatca catctctact aaaaatacaa aaatcagctg gacatggtgg caggcaccta 172320

tactctcagc tattcaggag ctgagagaat cacttgaact ctggaggcag aggttgcagt 172380

gagccgagat tgtgccactg cactccagcc tgggtgacag agtaagactc cgtctcaaca 172440

aactatttta ttttcttcat agccgctacc agtatctaaa tttctaagtt ctctctctct 172500

ctttatttac ttacatgttt aaaaaaattg tctccaccaa cactcccaca ataaaacaat 172560

agggccgtaa gagcagagac tttgttttgt ttccttctct atcttcagct attgatacat 172620

aatgggcttt taaaaagttt attctgttta cattactgac attaaaggtt taacaaattg 172680

aagctatctg agaaattgtt tgtattgcta atctatatag ccattctttt ctatcgctgt 172740

tttgtttgtt tgtttgtttg tttgcttgtt ttgcgacaag gtctcactac atcactcagg 172800

ctggaatgca gtggcacaat cccagctcac tgcaagttct gcctcccggg ctcaagtgat 172860

tctcccacct cagcctcctg agtagctggg accacaggcg cacgccacca cacctggctt 172920

ttgttttttt gtttgtttgt tttggtaaag acagagtttt gccatgttgg ccaggctggt 172980

ctcaaactcc taacctcaag tgatctgccc gtctcagcct cccaaagtgc tgggattaca 173040

ggtgtaagcc accgcacttg gccccaattt tcttttcttt tctttctttc tttttttttt 173100

tttttttttg agacagagtc tcactctgtc acccaggatg cagtgcaatt gcatgatctc 173160

agttcactgt aacctccacc tcctgggttc aagtgattct cctgcctctg cctcccaaat 173220

agctgggatt acagacatga accaccacgc caggctaatt ttttgtattt ttagtagaga 173280

agggggttca ccatgttggc caggctggtc tcgaactcct gacctcagtt gatctgcctg 173340

actcagcctc ccaaagtgct ggattgtagg catgagccac catgcccggc ccacctggcc 173400

ccatttttaa gtgtagagtt gagcgggagt aagtacattc acatccacac tattgtgcaa 173460

ccgtcactac cagccatatc cagaatattt tgaaccttgc aaaatcgaaa ctgtaccacc 173520

attctaattt ctgtctctat gaatttgact gctctaggac acaaggaggc tttaagcaac 173580

atccatcaaa taaataaatc ctgacctacc tgctttggta gccaggacgc tcagagttaa 173640

atttctgtgt tttattgcct agatttccta ataaaattct cttctgtctc tctctctttt 173700

tttttttttc gagacggagt ctcgctctgt cacccaggcc ggagtgcagt ggtgcaatct 173760

cagctcactg caatctctgc ctccccggtt gaagcgattc tcctgcctca acctcctgag 173820

tagctgggat tacaggtgcc cgccaccacg ctcggttaat ctttgtattt ttagtagaga 173880

tgtggtttca ccatgtcggt caggctggtc tcaaactcct gacctccggc aatcctcctg 173940

ccttggcctc ccaaagtgct gggattacag gcgtgagcca ccacgcccag cctcttctct 174000 cattttaata tagattttt tatacagatt ttgaacttgt acccctactc aaggttcaca 174060

ttgaatctaa gttaccatca ttatttatat tccactgaaa aactaggaca caacatagat 174120

tatgaaatac aaacacatca ttatttcaga cttccacatg tgcgaattgc attgacgaaa 174180

tggagtatgt gttttactcc ataagccaca tcaaactgtt tttcgaagca gtgggaggtg 174240

gtaaaaatca gaagagaaca acatcaaatc catgttttca gagatgaaac acctcttctg 174300

aagatattct tcccccaaaa agtgacccca cctccaagta caggctcacc agtggcattg 174360

gaggatgtct gcagtattcc agtcatccag gaatagagag ctctctggga atgagccgag 174420

gttttgtcgt agagatcttt gaacactgca gatctaaatg acacaaacac aggcatcagc 174480

taataacaag caggtggggg ctgcattaaa aagggagtca caacagcaaa tgtgacccaa 174540

agctgtcttg tgttaaaacc cttccctctc ctggatgtgg agaaatagaa acgcttttac 174600

actgttggtt ggaatgtaaa ttgattcaac cattgtggaa gacagcgtgg tgattcctca 174660

agaatctagg actagaatta ccatttgacc cagcaatccc atttctgggt atgtacccaa 174720

aggattataa atcatgctac tataaagaca catgcacacg tatgtttatt gtggcactat 174780

tcacaatagc aaagacttgg aaccaaccca aatgtcctcc aatgatggac tggattaaga 174840

aaatgtgaca catatacacc atggaatact atgcagccct aaaaaaggat gagttcgtgt 174900

cctttgcagg gacatggatg aagctggaaa ccaccattct cagcaaacta tcacaaggac 174960

agaaaaccaa acaccgcatg ttctcactca taggtgggaa ttgaacaatg agatcacttg 175020

ggcacagcaa ggggaacatc acacaccggg gcctgttggg gggtgggggg aggggggtggg 175080

gatagcatta ggagatatac ctaatgtaaa tgatgagttg atgggtgcag caaaccaaca 175140

tggcacatgt atacctatgt atcaaacctg cacgttgtgc acatgtaccc tagaacttaa 175200

agtatattta aaaaaaaaaa accttccctt tcttgaatgt aaattggttc aaccattgtg 175260

gaagacagtg tagcgattcc tcagagatct agaactagaa ataccatttg acccagcaat 175320

cccattatcg ggtatatacc caaaaatata taaatcattc tgtcacaaag ataaatgcac 175380

acatgatcat tgcagcacta atcacaatag taaagacatg tagtcaaccc aaatgcccat 175440

caataataga ctggataaag aaaatgtggt acatatatac catggaatac tatgcagcca 175500

taaaaatgaa caagattatg tcttttgcag ggacatgaat ggacctggaa gccattatcc 175560

tcagcaaact aacgcaggaa cagaaaatga aacaccccat gttctcactt gtaagtggaa 175620

gctgaacgat gagatcacat ggacacaggg aggggaacaa cacacactgg gtcctattgt 175680

gggggtgggg tgggggaggg agagcattag gaaaaatatc taatgcatgc tgggcttgat 175740

acctaggtgg tgggttgata ggtacagcaa accaccatgg tacacgttta cctatgtaac 175800

aaacctgcac atcctgcacg tgtaccccag aacttaaaaa taaaaaatac ccccaaacac 175860

actccttagg tatatgtaac tatttttccc attttcctct tccccttcac agctaaacac 175920 cttccaaaga atattccata catattgtct ccacttcctc acctcttggt cctttttgg 175980

ggaagtgggg gcagctctgt tgagatataa ttcccacacc ttacaattca cccatttaaa 176040

gtacacaatt gggccgggtg cagtggctca cacctgtaat cccagcactt tgggaagcca 176100

aggcaggcgg atcacctgag gtcaggagtt tgagaccagc ctgaccaata tgatgaaact 176160

ctgtctctac taaaaataca aaaagtagcc aggcgtggtg gcatgcacct gtaatcccag 176220

ctacttggga ggctgagaca ggagaatcac ttgaacctgg gaggtggagg ttgcagtgag 176280

ccgagatggg gccattgcac tccagcctgg gcaacaagag tgaaactctg tctcaaaaaa 176340

aaaaaaatta cataaataaa gtgcacaatt cagtggtttt cagcatataa gacaaagagc 176400

caaagatgtt tctgatggca tcatttgagt ctctggatcc agctgcgtct gaagtcagcc 176460

ctacacctaa aactattcaa ttacatgttc caataattcc ctttttgctt caagccagtt 176520

tgaatgggtt tttctgccac ttgggcttac tgcttactca ggtgagcagt aaccaataaa 176580

ctctctctcc caatgatgtt ctcttccaca atctgcatga acggaggagg gaagagatgc 176640

tgaatttaaa aggaatagta taatagcctg cagagcaaaa gggtttcatt tatatacatc 176700

ttagtgaaaa aaactggcaa cctagagaaa cgatgaggtc agcatactgt ctgttttcct 176760

tactgtcatt tggctacaaa agataaaagg aatctcccac attgttatca tcatattggt 176820

cttttccttc ttctaaatca tttttagatt tagaaagagt gctttatggc cataaagcac 176880

tacgattagt aaaaggaggg agttggcggt ggttgcaatg cctatggccc ttcaagtcct 176940

agacacaatc ctcgaatttg gtttacatca aagcgcaatt tcatgcaggg ctaggtgtaa 177000

ccaatctaag ctggcatggc atctcaaaag gagaagtcct cctttttcag tgctctgtca 177060

ctccttctgt tgtccctcaa gaacaaagtc tcagtttagt gccagtgaat ctttaaaaca 177120

cactcagatt gtttttaaat caatggaaca agacccaggc tcagagcctt tgtgctgtgg 177180

actgaatgtt tgcgtttccc caaatttcat gtgttgaaat cctaattccc aacatgaagg 177240

tattaagagg tggagttttg aggaggtgat taggtcatga ggctggatac ctcatgaatg 177300

ggattagtgc ccttataaaa gagatcccag agggcttgcc cacccccttcc accacaaggg 177360

gacacagcga gaaggtgcca cctataaacc agaaagcgag ccctcaccag acaccaagtc 177420

tgtcagcacc ttgatcttgg acttcccagc ctccagaacc atgagaaata aatgcttctt 177480

gtttactagc cacacagtct atggtatttt tgttacagca gcccaatctg actaacacac 177540

ttaagcaagc aacatatcta gctaggattt agtaacactt tttttcccac tgtatttatt 177600

tatgtgggtt tttttgtttt tgtttttgt ttgtttgttt ttgtttttga gatggagtct 177660

cactctgtcg cccaggttgg agtgcagtgg catgatccca gctcactgca acctccacct 177720

cccaggttca agcaattctc ctgcctcaag cctcccaagt agctgagatt acaggcgccc 177780 accaccatgc ccaactaatt tttgtatttt tagtagagac gtggtttcat tatgttggcc 177840

aggctggtct cgaactgctg acatcatgat ccacccacct ctgcttccca aagtgctggg 177900

atcacaggtg tgagctacca cacccagcct aactctacag atgttaagct tttgctccaa 177960

aacgtgcata tcaacctgtc cccacttgtc ttctccctgc tgccaggtga gatctctttg 178020

cattctcacg gcagcccctg aaggatactg tgatgagtta agaaggtggg agtggctggg 178080

tgcagtggct catgcccgta atccaagcac tttgggaggc caaggcaggt ggatcacttg 178140

atgtcaggag ttcgaaacca gcctggccaa catggtgaaa ctccatctgt acaaaaatac 178200

aaaaattagt cgggtgtggt aggcgcccct gtagtcccag ctacttggaa ttgaggcagg 178260

agaatcactt gaacctggga ggcagaggtt gcagtgagcc gagactgcac cactgcactc 178320

caacctgggc gacagagtga aactccatct caaaaaaaaa aaaaaaaaag ttcggaggaa 178380

ccactcacag gttcttaaag gcatcctcgc gcaggtcccg agggagccgc tctgtgatgt 178440

ctggctggag aaagctccct gaggaccccc tcagcacctt gcccaggatc tccacacact 178500

ccagggagtt cagcatctgg aaacacctct ccagtgtgat gagaaacagg ctgcggttca 178560

cgccagggct ctgcaaggct cgttctcgaa tctctcctaa gtcgatgatg atgtctttca 178620

ggtcctaaag caatgacaga agccagggtc agtttgagag ctgcctctca atttggaagc 178680

cctgggaggg accacacatt gcaacagtgg ggaaagtcag gatcactgtt taacaccatg 178740

ctgctatgtt gaaacttcta gaactttcca ttccctgcag aacaaagtca aggcctttac 178800

ctgctccact cactgtggag cctccccact ctccagatac acctgaccag catacatcag 178860

gcccttggtc tacctggaat tctgcatctt tcctttttct ggtgaaatct agctcatcct 178920

ttaagtctgg gctccattgc catcaccttc gtgaagtctc tgatccctcc aacttcctac 178980

ttcagatcag atttccctcc cttatgtggc cccagggcat cttgaaccaa actctataat 179040

gcattaaagg gactcatgcg tctttcctcc aagcccgtgg ttctcatctc tgggtgcaca 179100

tcagaatcac ctggggagct tttttaaagt tttattttat ttttgagaca gggtctcgct 179160

ctgtcaccca gtccgcagtg cagtggtgtg attatagctc actgcagcct caacctcctg 179220

ggattgagcg atcctcccac ctcagcctcc caagtagctg ggactacagg cacacaccac 179280

catgctcagc tgttttttgt agagatggag ttttgccatg ttgtccaggc tggtctcaaa 179340

ctactgggct caagcaatct gcccacaacc tcccaaagtg ctagggttac aggtatgagc 179400

caccatgccc agcccatctg gggagatttt gaaaaggact gattggcctg gaacagtggc 179460

tcacatctat aaccccagca tgttgggagg ctgaggcagg ttgatcgctt gagcctagga 179520

gtttgagacc agcctaggca acacagggag accccgtccc taaaaaataa attttttaaa 179580

attagctggg tgtgctaatg cgcacctgta gtcctagctg cttgagaggc tgaggtggga 179640

ggatcacttg ggctcaggat atcaaggctg cagtgagcta taattatacc actgcactcc 179700 agcctggatg acagagcaag cccctgtctc taagaaaaaa aaaaaaagcc tccattttac 179760

tgagcattga ctctactgtg ccaagcttgg agcaaacgca tcatctgatt aattctcatg 179820

acaatcctac cggttaagcc ccatgggtac tagggtcagg cattcccagg ttccagtccc 179880

agggcctcca ttatctgtgt gtcctgagca aatgacttac ctttttaag tcccagtttt 179940

taaggtctca gaggccttgc tttgtgatat tcttggatca ctaaactcat tttgtggaaa 180000

ttaactgttc ctgtttttg tctcccctcc agaccgtgag ctccttgagt gcaggaggat 180060

ctctgtgtcc actggtgcat agtaatttt actagtggat gtttattgac ccaaacaacc 180120

aggagattct cagatcctta attaggaccc gtcatttctt agtactgaca aaaatatgaa 180180

ttaggccagg catggtggct cacacctgta atcccagcac tttgggatgc cgaggcaggc 180240

aaatcacttg aggccaggag tttgagacca gcctggcaaa atctcgtctc tactaaaaat 180300

acaaaaatta gccaggcatg gtggtgcaca cttgtaatcc cagctacttg ggaggctgag 180360

gcacaagaat cacttaaacc caggaggcag aagttacagt gagctgagat tgtgccactg 180420

cactccagcc tgagtgacag agtgagactc taaaacaaaa ataaaataaa aaattgtaag 180480

gctcttcatt ttaactttgg agaaagtaag tgagaaaact taggtcctaa agcccccagg 180540

gggttggagg aggcagaagg agagaggaag agacaatgaa gtgccaaggc ctaagtcagg 180600

ctccatttat tcaattcaac aaacaccacg tggtgctgga cttcctctat ccccgggatg 180660

ccaagggctc ctcaccaagc cgtccttctt gtcttctaag aggcatttca tggcagtgcg 180720

gaactgctgg gcgtctgtct tcctcaggtc ctccagcagc ttctggggct ggtgcagacg 180780

ctgctccagg tggttttcca cggctgcctg tatttggggt gttgggagag tcggtgagaa 180840

cagaaagagc aggagcacat ggtcagcacc atggccagcg gcccagcccg gccccctcat 180900

tccctgcaga gaccacaact agtggccagc agtctggttt ggccacctcc aaatttcctc 180960

attagccgcc aacatttaaa aatgaggggc cgggcgcggt ggctcatgcc tgtaatccca 181020

gcactttggg aggccgaggc gggtggatca cgaggtgaag agatcaagac catcctggcc 181080

aacatggtga aaccccgtct ctactaaaaa tacaaaaatt agctgggtgt ggtggcgcac 181140

gcctgtagtc ccagctactt gggaggctga ggcaggagaa tcgcttaaac ccgggaggcg 181200

gaggttgcag tgagccgaga tcacgccact gcactccagc ctggtgacag agcgagactc 181260

cacctaaata ataataataa taataataat aataataata aggggattcc tcacaagatc 181320

aagatttcca acttctcttg aaaaagtaaa ggtttggcaa caccgggcgc acatcccctt 181380

ttaggcaata attggcaagc caagggacag ctgcttgttg aagtaactct caggcccagc 181440

cagcccgatt cccccattcc atggcctgcc tggtccctgt aggtgattga gtttcccatc 181500

agcagcagct ggcctttgtc caggaagttt agaaccccag atggcactgt gcccaaaccc 181560

aagctaccag cgcaggagca aaatacaggc gaaatgaaca tactctgtgg cgccccctac 181620
agggcaagga gcagaagcca tcagcctctc ctgaaagaga tttcctctgc tgtctgggga 181680
attgttctct aaatctccgt taaacagaca tgcccagtgt gatgtcatgg gcaaaaggtg 181740
aactctagac ccagacaagc cacatgcaaa tccccacctt gacacttact agctgtgtgg 181800
cttggacgag ttacttaact agggttgctg ggtaaaatac aggatgctca gttaaatttg 181860
aatttcggga aaaactacac attttttttt ttttagtata agtatgtctc aaatgttgca 181920
taggacatac ttaacttaaa aaatcactca ttgtttgcca ggcaaggtgg cccacacctg 181980
taatcccagc actttgggag gccaagacag gcggatcatg aggtcaggag tttgagacca 182040
gcctggccaa catagtgaaa ccccatctct actaaaaata cacaaattag ccaggcatga 182100
tggcatgcac ctgtagtccc agctactcag gaggctgagg caggagaatt gcttgaacct 182160
gggaggagga ggttgtggtg agctgaggtg gcaccactgc actccagcct gggcaacaga 182220
gcaagactct gtctcaaaaa aaaaaaaatc actcattatt tatctgaaat tcaaacgaaa 182280
ttggacatct gttttttgtt tttgtttttg ttttagagac ggggtcttgc tctgttgccc 182340
aggctgaaat gcagtggtat gatcatagct caccgtaacc tcaaactcct gggctcaagc 182400
aatcctcctg cctcagcctt ccaagtagct gggactacag gccacaccac cacactcagc 182460
taattttttg tttgtttgtt tgtttgtttt tgtagagaca aggtctcgct atgttggcaa 182520
ggcttgtctc caactcctgg tatcaagcaa tcctcccacc tatgcctctc aaagtgctgg 182580
gattacaggc atgagccacc acgcagagtc acatctgtat ttttatttgc taaatcagca 182640
acatttcttc tgggcctcag ttttcttatc tgcaaaatgg aggcaataaa aggacttacc 182700
acatagagtt attacgagga ttcagtgtgt aaagtgtttt caatagttgc tagcacacag 182760
taagtgctca atagtgccag ctattattct ctatcttgac caaggtgaat aatggcttaa 182820
ttaaagcctc accccctaaaa tgaaaggtcc ttgtgggtag tatcatgact gccttgttcc 182880
ctactgagtc ctcagtgtct agagctatgc ctggcagaaa gcagatgtta actcagtaga 182940
tattaaatga ataaataaat atggtatata cacaggaaga aacgtatcag attaaagtgg 183000
atacatctca tgagtatcca ttactgctag agagaaaagt ggcaagcatt gcttctcttt 183060
cagtttccct gactgatgaa gaagaaaaga aaagcagatg ccatgtgggg ccaacagcca 183120
cctccagcta gtggctgtcc tgggacctgg cagacaacaa gctctctgag tatatttgtg 183180
atatcaatgt tggttgatgt ttccattttc attcaaacat aagtaaaaat ggaaacaatg 183240
aagacatatg tcagaacttt attcattaat taatgacata aatggtcttt tgctaaactg 183300
aataactttc tcaatactga aagaattttt ctccaatttt ctgtgttatc cacagtgtga 183360
tgataaagac acaacatgcc acactaggct aacagtcagg caggtgctgt gcagttcacc 183420
gcagtcacca ccacaggcac gagccctatt gccccttaag gcaaatatcc tttgccacct 183480

atcaacacgc ttgtgctatt gttttcatta cagtagtaca tttcttggac ccatggtaga 183540

aagactgatc actatatcca aatatccact ctctcactgc tctttatcaa ttgaaccctt 183600

caacttttag ttgggtacat ggctcctttg cagtaagatg tgactatggt gtgtgagcca 183660

atctaatgtg tgcagaggga attgtgtaac ttgcgtcata tcctcagaag agaagatgct 183720

acccttgtct tcttcttccc acccatcccc tggattggag aaaggaagag gtgtttccac 183780

cacatgggca ataactcccc aggagatgac aaagcaaaaa gataaaagga acctgggtcc 183840

ttggataacc tcatggaata gagctgcctt ttctctgtgg ctttcccact tagctctgga 183900

ctattgtacg agaaagaaat aaactaaatt gtttgagctg ctgtcatctg gggtcttttt 183960

tgttatagca gctcagccta tatcctaata taccatgtct ccatcaaagg tgggaaaatg 184020

aaagaaagac aaaatagctt atatcatgtt tcaagaaaaa ctggacagaa ccctttcct 184080

tgcagaagca aagactatct ctacatccag cccacttctc caacttacct ggcccctgag 184140

tttgcaatcc ctgagcactg agatgggaac atatagatgg gtctcaggta cacacctgca 184200

ggctggggat ggtgaaggca acattccggg aattcagata ggccaggact ctgtgggaca 184260

ggtcatccgt ccacacgtgg gagcttcagt tgaagacaga caggaaaaga tcacaatgac 184320

agattctcct acaagcacta ctgtactagc taagtgccca ggggacaggt agggatggac 184380

caggggtgtt aggactttgt acttggaagt gggaggtttc tcttttcttt tctttccttt 184440

tttcttttct cttttttga aacagggtct tgctctgttg cgcgatcacg gctcactgca 184500

gcctcaatct ccccagccca agtgatcttc caacctcagc cacccaagca gctgggatca 184560

caggtgcatg ccacaacacc cagctaattt tttgtagaga tggggtctca ctatgttgcc 184620

caggctggtc tcaaactcct gggctcaagc aatcctccca cctctgcctc ccaaagtgct 184680

gggattacag gagtgagctg ctgcacccag cctgaagtaa aaaatttctt aaccaggcac 184740

agtgatagga tagtttccaa ttctaggaat ctgcctggat cccattctct caaagccaat 184800

tcccaaattt ctaagctgta tgcaatattc taattcccgt aacaatctgc ttagattgac 184860

tacaatccaa actgattgtg ttagaaagat gtacatttaa aagcagacgc taggtacacc 184920

atgagaggct ggaatagcat agttaggagt gtgggctcca aactggattt gaatcctagt 184980

tccatcactt agttgtgtgg cttgagacaa tttgataaat tttcttgtgc ctcagtttcc 185040

ctttatatga aatatggtta acaactgtga gattaaaatt tgttcacaca tgaaaattgc 185100

gtaagactgt gcccaacaca cagtaaatgc ccatgaatag ccttttctca tttttttttt 185160

tttttttgga gacagagtct cactctgtta cccaggctgg agtgcagtgg tgcaatctca 185220

gctcactgca acctccgcct cccaggttca agcgattctc ctgcctcagc cttccaagta 185280

gctggaatta caggcgtgca ccaccacatc cagctcattt ttctattttt agtagatact 185340 gagttttgcc atgttggccg ggctgggctg gaactcctgg ccttaagcga tcctcctacc 185400

ttggcctccc aaagtgctgg gattacagga taagccacca tgcccagcct atgaaaagcc 185460

ttttgtaatc ttacgtttgc ttctttgttt gtttgtttgt ttgttttgcg atggagtctc 185520

actctgttgc ccaggctgga gtgcagtggc tcaatcttgg cttatcacaa cctcagcctc 185580

ccgcgttcaa gtgatcctcc tgcctcagcc tcctgagtag ctgggactac aggtatgcac 185640

caccatgcct agctaattct tttgtacttt tagtagggac agggtttcac tatgttggcc 185700

aggctggtcc cgaactcctg acttcatgat ccgcccacct tggcctctca aagtcctgga 185760

attataggca tgagccaccg cgcccggcct gtaatcttat aaagagatgg atggatggat 185820

ggatggatgg atggatggat aaattaataa acaaataaaa tacttagact gaaagaatat 185880

atccaaaagt acccattggt gttatcttag ggaaaggagt ggttatggga gtctttcact 185940

ttaacataac tgggtatccc tgatatgagg ccccaagacc cctatttctt atcgatcata 186000

gtactcatca tattagaatt gtttattaat attggcgttt ccacactacc tagttccctg 186060

ccccatgtcc ctggtatctg tctgtatgca gcttataatg caacagtaaa gggaatgggt 186120

cctggagcca ggccactggt tcacatccca gttgtgtgtc cttgatttac ccttcctgga 186180

cctcagtttc accattgtgt gcaatgggat aacaatggga actacattgt gtggcacttg 186240

tgaggattag attgtttcta taccaagcgc ttagtagaat acccactaca tagaaagcac 186300

tcaataaatt tcagctctta ggccggttgc ggtggctcac acctgtaatc ccagcacttt 186360

gggaggctga ggccggcaga tcacctgagg tcaggagttt gagaccagcc tggccaacat 186420

ggtgaaaacc catctctact aaaaatacaa aaattagccg ggcatggtga caggtacctg 186480

ttaccccagc tactcaggag gctgaggcag gaaaattgct tgaaccccgg aggcagaggt 186540

tgcagtagtg agccaagata gcaccactgc actccatcct gggcaacaag agcgaaactc 186600

cattcaaaaa aagaaaaaaa attggccctt atgctagagt gaggaggaat gatgcccgtg 186660

gagcaagggt gtttgttgac ttttaacaca ttgtgcattg attgacattt tttgtaatga 186720

gcacatatta ctttttacaa ttaagagact tccattaaga atttttatta aaaatctata 186780

ccatcccttc ccctctcctc tagggcattt tacctttgaa actgtatgag atccagcagt 186840

gctacaatgg gagtaaaatg aaaagacatc tttaaaagga atcattccag agagatcaca 186900

gctacattct gtttgaaatt cttagaagaa gcaagtggag atcaaggggg cacattacca 186960

ttcagatagc ttccatctat tatttcttcc taaaaggcaa agagataaca tcaagtataa 187020

caaccaaaag ttggcaattc cagtagaggg gactctgatt aacagaaaat agaatactca 187080

ccgccatgtt ttgcaacaat ggatgcaaat ctgtaagaaa gccattgaat gtataactag 187140

aattgaattg tgtgaccttc aaaactacaa atatgggtta gaggaaaagc cccagaaagc 187200

cactcatgaa ggaaattttt agtgaatttt aattataact gtccactctg agaaaaccgt 187260

ggcttgaaaa atcctcttag aataatcctg aaaaatcata catgcacaca cacacagccc 187320
aaaaacaact tacagcatca tgtacctaat gtatctatcc cacctcattc cctaaactgt 187380
gtggaaacca caaagggtga ttcttccctc aggacttacc ctgcctggaa tttggcactg 187440
tataactcga cactccatgg ctcagaaaaa ggaataggaa aagggagtat gtcgtaggtt 187500
cctgagacat tctcctgtag ttaatttaag catagaagtg attcatgttt tttttatata 187560
tatatatata cacacacaca catatatata catgtatata aacatatata tatatatgga 187620
cattcagaag acaaaaggaa aaaccaagta cacccagaat cccttgaagt ctttttttaaa 187680
aataatttta aaactcacct actttcctcc atctaaaccc ccattcttgt ctaagacact 187740
atcatctctc tccttggtga ctacaatggc ctgtttatct taccactacc cccaactcct 187800
ctcaatccag caaaaaggta tagtcaaaag tagactctgg gccaagcatg gtggctcaca 187860
cctgtaatcc cagcactttg catggccaag gtgggtggat ctcttgaggt caggagttcg 187920
agaccagcct ggccaacata gtgaaaccct gtctctacta aaaataaaaa attaactggg 187980
tatggtggtg catgtctgta gtcccagcta cttgggaggc taagacagga gaagtgcctg 188040
aatctgtgag gtagaggttg cagtgagcca agattgtgcc actgcactcc agcctgggca 188100
acagagcgag actctggaaa aaaaaaagta tagactctat tttatctgca agaaatttat 188160
atccagaata caaaaactac tcttataatt caataagaag atagacaagc caaggggaaa 188220
aaatgttaaa aagatatgaa gttggccagg cgcagtggct cacgcctgta atcccagcac 188280
tttgggaggc tgaggtgggc ggatcatgag gtcaggagat ccagaccatc ctggctaaca 188340
cggtgaaacc ccgtctctaa taaaaataca aaaaaattag acgggcgtgg tggcgggcgc 188400
ctgtagtccc agctactcag gaggctgagg aagaagaatg gcgtgaaccc tggatgcgga 188460
gcttgcagtg agccgagatc acgccactgc actccagcct gggcgacaga gcaagactcc 188520
gtctcaaaaa aaaaaaaaaa aaaaagatat gaagctaaac ttcataaaaa cagatgtaaa 188580
aatggccaat aagcacacaa atgattatca cattagctaa caagaactag caaattaaac 188640
tcacaacgag accactacaa agtcactaga atgggcaaaa atttcaaaga ctgactatca 188700
aatgttggtg atgatatgga acaacaggaa ctctgataca gctgcttgga aaaactgttt 188760
gacagtgtct tacagagtta aatatacaat accctatgat caagaaatgc cacccctagg 188820
tatttaagca agagaaatga aaacatgtct acacaaagat ttatatagct tcattcatag 188880
tagccagata ttggaaacaa tcacatatcc aataacaggt gaaaggttaa acatggatct 188940
ccattcagtg ggataccatt cagtaataac aataaaaaga aattcatcat aattttattc 189000
tagctatgat ggagtaattg gcaatggata aaccgtcctg ccttaaacaa ttaaaaagct 189060
ggctgaaata tgtgcatcca aaggtttttc agacattggg cagcaggtag cacaagactg 189120

agatccctga gcaaggaagg caaacaagct gagctctata attgctcctg ctcactgcct 189180
ggagtctcca tccaacagca caaggaggga gaatccaaac agagcctaca ggtctcactg 189240
agcagaagag acggagtgga acttcaggga aatcaaggca gctagaatcc ataagaagaa 189300
tacttggaga ggaggaagtt gcacagagag aaagttctgg agatctgaag aagggtcttt 189360
tgagactttg gctgaatatt tatctacctt tgcatgtaag aaacctcctg aggctggaga 189420
atgaaccacc agtaggcaga acaatccttg gacctcacag gggaatgaga atagctcctg 189480
caagctgcaa tggaaaaacc tccaaacaca ttgggcatca gggtcaatca tcaaagaaca 189540
attgcctcca tgataggcca aattagccct aggctaaagt ttattccaga tctgccctaa 189600
caaatctcaa aagcaagttc tggaaggatc taattgattc caagtaactg aattgcattc 189660
cagaacaaaa cccaacaata tttaaagcag tatcataaaa cccagaaact tcagaaaacc 189720
agaagatgga gaaagaaaaa aaagaaaaaa aaaatccagc aatgtaaaat ccgaagtgtc 189780
cagatctaat caaaaattat caggcaggcc aagcacagtg gctcatgcct ataatcccag 189840
ccctttggga ggccaaggaa ggtgaattgt ctgagctcag gagtttgtga ccagcctggg 189900
caacatggtg aaactctgtc tttaccaaaa atgcaaaact ttgccaggga tgtggcaggt 189960
gcccaactac tcaggaggct gaggtgggag ggtcactgaa gcctggaaag ttgaggctgc 190020
aatgagccat gattatgcca ctgcactcca gccagggtga cagagcgaga acctgtctca 190080
aagaaatttt ttaataaata aataaaaatt accaggcata ctaggaagca ggaaaatatg 190140
gctgattatc aggagaaaaa tccaatcaat agaacagatc cagaaatgac acagaagatc 190200
caattaatgg acaaggatgt ttattataaa tacactctgg aagttcaaga aagtagggaa 190260
aactgcgagc atgttaaagg gagagacatg gtaggtgcga aaaaaacccc aagaggaact 190320
tctagagata aaaatacaat atctgaaatg gaaaatacac gggatagagt taacagcata 190380
ttagacactg cagaagaaaa attagtgaac ttgaagtcat agcaaaataa attatccaaa 190440
gtaaaatgca gggggaaaaa gactgaaaaa attaagaggg ctctggagca atatgaaact 190500
atttgaaata tgtgtaactg gagtcccagg aaaggagggg gtggcacaaa aatatctgaa 190560
taaatcatta gcagggcaca gtggctcacg cctgtaatcc cacattttgg aaggttgagg 190620
tgggcaaatc acttgaggcc aagagttcaa aaccagcctg gccaacatgg tgaagcccca 190680
tctctataaa aaaaaaaaaa aaaaaaaaaa aaaaaatagc caggtgtggt ggctgtaatc 190740
ccagctactc aggaggctgg gcacaaaaat cacttgaacc tgggaggcgg aagttgtagt 190800
gagccaagat tgcaccactg cactccagcc taggtgacag aatgagactg aaaaaaaaaa 190860
gagagagaga gaaagaaaaa gaaagaagga aagaaagaga gaaagaaaaa gaaagaaaga 190920
aagaaagaaa gaaagaaaga aagaaagaca gaaagaaaac atgacctcat gaccaaattt 190980
gtgtaaagtt tgatagaact acagatccac agatccaaga agctcaacca accccaagca 191040

gaggaatcat gaataaagcc acgccaaggc atataataat caaattgcta aaaaaaaaaa 191100
aaaaaaaaaa aaaaagattt ttaaaaaaaa tccagcaaaa aggagttgca tctagactat 191160
attgaaaaac tcttaaaact aaaaaataaa taagcaatcc aatcagaaaa cgagcagaag 191220
atatgtgcag atatatcatc aaagtatata tacagatagc aaataaacac atgggaagat 191280
actcagcact gttagccatt aggaaaatga ccactgaaac cacagtataa ttataggcct 191340
atcagatggc taaaataaaa aaaaaaaaac cataatgaag ccaggcgcag tgaatcatgc 191400
ctgtaatccc agcactttgg aaggccgagg tgggtgaatc acctgagatc aggagtttga 191460
gaccagcctg gccaacatgg tgaaacccca tctctactaa aaatacaaat attagccggg 191520
catggtggct catgcctgta gtcccagcta ctcaggaggc tgaggcagga gaattgcttg 191580
aaccgaggag gcggaggttg cagtgactgg gaggcagagg ttgcagtgag ccaagattgc 191640
accactgcac tccagcctgg gcaacagagt gagacttcaa ctcaaaaata aataaataaa 191700
tgtaaccata atgccattaa ctagaaagaa tgaggagaaa ctggatcact cacacattgc 191760
tgatgagaat ataaggtggt acagctattc tggaaaatag ttttgttgtt tcttataaaa 191820
ttaaatgtgt acttaacgta cagcccaaca attacactct tgggcattta tcccagaaaa 191880
atgaaaattt atgttcacat aaaaacctgt acacaggctg ggcatggtgg ctcacgccta 191940
taatcccagc actttgggag gctgaggcgg gtggatcacg aggtcaggag atcaagacca 192000
tcctggctaa catggtgaaa ccccatctct actaaaaata caaaaaatta gctaggcgtg 192060
gtggtgggcg cctgtagtcc cagctacttg ggaggctgag gcaggagaat ggcatgaacc 192120
caggagacag agcttgcagt gatgcagtga gccaagatca cgccactgca ttccagcctg 192180
ggtgacagtg tgagactcca actcaaaaaa caaacaaaca aacaaacaaa acctgtacac 192240
aaatgttcat agcatcttta ttcatagtag ctgaaaaatt gaaccaatct aaatgtcatt 192300
ccgtaggtga atggttaaac tcactgtgtg tggtatgtcc acgccattga atactactca 192360
gcaatgaaaa ggaatgaact attgatacag gcaacaactt tgatgcacct aaagggaatt 192420
atgcagaggg aagaaagcca atcactgaaa gttatatact aatgattcca attataaagc 192480
attcttgaaa taaaattata gaaatgcaga ctagattggt atttgccatg gggagaggga 192540
gataaggctg tcaagggatg catgaaagag gcttgtgatc accaggagag ttctgggtcc 192600
tgattagggt agctgataca ttagtctaca catgtcacaa aattgaagaa aacgatataa 192660
aatatacaca caaaggagtg catgtaaaac tagtgaaatc tgaataagcc ctgaatttgt 192720
accatgattt cctgaatttt atatggtact attgttacat aacatgtaac caatggggga 192780
aactgggtga agagtacaca ggatatttct tttgcaactt cctatgaatc tgtagttatt 192840
tcaaattctt ttttattttc ttgagagaca gagtctttct ctgtcaccca gcctgcagtg 192900

cagtggcatg atcatggctc actgcagcct tgaactcctg tgttcaagtg atcctctggt 192960

ctcagcctcc tgagaagctg agactacagg catgcactac catgcccagc taattctttt 193020

agttcttgta gaaatgggtt cttgctatgt ttcccaggct gatctcaaac tcctggcctc 193080

aagcaatcct cccatctcgg cctcccaaag tgctaggaat acaggcatgg gccaccatcc 193140

ctggccacac aattgttttt taatttagtt atagtagtct gtaccactgt aggatgacaa 193200

tagttaacaa taatatatag tttcaaatag ctagaaggaa gatactgaac agaaagaaat 193260

gagaaatgtt tgagatggta gacatgctaa ttaccctgac tgatcaccat acattataca 193320

catcaaaaca tctttatgta ccccataaat atgtacaatt attatatgtc aatttttttt 193380

tttttttttg agatggagtc tcactctgtc acccaggctg gagtgtagtg gcgcaatctc 193440

ggctcactgc aacctccgcc tcccaggtgc aagcgattct cctgcctcag cctcctgagt 193500

agctgggata acaggcatgc gccaccacac ccagctaatt tttgtatttt tagtagagat 193560

ggggtttcgc catgttggtc aggctggtct ggaactcctg acctctggtg atcagcccac 193620

ttcagcctcc caaagtgctg ggattgctgg cgtgagccac cgtgcccggc gtatatgtca 193680

attttttaaa taaataaaat aataaatttt taagcaccca aagggaaaaa acccatatta 193740

tatgcatagg aaaaggaata cactactgat ataggcaaca acacagatga acctcagaaa 193800

cattatgctt atgctgagca aaagaagcca gtcaaaaaag accacataac atatgattcc 193860

atttctgtga aactctaagt tatctagaat atctagattc taggttatct aggttaccta 193920

gaataatctt cagtgacagg aaacaaatta ggattggtct ggggcattat tagggtacaa 193980

tgggaaaagt ttattgcaac agggcacaag ggaacttctt ggagtgatgg aaaaattttt 194040

tttgtagata tagggtctca ctatagtgcc caagctggtc tcaaactcct ggtttcaagc 194100

aatcctccct tcgagcttcc caaagtactg ggattacagg tggatgccat tgcacccagc 194160

ccccattttt tgtatcttga ttgtggtggt tgtcacacag atacatacat ttatcagatc 194220

tcactgaact gtacacttag tgtgcattgt attgcctgta aatttatacc tcacaaaagt 194280

cagtatcaaa cagaaatgga aaagcatgga ctgtggagcc agtcacctgt tctgaatcac 194340

gtctccgcca cttactagct gtgcaatgtt ggacatattt ctttcttttt ttttttcttt 194400

tttcttttttt cttttttttt tttttgatat ggagtcttac tctcttgccc aggctggagt 194460

gcagtggtgc aatgtaggct cactgcaacc tctgcttccc aggttcaagc gattcttctg 194520

cctcagcctc ctgagtagct aggattacag gtacccacca ccacaccagg ctaagttttg 194580

tatttttagt agagacaggc tcccaccatg ttggccaggc tgatctcgaa ctcctgacct 194640

caagtgatcc acccactttg gcctcccaaa gtgctgagat tacaggtgtg agccactgag 194700

cccggccttg gacgagtttc ttaatctctc tgcactccaa ttttctcgtc tgtaaaatga 194760

gattaatgat ggtaccagca tcactgtgtc gtatgagggt aaaatgagtt gttaatagtg 194820 aatcacttca aacagggctt gacacaaagt cagctctgta tacaagttta ttaaataagc 194880

aaagaaacca gattgctgtt ttcaaacaga tgctgttttg aaaaacacca tctcaataaa 194940

ctcatgtcac cctccgcctc tccccattac cctcaagata aagcccaaag cttttccagt 195000

ggctgtatca ccctcctctt tgcccctcat tccccaaggt tcctgcccgg gccatctgtc 195060

tgtccctcca aggtgctatc ctccccagat taagaccctc tcacaggctg ttcctcggcc 195120

cagaggattc ttccccttcc tctttgcccg gctaagatca attcattgta aatttcaagt 195180

cttcaatgaa gccttccctg aattctctga ccaggccaaa gcaattagcc agaagaaaag 195240

gccaatactt ctcttttata gcaccaagca cagctataat acatatttat gtttgtgtaa 195300

ttatttcatg ctttcattct tcaaatattt gttgagtagt gctgtgagcc aggcattatt 195360

ctatgctctg agaatacagc agtttacaaa actaactccc tgctggtgtg ttggaatgaa 195420

ccaggcagaa aatagataat caaataacta tattatgtca gggagtgatg agtgccatgg 195480

agaaaaatac aacagggtag gaaaaaggag ccaaagaagg agcaggtact attatagaca 195540

gtttggtcag ggaagacctg aggaggtgac atttaagcag agacctcagg aagtgaagga 195600

atgaaccacg ggaatatctg gagaaagaaa gttccagaaa gaaaagaaaa tgacagtgca 195660

aaggctttga tattaggagc ataatttgca tgttcttcag taaattgcac gaagaggata 195720

taacagtagg acatggggtt ggagaggtat cagggccaga gtgagtaggg tctttcaggc 195780

cattgtgagc gctctgactt ttacttggag ggaggtggga gccactgaag ggatattagc 195840

caggatggat gctctgactt gtttttgttt ttgttttgtt ttgtttgaga tggtgtctca 195900

ctctgtcgcc caggctagag tgcagtggca caatctcagc tcactgcaac ctctgcctcc 195960

ccggctcaag tgattctcct gcctcagcct ctcaagtagc tgggattata ggtgcccacc 196020

accacaccca gctaattttt atatttttgg tagagacgag atttcaccct gttggccagg 196080

ctggtctcga actcctaagc tcaagtgatc cacccacctc agcctcccaa agctctggga 196140

tgacaggtgt gagccactgt gcctggccct gacttttatt ttaactggat caagcacatt 196200

gattgctgtg ttgagaatac cctggagtgg aacaatggct gtagcaggca gaccaatcta 196260

ggaaagagat gacaataact tggtcattga tagcagtgaa ggtggtgaga agtggttgga 196320

ttttggaggt atttggaagg cggaaccaaa agatcgaatg tagggaataa aagaaataga 196380

caggtcaaag acaacgccaa ggatttggcc tgagcatgta gagagggaga tgggaaaacc 196440

ataggagttc caagtttagg gaaggatccc tggagctcag ttttggacat actaagcttg 196500

agaagccttt tagagtgcag atagcatagc aagtaggcag tttgatgttt gagctgcagg 196560

ttaggggaga ggtgctgtca gaagcagcta tctaaacaga gatagtcagt ggttttttgt 196620

gtttgctttt ttgtttgttc ttctgttttg ggggtttttt tggggtgggg gacagagtct 196680 cactctgtca cgcacgctgg agtgcagtgg cacgatctca gctcactgca acctctgcct 196740

cccaggttca agtgattctt gtgtctcaac ctcccgagta gctgggatta caggcttgtg 196800

ccaccgcacc ggctaatttt tgtattttta gtagagacgg ggtttcacca tgttggccag 196860

gctggtctca aactcatgac ctcaagggat ccacctgaca caggctccca aagtgctggg 196920

attacaggca tgtaatccca ctgcacccgg ccccagatag tcagcatgta tttaaaccct 196980

gaatgatatc accaagaaag taaatttacg gagaaattag gtgagtttta aggactgaga 197040

cccatggcac ccccattgcc aagacgttga gtagactagg tgcagccaga gaaatcaaga 197100

ctgtggaaaa ctccccaaaa caaacaacct agtttcttta gccaatatat tacagagaga 197160

gagagaaaga gaacctatta actcaaaaac aaaagattga tcacttgtag aatctagact 197220

ttatttggat cctgattcaa acacgtcatg tctgggcacg gtggctcaca cctgtaatcc 197280

cagcactttg ggaggccgag gcgggcagat cacttgaggt caggagtttg agaccagcct 197340

ggccaacatg gtgaaaaccc atctctacta aagatacaaa aacttagctg ggcatggtgg 197400

cgcatgctaa taatcccagc tactcgggag gctgaggcag gagaatctct tgaacccggg 197460

aggcagaggt tgcagtgacc cgagatcatc ccattgcact ccaacctgga tgacaagagt 197520

gaaactctgt ctccaaaaaa aaaaaaaaaa agaaagaaac aaagcaaaca aacaaaaaat 197580

accataaaat aaaataaaaa attaggccag gcacagtggc tcacacctaa tcccagtgtt 197640

ttgggaggcc aaggcggagg attgtttgag gccaggagtt tgagaccagc ctgggcaaca 197700

cagcatgact ccatctctat aaaaacattt aaaatttagc ctggcgtggt gatgcacacc 197760

tgtagtccta gccactcagg aggctgagat gggaggatct cttgagctcc agaattcaag 197820

gctgcagtga gctatgatca tgccacagca ctccagcctg agtgacagaa caagatcctg 197880

tctcaaaaaa acccaaaaaa ttgtgagaca attgaggaaa tttgaacccc tgattagata 197940

tttgaagatt ttaaagaatt attgttaata gttttaagtt tgataatgaa actgtggtta 198000

tatttttaaa ggagtcctta ttgtttcaag gaggggttag taaagtacca cccacggacc 198060

aaatggaaat ggccacctga ttttgcgtgg ctcatgagct taaaatgatt ttttaggttg 198120

tttaaatagt tttttaaaac caaaagaata atatttattg acatatgaaa atttcctgtg 198180

tgaaatttaa atttaagtgt ctataaataa agttttattg gcacatgtcc atgcccattc 198240

attggcctat tgtccatggc agcttcctgc taccatagca gagttgagta cttgggacag 198300

ggaccacacg gcctccaaag ctcaaaatat gtattccatg ttcccttact gaaaaagctt 198360

gccagccctt cttttatttt atgtatttat ttaattgttt tttttttttt aagacagagt 198420

ctctctctgt tacccacgct ggggttacca tgcctcagcc tcccaagtag ctgggactac 198480

aggcgcccgc cactacgccc ggctaatttt ttgtattttt agtagagaca gggtttcgcc 198540

gtgttggcca ggatggtctc gatctcctga tcttgtgatc tgcccgcctc ggcctcccaa 198600 agtgctggga ttacaggcgt gagccaccgc gcccagccgc catcccagca ttctttaata 198660

cagatttccc tcgtattgtc taacttgtta tgcaaatgga ttatttttcc cccaagggga 198720

cctccacaac cagctttgaa aacaattatg cttactttac tgcccaggac aaaaacagta 198780

aaacttttga agttcccaga caccaaaagc cctgcattct aacaggtttt gagcaaacct 198840

gatcccaaag cactttgtag gtgtgttggc ccaccatcac catcctgtaa ctggaatgct 198900

caactccaat tagaagtctt cctggtttgg gtcatttggg gctttgggga acctttgacc 198960

ctttttctc ccttcccttt gggcagctgc cctgggacat tggccccatt cacaattctc 199020

cagtttccca gacaatgtgg cttcttgtgc tctttgctcg ggcccctcta ttaaccctac 199080

agagaggttt caggttactg aaacagacgc cttgctttct gctgtgcata atgtcctttt 199140

tccagcgaga aactgcagcc catcaggatc tggttttcat taaaggcact ttgggtcact 199200

tttttagcag attggtcaaa aggactgaaa ggactgggca gggcgatgat gatttggagg 199260

tcaatagctt tctctatggg ccatacccct tccccactga aagattcccc cactgcagac 199320

tggaggaaat cagtcaggca aggatccctg tggtgaaaac tactcgaaaa cagcaacaat 199380

aaccacaaac ctgaaatgaa gacaatttcc tgagaaacta tgaatgtttg gatgtgcaag 199440

gtgatcaact ttcccagttg cccgggacca aaaggtttcc caggactcaa aactttccat 199500

tttgaaacca gaaagtcaca ggcaaacccg gaggagttgg tcaacccatg gatcagagtc 199560

acttcatctt ctgtgaaatt tgcaaagatg ctaggaggtt cccctcctgc tgggacaccc 199620

cagcccagac acaaaccatt aattcacaat tacatggagt ttcactgtct gcaaggctgc 199680

tccatttaag ctctgggtca tgaacacata actctaggca tactgacact agctgggaga 199740

ttttccacca aaaaaaaaaa aaaaaaatgc catttcatga ctattaatcc aaaataggta 199800

aatgtgtctg gcttatagaa taccagcctg attacaaatg cttggtgttg gaatggccca 199860

gctcacagtg gttgtagaag tccagtaggc ccaggctttg tggctcactc ctgtaacccc 199920

agcactttgg gaggccaagg tgggaggatc acttgagtcc agaagttcaa aaccagcctg 199980

ggcaacatag ggagactcca tttctacaaa taatttaaaa attagctggg tgcatgcttg 200040

tagccctagc tactcaggag gttgaggcag gagaattgcc tgagcctggg aggtcgaggc 200100

cacagtgagc cgcgatcaca tgtcactaca ttccagcctg ggtgacagag caaaaccctg 200160

gctggaaaaa aaaaaaattc aagcagtcaa ggggatggga gcttggtgga ctagagatgc 200220

ccagacttga gttcctgtcc cagccctgcc actaatagtg agattttagg cacagctatt 200280

ccccttcttc tctcctaagt ctccggagaa caaatggctt tggattcaat gaggaaaaga 200340

aggaaagaaa ggaagaaact ggtctcactg agcatcctct atgagccggg catcacacca 200400

aaccctcctc cctgccattt actctttcca ccagccttga agggttaccg tactcagttt 200460 cacagatgag ttcagtgcac atgaacgatc acacagcaag cgaatggcag ggaggggggtt 200520

atgatgcagg gctgtggggc tccctagtaa ccttaatccc agcagtcatt taatgagaca 200580

gaactaagta cacgcctctc taacatacat tgtctcgctt aattcttgta gaatctttga 200640

ggcaagcatt accattatct ctacttcaca gatgagcaaa tgggttctga gaggtcaagt 200700

gacctgccca aggtcacaca gctatgcatg cacagccagg tggcaaaccc agagctaaat 200760

gattccaacg ctcactggct ttgctctcca gtgtcagctc agcagccctg aacctaaaac 200820

tccaggtggc aggtgccacc ccagggataa gaccccagct ttaacctgaa ccaaacctta 200880

accagctgtg tgctattggg caaattcctt aacctctata ggcttgcatt tcctcctctg 200940

catagtaggc atggcactca tacctccctg gcaaaaaact atcaacttta tttgtttatt 201000

tatttattta tttatttatt tatttattta ttcattcatg agacagagtc tccctgtgtc 201060

gcccagaatg gagggcagtg gcctgatctc ggctcactgc aacctccacc tgccggattg 201120

atgcgattct cctgcctcag cctccggagt agctgagatt acaggcgccc gccaccacac 201180

ctggctaata tttgtatttt tagtagagat ggaatttcac cactttggcc aggctggtct 201240

cgaactcctg gcctcaagtg attctcctgc ctcagcctcc caaagtgctg ggattacagg 201300

tgtgagcccc ggcctttacc tgcatttttc catttcttcc acacaactgt tatgaggtag 201360

gcattacttt atacataaaa tacataaaga actaagccca gcactggcac atataagaat 201420

gcaatgaatc gtagctatta ttactaatgc tattgttttt actgctgcta tattattata 201480

acaaataatc attgtgatag ttattgcaac agttgtcatc gctgttatta tattactaag 201540

tagttttgga tccctttttct cgaagtttca tcgtcccccc cctccaagtg ccctccaatc 201600

ccaggagccc ttaaagccgc tgcacccaca ctttgcccac cctctttctc tcggtctgtc 201660

ccccacgcca tccatcacct gcaccccttc cctagggagg cccagcggtg ggcgcccacc 201720

cgctccccca gcgctttgcc ccgtgagtcc tccgcccagg cccccgcgcg cgcctcacct 201780

gcggagcccg gcttggccgc actgagtccc acgggcgggc gggtggcgca gggcggggcc 201840

gcggggctca tgcggggagc gggcaggcag gagagcggcg gggcggcgcc agccggcagc 201900

tctgcgacct cctccctgca gcggcccagg tgggaactca gccagggcag cggcgggggt 201960

cacagtcccc gcctgggact tcctatctgt cgaagctt 201998

<210> 41
<211> 3762
<212> DNA
<213> Homo sapiens

<400> 41
acagaggctg tggctggaag gagctgggca tccggcctga ggcgcagcgg tcgcgttagt 60

tcggcccaat ggcggcaccg ctgcttcaca cgcgtttgcc gggagatgcg gccgcttcgt 120

cctctgcagt taagaagctg ggcgcgtcga ggactgggat ttcaaatatg cgtgcattag 180
agaatgactt tttcaattct cccccaagaa aaactgttcg gtttggtgga actgtgacag 240
aagtcttgct gaagtacaaa aagggtgaaa caaatgactt tgagttgttg aagaaccagc 300
tgttagatcc agacataaag gatgaccaga tcatcaactg gctgctagaa ttccgttctt 360
ctatcatgta cttgacaaaa gactttgagc aacttatcag tattatatta agattgcctt 420
ggttgaatag aagtcaaaca gtagtggaag agtatttggc ttttcttggt aatcttgtat 480
cagcacagac tgttttcctc agaccgtgtc tcagcatgat tgcttcccat tttgtgcctc 540
cccgagtgat cattaaggaa ggcgatgtag atgtttcaga ttctgatgat gaagatgata 600
atcttcctgc aaattttgac acatgtcaca gagccttgca aataatagca agatatgtac 660
catcgacacc gtggtttctc atgccaatac tggtggaaaa atttccattt gttcgaaaat 720
cagagagaac actggaatgt tacgttcata acttactaag gattagtgta tattttccaa 780
ccttgaggca tgaaattctg gagcttatta ttgaaaaact actcaagttg gatgtgaatg 840
catcccggca gggtattgaa gatgctgaag aaacagcaac tcaaacttgt ggtgggacag 900
attccacgga aggattgttt aatatggatg aagatgaaga aactgaacat gaaacaaagg 960
ctggtcctga acggctcgac cagatggtgc atcctgtagc cgagcgcctg gacatcctga 1020
tgtctttggt tttgtcctac atgaaggatg tctgctatgt agatggtaag gttgataacg 1080
gcaaaacaaa ggatctatat cgcgacctga taaacatctt tgacaaactc ctgttgccca 1140
cccatgcctc ctgccatgta cagtttttca tgttttacct ctgtagtttc aaattgggat 1200
tcgcagaggc atttttggaa catctctgga aaaaattgca ggacccaagt aatcctgcca 1260
tcatcaggca ggctgctgga aattatattg gaagcttttt ggcaagagct aaatttattc 1320
ctcttattac tgtaaaatca tgcctagatc ttttggttaa ctggctgcac atataccta 1380
ataaccagga ttcgggaaca aaggcattct gcgatgttgc tctccatgga ccattttact 1440
cagcctgcca agctgtgttc tacacctttg tttttagaca caagcagctt ttgagcggaa 1500
acctgaaaga aggtttgcag tatcttcaga gtctgaattt tgagcggata gtgatgagcc 1560
agctaaatcc cctgaagatt tgcctgccct cagtggttaa cttttttgct gcaatcacaa 1620
ataagtacca gctcgtcttc tgctacacca tcattgagag gaacaatcgc cagatgctgc 1680
cagtcattag gagtaccgct ggaggagact cagtgcagat ctgcacaaac ccgctggaca 1740
ccttcttccc ctttgatccc tgtgtgctga agaggtcaaa gaaattcatt gatcctattt 1800
atcaggtgtg ggaagacatg agtgctgaag agctacagga gttcaagaaa cccatgaaaa 1860
aggacatagt ggaagatgaa gatgatgact ttctgaaagg cgaagtgccc cagaatgata 1920
ccgtgattgg gatcacacca agctcctttg acacgcattt ccgaagtcct tcaagtagtg 1980
tgggctcccc acccgtgttg tacatgcaac ccagtcccct ctgacggcag aaatttgtga 2040

ctgagatgtg acatttggga ttccccatca cttgtcatgc cctcagcacc cagcttgtgc 2100

cattgggcat tgatggcatt gaactagagc gagtgcctgc ctcggctgtg gcacttccag 2160

gttcgactga atcaagcatc tgaagactgg gtttttttgt tgttgttgtt ccccttacag 2220

acaaaatgaa gactatcatg tgcaatcttt tacagtgggg ttgatgatac atttggaagg 2280

atttgcttgt ttaatatgta catttttttgt gttaacagct ttttgacaca attactgggt 2340

aatttctaat ataggcagca gactgtttta cgggttgctg ttttaacatg ggtttttgtc 2400

agatccatgg tcttaggact tgactgatga gctttcagtg aagaatcctc taagataaaa 2460

cttctattta aagactttaa ctagaaagtg tttattttgg ctacattgtt caccttctgc 2520

tgtattggta tttgtctgtt gggatttcaa gggagtgtag agaagacaga aggaaagctg 2580

agagctggcc cgacatggtc tgggacacag agttggagct ggcactgaag atctccaggg 2640

acttcagaga ccaataaaag cccataggga agagagagag gatataggga aacagaatca 2700

gatgtgtaat atacttggca cagcgaaaaa atggatttaa aagacaaaaa tggaggtcca 2760

ggtagatgta attcacacag actgaaagtg agttcgggct tgtgtaaaac acatgagatt 2820

ggatttgacc ccttggctct caagtgtccc cttagatcta gaactgctcc ttggtggcca 2880

ttagatcgag tcagttttga tctgcatcac ttagttattg ggaatttctt tgttggaaac 2940

aggaaaattt ttttagatta tttggtgtac ggttttgctc acaacaatag gtggaagttg 3000

ctagtgcagt cttggtctga tggctgtgtg catcgcacat tcggcttggt gaaatccttc 3060

tctaaagcct ctttttgtat ttttataact aaacagagga agtcttcaga agacctcgct 3120

ttaaaacaaa tttgtgcaaa cactgctaga gtcattttga agctcaagca ttttcacttt 3180

gtttcttaca tgtgtacttt tttgtttact tgtgaaaatg gccatcttta agcatattta 3240

ttttctgcca ctttatttaa aggcaagcaa tattttcttg atcataaata ttttgtaatg 3300

aaatacttcc tcttttccag ggctttgtat gcacttgtat aattacattg atggcaatgt 3360

agagtttgaa tttcagtctg taaatacttt tttggaaaat agaaattttt attgctttaa 3420

agttttggat atgggtggtt ttcttttccg ggtttggtgg aaagtaattt gagaacttta 3480

aggttgtctt tttaactgct ggcaaaatgt tgattttttta atattagata aaacgagtaa 3540

acgaaattcc ccagaaatta gtagtaagtg gggtctttgt gggttgggaa gtagttttaa 3600

tgtagaaaga catttacata taagtctgtt taatttcaaa ggagtttgtg aaaaaaaatc 3660

catggtgaaa atgaaacaat gacatggtta atctggaact tacgttctta taccaataaa 3720

aggtacctca atacatgttc tttcaaaaaa aaaaaaaaaa aa 3762

<210> 42
<211> 719
<212> DNA

<213> Homo sapiens

<220>
<221> misc_feature
<222> (520)..(520)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (645)..(645)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (682)..(682)
<223> n is a, c, g, or t

<400> 42
acataaagaa aattttcatt tatttttccc tcagatatac ttcaaaataa catgtagaca 60

cagaatcaca aatatataca aaacaaagta tcgcctggtt atttcaagaa gagttttccc 120

tcaattttct tagctgctgt gcataaaatt caaagctttc aggggggttca atgaaaggga 180

caggctctcc agcatacttc agcaaagatg cgtaacgctt tagaaacaga tcatccaatt 240

ttgtattctt tgcagtcaaa tgttcataca gtgctttagc agatgtgaca tctttctctg 300

agacatagct tttcatgagg gaattgtatg cttcttcctt ttcatttaat tcaggaatca 360

attctaacac agatttcaca gttgatgcct ttccttgttt cctagaattc ctaaggagga 420

acaacaacaa aatcggggtt tgttcagcaa ttgcaccaca tctctgtagg agagctctgg 480

catcatccac cttgcctgca tccacaagtt gaaggaaaan atcagtgaca ggtttataaa 540

ttgcaaactg attggccaat ctctccgcca tgatggctta tctttcaaac tgctggtcca 600

actgctcctc tattactttc tggataagat gccaaggccg agtantcggg tcaatgactt 660

tatcctctga agtaagctat tntcatggtt tctattgcgg catctatgta atcattctt 719

<210> 43
<211> 5095
<212> DNA
<213> Homo sapiens

<400> 43
acatgctcct cctgtccttc tggcggagcg tgcttcccgc tgcggggacg ttcgagcaat 60

ggcagccctg ctgagatccg cgcgttggtt gctgcgtgcc ggggcggccc cgcgcctccc 120

gctctccctg cgcctcctcc ctggcggccc gggccggctg catgccgcct cctatctgcc 180

cgccgctcgc gccgggcccg tggccggagg actactgagc ccagccaggc tgtatgccat 240

tgctgccaaa gaaaaagata ttcaagagga gtccactttt tcttctagga agatttccaa 300

tcagtttgat tgggctctaa tgagactaga tctttctgtt cgaagaactg gccgcattcc 360

aaagaagctt ctacaaaaag tttttaatga tacctgccgc tcaggtggcc taggtggtag 420 tcatgccttg cttctactac gtagttgtgg ttctctcttg cctgaactaa agcttgaaga 480

gagaacagaa tttgctcata ggatatggga cacacttcag aaattaggtg ctgtgtatga 540

tgtgagtcac tataatgctt tacttaaagt ctatcttcaa aatgaatata aattctcacc 600

aactgatttc ctggcaaaaa tggaggaagc aaacattcaa ccaaatcgag tgacatacca 660

gagattgatt gcttcttatt gtaatgtagg agatattgaa ggtgccagca agattcttgg 720

atttatgaaa actaaggatc tcccagttac agaggcagta ttcagtgccc ttgtgacagg 780

gcatgccaga gctggtgata tggagaatgc agaaaacatt ctcacagtga tgagagatgc 840

cggaattgag cctggtccag acacatacct cgcattattg aatgcatatg ctgagaaggg 900

cgacattgac catgttaagc agactctgga gaaggtggag aagtccgagc ttcaccttat 960

ggaccgtgat ttactgcaaa ttatttttag cttcagtaaa gctgggtatc ctcagtatgt 1020

ctcagaaatt ttggaaaaag ttacatgtga aagaagatat attccagatg caatgaacct 1080

cattttactt ttagtcactg aaaaattgga agatgtagcg ttgcaaattt tactagcatg 1140

ccccgtatca aaggaagatg gcccaagtgt ctttggcagt ttctttttac aacactgtgt 1200

gactatgaat acgcctgtgg agaagctaac agactactgt aagaagttaa aggaagtcca 1260

gatgcactcc tttcctctgc agttcaccct ccattgtgct ttactcgcca ataaaactga 1320

tttggcaaaa gccttaatga aggctgtgaa ggaggaaggt tttcctatca gacctcacta 1380

tttctggcca ttgctagttg gacgtcggaa ggaaaaaaat gttcaaggta taattgaaat 1440

cctcaaagga atgcaagaat tgggagtaca tcctgatcag gaaacatata cagattatgt 1500

gattccatgc tttgatagtg taaactcagc acgagccatt ttgcaggaaa atggatgtct 1560

gtctgatagt gatatgtttt ctcaagctgg attgagaagt gaagcagcaa atgggaactt 1620

agactttgta ttatcatttt tgaaatcaaa tacattgccc atctcgctgc agtctataag 1680

aagtagccta ctgctaggct tcaggaggtc tatgaatata aatctttgga gcgagataac 1740

agaattgttg tacaaggatg gacgttattg ccaggagcct cgaggaccga cggaagctgt 1800

tggctatttt ctttataact tgattgacag catgagtgac tcagaagtac aggccaagga 1860

ggagcatttg agacaatact tccatcagct ggagaagatg aatgtaaaaa ttcctgaaaa 1920

tatctacaga ggcattcgta atctcctgga aagctaccat gttcctgaat tgattaagga 1980

tgctcacttg ttggttgaga gtaagaattt agactttcaa aaaactgtgc aacttacatc 2040

atctgaattg gagtccacac ttgaaacact aaaagctgaa aatcaaccta taagagatgt 2100

cctaaagcaa ctcatattag tgctttgttc agaagagaat atgcaaaaag cccttgaatt 2160

gaaagcaaaa tatgaatccg acatggttac tggtggctat gcagctttaa taaatttatg 2220

ctgtcgacat gataaagtag aagatgcctt gaacttgaaa gaagaatttg accgcttaga 2280 ttcatctgct gtccttgaca ccggcaagta tgtaggcctt gtaagagtat tggcaaagca 2340

tggcaagctc caagatgcta ttaacattct gaaggagatg aaagagaagg atgttcttat 2400

caaagataca acagccttgt ccttttttcca catgctaaat ggcgcagctt taagaggtga 2460

aattgaaaca gtaaaacagt tgcatgaagc catcgtgact ctagggttag cagaaccatc 2520

caccaacata agtttcccat tggtcactgt acacttggaa aagggcgacc tatctactgc 2580

tcttgaggtc gccattgact gctatgaaaa gtataaagta ttaccaagga ttcatgatgt 2640

cttgtgtaaa ctggtagaga aaggcgagac tgatctaatt cagaaagcaa tggactttgt 2700

gagccaagaa caaggtgaaa tggtgatgct ctatgatctc ttctttgcct tcctacaaac 2760

aggaaattac aaagaggcca agaagatcat tgagactcca gggattagag ctcgatctgc 2820

aaggcttcag tggttttgtg acagatgtgt tgcaaataat caggttgaaa ctctggaaaa 2880

attagtggag ctgacacaga agctatttga atgtgataga gaccagatgt actacaatct 2940

gctaaaactg tataaaataa acggtgactg gcaaagagct gatgcagtct ggaataaaat 3000

ccaagaagaa aatgttattc ctcgtgaaaa gacattaaga ttattagcag aaatccttag 3060

agagggtaac caggaagttc cgtttgacgt acctgagttg tggtatgaag atgaaaaaca 3120

ttccctgaat tcttcgtcag cctcaaccac agaacctgat ttccagaaag atatattgat 3180

tgcctgccga ttgaaccaaa aaaaaggggc atatgatatt ttcctgaatg caaaagagca 3240

aaacattgtg tttaatgctg aaacctacag caatctcatt aaattactga tgtcagaaga 3300

ttattttaca caagcaatgg aagtgaaagc attcgcggag acccacatca agggcttcac 3360

actgaacgat gctgccaaca gccgcctcat cataacgcaa gttaggcggg attatttgaa 3420

agaggctgtg acaacactga aaacagtatt ggatcagcag cagacccctt ctaggttagc 3480

agtgacccgt gtcatccagg cattggccat gaagggtgat gttgaaaaca tagaagtagt 3540

tcagaagatg ttaaatggac tcgaagactc cattggactt tcaaaaatgg ttttcatcaa 3600

taacattgct ttggctcaaa taaagaataa taacatagat gccgcaatag aaaacattga 3660

aaatatgctt acttcagaga ataaagtcat tgaaccccaa tacttcggct tggcatactt 3720

attcagaaaa gtaatagagg agcagttgga accagcagtt gaaaagataa gcatcatggc 3780

ggagagattg gccaatcagt ttgcaattta taaacctgtc actgattttt tccttcaact 3840

tgtggatgca ggcaaggtgg atgatgccag agctctccta cagagatgtg gtgcaattgc 3900

tgaacaaacc ccgattttgt tgttgttcct ccttaggaat tctaggaaac aaggaaaggc 3960

atcaactgtg aaatctgtgt tagaattgat tcctgaatta aatgaaaagg aagaagcata 4020

caattccctc atgaaaagct atgtctcaga gaaagatgtc acatctgcta aagcactgta 4080

tgaacatttg actgcaaaga atacaaaatt ggatgatctg tttctaaagc gttacgcatc 4140

tttgctgaag tatgctggag agcctgtccc tttcattgaa ccccctgaaa gctttgaatt 4200 ttatgcacag cagctaagaa aattgaggga aaactcttct tgaaataacc aggcgatact 4260

ttgttttgta tatatttgtg attctgtgtc tacatgttat tttgaagtat atctgaggga 4320

aaaataaatg aaaattttct ttatgtactt atgtatgtgt gatgcatgtt caaagtctta 4380

ttgaccataa ctctgtgcac ttggttattg gacattttg gagtttttt ctctgggaaa 4440

aatcgatagt gttttcttca atgctgctgc tgtgtgaagc catacttttt caggattctt 4500

cccctaattg gctctttggt ttccctgctc tgtttcattt atttcattaa aatgttattc 4560

ctttatttaa gattcactta ttagtctgct gtttctctga aaaattttag agctaggtat 4620

agtgaccgtg aactttctaa cgcataatat tctgtgatac agccattccg tacatgtgtg 4680

aagtcctgca taactttcga actttgttaa atgttggcac taggagtcat cagatctagg 4740

cttcatcatt ttccagtgag aagcagagac ccaaagggcc tgttacttgt gcttggtcag 4800

gggactgtct gtcatgcctg gaggctcttc ggcacacttc cccatctttc ccttctgcca 4860

ctgtggcttc aagcacctct gttcatagag cgtctctgaa attgagtctc ggtcatgact 4920

tatcccgaag tagagcaatg tgtttcctct cattgtagtt tcaggacttt gtcagtacaa 4980

gctctgccct aggcttgtta ctttatactc atatcctgaa aagatgtgat ttcatctatg 5040

aaggggtaaa atattggttt gtatttaatt gtttgaaata aaagtgatcc ctata 5095

<210> 44
<211> 5427
<212> DNA
<213> Homo sapiens

<400> 44
agatgcgacg aggttagcgg ggcccgacca ctccttggct tcccaggggt gagcctcgcg 60

agttaggagt tgggtagaga gtcagcccgg ggcccggcat ccgctttttc gttgaagcaa 120

cgactttggc cggatgactc cccagggtcg gcatcagcgt gggactggga acaggtgaag 180

ggaaacagaa gagagcctga gaagaccgct ctcctccgat cctaattgac tgagccaatg 240

agagccaaag aggtcacatg ctcaactggg cggggagcgg gtttccacct gcggcatctt 300

tcgcgagcgg ggagatgagt ggggcggaat atgggagaaa gggagggccc gccacgctct 360

ggctggacac ggccttaatc ggcccgttca ctcgacgttt ttggttctac gttgaccccg 420

agaaaccgaa accgcagggc ctagggcggg tggaacgaga gggaactaca tttcccagca 480

ggctgcggaa acgggactgc ggccactact tccggcgtgt accgagagac tggcgtccgg 540

cgtgtaccga gagactggcg tccggtgtgc aggtggccac atggatcctg gcagccggtg 600

gcggaacctg cccagcgggc ctagcctaaa gcacttgact gacccctctt atggaatccc 660

gcgggaacag caaaaggcag cgttgcagga gctgacgcgg gcgcacgtgg agtccttcaa 720

ctacgctgtg cacgagggtc tcggcctcgc ggtgcaggct atacctccct ttgaatttgc 780

tttcaaagat gagcgtatct cttttactat tctggatgct gttatcagtc cacctacagt 840
tccaaaaggg accatctgca aagaggccaa tgtttatcca gcagaatgcc ggggccgaag 900
gagtacctac cgtgggaagt tgacagctga tatcaactgg gcagtgaatg gaatctcaaa 960
aggaatcatt aagcagtttc ttggctatgt tcccatcatg gtgaaatcca agctttgcaa 1020
cttacgtaac cttcccccac aagccctcat tgagcaccat gaggaggcag aggaaatggg 1080
gggctatttt ataatcaatg gcattgaaaa agtcatccga atgttgatta tgcctcggag 1140
aaatttttccc attgcaatga taagaccaaa atggaaaacc agagggcctg gttatactca 1200
gtatggagtt tcaatgcact gtgtgaggga agaacattcc gctgtcaata tgaacctcca 1260
ctacttggaa aatggcactg ttatgttgaa ctttatttac cgaaaagaac tgttctttct 1320
tcctttggga tttgcactta aggcacttgt cagcttttct gattatcaga tctttcagga 1380
gctcatcaaa ggaaaagagg atgattcttt ccttaggaac tctgtttctc agatgttaag 1440
gattgtaatg gaagagggtt gttcgacaca aaaacaggtc cttaactacc taggtgaatg 1500
cttcagagta aaactcaatg ttcctgactg gtacccaaat gagcaagctg cggagttcct 1560
gtttaaccag tgcatctgta tccacttgaa atccaatact gaaaagtttt atatgctttg 1620
tctcatgacg cgaaagctct ttgctttagc caaaggagag tgcatggagg acaatcctga 1680
tagtttggtg aaccaggaag tcctcacacc gggtcagctc ttccttatgt tcctgaagga 1740
aaaactggaa ggttggttag tgtctattaa aatagctttt gataagaagg ctcagaagac 1800
cagtgtttcc atgaacactg acaatttgat gaggattttt acaatgggca tagaccttac 1860
aaaaccattt gaataccttt ttgctactgg gaatctgcgt tctaaaacag gtcttggcct 1920
cctacaagat tctggacttt gtgttgtggc tgacaagctg aacttcatac gctacctctc 1980
ccatttccgc tgcgtgcaca gaggggctga ttttgccaag atgaggacca ccacagtacg 2040
caggctgctg ccagagtcct ggggcttcct ttgtcccgtg cataccccag acggggagcc 2100
ctgtggcctg atgaaccacc taactgccgt atgtgaggtt gtcacacagt ttgtgtatac 2160
ggcatctatt ccagctttac tgtgcaactt gggggtcact cccattgatg gagctcccca 2220
ccgatcatac agtgagtgct accctgtcct gctggacggt gtcatggttg gctgggtgga 2280
taaggatctt gctccaggca tcgcagattc tcttcgtcat tttaaggtgt tgagagagaa 2340
aagaattcct ccctggatgg aagtggtcct tatacccatg acaggaaaac caagtctgta 2400
cccaggattg ttccttttta ccactccttg tagactggta cggcctgtgc agaacttagc 2460
attgggcaaa gaagagctaa ttggaactat ggaacagatc ttcatgaatg tcgctatctt 2520
tgaggatgaa gtttttgctg gagttaccac acaccaggaa ctctttccac acagcctgct 2580
gagtgtgatt gccaacttca tccctttctc tgatcacaac cagagtccac ggaacatgta 2640

ccaatgccag atgggtaagc aaactatggg ctttccactt ctcacttatc aagaccgatc 2700

ggataacaaa ctgtatcgtc ttcagactcc tcagagtccc ttggtgagac cctccatgta 2760

tgattattat gacatggata actatccaat tgggaccaat gccatcgttg ctgtgatttc 2820

ttacactggc tatgatatgg aagatgccat gattgtgaat aaggcctctt gggaacgagg 2880

ctttgcccat ggaagtgtct acaagtctga gttcatagac ctctctgaaa aaattaaaca 2940

aggagatagt agcctggtgt ttggcatcaa acctggtgac ccacgcgttc tgcagaagtt 3000

agatgacgat ggattgccgt ttataggagc aaaactgcag tacggagatc cgtattacag 3060

ctacctcaac ctcaacaccg gggaaagttt tgtgatgtac tataagagta aagaaaattg 3120

tgttgtggat aacatcaaag tgtgcagtaa tgacactggg agtggaaaat tcaagtgtgt 3180

ttgcatcact atgagagtgc ctcggaaccc aactatcgga gataaatttg ccagtcgcca 3240

tgggcagaag ggcattttaa gcagattgtg gccggctgag gacatgcctt ttactgagag 3300

tgggatggtc ccagacattc tgttcaatcc ccatggtttt ccatcccgca tgaccattgg 3360

gatgttaatt gagagtatgg ccgggaagtc tgcagctttg catggtctct gccatgatgc 3420

tacacccttc atcttctcag aggagaactc ggccttagaa tactttggtg agatgttaaa 3480

ggctgctggc tacaatttct atggcaccga gaggttatat agtggcatca gtgggctaga 3540

actggaagca gacatcttca taggagtggt ttattatcag cgcttacgcc atatggtctc 3600

agacaaattt caagtaagga caactggagc ccgagacaga gtcaccaacc agcctattgg 3660

gggaagaaat gtccagggtg gaatccgttt tggggagatg gaacgggatg cgcttttagc 3720

tcatggtaca tcttttctcc ttcatgaccg cctcttcaac tgctcagatc ggtcggtagc 3780

ccatgtgtgt gtgaagtgtg gcagtttact ctctccactg ttggagaagc cacccccttc 3840

ttggtctgcc atgcgcaaca gaaaatacaa ctgtactctg tgtagtcgca gtgacactat 3900

cgatactgtt tctgtgcctt atgtttttcg gtattttgta gctgaactgg cagctatgaa 3960

catcaaagtg aaactggatg ttgtttaact tgatgttgac cttttggatt aagaggacta 4020

tcagattaaa gcaaaatgta attttaattc aatgaagata tcattaccag gttactcttg 4080

agatttttca acggtgttag aactctcaac caagacctga aaaccaagta tgcaaggttt 4140

ctgaatctct ctggtagatt aactattgac aatgatttc tgttatcttt gttcaaaaag 4200

ttcatgtctt ctcaaaatat gaaatattga taaatggaag agcatacggt gacaagtctc 4260

ctttccaacc ccaggttccc tacaccctgc tctcagcagg cagtgagtgt cacacacctg 4320

ttaatccatc ttgagcagga cagtactata caaatagaat gcaagctgta atgtaatttt 4380

atattttctt atagccacgt tgaagtaaaa acaaacaggt acagtgtttt ttaccagctt 4440

tatagaagta cagttgttac atatttaatg aatacaattt gatgggtctg actatatgca 4500

cacacctttg ataccatcac cacaatcagg gtaataaaca tacctgtcat ctccacaagt 4560 ttcctcctgc ccctttgttt tttgcttttt ggttgctgtt gagtttttgt tttgtcttct 4620

gtggtaagaa cacttaactc aagacctacc ctcttaacaa atctttaagt gcacgatata 4680

gtattgttaa ttccaggcac catgttgtac aacagatctt tagaccttac ttgtcttgca 4740

taactgaagc tttatacctg ttgaacaact ctccatttcc ctggccccta gcaaccaccc 4800

ttctaccctg tttctatgag tttgactatt acagatatct catatagtgg gatcatgcaa 4860

tatttgtcct gtgactggct tatttcactt agcatagtga aataagattc atccattttg 4920

gaagccaggc atggtgctgt gcatctatag tccctgctat ttgagaggct gaggtgggag 4980

gatcatttga gtgcaggagt tcaaggacag cctgggtaat ataggaagac cctgtcttga 5040

agaccctgac ctcaagtgat ccacccacct cggcctccga aagtgctagg attacaggtg 5100

tgagccactg tgcctggcct ccggtgagta ttttatattt agtctacact tccatacttg 5160

gctttttct gcttttatat tgatctgctt tcatagcagt gtgtagagtg ccacttatgt 5220

tttctttctt gtgtacagta ttttattgta tggatttacc atcccctgtg tatttaagtt 5280

gttccattct ttggccatta taactttttt ctgcaaatat tctggtgact tatctttggc 5340

cattataaac tgttgataat agatcatctt gtatatactt ctgcaattat aagatgtttt 5400

ttgatgatgt aaaaaaaaaa aaaaaaa 5427

<210> 45
<211> 1138
<212> DNA
<213> Homo sapiens

<400> 45
agggctacta ggacgcacgc gcgagataga acctctagtc tcgtggagag attgaagatg 60

gcggcttctc aggcggtgga ggaaatgcgg agccgcgtgg ttctggggga gtttggggtt 120

cgcaatgtcc atactactga ctttcccggt aactattccg gttatgatga tgcctgggac 180

caggaccgct tcgagaagaa tttccgtgtg gatgtagtac acatggatga aaactcactg 240

gagtttgaca tggtgggaat tgacgcagcc attgccaatg cttttcgacg aattctgcta 300

gctgaggtgc caactatggc tgtggagaag gtcctggtgt acaataatac atccattgtt 360

caggatgaga ttcttgctca ccgtctgggg ctcattccca ttcatgctga tccccgtctt 420

tttgagtatc ggaaccaagg agatgaagaa ggcacagaga tagatactct acagtttcgt 480

ctccaggtca gatgcactcg gaacccccat gctgctaaag attcctctga ccccaacgaa 540

ctgtacgtga accacaaagt gtataccagg catatgacat ggatcccccct ggggaaccag 600

gctgatctct ttccagaggg cactatccga ccagtgcatg atgatatcct catcgctcag 660

ctgcggcctg gccaagaaat tgacctgctc atgcactgtg tcaagggcat tggcaaagat 720

catgccaagt tttcaccagt ggcaacagcc agttacaggc tcctgccaga catcaccctg 780 cttgagcccg tggaagggga ggcagctgag gagttgagca ggtgcttctc acctggtgtt 840

attgaggtgc aggaagtcca aggtaaaaag gtggccagag ttgccaaccc ccggctggat 900

accttcagca gagaaatctt ccggaatgag aagctaaaga aggttgtgag gcttgcccgg 960

gttcgagatc attatatctg taagaaagat ttgctggctg cggtggctca cacctgtaat 1020

cccagcactt tgggaggcca aggcgagtgg atcacggggt caagagagcg agaccatcct 1080

ggctaacatg gtgaaacccc gtctctataa aaaaaaaaaa aaaaaaaaaa aaaaaaaa 1138

<210> 46
<211> 1277
<212> DNA
<213> Homo sapiens

<400> 46
gagagagaga agatggcggc ttctcaggcg gtggaggaaa tgcggagccg cgtggttctg 60

ggggagtttg gggttcgcaa tgtccatact actgactttc ccggtaacta ttccggttat 120

gatgatgcct gggaccagga ccgcttcgag aagaatttcc gtgtggatgt agtacacatg 180

gatgaaaact cactggagtt tgacatggtg ggaattgacg cagccattgc caatgctttt 240

cgacgaattc tgctagctga ggtgccaact atggctgtgg agaaggtcct ggtgtacaat 300

aatacatcca ttgttcagga tgagattctt gctcaccgtc tggggctcat tcccattcat 360

gctgatcccc gtctttttga gtatcggaac caaggagatg aagaaggcac agagatagat 420

actctacagt ttcgtctcca ggtcagatgc actcggaacc cccatgctgc taaagattcc 480

tctgacccca acgaactgta cgtgaaccac aaagtgtata ccaggcatat gacatggatc 540

cccctgggga accaggctga tctctttcca gagggcacta tccgaccagt gcatgatgat 600

atcctcatcg ctcagctgcg gcctggccaa gaaattgacc tgctcatgca ctgtgtcaag 660

ggcattggca aagatcatgc caagttttca ccagtggcaa cagccagtta caggctcctg 720

ccagacatca ccctgcttga gcccgtggaa ggggaggcag ctgaggagtt gagcaggtgc 780

ttctcacctg gtgttattga ggtgcaggaa gtccaaggta aaaaggtggc cagagttgcc 840

aacccccggc tggatacctt cagcagagaa atcttccgga atgagaagct aaagaaggtt 900

gtgaggcttg cccgggttcg agatcattat atcttctctg ttgagtcaac gggggtgttg 960

ccaccagatg tgctggtgag tgaagccatc aaagtactga tggggaagtg ccggcgcttc 1020

ttggatgaac tagatgcggt tcagatggac tgagcttgga tgcttctgag gcaagctgaa 1080

gctttgggtt ctgactgacc caccctacag gactgctgaa cagagagccc agtgtgacta 1140

gggatcctga gttttctggg acaattccag ctttaatcaa tacattttgt taaatgtgcc 1200

ataaaatgag actttttacg cctttataag gccttagatg taaataaact cacccaaaca 1260

aaaaaaaaaa aaaaaaa 1277

<210> 47
<211> 725
<212> DNA
<213> Homo sapiens

<400> 47
ccttccttcc tgcttcgcct ccgcgcctcg cgctatggga cagagccccc gatccgccag 60

caccacctga ggatccagaa accgccccag cgatggaaga ggatcaggag ctggagagaa 120

aaatatctgg attgaagacc tcaatggctg aaggcgagag gaagacagcc ctggaaatgg 180

tccaggcagc tggaacagat agacactgtg tgacatttgt attgcacgag gaagaccata 240

ccctaggaaa ttctctacgt tacatgatca tgaagaaccc ggaagtggaa ttttgtggtt 300

acactacgac ccatccttca gagagcaaaa ttaatttacg cattcagact cgaggtaccc 360

ttccagctgt tgagccattt cagagaggcc tgaatgagct catgaatgtc tgccaacatg 420

tgcttgacaa gtttgaggcc agcataaagg actataagga tcaaaaagca agcagaaatg 480

aatccacatt ctagtccttt atgcagtata caaggagaac tgtcctgtag gatattctct 540

tcctgatggt gcagaaccca gaattagaag tttgtggtta cagcatactc tgtccttcag 600

aaaggcgtga ttctagctgt tgaccccttg cagctgttgg aatctctgca agaacctctg 660

tattcttcta ataaattccc tcttttattt aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 720

aaaaa 725

<210> 48
<211> 6584
<212> DNA
<213> Homo sapiens

<400> 48
tttttttttc ctttttggga gctgaaaaat ttccggtaag ggaaagaagg gctcctttcg 60

ctccttattt ccccgcctcc ttccctcccc caccttcccc tcctccggct ttttcctccc 120

aactcgggga ggtccttccc ggtggccgcc ctgacgaggt ctgagcacct aggcggaggc 180

ggcgcaggct ttttgtagtg aggtttgcgc ctgcgcagcg cgcctgcctc cgccatgcac 240

gggggtggcc ccccctcggg ggacagcgca tgcccgctgc gcaccatcaa gagagtccag 300

ttcggagtcc tgagtccgga tgaactgaag cgaatgtctg tgacggaggg tggcatcaaa 360

tacccagaga cgactgaggg aggccgcccc aagcttgggg ggctgatgga cccgaggcag 420

ggggtgattg agcggactgg ccgctgccaa acatgtgcag gaaacatgac agagtgtcct 480

ggccactttg gccacattga actggccaag cctgtgtttc acgtgggctt cctggtgaag 540

acaatgaaag ttttgcgctg tgtctgcttc ttctgctcca aactgcttgt ggactctaac 600

aacccaaaga tcaaggatat cctggctaag tccaagggac agcccaagaa gcggctcaca 660

catgtctacg acctttgcaa gggcaaaaac atatgcgagg gtggggagga gatggacaac 720
aagttcggtg tggaacaacc tgagggtgac gaggatctga ccaaagaaaa gggccatggt 780
ggctgtgggc ggtaccagcc caggatccgg cgttctggcc tagagttgta tgcggaatgg 840
aagcacgtta atgaggactc tcaggagaag aagatcctgc tgagtccaga gcgagtgcat 900
gagatcttca aacgcatctc agatgaggag tgttttgtgc tgggcatgga gccccgctat 960
gcacggccag agtggatgat tgtcacagtg ctgcctgtgc cccgctctc cgtgcggcct 1020
gctgttgtga tgcagggctc tgcccgtaac caggatgacc tgactcacaa actggctgac 1080
atcgtgaaga tcaacaatca gctgcggcgc aatgagcaga acggcgcagc ggcccatgtc 1140
attgcagagg atgtgaagct cctccagttc catgtggcca ccatggtgga caatgagctg 1200
cctggcttgc cccgtgccat gcagaagtct gggcgtcccc tcaagtccct gaagcagcgg 1260
ttgaagggca aggaaggccg ggtgcgaggg aacctgatgg gcaaaagagt ggacttctcg 1320
gcccgtactg tcatcacccc cgaccccaac ctctccattg accaggttgg cgtgccccgc 1380
tccattgctg ccaacatgac ctttgcggag attgtcaccc ccttcaacat tgacagactt 1440
caagaactag tgcgcagggg gaacagccag tacccaggcg ccaagtacat catccgagac 1500
aatggtgatc gcattgactt gcgtttccac cccaagccca gtgaccttca cctgcagacc 1560
ggctataagg tggaacggca catgtgtgat ggggacattg ttatcttcaa ccggcagcca 1620
actctgcaca aaatgtccat gatggggcat cgggtccgca ttctcccatg gtctaccttt 1680
cgcttgaatc ttagtgtgac aactccgtac aatgcagact ttgacgggga tgagatgaac 1740
ttgcacctgc cacagtctct ggagacgcga gcagagatcc aggagctggc catggttcct 1800
cgcatgattg tcaccccca gagcaatcgg cctgtcatgg gtattgtgca ggacacactc 1860
acagcagtgc gcaaattcac caagagagac gtcttcctgg agcggggtga agtgatgaac 1920
ctcctgatgt tcctgtcgac gtgggatggg aaggtcccac agccggccat cctaaagccc 1980
cggcccctgt ggacaggcaa gcaaatcttc tccctcatca tacctggtca catcaattgt 2040
atccgtaccc acagcaccca tcccgatgat gaagacagtg gcccttacaa gcacatctct 2100
cctggggaca ccaaggtggt ggtggagaat ggggagctga tcatgggcat cctgtgtaag 2160
aagtctctgg gcacgtcagc tggctccctg gtccacatct cctacctaga gatgggtcat 2220
gacatcactc gcctcttcta ctccaacatt cagactgtca ttaacaactg gctcctcatc 2280
gagggtcata ctattggcat tggggactcc attgctgatt ctaagactta ccaggacatt 2340
cagaacacta ttaagaaggc caagcaggac gtaatagagg tcatcgagaa ggcacacaac 2400
aatgagctgg agcccacccc agggaacact ctgcggcaga cgtttgagaa tcaggtgaac 2460
cgcattctta acgatgcccg agacaagact ggctcctctg ctcagaaatc cctgtctgaa 2520
tacaacaact tcaagtctat ggtcgtgtcc ggagctaaag gttccaagat taacatctcc 2580

caggtcattg ctgtcgttgg acagcagaac gtcgagggca agcggattcc atttggcttc 2640

aagcaccgga ctctgcctca cttcatcaag gatgactacg ggcctgagag ccgtggcttt 2700

gtggagaact cctacctagc cggcctcaca cccactgagt tctttttcca cgccatgggg 2760

ggtcgtgagg ggctcattga cacggctgtc aagactgctg agactggata catccagcgg 2820

cggctgatca agtccatgga gtcagtgatg gtgaagtacg acgcgactgt gcggaactcc 2880

atcaaccagg tggtgcagct gcgctacggc gaagacggcc tggcaggcga gagcgttgag 2940

ttccagaacc tggctacgct taagccttcc aacaaggctt ttgagaagaa gttccgcttt 3000

gattatacca atgagagggc cctgcggcgc actctgcagg aggacctggt gaaggacgtg 3060

ctgagcaacg cacacatcca gaacgagttg gagcgggaat ttgagcggat gcgggaggat 3120

cgggaggtgc tcagggtcat cttcccaact ggagacagca aggtcgtcct cccctgtaac 3180

ctgctgcgga tgatctggaa tgctcagaaa atcttccaca tcaacccacg ccttccctcc 3240

gacctgcacc ccatcaaagt ggtggaggga gtcaaggaat tgagcaagaa gctggtgatt 3300

gtgaatgggg atgacccact aagtcgacag gcccaggaaa atgccacgct gctcttcaac 3360

atccacctgc ggtccacgtt gtgttcccgc cgcatggcag aggagtttcg gctcagtggg 3420

gaggccttcg actggctgct tggggagatt gagtccaagt tcaaccaagc cattgcgcat 3480

cccggggaaa tggtgggggc tctggctgcg cagtcccttg gagaacctgc cacccagatg 3540

accttgaata ccttccacta tgctggtgtg tctgccaaga atgtgacgct gggtgtgccc 3600

cgacttaagg agctcatcaa catttccaag aagccaaaga ctccttcgct tactgtcttc 3660

ctgttgggcc agtccgctcg agatgctgag agagccaagg atattctgtg ccgtctggag 3720

catacaacgt tgaggaaggt gactgccaac acagccatct actatgaccc caacccccag 3780

agcacggtgg tggcagagga tcaggaatgg gtgaatgtct actatgaaat gcctgacttt 3840

gatgtggccc gaatctcccc ctggctgttg cgggtggagc tggatcggaa gcacatgact 3900

gaccggaagc tcaccatgga gcagattgct gaaaagatca atgctggttt tggtgacgac 3960

ttgaactgca tctttaatga tgacaatgca gagaagctgg tgctccgtat tcgcatcatg 4020

aacagcgatg agaacaagat gcaagaggag gaagaggtgg tggacaagat ggatgatgat 4080

gtcttcctgc gctgcatcga gtccaacatg ctgacagata tgaccctgca gggcatcgag 4140

cagatcagca aggtgtacat gcacttgcca cagacagaca acaagaagaa gatcatcatc 4200

acggaggatg gggaattcaa ggccctgcag gagtggatcc tggagacgga cggcgtgagc 4260

ttgatgcggg tgctgagtga gaaggacgtg gaccccgtac gcaccacgtc caatgacatt 4320

gtggagatct tcacggtgct gggcattgaa gccgtgcgga aggccctgga gcgggagctg 4380

taccacgtca tctcctttga tggctcctat gtcaattacc gacacttggc tctcttgtgt 4440 gataccatga cctgtcgtgg ccacttgatg gccatcaccc gacacggagt caaccgccag 4500

gacacaggac cactcatgaa gtgttccttt gaggaaacgg tggacgtgct tatggaagca 4560

gccgcacacg gtgagagtga ccccatgaag gggtctctg agaatatcat gctgggccag 4620

ctggctccgg ccggcactgg ctgctttgac ctcctgcttg atgcagagaa gtgcaagtat 4680

ggcatggaga tccccaccaa tatccccggc ctgggggctg ctggacccac cggcatgttc 4740

tttggttcag cacccagtcc catgggtgga atctctcctg ccatgacacc ttggaaccag 4800

ggtgcaaccc ctgcctatgg cgcctggtcc cccagtgttg ggagtggaat gaccccaggg 4860

gcagccggct tctctcccag tgctgcgtca gatgccagcg gcttcagccc aggttactcc 4920

cctgcctggt ctcccacacc gggctccccg gggtccccag gtccctcaag cccctacatc 4980

ccttcaccag gtggtgccat gtctcccagc tactcgccaa cgtcacctgc ctacgagccc 5040

cgctctcctg gggctacac accccagagt ccctcttatt cccccacttc accctcctac 5100

tcccctacct ctccatccta ttctccaacc agtcccaact atagtcccac atcacccagc 5160

tattcgccaa cgtcacccag ctactcaccg acctctccca gctactcacc cacctctccc 5220

agctactcgc ccacctctcc cagctactcg cccacctctc ccagctactc acccacttcc 5280

cctagctact cgcccacttc ccctagctac tcgccaacgt ctcccagcta ctcaccgaca 5340

tctcccagct actcgccaac ttcacccagc tattctccca cttctcccag ctactcacct 5400

acctctccaa gctattcacc cacctccccc agctactcac ccacttcccc aagttactca 5460

cccaccagcc cgaactattc tccaaccagt cccaattaca ccccaacatc acccagctac 5520

agcccgacat cacccagcta ttcacctact agtcccaact acacacctac cagccctaac 5580

tacagcccaa cctctccaag ctactctcca acatcaccca gctattcccc gacctcacca 5640

agttactccc cttccagccc acgatacaca ccacagtctc caacctatac cccaagctca 5700

cccagctaca gccccagctc gcccagctac agcccaacct cacccaagta caccccaacc 5760

agtccttctt acagtcccag ctccccagag tataccccaa cctctcccaa gtactcacct 5820

accagtccca aatattcacc cacctctccc aagtactcgc ctaccagtcc cacctattca 5880

cccaccaccc caaaatactc cccaacatct cctacttatt ccccaacctc tccagtctac 5940

accccaacct ctcccaagta ctcacctact agccccactt actcgcccac ttcccccaag 6000

tactcgccca ccagccccac ctactcgccc acctccccca aaggctcaac ctactctccc 6060

acttcccctg gttactcgcc caccagcccc acctacagtc tcacaagccc ggctatcagc 6120

ccggatgaca gtgacgagga gaactgaggg cacgtggggt gcggcagcgg gctagggccc 6180

agggcagctt gcccgtgctg ctgtgcagtt cttgcctccc tcacggggcg tcacccccag 6240

cccagctccg ttgtacataa atgccttgtg gcagagctcc cggtgaactt ctggatcccg 6300

tttctgatgc agactcttgt cttgttctcc acttgtgctg ttagaactca ctggcccagt 6360 ggtgttctca ctcctacccc acccaccccc tgcctgtccc caaattgaag atccttcctt 6420

gcctgtggct tgatgcgggg cgggtaaagg gtattttaac ttaggggtag ttcctgctgt 6480

gagtggttac agctgatcct cgggaagaac aaagctaaag ctgccttttg tctgttattt 6540

tatttttttg aagtttaaat aaagtttact aattttgacc aaaa 6584

<210> 49
<211> 342
<212> DNA
<213> Homo sapiens

<400> 49
atccggcgtt ctggcctaga gctgtatgcg gaatggaagc acgttaatga ggactctcag 60

gagaagaaga tcctgctgag tccagagcga gtgcatgaga tcttcaaacg catctcagat 120

gaggagtgtt ttgtgctggg catcgagccc cgctatgcac ggccagagtc gatgattgtc 180

acagtgctgc ccgtgccccc gctgtccgtg cggcctgctg ttgtgatgca gggctctgcc 240

cgtaaccagg tcagtggctc cagggctctg cctcttagct ggagggcgag gtgtctaaag 300

ggaggaagac ctgtaggtcg ggatggtgat gacctggttt ga 342

<210> 50
<211> 3748
<212> DNA
<213> Homo sapiens

<400> 50
cttttgattt caagagttag gagctcgaga accgtttggc aatatgtacg acgcggatga 60

ggatatgcaa tatgatgagg atgatgatga aatcaccccg gatttgtggc aagaagcatg 120

ctggattgta atcagttcct attttgacga gaaaggcttg gttagacaac agctggattc 180

ttttgatgag tttattcaga tgtctgttca aagaattgtg gaagacgctc ctcctataga 240

cctacaggct gaagctcagc atgctagtgg agaagttgaa gaaccgccac gatatttgct 300

gaagtttgaa caaatttatc tttccaagcc tacccattgg gaaagagatg gtgctccttc 360

accaatgatg cccaatgaag ctagattaag gaatctcacg tattctgctc cgctttatgt 420

tgatataaca aaaacagtca ttaaagaagg tgaagaacaa cttcagactc agcatcagaa 480

aacttttata ggaaaaattc caattatgtt gcggtcaact tactgccttt tgaatggctt 540

gacagatcgt gatctttgtg agttaaatga atgccctttg gatcctggtg gctatttcat 600

tattaatgga tcagaaaagg ttctgattgc ccaagagaaa atggcaacaa acacagttta 660

tgtgtttgcc aaaaaggatt ctaaatatgc ctacacagga gagtgtagat catgtcttga 720

gaattcttcc cgacccacca gtactatatg ggttagcatg ctggcaagag gaggacaggg 780

tgccaagaag agtgctattg gtcagcgcat tgtggcaact ctaccatata tcaagcaaga 840 agttcccatc attattgtgt tcagagcatt aggttttgtg tccgacagag atattttaga 900

acatattatt tatgattttg aagatccaga gatgatggaa atggttaaac cttctctcga 960

tgaagctttt gtcatccaag aacagaatgt tgcactaaat ttcattggtt cacgaggagc 1020

aaagcctggt gttactaaag agaaaagaat taaatatgca aaggaagttt tacaaaaaga 1080

aatgctccct catgttggtg tcagtgattt ttgtgagacc aaaaaagcct atttcttggg 1140

atacatggtt cataggttac ttctggcagc tttgggtaga agagaactag atgacagaga 1200

tcactatgga aacaagagat tggatcttgc tgggccgctg cttgcattct tatttagagg 1260

tatgtttaag aatttgctta aagaagtgcg gatctatgca cagaaattta ttgatcgagg 1320

aaaggatttt aacttggagt tggcaattaa aacacggatc atatctgatg gcctaaaata 1380

ctctttagct actggaaact ggggtgatca aaagaaagct catcaagcca gagctggagt 1440

atctcaggtg ttaaaccgcc tgacttttgc gtctactctt tctcacctgc gtcgtttaaa 1500

ttctcctatt ggtagagacg gcaagctagc aaaaccaaga cagttgcata atacgttgtg 1560

gggaatggtg tgtcctgccg agaccccaga gggccatgct gtaggacttg tgaagaattt 1620

agccttgatg gcgtatattt cagttggatc tcaaccatct ccaattctgg aatttttaga 1680

agaatggagt atggaaaatt tagaagaaat ttctcctgca gctattgctg atgcaaccaa 1740

gatttttgtt aatggctgct gggttggaat acataaagat cccgaacaac ttatgaacac 1800

cctaaggaaa ttgagacgtc agatggacat cattgtgtct gaagtttcta tgatcagaga 1860

tattcgagag agggagattc ggatctatac ggatgcaggc cgtatttgta gaccacttct 1920

gattgtggaa aaacaaaagc tacttttgaa gaagaggcat attgaccaat tgaaagagag 1980

agaatataac aactatagtt ggcaggatct tgtggccagt ggggtagtgg agtatattga 2040

taccctggaa gaagaaacag tgatgcttgc aatgactcca gatgatttac aggagaaaga 2100

agtagcttat tgttccacat atacacactg tgagattcat ccctcaatga tccttggtgt 2160

ctgtgcatct attattccct ttcctgatca taaccagtcc cctagaaaca cataccagtc 2220

tgctatgggt aagcaggcta tgggagttta catcaccaac ttccatgttc gcatggacac 2280

attggcccat gttctctatt atcctcaaaa gccacttgtg actacacggt ctatggaata 2340

tctacgattt agagagctgc cagcaggcat caactcaatt gtggccattg catcatacac 2400

tggatataat caggaagact ctgttatcat gaatcgttca gctgtagacc gcggcttctt 2460

caggtctgtt ttctatcgct catacaaaga acaggagtct aaaaaaggat ttgatcaaga 2520

agaagttttt gagaagccta cacgtgaaac atgccagggc atgaggcatg ccatttacga 2580

caagctggat gatgatggtt tgatagctcc aggggttcgt gtatcaggag atgatgttat 2640

tataggcaaa acagtcacct tgcctgaaaa tgaagatgaa ttggagagca ccaatagacg 2700

ctataccaag agagactgta gcacttttct cagaactagt gagacgggca ttgtggatca 2760 ggttatggta actctcaatc aggaaggata taaattttgt aaaataaggg tacgctctgt 2820

taggattcca cagattggag acaaatttgc tagtcgacat ggtcaaaagg gtacttgtgg 2880

tattcagtat agacaagagg atatgccttt cacctgtgaa ggtatcaccc ctgatatcat 2940

catcaatccc catgccatcc cctctcgtat gactattggt cacttaattg aatgccttca 3000

agggaaggta tcggctaaca agggtgaaat tggtgatgcc actccattta atgatgctgt 3060

taacgtgcag aagatttcta atcttttatc tgattatggc tatcatctca gaggaaatga 3120

ggtcctgtac aatgggttca ctggtcgaaa aatcacatca caaatattta ttggccccac 3180

ttattaccag cgtttgaagc atatggtgga tgataagatt cactctcgtg ctaggggacc 3240

tattcagatc ctcaatagac agcccatgga gggtagatct cgtgatggtg gcctgcgttt 3300

tggagaaatg gaacgagatt gtcagattgc ccatggagca gcccagtttt taagggaaag 3360

attgtttgag gcatcagatc catatcaggt tcatgtttgc aatctttgtg gaataatggc 3420

gattgccaac accaggaccc atacatatga atgcaggggc tgccgcaata aaacccagat 3480

ttctttggtg cgaatgcctt acgcatgcaa actattgttt caggaactta tgtctatgag 3540

tattgcaccg cgaatgatga gtgtttagct attttacagg agtcaacaag ataattaaat 3600

atcttggtgt cttgtttcta ttgtgtggct ttataaaaat gacaaatatg tactgtgttg 3660

tgataaaaag tattttattt gtttaatgat atgcatgctt ttcttctgta aatatataat 3720

aaatttttgt agatagtctt gatgtgtg 3748

<210> 51
<211> 1422
<212> DNA
<213> Homo sapiens

<400> 51
ggtggcccct gggcgagatg ccgtacgcca accagcctac cgtgcggatc acggagctca 60

ctgacgagaa tgtcaagttc atcatcgaga acaccgacct ggcggtggcc aattcgattc 120

ggagggtctt catcgctgag gttcccataa tagccattga ctgggttcag attgatgcca 180

attcctcagt ccttcatgat gaattcattg ctcacaggct tggattaatt cccctcatta 240

gtgatgacat tgtggacaag ctgcagtact ctcgggactg cacatgtgag gagttctgcc 300

ccgagtgctc ggtggagttc accctcgatg tgcggtgcaa tgaagaccag acgcgacatg 360

tcacgtctcg agacctcatc tccaacagcc cccgggtcat tccggtgaca tcccggaacc 420

gagataatga ccccaatgac tacgtggagc aggatgacat cctcatcgtc aagttgagaa 480

agggccagga gctgagactt cgagcctatg ccaaaaaggg ctttggcaag gagcatgcca 540

agtggaaccc tactgcaggg gtggcttttg aatacgatcc agacaatgcc ctgaggcaca 600

cagtgtaccc caagcccgag gaatggccaa agagtgagta ctcggagctg gatgaggatg 660 agtcgcaggc tccctatgac cccaacggca agccagaaag gttttactac aatgtggagt 720

cctgtggctc tctgcgtcct gaaaccattg tcctgtcagc cctctcagga ttgaagaaga 780

aactgagtga tttacaaact caattaagcc acgagatcca gagtgatgtg ctaaccataa 840

attaactgca gcttgcctgc ttcagcaaaa acggagattc aggccagcag ctggatatgg 900

gggtctctct tcagactctt ctcgtttctg agaatctagt ctactgttgg ttgagcttct 960

tggcaggaca tcagtaccaa ctagaagtgg gtcatagata gattaccagg gatgcagtgg 1020

tgtttaggca ggataggtct ttactggccc tgactgctgt taataattgg cagcagtgct 1080

ccccagatcc cagaaggtcc ctgctggagt gtttccagtg cacctgtagg gaaccaacta 1140

gacttctctc ctggttagtc cagctcttta ctctaaaccc tttctgtcca aaatgagtca 1200

ttttcagttg taccttagat gtctggtgtt gaggatcaag tgccatagcc tttattcagg 1260

gggcctataa acccttccag ttcttgcccc agggctggcc tgctagcgct tcaaattccc 1320

aggtgtccct aatttgagaa gtaacctttt ggaatagcat tagaccctgg ctgtcccctc 1380

cccaccaaat aaacatgata tttcaaaaaa aaaaaaaaaa aa 1422

<210> 52
<211> 778
<212> DNA
<213> Homo sapiens

<400> 52
tatgttgtgc tctttattgc caaaaataaa cacttttaaa aagacaactt ctgttaatca 60

ataaatatcc tttcctttct gtactggatc tctggttttc tccaggctgc ctaattaact 120

gctttgtgca tgttcctatc tggagacagt gctgaggggt tagggccaga gccttcatta 180

aggagaaatg aaatactcct cctgtgtatg tgggaggggc ccttgctcct gcggctgctt 240

cacccccacc cccaccccca cccccaccaa actaaatgat cacggcattg gaggtgaagg 300

ggctcatgac tgctggacag acaatgaaat atcatgttta tttggtggcg agggcacagc 360

cactgtctaa tgctattcca aaaggttact gatcacacta cggacacctg cgaatgctaa 420

gcgctagcat gccacccctg ggtcaagaac tgcaagggct tataggcccc tgaataaagg 480

ctatggcact tgatcctgaa caccagacat ctaaggtaca actgacaatg actcattccg 540

gacagaaatg gcttagacta aagagctgga ctaaccagcg aaagaagtct agttgatccc 600

ttcaggtgca ctggaaacac ttcagccggg acctttctgg atctggggac cactgctgcc 660

caacataaac tgcagaccag gccatttaag acctatgctt tgctaacaca actggtgccc 720

tgatatccat ttatgaccca ctttttattg ccactgtgac cttgccagaa agctaacc 778

<210> 53
<211> 1616

<212> DNA
<213> Homo sapiens

<400> 53
accgcggagc agacgcggag gctggtggcc ctgggcgaga tgccgtacgc caaccagcct   60

accgtgcgga tcacggagct cactgacgag aatgtcaagt tcatcatcga gaacaccgac   120

ctggcggtgg ccaattcgat tcggagggtc ttcatcgctg aggttcccat aatagccatt   180

gactgggttc agattgatgc caattcctca gtccttcatg atgaattcat tgctcacagg   240

cttggattaa ttcccctcat tagtgatgac attgtggaca agctgcagta ctctcgggac   300

tgcacatgtg aggagttctg ccccgagtgc tcggtggagt tcaccctcga tgtgcggtgc   360

aatgaagacc agacgcgaca tgtcacgtct cgagacctca tctccaacag cccccgggtc   420

attccggtga catcccggaa ccgagataat gaccccaatg actacgtgga gcaggatgac   480

atcctcatcg tcaagttgag aaagggccag gagctgagac ttcgagccta tgccaaaaag   540

ggctttggca aggagcatgc caagtggaac cctactgcag gggtggcttt tgaatacgat   600

ccagacaatg ccctgaggac cacagtgtac cccaagcccg aggaatggcc aaagagtgag   660

tactcggagc tggatgagga tgagtcgcag gctccctatg accccaacgg caagccagaa   720

aggttttact acaatgtgga gtcctgtggc tctctgcgtc ctgaaaccat tgtcctgtca   780

gccctctcag gattgaagaa gaaactgagt gatttacaaa ctcaattaag ccacgagatc   840

cagagtgatg tgctaaccat aaattaactg cagcttgcct gcttcagcaa aaacggagat   900

tcaggccagc agctggatat gggggtctct cttcagactc ttctcgtttc tgagaatcta   960

gtctactgtt ggttgagctt cttggcagga catcagtacc aactagaagt gggtcataga   1020

tagattacca gggatgcagt ggtgtttagg caggataggt ctttactggc cctgactgct   1080

gttaataatt ggcagcagtg ctccccagat cccagaaggt ccctgctgga gtgtttccag   1140

tgcacctgta ggaacaacta gacttctctc ctggttagtc cagctcttta ctctaaaccc   1200

tttctgtcca aaatgagtca ttttcagttg taccttagat gtctggtgtt gaggatcaag   1260

tgccatagcc tttattcagg gggcctataa acccttccag ttcttgcccc agggctggcc   1320

tgctagcgct tcaaattccc aggtgtccct aatttgagaa gtaacctttt ggaatagcag   1380

gagcaagggc cctcccacca tacacaggag gagtatttca tttctcctta atgaaggctc   1440

tggccctaac ccctcagcac tgtctccaga taggaacatg cacaaagcag ttaattaggc   1500

agcctggaga aaaccagaga tccagtacag aaaggaaagg atatttattg attaacagaa   1560

gttgtctttt taaaagtgtt tatttttgca ataaagagca caacataaaa aaaaaa   1616

<210> 54
<211> 2338
<212> DNA
<213> Homo sapiens

<400> 54
gctccgcgct tctgcttccg gcggcgctgg tggtggcgcg cggcgcggcg cggcgatggc 60

ggcgggtggc agcgatccgc gggctggcga cgtagaggag gacgcctcac agctcatctt 120

tcctaaagag tttgaaacag ctgagacact tctaaattca gaagttcata tgcttctgga 180

acatcgaaag cagcagaatg agagtgcaga ggacgaacag gagctctcag aagtcttcat 240

gaaaacatta aactacacag cccgtttcag tcgtttcaaa aacagagaga ccattgccag 300

tgttcgtagc ttgctactcc agaaaaagct tcataagttt gagttggcct gtttggccaa 360

cctttgccca gagactgctg aggagtccaa ggctctaatc ccaagcttgg agggacggtt 420

tgaagatgag gagctgcagc agattcttga tgatatccag acaaagcgca gctttcagta 480

ttaatctcca aacatcactg ctgctcggag aaaccacatc cccaggcata acaccacctt 540

cccactgtct ggggctgact tgcacagaaa ttctgttgaa gacagttgag aattcctttg 600

gagaaaacag cccagcttgg cgtggggtta ggttgctgtt tcaaataact cacaggccca 660

ggtgacatgg aatcttggag cagccttgtg cagtggcagc cagtggcttc ctgaacgtgc 720

ctctgcgaag tgtgagatga ggggtcacat aaccacactg ttgactacct cattcctggt 780

ttttggcctc cacatcatct tttttcttaa tatttcatgt tttaatttca gggtgtttat 840

actttttgaa actagaccag aagatagtag actttataga gaaagaccag ttttacctag 900

atactaaagg aagaattaaa ccgctgttag tttgaaatgc tttttttttt ttttttttaaa 960

tggagatagg gtcttaactc ttgtccaggc tggaggagtg cagtcgtaca gtcatggctc 1020

actgaagtct tgaccccgct gcctcagcct cccaaataac tggggccaca ggtgtgcacc 1080

acaactctca gctaattttt aaaatttttt atagaggtgg ggttttacta tgctgtccag 1140

actggtctta aactcctggg ctcaagtgat ccccctgcct tggcctccca aactggtgag 1200

attacaggca tgagccacca caactggcct gaaattctta aaggatggga gtgtcgatga 1260

cagcaccttg gcatcgttgt gcctaacctg ggagacggaa gaagcacgcc atgggaagtg 1320

tttacacttg gggacaagtg ctaagtattg tggagcccat agccccttga gatagatggc 1380

tactttgcct ttcttcttga actgtcttgc agaatgtgga tttggggtaa gtggtcttga 1440

aggattcatt tagtcaccct caaattaaga tttttacttc atctttcttg ggcctgcacc 1500

tccaagataa caaagaagaa gcaatggtcg tgccaaagag gtccacaacc aggtgtgcac 1560

tgttcactgc agcccatttg ctgtatgaac tgtggttgtt gtgtgcccaa tgacaaggct 1620

actaagaaat tcatcatttg aaacgtagag gccgcagcag tcagcgatgt ttctgaaatg 1680

agcatccttg acgcctgtgt acttcccagg ctggatgtga agctacatta ccatgtgagt 1740

tgtgccattc acagcacagt ggtgaggaat tgagctcatg aagcaggcaa ggaccgaaca 1800

cctccacccc aacgtagacc tgcaggtgct gccccatgac ctccaccaaa gcccatataa 1860 ggagcggagt tgttaaggac tgaagaaaaa cttctctgga gaaaaataaa attgcaattc 1920

tacttaaaaa aatttttttt tttttttact tcataggcca ggcttgaagt tctgaacact 1980

ttgaagtctc caattatgag agatccagtc taagcctctg gcctgctaat tagcaataag 2040

tgctttattt ggaaggaggg agtcatccac tcttgagcca ctgcagtgaa gtcacttgat 2100

ctcagtctgg gggaaaacac ttcaatagct aaacattcta gctttgattt ttctgaaggg 2160

aatacacttg ttttcaattt tggggttttt ctttggggca cttgcttgac tctgtatgaa 2220

cttgtgatcc aaggaaaaag gagaaagaac agtgttggct tttaaaatca ggatggtttt 2280

atgtttgcta cgaaataagg caagaataaa aaattcttat ttttaaaaaa aaaaaaaa 2338

<210> 55
<211> 566
<212> DNA
<213> Homo sapiens

<400> 55
aaaatacgaa ttttttattc ttgccttatt tcgtagcaaa cataaaacca tcctgatttt 60

aaaagccaac actgttcttt ctcctttttc cttggatcac aagttcatac agagtcaagc 120

aagtgcccca aagaaaaacc ccaaaattga aaacaagtgt attcccttca gaaaaatcaa 180

agctagaatg tttagctatt gaagtgtttt cccccagact gagatcaagt gacttcactg 240

cagtggctca agagtggatg actccctcct tccaaataaa gcacttattg ctaattagca 300

ggccagaggc ttagactgga tctctcataa ttggagactt caaagtgttc agaacttcaa 360

gcctggccta tgaagtaaaa aaaaaaaaaa aattttttta agtagaattg caattttatt 420

tttctccaga gaagtttttc ttcagtcctt aacaactccg ctccttatat gggctttggt 480

ggaggtcatg gggcagcacc tgcaggtcta cgttggggtg gaggtgttcg gtccttgcct 540

gcttcatgag ctcaatttct caccac 566

<210> 56
<211> 903
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (359)..(359)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (518)..(518)
<223> n is a, c, g, or t

<220>
<221> misc_feature

<222> (594)..(594)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (599)..(599)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (602)..(602)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (625)..(625)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (647)..(647)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (650)..(650)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (664)..(665)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (724)..(724)
<223> n is a, c, g, or t

<400> 56
cctcaccgtg ctggtgtccc atacgatgac cccaccgacc agatgtttgt gttctttcca 60

gaggagccca aggtgggcat caagaccatc aaggtgtact gccagcgcat gcaggaggag 120

aacatcacac gggctctcat cgtggtgcag cagggcatga caccctccgc caagcagtcc 180

ctggtcgaca tggcccccaa gtacatcctg gagcagtttc tgcagcagga gctgctcatc 240

aacatcacgg agcacgagct agtccctgag cacgtcgtca tgaccaagga ggaggtgaca 300

gagctgctgg cccgatacag taagctccga gagaaccagc tgcccaggat ccaggcggng 360

gaccctgtgg cgcgctactt tgggataaag cgtgggcagg tggtgaagat catccggccc 420

agtgagacgg ctggcaggta catcacctac cggctggtgc agtagctacc gcctgacagc 480

ccctagaggc ggacacacag cgacccccat ccctgcanga caaacgcccc tgccctgcca 540

gaatccggcc cccacagctc tcacggctgc tgctcctctg gactccccag gcangtggnc 600

tncaccccac gtctcccgtc ctggngtgag ggcttctgtg ggccagnccn gccccaatca 660

cctnngtgat tgtgcgagat gcagcctcag aaggaaacaa gcccccagag ggaggtcacc 720 tggnggcagc tggtgccggg tctcacccaa accacgcttg gtcccctctg tgggggtttg 780

ggggcggggc tccaccaagc actgttttct cctggccctt ggctttcacc ctttttttcc 840

ttctcggggg cctgatggtg ggggccccgc ccgctcgggt tttattcctt caaaccggcc 900

cgc 903

<210> 57
<211> 1248
<212> DNA
<213> Homo sapiens

<400> 57
ggcggcggcg gcggaggctg ccatggacga cgaggaggag acgtaccggc tctggaaaat 60

ccgcaagacc atcatgcagc tgtgccacga ccgtggctat ctggtgaccc aggacgagct 120

tgaccagacc ctggaggagt tcaaagccca atttggggac aagccgagtg aggggcggcc 180

gcggcgcacg gacctcaccg tgctggtggc ccacaacgat gaccccaccg accagatgtt 240

tgtgttcttt ccagaggagc ccaaggtggg catcaagacc atcaaggtgt actgccagcg 300

catgcaggag gagaacatca cacgggctct catcgtggtg cagcagggca tgacaccctc 360

cgccaagcag tccctggtcg acatggcccc caagtacatc ctggagcagt ttctgcagca 420

ggagctgctc atcaacatca cggagcacga gctagtccct gagcacgtcg tcatgaccaa 480

ggaggaggtg acagagctgc tggcccgata taagctccga gagaaccagc tgcccaggat 540

ccaggcgggg gaccctgtgg cgcgctactt tgggataaag cgtgggcagg tggtgaagat 600

catccggccc agtgagacgg ctggcaggta catcacctac cggctggtgc agtagctacc 660

gcctgacagc ccctagaggc ggacacacag cgacccccat ccctgcagga caaacgcccc 720

tgccctgcca gaatccggcc cccacagctc tcacggctgc tgctcctctg gactccccaa 780

ggcaggtggc ctccacccac gttctcccgt cctggggtga ggcttcctgt ggcccagccc 840

gccccattca cctgtggatt tgtgcgagat gcagcctcag aaggaacaag gcccccagag 900

ggaggtcacc tgggggcagc tggtgccggg tcttcaccca gaccacgctg ggtcccctct 960

gttgggggtt tggggtccgg gtctcccacc agccactgct tcctcctggg ccctcggcct 1020

tccacccctc gtcttccctc cctcgggggc cctgatgcgt ggcggccccc acccggcctc 1080

ggctctttac tccattcaca gccgtgcacg cgctcaagcc accagggtgc gagatgccag 1140

ctctggagtt ctcggttgtt gtaggaggtt gggtgttttc aaatggtaaa gatgttttga 1200

gcaaataaat ttgcttgata cagaaaaaaa aaaaaaaaaa aaaaaaaa 1248

<210> 58
<211> 588
<212> DNA
<213> Homo sapiens

<400> 58
cgcaggcgca agataagcta ggagccgcgc gagtcgtagt gtcgctgttt gcgggtctcc   60

gcgcgggacc ggggcgcagc ggggtcgctg aggcgagggt gtcatgtcag acaacgagga   120

caattttgat ggcgacgact ttgatgatgt ggaggaggat gaagggctag atgacttgga   180

gaatgccgaa gaggaaggcc aggagaatgt cgagatcctc ccctctgggg agcgaccgca   240

ggccaaccag aagcgaatca ccacaccata catgaccaag tacgagcgag cccgcgtgct   300

gggcacccga gcgctccaga ttgcgatgtg tgcccctgtg atggtggagc tggaggggga   360

gacagatcct ctgctcattg ccatgaagga actcaaggcc cgaaagatcc ccatcatcat   420

tcgccgttac ctgccagatg ggagctatga agactggggg gtggacgagc tcatcatcac   480

cgactgagct ggagtcatct tcctgccctt gccccatgcc caattttcat tctcacttta   540

tatgtgtaaa taataaaata ttcaactttc aaaaaaaaaa aaaaaaaa   588

<210> 59
<211> 852
<212> DNA
<213> Homo sapiens

<400> 59
ctcctagtcg cagcaagcgc ggaactgggg ttgcggcgtc taagtgtttc cggtggattc   60

ccagggactg tcggaggtgt ggactctgcc tgcctacctg gtctgggaag atgttctacc   120

atatctccct agagcacgaa atcctgctgc acccgcgcta cttcggcccc aacttgctca   180

acacggtgaa gcagaagctc ttcaccgagg tggaggggac ctgcacaggg aagtatggct   240

ttgtaattgc tgtcaccacc attgacaata ttggtgctgg tgtgatccag ccaggccgag   300

gctttgtcct ttatccagtt aagtacaagg ccattgtttt ccggccattt aaaggggagg   360

tcgtggatgc tgttgtcact caggtcaaca aggttggact cttcacagaa attgggccca   420

tgtcttgctt catctctcga cattccatcc cttcagagat ggagtttgat cctaactcca   480

acccaccatg ttacaagaca atggatgagg atattgtgat tcagcaggac gatgagatcc   540

gcttaaagat tgtggggacc cgtgtggaca agaatgacat ttttgctatt ggctccctga   600

tggacgatta cttggggctt gtaagctgag cctggtggcc tcctaccctt ggtcctactc   660

taggaagtgt gattgtcaca cttatcatgt tgtccagagg tccagtctgg ctgctgttgt   720

ggaggcaagg aaggcaactc atcccagaag gcatctggtg cttcttgtag cttaactact   780

gcctcctcat tttcagtat gtgttctaag tataaaaagt ccttggttct catggaaaaa   840

aaaaaaaaaa aa   852

<210> 60
<211> 867
<212> DNA
<213> Homo sapiens

<400> 60
ccctcactaa agggacaaaa gctggagctc caccgcggtg cggccgctct agaactagtg 60

gatcccccgg gctgcaggtc cggtcttgtc cacgctaggg ggtgcacgta ctcccaactg 120

tggtcgcgct ctcacccctt ctgctgctct cgtggccccc tcgcgatggc gggcatcctg 180

tttgaggata ttttcgatgt gaaggatatt gacccggagg ccaagaagtt tgaccgagtg 240

tctcgactgc attgtgagag tgaatctttc aagatggatc taatcttaga tgtaaacatt 300

caaatttacc ctgtagactt gggtgacaag tttcggttgg tcatagctag taccttgtat 360

gaagatggta ccctggatga tggtgaatac aaccccactg atgataggcc ttccagggct 420

gaccagtttg agtatgtaat gtatggaaaa gtgtacagga ttgagggaga tgaaacttct 480

actgaagcag caacacgcct ctctgcgtac gtgtcctatg gggcctgct catgaggctg 540

caggggggatg ccaacaacct gcatggattc gaggtggact ccagagttta tctcctgatg 600

aagaagctag ccttctgaac ctcgcctgaa gccagcctct ctgccaagtc actcaggtca 660

tgggcattgt tcaagcctga gtggcagccg ctcttgctca cctgttgagg aagggctggc 720

tcactgtcca ccgtggcggc atctttaact ggcctccact caatgggaaa ctgactcgcc 780

tgtgaaagac acagtgggag agctgaaaat gaatcagaag ctttatgtat atgattttta 840

aattaaactt tactttttca gactgcc 867

<210> 61
<211> 960
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (631)..(631)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (646)..(646)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (663)..(663)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (670)..(670)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (740)..(740)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (742)..(742)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (803)..(803)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (806)..(806)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (821)..(821)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (838)..(838)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (846)..(846)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (870)..(870)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (874)..(874)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (892)..(892)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (901)..(901)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (910)..(910)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (940)..(940)
<223> n is a, c, g, or t

<220>
<221> misc_feature

<222> (943)..(943)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (957)..(957)
<223> n is a, c, g, or t

<400> 61
gggggcactg gccttcaacc aagccaagcg ccttggcttc tgtcctggat gctcagtgtt   60

ttgcggttta cgatgcgcat cgttttcccg gctctggcac cagatccata ctattctgta   120

aactgaggcg gcgatgccca aaaattggaa gcaattactc taagatatca aaagattgaa   180

gcaagccatg aaacctctgc gcggagagag ccgagcccga ggcggcgggg agccagcacg   240

tgtgcgcagc tccgccctct tttggcgaag cacctaggcc tggcccctcc cgcgacctgt   300

agcgcggcgg agcaagcgca gaaggctggg agggctgcgc gggctgcgcg tcgccatgga   360

gcccgacggg acttacgagc cgggcttcgt gggtattcgc ttctgccagg aatgtaacaa   420

catgctgtac cccaaggaag acaaggagaa ccgcattctg ctctacgcgt gccggaactg   480

tgatttccag caggaggccg acaacagctg catctatgtc aacaagatca cgcacgaagt   540

ggacgaactg acccagatta tcgccgacgt gtcccaggac cccacgttgc cgcggaccga   600

ggaccacccg tgccaaaagt gcggccacaa ngaggctgtg ttcttncagt cacacagtgc   660

gcnggccgan gacccatgcg cctttactac gtgtgcacaa gccccacact tgcgggcacc   720

gctggaccga atgacctctn tnttccccga tgtaataaac accaaattca tgcgtgaaaa   780

aaaaaaaaaa aactttcatc cgncanctca tgatatattc naccggaatt ccgaccgnac   840

ctgagngtac actttcttta tggcgataan ctgngatatg gctactgtct gngaatgttc   900

ntcatcccan atcaccgact aaggaaactg ggcatagacn atnatatggt ggtatcnttc   960

<210> 62
<211> 652
<212> DNA
<213> Homo sapiens

<400> 62
ggtggcggcg gcggcggacc cttggggtct ggacgcaacg gcggcgggag catgaacgcc   60

cctccagcct tcgagtcgtt cttgctcttc gagggcgaga agaagatcac cattaacaag   120

gacaccaagg tacccaatgc ctgtttattc accatcaaca aagaagacca cacactggga   180

aacatcatta aatcacaact cctaaaagac ccgcaagtgc tatttgctgg ctacaaagtc   240

ccccacccct tggagcacaa gatcatcatc cgagtgcaga ccacgccgga ctacagcccc   300

caggaagcct ttaccaacgc catcaccgac ctcatcagtg agctgtccct gctggaggag   360

cgctttcggg tggccataaa agacaagcag gaaggaattg agtaggggcc agagggggct   420

ctgctcggcc tgtgagcccc gttcctacct gtgcctgacc ctccgctcca ggtaccacac   480 cgaggagagc ggcggtccca gccatggccc gccttgtggc acccctcacc ctgacaccga 540

cgtgttggca cccctcaccc tgacaccgac gtgtcctgta catagattag gttttatatt 600

cctaataaag tatagcggga gagacctgga aaaaacaaca acaaaaaaac ct 652

<210> 63
<211> 433
<212> DNA
<213> Homo sapiens

<400> 63
cacgaggcga gtcgttcttg ctcttcgagg gcgagaagat caccattaac aaggacacca 60

aggtacccaa tgcctgttta ttcaccatca acaaagaaga ccacacactg ggaaacatca 120

ttaaatcacg tgcctgcttc cccttcgcct tctgccgtga ttgtcagttt cctgaggcct 180

ccccagccac gcttcctgta cagcctgcag aactctgccc cagagcacat cagctatgtg 240

ccccagctct caaacgacac cttgtccggg gaggctcacc ctgtccacct tcacgctgga 300

gcagcctcta ggccagttca gcagccacaa catctctgac ttggatacca tctggctggt 360

ggtggccctc agcaacgcca cccagagctt cacggcccca cggacaaacc aggacatccc 420

tgctcctgcc aac 433

<210> 64
<211> 494
<212> DNA
<213> Homo sapiens

<400> 64
tggggggcga gggtctcact ctgtcaccca ggctggagtg cagtggtgcg atcatagctc 60

actgcggcct cgacctccca ggctcaaggc tctcagattc tcagacccca ccctggggaa 120

ggatggagga tgaggacaag cttcccaggt ccacatgaca gagggactca ggattggcaa 180

tttggcctgt gtccagcttt gctgctcaga gagctcagat ccccacccct caattcactc 240

ccttgtgtgg aagggagtga ccctcacttt ctagcacaca cagggaagag ggcttggcta 300

agagcaccca ggaggtcagg gcaggactgg ggattagatc acacccaaat gcaagctctg 360

tggcactcct ggtatagccc tgctgtcccc ctgtagcctg gccctgggaa gggggctccc 420

agacttcaca ccccctcacc tctttatcca ggggaccggc tgtctcttcc tccttcatcc 480

ccagcctctg tcag 494

<210> 65
<211> 959
<212> DNA
<213> Homo sapiens

<400> 65
gtaccggtcc ggaattcccg ggatctcgac acctggacta gccgggttgt atttggaaac 60

gcggagtgag tttttccgtg ctgtgtaggg gctaacaatg gacacccaga aggacgttca 120

acctccaaag cagcaaccaa tgatatatat ctgtggagag tgtcacacag aaaatgaaat 180

aaaatctagg gatccaatca gatgcagaga atgtggatac agaataatgt acaagaaaag 240

gactaaaaga ttggtcgttt ttgatgctcg atgaatgctg ggaattcaga ggaatgtctt 300

cacttatact tggatttgct ctcttcccat ttctgattgt tgtatagctt tcgattttgc 360

ttacagtagt tccccctttat cttcgggaga tacattccaa ggcccccagt gaactcctga 420

aacctcaaac agtaccaaac ctttatacac tgtttttttcc atatatatat acctatgata 480

aagtataatg tataaattaa gcatagcaag agattaataa taatgtaata gaacaatgat 540

aacatactat aataaaagtt atgtgaatgt ggttggtctc tcttgctttc aaaatatctt 600

cttgtacagt actcacctat tttagaatgt ggttgactac aggtaaccaa aaccacagaa 660

agggaaactt tggatgmggg gggcactact gtacttagga atacaactat atacatatga 720

ttttattttt aagaccatat tatatttggg tatctactaa tattttgtat aaagcaattt 780

tttgttccat tacgtgactt tttgttttat tgtatatgta atttaacaca caataaaggg 840

taaagttgct tcccmaacca cagttttaat caaaacctag aatcatctgc agtccttgtt 900

aaaaatgcag gtttctagaa ccctctgaag ttctgattaa ataaatttwt tgcaaacca 959

<210> 66
<211> 971
<212> DNA
<213> Homo sapiens

<400> 66
gcactgtgcg tcatgtacca gcgccggaag ttggtctcga cacctggact agccgggttg 60

tatttggaaa cgcggagtga gtttttccgt gctgtgtagg ggctaacaat ggacacccag 120

aaggacgttc aacctccaaa gcagcaacca atgatatata tctgtggaga gtgtcacaca 180

gaaaatgaaa taaaatctag ggatccaatc agatgcagag aatgtggata cagaataatg 240

tacaagaaaa ggactaaaag attggtcgtt tttgatgctc gatgaatgct gggaattcag 300

aggaatgtct tcacttatac ttggatttgc tctcttccca tttctgattg ttgtatagct 360

ttcgattttg cttacagtag ttccccctta tcttcgggag atacattcca aggcccccag 420

tgaactcctg aaacctcaaa cagtaccaaa cctttataca ctgtttttttc catatatata 480

tacctatgat aaagtataat gtataaatta agcatagcaa gagattaata ataatgtaat 540

agaacaatga taacatacta taataaaagt tatgtgaatg tggttggtct ctcttgcttt 600

caaaatatct tcttgtacag tactcaccta ttttagaatg tggttgacta caggtaacca 660

aaaccacaga aagggaaact ttggatgagg ggggcactac tgtacttagg aatacaacta 720

tatacatatg attttatttt taagaccata ttatatttgg gtatctacta atattttgta 780 taaagcaatt ttttgttcca ttacgtgact ttttgtttta ttgtatatgt aatttaacac 840

acaataaagg gtaaagttgc ttccccaaac cacactttta atcaaaacct agaatcatct 900

gcagtccttg ttaaaaatgc aggtttctag aaccctctga agttctgatt aaataaattt 960

attgcaaacc a 971

<210> 67
<211> 469
<212> DNA
<213> Homo sapiens

<400> 67
tttttttttt gaacactgtg cccagctaat ttattaaatt ttgttgtaga gatggggtct 60

cattatgatg cccaggcggg tcttgaattt ctggactcaa gtgatcctcc tgccttggcc 120

tcccaaagtc ctgggattat aggggtgagc cctgttgccc ggcctaaatc tggctcttca 180

gattccgaga gagagggtac aggagcctct ctgctcagcc tgaagaacac ccagaaggag 240

ggaacctcaa agagccccga gatggaaggc tctcctggca gcgctcctgc aggggtgcct 300

gtggaacagt gtgccccagg gctaggagag ttcatggccc agtctgcttc actgggctgc 360

cacagtgtga aagctgggcc tagacctggc tcctgacatc ctagtgaggg ctgggactta 420

gtggttctga aagaccttta ctggatggtt cttccagagt ggggtatgg 469

<210> 68
<211> 423
<212> DNA
<213> Homo sapiens

<400> 68
gagtctggga cgcgccgccg ccatgatcat ccctgtacgc tgcttcactt gtggcaagat 60

cgtcggcaac aagtgggagg cttacctggg gctgctgcag gccgagtaca ccgaggggga 120

tgcgctggat gccctgggcc tgaagcgcta ctgctgccgc cggatgctgc tggcccacgt 180

ggacctgatc gagaagctgc tcaattatgc acccctggag aagtgaccac gctggaaccc 240

acccacccgc tgtgctgacc atgggccctg agcgtcctgc cccgaattca cgaggctgag 300

gcatccggga gctggcgtaa tgcctggccg cagtgtgtgt gtatcccata ccccactctg 360

gaaggaacca tccagtaaag gtctttcaga aacaaaaaaa aaaaaaaaaa aaaaaaaaaa 420

aaa 423

<210> 69
<211> 4237
<212> DNA
<213> Homo sapiens

<400> 69
agattcgggc agggagcggg cggaaccttt ctaccgcgtc tctagctaac acgcacggcg 60

gggacagttt aggcctccgc gcaccgttcg ccgggagtct tgcagtttgc ttggtgcagg 120
gaaggcgggc gcggaggttc tatctgtttc ttcctccttc gtgagcagca tggacgtgct 180
agcggaggag tttgggaacc tgactccgga gcagctggcg gcgccgatcc cgactgtaga 240
ggaaaaatgg aggctgcttc cagcattttt aaaggtgaaa ggccttgtga aacagcatat 300
agattcattt aactatttca ttaatgtaga gataaagaag ataatgaaag ccaatgaaaa 360
ggttacaagt gacgctgacc ctatgtggta cttaaaatat cttaatatct atgttgggct 420
tcctgatgtt gaagaaagct tcaatgtaac tagaccagtg tcccctcatg agtgccgttt 480
gagagacatg acatactctg cccctattac agtggatatt gaatataccc gaggcagcca 540
gaggatcatc cgcaatgcct tacctatcgg cagaatgccc ataatgctac gtagttcaaa 600
ctgtgttctt acaggaaaaa cgccagcaga atttgccaaa ctgaacgaat gtcccttaga 660
tccaggtggc tacttcattg ttaaaggagt agaaaaagtt attcttatcc aagagcagct 720
gtctaagaac aggatcatcg tggaggctga tagaaaaggg gctgttggag cttcagttac 780
cagctctacc catgagaaaa aaagcagaac caatatggct gtgaaacaag gacgatttta 840
tttgaggcat aatactttgt cagaagatat acccattgtc atcatattta aggccatggg 900
tgttgagagt gaccaggaaa ttgtgcagat gattggaaca gaggagcacg tgatggctgc 960
atttgggccc agtctggaag agtgccagaa agctcagatt ttcacacaga tgcaggcatt 1020
aaaatatata gggaacaaag taagaaggca aaggatgtgg ggaggtggac caaagaaaac 1080
caaaatagaa gaagcaagag agctcctggc ttccaccatt ctgacccatg tcccagttaa 1140
ggaattcaat ttccgagcca aatgtatcta tactgcagtg atggtgcgaa gagttattct 1200
ggcccaagga gataataaag ttgacgacag agattattat ggtaacaagc gactggaatt 1260
ggcaggacag cttttatctc ttctttttga agacttgttc aaaaaattta attctgaaat 1320
gaaaaagatt gccgaccagg tgattcctaa gcaaagagca gcccagtttg atgttgtcaa 1380
acacatgcgc caagaccaga tcaccaatgg catggtgaat gctatttcta ccggaaattg 1440
gtctttaaag agatttaaaa tggaccgcca gggtgtaacc caagtgctgt ctcgcttgtc 1500
atatatatcc gcactgggca tgatgacaag aatctcttcc cagtttgaaa aaacgagaaa 1560
agtgagtggt cctcgctccc tccagccatc tcagtgggga atgctgtgtc cttcggacac 1620
tcctgaagga gaggcatgtg gtttggttaa aaacttggcc cttatgacac acatcacaac 1680
tgatatggaa gatggaccca ttgttaaatt agccagtaac ttgggagtag aagatgtgaa 1740
tttattatgt ggggaagagc tctcttaccc aaatgtgttt cttgtctttc ttaatggtaa 1800
catcttaggt gtcattcgag accacaaaaa gctagtgaat acatttcgac tcatgagaag 1860
agcaggatat atcaatgaat ttgtttccat ctcaacaaat cttacagatc gatgtgtcta 1920

tatttcttct gatgggggaa ggctatgcag accctacata attgtcaaga aacagaagcc 1980
agcagtcaca aataaacata tggaagagct ggcccaaggg tacaggaatt ttgaagattt 2040
cttacatgag agtctggttg aatatttaga tgtgaatgaa gaaaatgatt gtaacattgc 2100
actgtacgaa cacacaatta ataaagacac cacccacttg gagattgaac ccttcactct 2160
tctcggcgtg tgtgctggac ttatcccata ccctcaccat aaccagtcac cgagaaacac 2220
ttatcagtgt gccatgggga aacaagccat gggtactata ggatacaacc agcgaaacag 2280
aattgatact ctcatgtatc tactagcata tccacaaaaa cccatggtta agacaaaaac 2340
cattgaattg atagaatttg agaaactgcc agctggacag aatgcaacag ttgctgtgat 2400
gagctatagt ggctatgata ttgaagatgc tcttgtttta aacaaggcct ctttagacag 2460
aggctttggg cgttgccttg tatataaaaa tgctaaatgt acgttgaaac gatacaccaa 2520
tcagactttt gataaagtga tggggcccat gttggatgct gctacaagga aacctatctg 2580
gcgacatgaa atcttagatg cagatggtat ttgttctcca ggtgagaaag tagaaaacaa 2640
acaagtgctt gtaaataagt ccatgcccac agtgactcag attcctttgg aaggaagtaa 2700
tgtaccacag caaccacagt acaaagatgt acccataacc tacaaaggag caacagactc 2760
atatattgaa aaagtgatga tatcttcaaa tgctgaagat gcttttctga tcaaaatgct 2820
gctgagacag acaaggcgtc cagaaattgg agacaaattc agcagtcgtc atgggcaaaa 2880
aggtgtttgt ggcttgatcg tcccccagga agacatgcca ttttgtgatt ctggcatctg 2940
tccggacatc atcatgaacc cacacggctt cccatcacga atgacggtgg ggaagctcat 3000
tgagctgctg gctggcaagg ccggtgtgct ggacggcaga ttccactacg gcactgcgtt 3060
tggaggcagt aaagtgaagg atgtgtgtga ggacctcgtt cgccatggtt ataactactt 3120
ggggaaagac tatgttacat ccggcatcac aggtgagccc ttcgaagcat acatctattt 3180
tggccccgtg tactatcaga agctgaaaca catggtgcta gataaaatgc atgcccgggc 3240
ccgggggccca cgagccgtcc ttaccaggca acccactgaa ggacggtctc gtgatggtgg 3300
cttgcgtctc ggggaaatgg aacgtgactg tttaatcggt tatggagcca gtatgctttt 3360
gctagagaga ctaatgattt caagtgatgc ctttgaggtt gatgtctgtg ggcagtgtgg 3420
acttctgggg tattctggct ggtgccatta ctgcaagtca tcctgccacg tgtcttccct 3480
ccgtattccg tatgcctgca agctgctctt ccaggaacta cagtctatga acatcatccc 3540
caggttaaaa ctgtccaagt acaatgaatg aggatggaaa aaatgattat taaagagaac 3600
aagtgataca tccaatgcaa cggaaagcag aagggattta ggactacgtc tcctcctgtg 3660
aagaattccc ttgcgtattc tctctctaaa acaaccaaaa aaaaatggag aggcttttta 3720
tatactctaa gactggctaa acaaccttga tcattgagcc tcgagccatg ggagagatgc 3780
tgaccatgtg gactgcaagg ctgcttgatt cacagatgga tgtgacctaa aggataaata 3840

agctattact tatgtgctga tctcttgaca ttcactcatt agaagacctt actccttcaa 3900

gcaaatgttt ggggtcaaat ttaccatatc ttctggctaa ccatattcaa gattcttctg 3960

aaacttggag gatgtaaaga atccatttga tttggtcagc ctggcttttg tcgtggtggc 4020

tggctcggat aaattttccc aacaattaaa tcttgccttt acacacccaa actttgtaat 4080

tttagtcttg gtgaaatata atgaatttgt tcctaccttg tcaagcaaga atgtcgtctt 4140

ctcctatgga ctcaattgct attattttaa acctgcatga ttgtaccatg caatactatt 4200

cgttaaatat ctgaatctag aaaaaaaaaa aaaaaaa 4237

<210> 70
<211> 1843
<212> DNA
<213> Homo sapiens

<220>
<221> misc_feature
<222> (578)..(578)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (599)..(599)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (601)..(601)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (630)..(633)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (685)..(685)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (830)..(830)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (1701)..(1702)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (1825)..(1825)
<223> n is a, c, g, or t

<400> 70

tcttcggtgg cgatccggtc ctagaaaggg cggtgggctc cacctcggcc tagaagcaga 60

cgggaccgta ggaagccgtc gcgggaagct cagccgaatt ggagttggag cccccggatt 120

gcgctgaccc tgagctctca gactccccag tacaatgact caagcagaaa ttaagctctg 180

ttctttgttg ctgcaagagc attttggaga gattgtagaa aaaattggag tccatctgat 240

aagaaccggc agccagccac taagagtaat tgcccatgac acaggaacat cactggatca 300

ggtgaagaaa gccctgtgtg tcctcgtcca acataacctg gtgagttatc aagtgcacaa 360

acgtggtgtg gtggagtatg aagcccagtg cagccgggta ttgcgaatcg ttagatatcc 420

ccggtacatc tatactacca aaactctgta cagtgacact ggagagctga ttgttgagga 480

gcttctgttg aacggcaaac tgacaatgtg cactgttgtg aagaaagtgg cagaccggct 540

cacagagacc atggaggatg gcaagaccat ggactatnct gaagtatcaa acacatttnt 600

ncgactggca gacacacact ttgtacaacn nnncccttcg gtacctacca ctgagaattc 660

agaccctggg ccaccaccac ctgcncccac acttgtcatt aatgaaaagg acatgtacct 720

ggttcctaaa ctcagcttga tagggaaagg taaaaggagg agatcatctg atgaagatgc 780

tgctggggag cccaaggcaa gagaccaaaa tatactacag ataacaaggn gcccattcca 840

gatgatggga tttattggca ggcccaacct tgacagattc cgccaacact tccgtgacca 900

agccattgtg agcgcagttg ctaacaggat ggaccagaca agcagcgaga ttgtgcgaac 960

catgctccga atgagtgaga ttaccacttc ctctagtgct cccttcaccc agccattgtt 1020

ttccaatgag atcttcagat ccctacctgt tggctataac atctctaagc aagttcttga 1080

tcagtatctc actctgctgg cagatgatcc actagagttt gttggaaagt ctggcgacag 1140

tggtggagga atgtatgtca tcaacctcca taaggcatta cgatccctag ccacagccac 1200

tctggagtcc gtcgtacagg agagatttgg gtctcgctgt gctagaatat gccgtctagt 1260

tttgcagaag aaacacatag agcagaagca agtggaagac tttgcaatga ttctgcaaga 1320

ggcaaaggat atgctatata agatgctctc aggaaatttc atgtcactcc aggaaattcc 1380

caaaacacca gaccatgccc catccaggac cttctattta tatactgtga acatcctgtc 1440

agctgcccga atgttgttcg acaggtgcta caagagcata gccaacttga tagaaaggag 1500

gcaatttgaa accaaagaga ataagcgtct actagaaaaa tctcagaggg tagaagccat 1560

cattgcatct atgcaggcta ctggtgcaga ggaagcacag ttacaagaaa tagaggagat 1620

gatcacagct cctgaacgtc agcagctaga gaccctgaaa cgtaatgtca acaagttgga 1680

tgccagtgag atccaggtgg nngaaaccat cttcctgctg gagtcttaca ttgagtgcac 1740

catgaagaga cagtgatcca gaagaagcat cttcctcaga agatctgggg ggatggaaag 1800

caaaataaag gaggtgcctg gatgnaaaaa aaaaaaaaaa aaa 1843

<210> 71
<211> 1621
<212> DNA
<213> Homo sapiens

<400> 71
cggaagccgc cgaaactctg ggtagagtg gcacgcgctt tttgcttttc cgcgtcttct 60

tcggtggcga tccgcgtcct agaaagggcg gtgggctcca cctcggccta gaaggccagc 120

gggagccgta ggaagccgtc gcgggaagct cagccgaatt ggagttggag cccccggatt 180

gcgctgaccc tgagctctca gactccccag tacaatgact caagcagaaa ttaagctctg 240

ttctttgttg ctgcaagagc attttggaga gattgtagaa aaaattggag tccatctgat 300

aagaaccggc agccagccac taagagtaat tgcccatgac acaggaacat cactggatca 360

ggtgaagaaa gccctgtgtg tcctcgtcca acataacctg gtgagttatc aagtgcacaa 420

acgtggtgtg gtggagtatg aagcccagtg cagccgggta ttgcgaatgc ttagatatcc 480

ccggtacatc tatactacca aaactctgta cagtgacact ggagagctga ttgttgagga 540

gcttctgttg aacggcaaac tgacaatgtc agctgttgtg aagaaagtgg cagaccggct 600

cacagagacc atggaggatg gcaagaccat ggactatgct gaagtatcaa acacatttgt 660

gcgactggca gacacacact ttgtacaacg ctgcccttcg gtacctacca ctgagaattc 720

agaccctggg ccaccaccac ctgcccccac acttgtcatt aatgaaaagg acatgtacct 780

ggttcctaaa ctcagcttga tagggaaagg taaaaggagg agatcatctg atgaagatgc 840

tgctggggag cccaaggcca agagaccaaa atatactaca gataacaagg agcccattcc 900

agatgatggg atttattggc aggccaacct tgacagattc caccaacact tccgtgacca 960

agccattgtg agcgcagttg ctaacaggat ggaccagaca agcagcgaga ttgtgcgaac 1020

catgctccga atgagtgaga ttaccacttc ctctagtgct cccttcaccc agccattgtc 1080

ttccaatgag atcttcagat ccctacctgt tggctataac atctctaagc aagttcttga 1140

tcagtatctc actctgctgg cagatgatcc actagagttt gttggaaagt ctggcgacag 1200

tggtggagga atgtatgtca tcaacctcca taaggcatta gcatccctag ccacagccac 1260

tctgaagtcc gtcgtacagg agagatttgg gtctcgctgt gctagaatat tccgtctagt 1320

tttgcagaag aaacacatag agcagaagca agtggaagac tttgcaatga ttcctgcaaa 1380

ggaggcaaag gatatgctat ataagatgct ctcagaaaat ttcatgtcac tccaggttgg 1440

atgccagtga gatccaggtg gacgaaacca tcttcctgct ggagtcttac attgagtgca 1500

ccatgaagag acagtgatcc agaagaagca tcttcctcag aagatctggg gggatggaaa 1560

gcaaaataaa ggaggtgcct ggatgcatta aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1620

a 1621

<210> 72
<211> 1946
<212> DNA
<213> Homo sapiens

<400> 72
gatggcggcg cccaggccct gccacgcaga cttccgcccg gcgcggagac cgaaggctgg   60

cggctggtcg cgttgcaggc aacatgtcgg aaggaaacgc cgccggcgag cccagcacgc   120

cgggagggcc ccgacctctc ctgactgggg cccgggggct catcgggcgg cggccggcgc   180

ctcccctcac cccggccgc cttccctcca tccgttccag ggacctcacc ctcgggggag   240

tcaagaagaa aaccttcacc ccaaatatca tcagtcggaa gatcaaggaa gagcccaagg   300

aagaagtaac tgtcaagaag gagaagcgtg aaagggacag agaccgacaa cgagaggggc   360

atggacgagg gcgaggccgt ccagaagtga tccagtctca ctccatcttt gagcagggcc   420

cagctgaaat gatgaagaaa aaagggaact gggataagac agtggatgtg tcagacatgg   480

gaccttctca tatcatcaac atcaaaaaag agaagagaga gacagacgaa gaaactaaac   540

agatcttgcg tatgctggag aaggacgatt tcctcgatga ccccggcctg aggaacgaca   600

ctcgaaatat gcctgtgcag ctgccgctgg ctcactcagg atggcttttt aaggaagaaa   660

atgacgaacc agatgttaaa ccttggctgg ctggccccaa ggaagaggac atggaggtgg   720

acatacctgc tgtgaaagtg aaagaggagc cacgagatga ggaggaagag gccaagatga   780

aggctcctcc caaagcagcc aggaagactc caggcctccc gaaggatgta tctgtggcag   840

agctgctgag ggagctgagc ctcaccaagg aagaggaact gctgtttctg cagctgccag   900

acaccctccc tggccagcca cccacccagg acatcaagcc tatcaagaca gaggtgcagg   960

gcgaggacgg acaggtggtg ctcatcaagc aggagaaaga ccgagaagcc aaattggcag   1020

agaatgcttg taccctggct gacctgacag agggtcaggt tggcaagcta ctcatccgca   1080

agtctggaag ggtgcaactc ctcttgggca aggtgactct ggacgtgacc atgggaactg   1140

cctgctcctt cctgcaggag ctggtgtccg tgggccttgg agacagtagg acaggggaga   1200

tgacagtcct gggacacgtg aagcacaaac ttgtatgttc ccctgatttt gaatccctct   1260

tggatcacaa acaccggtaa aatgagcagg tggaggagga cggcgcctgt gcccacggct   1320

gctgcctgct ccagacattt tgttcttgaa tctgtgagac ccagaagggg cccactgagc   1380

ccactcactc cagcctttgg caaccattgt tccaggtccc ccagggcttc ctcccacagc   1440

agctgtgaat ggcacagtga ccttcctgca gcgtggagat ggcacatcct tgctgctggg   1500

gacttggccc tgctatttat ttttgtattt atgtcttaat ctcttccact gatgcatcct   1560

ccaagggtag atggggaggg tctgtgtgaa ggggccggct tctcttggtg cctgctgggt   1620

tgcaggggca ggaagcgtgt ggactgcagc ttctgctggt gctccccccg tcctcctgga   1680

ggcagtatag gagagagagc aaggattgag tctgagactt aagcactcgg tcccagcttg   1740 ccagttcctg gttctgtgtc cttggacaaa ctacctaacc tttctgagcc tcctatacct 1800

catccgacac aaatggggat gatacctacc tccagggttg gcgtgaggat tcatgggcta 1860

ttatagatga aaactgcaca aggccagaac cagcaggcac tcaataaacg ttcatgtcct 1920

ttttctctaa aaaaaaaaa aaaaaa 1946

<210> 73
<211> 2640
<212> DNA
<213> Homo sapiens

<400> 73
ggagtttctc caccagcaac atggccgccg cctgagagga gagccgggcc gccgccgtct 60

ctgcagcccg cgggtaactg ggccgttgcc gccgtccgcg ctcggccccc gcggagagat 120

cgagctgaag gactgcgcgg ctggctctcc tctagtatgg ccaatgaaga ggatgaccca 180

gttgtacagg agatcgatgt gtacttggcc aagagtctgg cggaaaagct gtatctattt 240

cagtaccctg tgcgtccagc ctcgatgacc tacgatgaca ttccgcacct ctcagccaag 300

atcaagccca agcagcagaa ggtagagctt gagatggcca tcgacaccct gaaccccaac 360

tattgccgca gcaaagggga gcagattgcg ctgaacgtgg acggggcctg cgccgacgag 420

accagcacgt attcctcgaa gctgatggac aagcagacct tctgctcttc ccagaccacc 480

agtaacacat cccgttatgc cgctgcactc tacaggcaag gtgagctcca cctgacacct 540

ttacatggca tcctgcagct gcggcccagc ttctcctacc tggataaggc tgacgccaag 600

caccgggaga gggaggcggc caacgaggca ggggactctt cacaggatga ggcggaagac 660

gatgttaagc agatcacggt gcggttctcc cggccggagt cagagcaggc ccgccagcgc 720

cgtgtgcagt cctatgagtt cctgcagaag aagcacgcag aggagccctg ggtccacctg 780

cattactatg gcctgaggga cagtcgctct gagcatgagc gtcagtacct gctgtgcccc 840

ggctcaagcg gggtggagaa cacggagctc gtcaagtcac ccagtgagta cctgatgatg 900

ctgatgccac ccagccagga ggaggagaaa gacaagcctg tggcccccag caacgtcctg 960

tcgatggccc agctgcgcac gctgcccctg gccgatcaga tcaagatcct gatgaagaat 1020

gtgaaggtca tgccttttgc caacttgatg agcctcctgg gcccctccat cgattccgtg 1080

gctgttctgc ggggcatcca gaaggtggcg atgttggtcc aagggaactg ggtggtgaag 1140

agtgacatcc tatacccaa ggactcgtcc agccctcaca gcggcgtgcc tgctgaggtg 1200

ctctgcaggg gccgagactt cgttatgtgg aagttcacgc agagccgctg ggtggttagg 1260

aaagaggtgg caaccgtgac caaactctgc gccgaggatg tgaaggactt cctggagcac 1320

atggccgtgg tgaggatcaa caaaggctgg gagttcattc tgccttatga tggggagttc 1380

atcaagaagc acccggatgt ggtccagcgg cagcacatgc tgtggacggg tatccaggcc 1440 aaactggaaa aagtctataa tcttgtaaag gaaaccatgc caaagaagcc ggatgcacaa 1500

tcagggcctg ccgggctggt ctgtggggac cagcggatcc aagtagccaa aaccaaggcc 1560

cagcagaacc acgcgttgct ggagcgggag ctgcagcggc ggaaggagca gctgcgggtg 1620

cctgcggtcc cgcccggtgt gcggatcaag gaggagcccg tgagcgagga gggcgaggag 1680

gacgaggagc aggaggcgga ggaggagccc atggacactt cccccagcgg cctccacagc 1740

aagctggcca acgggctgcc tctcgggcgg gctgcgggca cagacagctt caacgggcac 1800

ccgccccagg gctgcgccag cacccctgtg gctcgggaac tgaaggcctt cgtggaggcc 1860

acctttcaga gacagtttgt gctcacgctg agcgaactca agcgcctctt caatctgcac 1920

ttggccagcc tgcccccccgg ccacacactc ttcagcggca tctcggaccg catgctacag 1980

gacacggtgc tggccgccgg ttgcaagcag atactggtgc cttttccccc ccagactgct 2040

gcttccccgg atgagcagaa ggtgtttgcc ctctgggagt ctggagacat gagtgatcag 2100

catcgacagg ttttgcttga atttttttcc aaaaattacc gggtacgccg aaacatgatc 2160

cagtctcggt tgactcaaga gtgtggagaa gatctcagta aacaggaggt ggataaagta 2220

ctaaaggact gctgtgtaag ctatggtggc atgtggtacc ttaaagggac agtacagtct 2280

tgacaatagt agcaaactac taacccagca aatctaagcc caaggaagaa gggcggaacc 2340

agaagtaggg cctcgacttg cttcagacga cacagagcaa gaggaactga ccatctcatg 2400

acctgtggca ttgcacggtg cagtggacag aagggattat cctcagccag tcgcagggtc 2460

agcttaagtt agttagatca ctcccagaag agaccagctg ggaccttctt tgcagtacaa 2520

tttgaaattc ctgatgtatt ttgcttatta tttggtttca ttctcataat aaagagagtg 2580

tatactgaca tgggcaggat gataaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 2640

<210> 74
<211> 2177
<212> DNA
<213> Homo sapiens

<400> 74
ggaagaaggc ccctcggcct cagcagagcg ctatcctcca ctggttaccc gggttccccg 60

gcttgctacc gggctgctcc gtgcatcttt ccccccaggc gtcaggaact gcgccctcat 120

ggcggaggtg aaggtgaagg tgcagccgcc tgacgcggat ccggtcgaaa tagaaaacag 180

gattatagaa ttatgtcacc agttccctca tggaatcaca gaccaagtaa ttcagaatga 240

aatgcctcat atagaagccc agcagcgggc agtagccatc aataggttgt tgtctatggg 300

tcagttggat ctcttaagga gcaatacggg ccttttatat agaataaagg actctcagaa 360

tgctggtaaa atgaagggat ccgataacca agaaaaacta gtatatcaaa tcatagagga 420

tgcaggaaat aaaggaatat ggagcagaga tatccgctat aaaagtaatt tgccattaac 480 agaaatcaac aaaattctga agaatctgga aagtaaaaag cttatcaaag ctgttaagtc 540

tgtagcagcc tcaaaaaaga aggtgtatat gctctataac ctgcagccag accggtctgt 600

gactggtgga gcctggtaca gtgaccagga ttttgaatct gaatttgtag aggtgcttaa 660

ccaacagtgt tttaaattcc tacagtccaa ggcagaaaca gcacgagaaa gcaaacagaa 720

cccaatgata caaagaaata gttcatttgc ctcatcacat gaagtgtgga aatatatctg 780

cgaattggga atcagtaagg tagagttatc catggaagac attgaaacca tcctgaatac 840

actcatttat gatggaaaag tggagatgac gattattgct gcaaaagaag gcacagttgg 900

cagtgtagat ggacacatga aactgtacag ggcagtcaat ccaatcatcc ctcccacagg 960

tttggtccgg gcaccctgtg gactctgccc ggtttttgat gactgccacg aaggtggtga 1020

gatttcacca tctaactgta tttacatgac agagtggctc gaattttaat agagagctat 1080

gaactttatt gacattttgc aaatgaagtt acttagggag cagataattt aattcatgat 1140

ggaacacgaa atctccttga aagcaaactt cacaataatg gacgtagact tgctgctatg 1200

aaaacatatt tttttatttt atgaagacta aatttatatt ggtaaaatag ccagtagaat 1260

atgaaagaaa taaggttagt agtgaaattc attcttcaat aaataaaaca ctttgaaact 1320

ccggaggacc acatctttca agacttctga tgggtcaagg aaaaagatgc caacataccc 1380

gtatttacca agtactatga taatggctag agtataaaaa tgttcttttt aaagttattt 1440

attaagttct tcattggaag tttttttta tatctggttc actaccacca ttttctgttt 1500

cctactttct cagtggtttc attgaaaaga aattagaagg ggttaaaggc aggaatagca 1560

aagagtgcaa acttggggta tgactggggg agagtggaac atgccttttc cgcacaatat 1620

taattccttt ttgtatcaga aatgttcttt taggagatta tgctaccata cttacttcaa 1680

acccaatgac tactgtcaag gacatatttt cagtacataa atactatcat tttcattcta 1740

aagaatattt tcactgttcc ttctttctta aagtcttatg tttcactctt taactcaaat 1800

gtattctttg ttagaattta ccctagattc ttatttaatg tctgcagtag actgaatgtt 1860

tgtgtgcccc cagaattcta atgttgaaat ctcatttcca atgtgatggt atttggaggt 1920

ggggcttttg gtaagtgata ggtcaggaga gtaacagcgc tcatgaatgg gattagtgcc 1980

cttatataaa gagacccaga gagctccatc acccсttctg ccatgtgaaa gggagaagac 2040

aaacatccac gaaccaggaa gtgggtcctc accagaaaac aaatctgtaa gcaccttgat 2100

cttggacttc ccagcctcca gaattgtgag aaataaattt ctgttgttga taaaaaaaaa 2160

aaaaaaaaaa aaaaaaa 2177

<210> 75
<211> 414
<212> DNA

<213> Homo sapiens

<220>
<221> misc_feature
<222> (336)..(336)
<223> n is a, c, g, or t

<220>
<221> misc_feature
<222> (406)..(406)
<223> n is a, c, g, or t

<400> 75
gtttgtggtg attgtttatt caagaatatt tggaacaaac atccctgctt aatagttact 60

agttaagatc tatattaaca aattagtttt ctctaacaaa taaacattta aataagttca 120

aaataaaaca tcttactata aaacaaacga acacattttg ttttgtaaga atttaatgtg 180

taaataaata cagaaaaaca agcattacag caaccagttt tgctcatgat agcaatttta 240

aagacacttc aacaaatact tttatgtaag aggctaattc tgcaaaataa tattcacttt 300

cctatggaat aatttcacag tcccagaatt ttaagncccc aaggggaaga gaaaagaaag 360

tgggaaaggg aaaccgtttc ccaatttggg gcccttttttg gggggntttt ttaa 414

<210> 76
<211> 3285
<212> DNA
<213> Homo sapiens

<400> 76
actccgcggt cctggtgccg cgtgcaggtc ggtgcgcgct tctcccgagg tggaacgggc 60

ggcagtcaag cgccggcgtt ctctgccgtc accctttcct tgccggccgg cacttcggct 120

gcagagtttt gcccacgctt cgagacttag ggagcagtgc ctttcagaat ttgcccactc 180

atctggtata actggttctg atggctggga ataaaggaag aggacgtgct gcttatacct 240

ttaatattga ggctgttgga tttagcaaag gtgaaaagtt acctgatgta gtgttgaaac 300

cacccccact atttcctgat acagattata aaccagtgcc actgaaaaca ggagaaggtg 360

aagaatatat gctggctttg aaacaggagt tgagagaaac aatgaaaaga atgccttatt 420

ttattgaaac acctgaagaa agacaagata ttgaaaggta tagtaaaaga tacatgaagg 480

tatacaagga agaatggata ccagattgga gaagacttcc aagagagatg atgccaagaa 540

ataaatgtaa aaaagcaggc ccaaaaccca aaaaggcaaa agacgcaggc aaaggcacac 600

cactcactaa tactgaagat gtgttgaaaa aaatggagga attggaaaaa agaggtgatg 660

gtgaaaaatc agatgaggaa aatgaagaga aagaaggaag caaagagaaa agtaaagaag 720

gtgatgatga cgatgacgat gatgccgcag aacaggagga atatgatgaa gaagagcaag 780

aagaggaaaa tgactacatt aattcatact ttgaagatgg agatgatttt ggcgcagaca 840 gtgatgacaa catggatgag gcaacctatt aggcatgaaa tttttcaaaa aatattttta 900

tgatgcagct tctgaacatt tggacagact tgatttgtat tttatttctg ataaggaata 960

agtacttgtt tctgttgttt tggacaaata gttgttacca aaatattcaa aaccactttg 1020

agtttacata ctagttacct taaaaattat tccctgacac tctgacattc cttctaaaca 1080

cctctgccat ctctcttatg tactctcatg ggtttttttg tatgatttga atatgaatgt 1140

gcctaaagaa tttttgctct cttaatctat gtatacatac ttgaacaaat cattcttgct 1200

taactgctga tcttttgtaa aactattgct agtcataatg gatttctata acctgaaagt 1260

tgatataata cagcacgtgg agcactttaa aaacaaagaa acttgaaaat aattttattt 1320

tttacctttta catggtgtgt tctagactac tgcattatga ctacataaac ctacctgagc 1380

tcttacaagt acagaattgt gagacagagg ttttgtgatg gaaaacatac acatagggaa 1440

aaataccaaa ctgagcaata taaatttcac agctcacatc ttagccagta caaagaaact 1500

ttcacatggg gtgaatatta taatacttga atttgagact taatactgta tatgaattaa 1560

tcatcccact gtaataactg acatcttcaa atagttgggc ttatgaaaaa tttgactaag 1620

acattttatt ttatacttta tgtaaatgtg aaataagttc tggtcatata aatatgtaga 1680

atgcatgtgt gtgaactcta tgaagtgaag tgtggtggcc ttttatgttt cagagtactc 1740

aagtgtacaa ttgatttata taaaaatgca aaatgaactc aggtcatttt gaaattaata 1800

ttaatatttt tgttgtttaa ctttaaagct agctccccta actttatata ttttttggga 1860

aaaaatacct aaaaacctca gcctacaaag actctcagaa atgccaaaga ttttcaagga 1920

aattcatatt atatatttca aaaatgattt atcaatgtta tctaccaaaa gaaataattt 1980

tatttttccc ctttggggag atattatcct ctaatatgag aatcagatcc ctagattcta 2040

tttctaccta tactattaaa ctcaaccttg aacctgagct tggttgtttc tgtgccttag 2100

tttctccact cgtaaacaat ttcctataaa aattgtttaa acaccaacca aaatttacca 2160

tatgtattca ttctcttgcc ttaaagatca cttcttaact tacatttggc tctcaccctc 2220

taaaaatttg tagtttggag agtaactgaa agcacaattt taagaaacat gttttcctca 2280

caaaatatat tgtaatctga ttttcctatc ttgtattcat gcagaaaatt ggagaaaatg 2340

tgtattgtct tgtgccattc tgcaatatga atttctcagg aaaaagatgt ttcaatggca 2400

atactttctt aaataattca agatgggtgt gaggacctat cttattagtt tagtgctgtg 2460

ttaaccattt gtgaattact tcttgtttaa aggaagttgt caaaaaattt taagattcta 2520

aatggaatac acgaataaag taagccttag aatagaaacc tcacaaaaag ttgtcatttg 2580

cggttgataa ttaacatagg tgttttattc tctcaactcc ttagtaccag agccaaatac 2640

aatttagttt tcagaaatct tatcaatatt gttctattta cctacttata aatctgcaga 2700

taaccacttg aaccaggcaa atacactgaa gataaaggtt atttcttttt ttagctttga 2760 atttgtcatg accattttag tcttgcagat agcagggcag ccccttgggg caaggattta 2820

ctctgggagg taccgttaag agccttcttt ccccctttga aagatccttt tacaatgtta 2880

aagtatacta gttgcaagaa caagcaggat ttgcaggttg ctttaccagc atgagtctca 2940

tttttctggc ttaaaatctg ggactgtgaa attattccat aggaaagtga atgttatttt 3000

gcagaattag cctcttacat aaaagtattt gttgaagtgt ctttaaaatt gctatcatga 3060

gcaaaactgg ttgctgtaat gcttgttttt ctgtatttat ttacacatta aattcttaca 3120

aaacaaaatg tgttcgtttg ttttatagta agatgtttta ttttgaactt atttaaatgt 3180

ttatttgtta gagaaaacta atttgttaat atagatctta actagtaact attaagcagg 3240

gatgtttgtt ccaaatattc ttgaataaac aatcaccaca aacaa 3285

<210> 77
<211> 1377
<212> DNA
<213> Homo sapiens

<400> 77
gggagcctgc ggagttcgag accatgctgc tgttctgccc cggctgcggg aacgggctga 60

tcgtggagga gggacaacgc tgccaccgct tcgcctgcaa cacgtgcccc tacgtgcaca 120

acatcacccg caaggtaaca aatcggaagt acccaaaact gaaagaagtg gatgatgtgc 180

ttggtggagc agctgcctgg gagaatgttg actctactgc agagtcgtgt cccaaatgcg 240

aacatcctcg tgcttacttc atgcagcttc agacccgctc tgcagatgag ccgatgacca 300

ccttctacaa gtgctgcaat gctcagtgtg gacaccgctg gagggattag ggccaggatg 360

gcccagctgc cctagtgtgt gcttgccttg tccctcgggg tagatgctta gctggcagta 420

tgagttgtgt gtcctgaggg tctttgctag tgtggtggaa agataaacct tttgaggtga 480

agagccaggg ggtcaggaaa tatggcctat ctgccaggca gggtggatga agtcatgaat 540

gtctgggagt tttctgtgt ggggaggaga cagagaccca taactaaata tgctctgtgt 600

aaagtcctat tctttcatct tccactttat tggcagttga cattccctta ctcccgatca 660

acactcttaa atatttgtac tgtttgtaaa acttagtaca tgtccctaaa atatttaact 720

gttacttgta aacttgtgta atttattatt tattttaatc aaaattctga atatttcatt 780

taaatgaaag ttggaatatt gccttctcag tgttttaaaa tagcttaaga ggtagaaagc 840

agaccccaaa tcacagtgat cccagaatag actacatact tacatttgaa caggctacga 900

aaatgttaat caccgggatt tcaatttctt tctttattcc tattcttact cccttattca 960

tttttttgaa cctcccctga tttgcttcct cgggttgctt taaatgtata agatactatt 1020

tatttattta tttattttat ttatttattt atttatttaa gacagagtct tgttctgtcg 1080

cccaggctgg agtgcagtgg tacgaattca gctcactgca acctctgcct tccgggttca 1140 agggattctc ctgtctcagc ctcccaagca gctgggacta caggcatgca ccaccacacc 1200

tgggtaattt tgcattttta gtagagatgg ggtttctcca tgttggtcgg gctggtctcc 1260

aactcctgac ctcagataat ctgtctgcct cggcctccta aagtgctggg attacaggcg 1320

tgagccactg tgtcccacca atctttacat ttaatatgac tattttctta ttgttcg 1377

<210> 78
<211> 3800
<212> DNA
<213> Homo sapiens

<400> 78
ggcgcgggcg catgcgcagg cgcgggccgg tggggtggcc tggagcggcg tgcgtaatgt 60

cggcactttg ctggggccgc ggagcggcgg ggctcaaacg agccctacgg ccttgcggcc 120

gcccgggact ccccggcaaa gaagggaccg ccggtggcgt ctgcggcccc aggaggagct 180

cgtccgccag cccccaggag caagaccaag accgcaggaa ggactggggc cacgtggagc 240

tgctggaggt gctccaggcg cgggtgcggc agctgcaggc tgagagcgtg tcggaggtgg 300

tggtgaacag ggtggatgtg gcgcggctcc cagaatgtgg cagtggagat ggtagcctcc 360

agccacccag gaaggtccag atgggggcca aggatgccac cccggtgccc tgtggccgct 420

gggcaaagat actggagaag gataagcgga cccagcagat gcgtatgcag cggttgaagg 480

cgaagctgca gatgccattc cagagcgggg agttcaaggc gctgaccagg cgcctgcagg 540

tggagccccg gctcctgagc aagcagatgg ccgggtgcct ggaggactgc acgcgccagg 600

cccccgagag cccctgggag gagcagctgg cccggctgct gcaggaggcc cctgggaagc 660

tgagcctcga tgtggagcag gccccgtcgg ggcagcactc gcaggcccag ctctcaggtc 720

agcagcagag gctcctggcc ttcttcaagt gctgcctgct cactgaccag ctgcccctcg 780

cccaccacct gctggtcgtc caccacggcc agcggcagaa gcggaagctg ctcacgctgg 840

acatgtacaa cgccgtgatg cttggctggg cgcggcaggg tgccttcaag gagctggtat 900

atgtgttatt catggtgaag gatgccggct tgactccgga cctgctgtcc tatgcggctg 960

ccctccagtg catggggagg caggaccagg acgccgggac catcgaaagg tgtctggaac 1020

agatgagcca ggaggggctg aagctgcagg cactcttcac cgccgttctg ctgtctgagg 1080

aggatcgggc cactgttctg aaggccgtgc acaaggtgaa gcccaccttc agcctcccgc 1140

cgcagctgcc gcccccggtc aacacctcca agctgctcag ggacgtgtat gccaaggatg 1200

ggcgtgtgtc ctacccgaag ctgcacctgc ccttgaagac cctgcagtgc ctctttgaga 1260

agcagctcca catggagctg gccagcaggg tgtgcgtggt gtccgtggag aagcccacgt 1320

tgccaagcaa ggaggtcaag cacgcgcgga agaccctgaa gaccctgcgg gaccaatggg 1380

agaaagcact gtgccgggcg ctgcgggaga ccaagaaccg cctagagcgc gaggtgtacg 1440 agggccggtt ctcactttac cccttcctgt gcctgctgga cgagcgcgag gtggtgcgga 1500

tgctcctgca ggtcctgcag gcgctgcccg cccaaggtga gtccttcacc accctggccc 1560

gggagctgag tgcgcgcact ttcagccggc acgtggtgca gaggcagcgg gtcagtggcc 1620

aggtgcaggc gctgcagaac cactacagga agtacctctg cttgctggcc tccgacgccg 1680

aggtgcccga gccctgcctg ccgcggcagt actgggagga gctgggggcg cccgaggccc 1740

tgcgggagca gccctggccc ctgccagtgc agatggagct gggcaagctg ctggcggaga 1800

tgctggtgca ggctacgcag atgccatgca gcctggacaa gccgcatcgt tcctctcggc 1860

ttgtccccgt gctctaccac gtgtattcct tccgcaacgt ccagcagatc ggcatcctga 1920

agccgcaccc ggcctacgtg cagctgctgg agaaggccgc ggagcccacg ctgaccttcg 1980

aggcggtgga tgtacccatg ctttgccccc cgctgccctg gacatcgccg cactctggtg 2040

ctttcctgct cagccccacc aagctgatgc gcacggtgga aggcgccacg cagcaccagg 2100

agctgctgga aacctgcccg cccaccgcgc tgcatggcgc actggacgcc ctcacccaac 2160

tgggcaactg cgcctggcgc gtcaacgggc gcgtgctgga cctggtgctg cagctcttcc 2220

aggccaaggg ctgcccccag ctaggcgtgc cggccccgcc ctccgaggcg ccccagccgc 2280

ccgaggccca cctgccgcac agcgccgcgc ccgcccgcaa ggccgagctg cgccgtgagc 2340

tggcgcactg ccagaaggtg gcccgggaga tgcacagcct gcgggcggag gcgctgtacc 2400

gcctctcgct ggcgcagcac ctgcgggacc gcgtcttctg gctgccgcac aacatggact 2460

tccgcggccg cacctacccc tgcccgccgc acttcaacca cctgggcagc gacgtggcgc 2520

gggccctgct ggagttcgcc cagggccgcc cgctcggccc gcacggcctg gattggctca 2580

agatccacct ggtcaatctc acggggttga agaagcggga gccgctgcgg aagcgcctgg 2640

cctttgcgga ggaggtgatg gatgacatcc tggactccgc ggaccaaccc ttgacgggcc 2700

gaaagtggtg gatgggcgcg gaggaaccct ggcagacgct ggcctgctgt atggaggtgg 2760

cgaacgctgt gcgcgcctcc gaccctgccg cctatgtctc ccacctcccc gtccatcagg 2820

acggctcttg caacggcctg cagcattatg ctgctctggg ccgcgacagc gtgggcgccg 2880

cctccgtcaa cctggagccc tcggatgtgc cgcaggacgt gtacagcggc gtggccgcgc 2940

aggtggaggt gttccgtagg caggacgccc agcggggcat gcgggtggca caggtgctgg 3000

aaggtttcat cacccgcaag gtggtgaagc agacggtgat gacggtggtg tacggggtca 3060

cgcgctatgg cgggcgcctg cagattgaga agcgcctccg ggagctgagc gactttcccc 3120

aggagttcgt gtgggaggcc tctcactatc tcgtacgcca ggtcttcaag agtctacagg 3180

agatgttctc ggggacccgg gccatccagc actggctgac cgagagtgcc cgcctcatct 3240

cccacatggg ctctgtggtg gagtgggtca cacccctggg cgtccccgtc atccagccct 3300 atcgcctgga ctccaaggtc aagcaaatag gaggtggaat tcagagcatc acctacaccc 3360

acaacggaga catcagccga aagcccaaca cacgtaagca gaagaacggc ttcccgccca 3420

acttcatcca ctcgctggac tcctcccaca tgatgctcac cgccctgcac tgctacagga 3480

agggcctgac cttcgtctct gtgcacgact gttactggac tcacgcagct gatgtctccg 3540

tcatgaacca ggtgtgccgg gagcagtttg tccgcttgca cagcgagccc atcctgcagg 3600

acctgtccag attcctggtc aagcggttct gctctgagcc ccagaagatc ttggaggcca 3660

gccagctgaa ggagacactg caggcggtgc ccaagccagg ggccttcgac ctggagcagg 3720

tgaagcgttc cacctacttc ttcagctgac accccgtgag ccttgtcagt gtgtaaataa 3780

agctcttttg ccaccccag 3800

<210> 79
<211> 698
<212> DNA
<213> Homo sapiens

<400> 79
cggcagccag aagacgcggt cccgcaggtg ctgcgccagc gagaggcggt acagcgcctc 60

cgcccgcagg ctgtgcatct cccgggccac cttctggcag tgcgccagct cacggcgcag 120

ctcggccttg cgggcgggcg cggcgctgtg cggcaggtgg gcctcgggcg gctggggcgc 180

ctcggagggc ggggccggca cgcctagctg gggcagccc ttggcctgga agagctgcag 240

caccaggtcc agcacgcgcc cgttgacgcg ccaggcgcag ttgcccagtt gggtgagggc 300

gtccagtgcg ccatgcagcg cggtgggcgg gcaggtttcc agcagctcct ggtgctgcgt 360

ggcgccttcc accgtgcgca tcagcttggt ggggctgagc aggaaggcac cagagtgcgg 420

cgatgtccag ggcagcgggg ggcaaagcat gggtacatcc accgcctcga aggtcagcgt 480

gggctccgcg gccttctcca gcagctgcac gtaagccggg tgcggcttca tgatgccgat 540

ctgctggacg tagcggtacg aatacacgag gtagagcacg gtgacaagcc gagaggaacg 600

atgcggcttg tccacgctgc atggcatctg cgtagcctgc accatcatgt acgccagcat 660

cttggtcaca tcatcagcac tgacagaggc catggctg 698

<210> 80
<211> 3861
<212> DNA
<213> Homo sapiens

<400> 80
gcggccgctc tcagcggagg gggcggcccc ggggcccagg cgccgcggcc cggcaccccg 60

tggaagagcc gcgcgtacag cccgggcatc cagggactac atgaggaaat aattgacttt 120

tataacttca tgtccccttg tcctgaagaa gcagctatga gaagagaggt ggtgaaacgg 180

atcgaaactg tggtgaaaga cctttggccg acggctgatg tacagatatt tggcagcttt 240

agtacaggtc tttatcttcc aactagcgac atagacctgg tggtcttcgg gaaatgggag 300
cgtcctcctt tacagctgct ggagcaagcc ctgcggaagc acaacgtggc tgagccgtgt 360
tccatcaaag tccttgacaa ggctacggta ccaataataa agctcacaga tcaggagact 420
gaagtgaaag ttgacatcag ctttaacatg gagacgggcg tccgggcagc ggagttcatc 480
aagaattaca tgaagaaata ttcattgctg ccttacttga ttttagtatt gaaacagttc 540
cttctgcaga gggacctgaa tgaagttttt acaggtggaa ttagctcata cagcctaatt 600
ttaatggcca ttagctttct acagttgcat ccaagaattg atgcccggag agctgatgaa 660
aaccttggaa tgcttcttgt agaatttttt gaactctatg ggagaaattt taattacttg 720
aaaaccggta ttagaatcaa agaaggaggt gcctatatcg ccaaagagga gatcatgaaa 780
gccatgacca gcgggtacag accgtcgatg ctgtgcattg aggacccccct gctgccaggg 840
aatgacgttg gccggagctc ctatggcgcc atgcaggtga agcaggtctt cgattatgcc 900
tacatagtgc tcagccatgc tgtgtcaccg ctggccaggt cctatccaaa cagagacgcc 960
gaaagtactt taggaagaat catcaaagta actcaggagg tgattgacta ccggaggtgg 1020
atcaaagaga agtggggcag caaagcccac ccgtcgccag gcatggacag caggatcaag 1080
atcaaagagc gaatagccac atgcaatggg gagcagacgc agaaccgaga gcccgagtct 1140
ccctatggcc agcgcttgac tttgtcgctg tccagccccc agctcctgtc ttcaggctcc 1200
tcggcctctt ctgtgtcttc actttctggg agtgacgttg attcagacac accgccctgc 1260
acaacgccca gtgtttacca gttcagtctg caagcgccag ctcctctcat ggccggctta 1320
cccaccgcct tgccaatgcc cagtggcaaa cctcagccca ccacttccag aacactgatc 1380
atgacaacca acaatcagac caggtttact atacctccac cgaccctagg ggttgctcct 1440
gttccttgca gacaagctgg tgtagaagga actgcgtctt tgaaagccgt ccaccacatg 1500
tcttccccgg ccattccctc agcgtccccc aacccgctct cgagccctca tctgtatcat 1560
aagcagcaca acggcatgaa actgtccatg aagggctctc acggccacac ccaaggcggc 1620
ggctacagct ctgtgggtag cggaggtgtg cggccccctg tgggcaacag gggacaccac 1680
cagtataacc gcaccggctg gaggaggaaa aaacacacac acacacggga cagtctgccc 1740
gtgagcctca gcagataatg gctcctggct gcgtcagcct cccccacccc tctgcagact 1800
gccccgcggc ctcggccacc ggcaggggaa ccgagaccag caccccgcac gtcagccggg 1860
ctcgcggcac gcccgccgct gatcactctg catgtttctt cgtgtggtgg tcgcgtccat 1920
cttcaagaac agctcgttgt gctcatctgt gaagccttat taaacgtgga cgttgttttc 1980
tgccttccca ggattcttcc ttcagtgctg aggcaggtcg ggctcaggaa ctgcagggac 2040
gtgaacatgc gcttgcggtt tgaggtagcc gtgtctgttc cttcgcggtt tgctattttc 2100

atttcctgtt cgtcaaagca gcagaggaga tcaaaccccg ttcgtgtgtc tttcctccac 2160

ggataagctt gggaggtcat tgttttactg ccctcacatt ttgtttgaaa tttcagaact 2220

gtttttctat gtaaatattg aaaacttatg atttgtgcaa taactcagat atttttttatt 2280

taatttccta ttttcacata agttatattt aagggaggag ggaatttttt ttaaacaagc 2340

ttaggtcctt tcccgagctg cattttctaa gttgggtcat cgtgtcggct ggttgtctga 2400

cgagcatcgt tacaaacacc atgatgaggg gtttggggtt ttattttgat gtcttttctt 2460

ttggtcggaa gtgagtgaag gagccaggtc gccctgaagg ttttccaaag ggcttggctc 2520

cagagccacc tggcagactg cccgtggccc tgctgtcggg ccccaggccg ttgtcctgct 2580

ctgaccacag agttttaatg ttttggtttt cacttctttt aaactggaca acaaatccag 2640

catttcaagt gccagaagta taactttcta aggagagaag ggttgtcaca ttataaaatc 2700

tttaggaaaa tgtgaactgg aaaacgcttc ggtcagtttt agtgacatag cctgtgatga 2760

tgggtctggt gactattatt gcggaccgtg gtacccagtt ttaggaatgt ggagaaagga 2820

attctgttga ttccgttgag gaatctgtag cgtatgcatt cgttctgtta agagcaaatc 2880

taggagaagt gcttcagctg cccagtgcgc cgtggggagt gttttaacgg atcgtgtcgc 2940

aggagagcac agcccagcgt tggggccggg accgctggcg cccgacgtcg gaagcataca 3000

ggtatactat gcaagtgtat tctgccacaa caaccactgt ctttgttacc tttttttgaa 3060

caagaatata tccatcctgc ctaaccctga gtttttggag caccacagtt gtcctgggag 3120

ttggttgcat cttgtaggcc atctgacttc ctgtttttaa aacgggggtc tggtcttgct 3180

aaacactaca ggtaggttgg tctttgaagt ccactagtgg agaatgtcaa gacaagatac 3240

ttattaccat gacatctgat gcatgtgcag cagtggggag ttctagattg atctctgaat 3300

gtgatcgacg cccagcaagg acaagcttta aaatgtctgc ggtctgccct tttgaagcag 3360

gactggctca ctctgtcatt gggagctgtc agctgcgact gcaggttctc taggaggcat 3420

tccagaatag agtagcacac tgtgtctgca gttctcgatg accgaaagtt atcaaaaata 3480

tttaaaatat ttaaattgtg aacctattga taaagaatat ttataaaaac tgatctgtag 3540

gcctgtacta atctctacgc attagcaata ttgactgtaa acccacatta aggaaaccac 3600

tacgggtctg gcagtgcgtg tcccgtgggg tgtgcatttt aaaactcgat tcatagacac 3660

aggtaccatg ttccatttcc gtcatggtga agcaaatgaa ttggcctggc taccactgtg 3720

gtcgcgtgct acaggtttga caaaaagata tcatgtttcg atttttttgt gtgtggacaa 3780

caatatggaa gctaaaattg acatattttt atgtaaagtt tttctattct ttgattttta 3840

ataaactttg gaaaccagtt t 3861

<210> 81
<211> 651

<212> PRT
<213> Homo sapiens

<400> 81

Met Ala Ala Pro Leu Leu His Thr Arg Leu Pro Gly Asp Ala Ala Ala
1   5   10   15

Ser Ser Ser Ala Val Lys Lys Leu Gly Ala Ser Arg Thr Gly Ile Ser
            20   25   30

Asn Met Arg Ala Leu Glu Asn Asp Phe Phe Asn Ser Pro Pro Arg Lys
        35   40   45

Thr Val Arg Phe Gly Gly Thr Val Thr Glu Val Leu Leu Lys Tyr Lys
    50   55   60

Lys Gly Glu Thr Asn Asp Phe Glu Leu Leu Lys Asn Gln Leu Leu Asp
65   70   75   80

Pro Asp Ile Lys Asp Asp Gln Ile Ile Asn Trp Leu Leu Glu Phe Arg
                85   90   95

Ser Ser Ile Met Tyr Leu Thr Lys Asp Phe Glu Gln Leu Ile Ser Ile
            100   105   110

Ile Leu Arg Leu Pro Trp Leu Asn Arg Ser Gln Thr Val Val Glu Glu
        115   120   125

Tyr Leu Ala Phe Leu Gly Asn Leu Val Ser Ala Gln Thr Val Phe Leu
    130   135   140

Arg Pro Cys Leu Ser Met Ile Ala Ser His Phe Val Pro Pro Arg Val
145   150   155   160

Ile Ile Lys Glu Gly Asp Val Asp Val Ser Asp Ser Asp Asp Glu Asp
                165   170   175

Asp Asn Leu Pro Ala Asn Phe Asp Thr Cys His Arg Ala Leu Gln Ile
            180   185   190

Ile Ala Arg Tyr Val Pro Ser Thr Pro Trp Phe Leu Met Pro Ile Leu
        195   200   205

Val Glu Lys Phe Pro Phe Val Arg Lys Ser Glu Arg Thr Leu Glu Cys
    210   215   220

Tyr Val His Asn Leu Leu Arg Ile Ser Val Tyr Phe Pro Thr Leu Arg
225   230   235   240

His Glu Ile Leu Glu Leu Ile Ile Glu Lys Leu Leu Lys Leu Asp Val
245 250 255

Asn Ala Ser Arg Gln Gly Ile Glu Asp Ala Glu Glu Thr Ala Thr Gln
260 265 270

Thr Cys Gly Gly Thr Asp Ser Thr Glu Gly Leu Phe Asn Met Asp Glu
275 280 285

Asp Glu Glu Thr Glu His Glu Thr Lys Ala Gly Pro Glu Arg Leu Asp
290 295 300

Gln Met Val His Pro Val Ala Glu Arg Leu Asp Ile Leu Met Ser Leu
305 310 315 320

Val Leu Ser Tyr Met Lys Asp Val Cys Tyr Val Asp Gly Lys Val Asp
325 330 335

Asn Gly Lys Thr Lys Asp Leu Tyr Arg Asp Leu Ile Asn Ile Phe Asp
340 345 350

Lys Leu Leu Leu Pro Thr His Ala Ser Cys His Val Gln Phe Phe Met
355 360 365

Phe Tyr Leu Cys Ser Phe Lys Leu Gly Phe Ala Glu Ala Phe Leu Glu
370 375 380

His Leu Trp Lys Lys Leu Gln Asp Pro Ser Asn Pro Ala Ile Ile Arg
385 390 395 400

Gln Ala Ala Gly Asn Tyr Ile Gly Ser Phe Leu Ala Arg Ala Lys Phe
405 410 415

Ile Pro Leu Ile Thr Val Lys Ser Cys Leu Asp Leu Leu Val Asn Trp
420 425 430

Leu His Ile Tyr Leu Asn Asn Gln Asp Ser Gly Thr Lys Ala Phe Cys
435 440 445

Asp Val Ala Leu His Gly Pro Phe Tyr Ser Ala Cys Gln Ala Val Phe
450 455 460

Tyr Thr Phe Val Phe Arg His Lys Gln Leu Leu Ser Gly Asn Leu Lys
465 470 475 480

Glu Gly Leu Gln Tyr Leu Gln Ser Leu Asn Phe Glu Arg Ile Val Met

485 490 495

Ser Gln Leu Asn Pro Leu Lys Ile Cys Leu Pro Ser Val Val Asn Phe
500 505 510

Phe Ala Ala Ile Thr Asn Lys Tyr Gln Leu Val Phe Cys Tyr Thr Ile
515 520 525

Ile Glu Arg Asn Asn Arg Gln Met Leu Pro Val Ile Arg Ser Thr Ala
530 535 540

Gly Gly Asp Ser Val Gln Ile Cys Thr Asn Pro Leu Asp Thr Phe Phe
545 550 555 560

Pro Phe Asp Pro Cys Val Leu Lys Arg Ser Lys Lys Phe Ile Asp Pro
565 570 575

Ile Tyr Gln Val Trp Glu Asp Met Ser Ala Glu Glu Leu Gln Glu Phe
580 585 590

Lys Lys Pro Met Lys Lys Asp Ile Val Glu Asp Glu Asp Asp Asp Phe
595 600 605

Leu Lys Gly Glu Val Pro Gln Asn Asp Thr Val Ile Gly Ile Thr Pro
610 615 620

Ser Ser Phe Asp Thr His Phe Arg Ser Pro Ser Ser Ser Val Gly Ser
625 630 635 640

Pro Pro Val Leu Tyr Met Gln Pro Ser Pro Leu
645 650

<210> 82
<211> 1394
<212> PRT
<213> Homo sapiens

<400> 82

Met Ala Ala Leu Leu Arg Ser Ala Arg Trp Leu Leu Arg Ala Gly Ala
1 5 10 15

Ala Pro Arg Leu Pro Leu Ser Leu Arg Leu Leu Pro Gly Gly Pro Gly
20 25 30

Arg Leu His Ala Ala Ser Tyr Leu Pro Ala Ala Arg Ala Gly Pro Val
35 40 45

Ala Gly Gly Leu Leu Ser Pro Ala Arg Leu Tyr Ala Ile Ala Ala Lys

50 55 60

Glu Lys Asp Ile Gln Glu Glu Ser Thr Phe Ser Ser Arg Lys Ile Ser
65 70 75 80

Asn Gln Phe Asp Trp Ala Leu Met Arg Leu Asp Leu Ser Val Arg Arg
85 90 95

Thr Gly Arg Ile Pro Lys Lys Leu Leu Gln Lys Val Phe Asn Asp Thr
100 105 110

Cys Arg Ser Gly Gly Leu Gly Gly Ser His Ala Leu Leu Leu Leu Arg
115 120 125

Ser Cys Gly Ser Leu Leu Pro Glu Leu Lys Leu Glu Glu Arg Thr Glu
130 135 140

Phe Ala His Arg Ile Trp Asp Thr Leu Gln Lys Leu Gly Ala Val Tyr
145 150 155 160

Asp Val Ser His Tyr Asn Ala Leu Leu Lys Val Tyr Leu Gln Asn Glu
165 170 175

Tyr Lys Phe Ser Pro Thr Asp Phe Leu Ala Lys Met Glu Glu Ala Asn
180 185 190

Ile Gln Pro Asn Arg Val Thr Tyr Gln Arg Leu Ile Ala Ser Tyr Cys
195 200 205

Asn Val Gly Asp Ile Glu Gly Ala Ser Lys Ile Leu Gly Phe Met Lys
210 215 220

Thr Lys Asp Leu Pro Val Thr Glu Ala Val Phe Ser Ala Leu Val Thr
225 230 235 240

Gly His Ala Arg Ala Gly Asp Met Glu Asn Ala Glu Asn Ile Leu Thr
245 250 255

Val Met Arg Asp Ala Gly Ile Glu Pro Gly Pro Asp Thr Tyr Leu Ala
260 265 270

Leu Leu Asn Ala Tyr Ala Glu Lys Gly Asp Ile Asp His Val Lys Gln
275 280 285

Thr Leu Glu Lys Val Glu Lys Ser Glu Leu His Leu Met Asp Arg Asp
290 295 300

Leu Leu Gln Ile Ile Phe Ser Phe Ser Lys Ala Gly Tyr Pro Gln Tyr
305 310 315 320

Val Ser Glu Ile Leu Glu Lys Val Thr Cys Glu Arg Arg Tyr Ile Pro
325 330 335

Asp Ala Met Asn Leu Ile Leu Leu Leu Val Thr Glu Lys Leu Glu Asp
340 345 350

Val Ala Leu Gln Ile Leu Leu Ala Cys Pro Val Ser Lys Glu Asp Gly
355 360 365

Pro Ser Val Phe Gly Ser Phe Phe Leu Gln His Cys Val Thr Met Asn
370 375 380

Thr Pro Val Glu Lys Leu Thr Asp Tyr Cys Lys Lys Leu Lys Glu Val
385 390 395 400

Gln Met His Ser Phe Pro Leu Gln Phe Thr Leu His Cys Ala Leu Leu
405 410 415

Ala Asn Lys Thr Asp Leu Ala Lys Ala Leu Met Lys Ala Val Lys Glu
420 425 430

Glu Gly Phe Pro Ile Arg Pro His Tyr Phe Trp Pro Leu Leu Val Gly
435 440 445

Arg Arg Lys Glu Lys Asn Val Gln Gly Ile Ile Glu Ile Leu Lys Gly
450 455 460

Met Gln Glu Leu Gly Val His Pro Asp Gln Glu Thr Tyr Thr Asp Tyr
465 470 475 480

Val Ile Pro Cys Phe Asp Ser Val Asn Ser Ala Arg Ala Ile Leu Gln
485 490 495

Glu Asn Gly Cys Leu Ser Asp Ser Asp Met Phe Ser Gln Ala Gly Leu
500 505 510

Arg Ser Glu Ala Ala Asn Gly Asn Leu Asp Phe Val Leu Ser Phe Leu
515 520 525

Lys Ser Asn Thr Leu Pro Ile Ser Leu Gln Ser Ile Arg Ser Ser Leu
530 535 540

Leu Leu Gly Phe Arg Arg Ser Met Asn Ile Asn Leu Trp Ser Glu Ile
545 550 555 560

Thr Glu Leu Leu Tyr Lys Asp Gly Arg Tyr Cys Gln Glu Pro Arg Gly
                565   570   575

Pro Thr Glu Ala Val Gly Tyr Phe Leu Tyr Asn Leu Ile Asp Ser Met
            580   585   590

Ser Asp Ser Glu Val Gln Ala Lys Glu Glu His Leu Arg Gln Tyr Phe
        595   600   605

His Gln Leu Glu Lys Met Asn Val Lys Ile Pro Glu Asn Ile Tyr Arg
    610   615   620

Gly Ile Arg Asn Leu Leu Glu Ser Tyr His Val Pro Glu Leu Ile Lys
625   630   635   640

Asp Ala His Leu Leu Val Glu Ser Lys Asn Leu Asp Phe Gln Lys Thr
                645   650   655

Val Gln Leu Thr Ser Ser Glu Leu Glu Ser Thr Leu Glu Thr Leu Lys
            660   665   670

Ala Glu Asn Gln Pro Ile Arg Asp Val Leu Lys Gln Leu Ile Leu Val
        675   680   685

Leu Cys Ser Glu Glu Asn Met Gln Lys Ala Leu Glu Leu Lys Ala Lys
    690   695   700

Tyr Glu Ser Asp Met Val Thr Gly Gly Tyr Ala Ala Leu Ile Asn Leu
705   710   715   720

Cys Cys Arg His Asp Lys Val Glu Asp Ala Leu Asn Leu Lys Glu Glu
                725   730   735

Phe Asp Arg Leu Asp Ser Ser Ala Val Leu Asp Thr Gly Lys Tyr Val
            740   745   750

Gly Leu Val Arg Val Leu Ala Lys His Gly Lys Leu Gln Asp Ala Ile
        755   760   765

Asn Ile Leu Lys Glu Met Lys Glu Lys Asp Val Leu Ile Lys Asp Thr
    770   775   780

Thr Ala Leu Ser Phe Phe His Met Leu Asn Gly Ala Ala Leu Arg Gly
785   790   795   800

Glu Ile Glu Thr Val Lys Gln Leu His Glu Ala Ile Val Thr Leu Gly
                805   810   815

Leu Ala Glu Pro Ser Thr Asn Ile Ser Phe Pro Leu Val Thr Val His
            820   825   830

Leu Glu Lys Gly Asp Leu Ser Thr Ala Leu Glu Val Ala Ile Asp Cys
        835   840   845

Tyr Glu Lys Tyr Lys Val Leu Pro Arg Ile His Asp Val Leu Cys Lys
    850   855   860

Leu Val Glu Lys Gly Glu Thr Asp Leu Ile Gln Lys Ala Met Asp Phe
865   870   875   880

Val Ser Gln Glu Gln Gly Glu Met Val Met Leu Tyr Asp Leu Phe Phe
                885   890   895

Ala Phe Leu Gln Thr Gly Asn Tyr Lys Glu Ala Lys Lys Ile Ile Glu
            900   905   910

Thr Pro Gly Ile Arg Ala Arg Ser Ala Arg Leu Gln Trp Phe Cys Asp
        915   920   925

Arg Cys Val Ala Asn Asn Gln Val Glu Thr Leu Glu Lys Leu Val Glu
    930   935   940

Leu Thr Gln Lys Leu Phe Glu Cys Asp Arg Asp Gln Met Tyr Tyr Asn
945   950   955   960

Leu Leu Lys Leu Tyr Lys Ile Asn Gly Asp Trp Gln Arg Ala Asp Ala
                965   970   975

Val Trp Asn Lys Ile Gln Glu Glu Asn Val Ile Pro Arg Glu Lys Thr
            980   985   990

Leu Arg Leu Leu Ala Glu Ile Leu Arg Glu Gly Asn Gln Glu Val Pro
        995   1000   1005

Phe Asp Val Pro Glu Leu Trp Tyr Glu Asp Glu Lys His Ser Leu
    1010   1015   1020

Asn Ser Ser Ser Ala Ser Thr Thr Glu Pro Asp Phe Gln Lys Asp
    1025   1030   1035

Ile Leu Ile Ala Cys Arg Leu Asn Gln Lys Lys Gly Ala Tyr Asp
    1040   1045   1050

Ile Phe Leu Asn Ala Lys Glu Gln Asn Ile Val Phe Asn Ala Glu

1055 1060 1065

Thr Tyr Ser Asn Leu Ile Lys Leu Leu Met Ser Glu Asp Tyr Phe
1070 1075 1080

Thr Gln Ala Met Glu Val Lys Ala Phe Ala Glu Thr His Ile Lys
1085 1090 1095

Gly Phe Thr Leu Asn Asp Ala Ala Asn Ser Arg Leu Ile Ile Thr
1100 1105 1110

Gln Val Arg Arg Asp Tyr Leu Lys Glu Ala Val Thr Thr Leu Lys
1115 1120 1125

Thr Val Leu Asp Gln Gln Gln Thr Pro Ser Arg Leu Ala Val Thr
1130 1135 1140

Arg Val Ile Gln Ala Leu Ala Met Lys Gly Asp Val Glu Asn Ile
1145 1150 1155

Glu Val Val Gln Lys Met Leu Asn Gly Leu Glu Asp Ser Ile Gly
1160 1165 1170

Leu Ser Lys Met Val Phe Ile Asn Asn Ile Ala Leu Ala Gln Ile
1175 1180 1185

Lys Asn Asn Asn Ile Asp Ala Ala Ile Glu Asn Ile Glu Asn Met
1190 1195 1200

Leu Thr Ser Glu Asn Lys Val Ile Glu Pro Gln Tyr Phe Gly Leu
1205 1210 1215

Ala Tyr Leu Phe Arg Lys Val Ile Glu Glu Gln Leu Glu Pro Ala
1220 1225 1230

Val Glu Lys Ile Ser Ile Met Ala Glu Arg Leu Ala Asn Gln Phe
1235 1240 1245

Ala Ile Tyr Lys Pro Val Thr Asp Phe Phe Leu Gln Leu Val Asp
1250 1255 1260

Ala Gly Lys Val Asp Asp Ala Arg Ala Leu Leu Gln Arg Cys Gly
1265 1270 1275

Ala Ile Ala Glu Gln Thr Pro Ile Leu Leu Leu Phe Leu Leu Arg
1280 1285 1290

Asn Ser Arg Lys Gln Gly Lys Ala Ser Thr Val Lys Ser Val Leu
    1295   1300   1305

Glu Leu Ile Pro Glu Leu Asn Glu Lys Glu Glu Ala Tyr Asn Ser
    1310   1315   1320

Leu Met Lys Ser Tyr Val Ser Glu Lys Asp Val Thr Ser Ala Lys
    1325   1330   1335

Ala Leu Tyr Glu His Leu Thr Ala Lys Asn Thr Lys Leu Asp Asp
    1340   1345   1350

Leu Phe Leu Lys Arg Tyr Ala Ser Leu Leu Lys Tyr Ala Gly Glu
    1355   1360   1365

Pro Val Pro Phe Ile Glu Pro Pro Glu Ser Phe Glu Phe Tyr Ala
    1370   1375   1380

Gln Gln Leu Arg Lys Leu Arg Glu Asn Ser Ser
    1385   1390

<210> 83
<211> 1079
<212> PRT
<213> Homo sapiens

<400> 83

Met Asp Pro Gly Ser Arg Trp Arg Asn Leu Pro Ser Gly Pro Ser Leu
1   5   10   15

Lys His Leu Thr Asp Pro Ser Tyr Gly Ile Pro Arg Glu Gln Gln Lys
            20   25   30

Ala Ala Leu Gln Glu Leu Thr Arg Ala His Val Glu Ser Phe Asn Tyr
        35   40   45

Ala Val His Glu Gly Leu Gly Leu Ala Val Gln Ala Asp Ile Asn Trp
    50   55   60

Ala Val Asn Gly Ile Ser Lys Gly Ile Ile Lys Gln Phe Leu Gly Tyr
65   70   75   80

Val Pro Ile Met Val Lys Ser Lys Leu Cys Asn Leu Arg Asn Leu Pro
                85   90   95

Pro Gln Ala Leu Ile Glu His His Glu Glu Ala Glu Glu Met Gly Gly
            100   105   110

Tyr Phe Ile Ile Asn Gly Ile Glu Lys Val Ile Arg Met Leu Ile Met
        115   120   125

Pro Arg Arg Asn Phe Pro Ile Ala Met Ile Arg Pro Lys Trp Lys Thr
    130   135   140

Arg Gly Pro Gly Tyr Thr Gln Tyr Gly Val Ser Met His Cys Val Arg
145   150   155   160

Glu Glu His Ser Ala Val Asn Met Asn Leu His Tyr Leu Glu Asn Gly
                165   170   175

Thr Val Met Leu Asn Phe Ile Tyr Arg Lys Glu Leu Phe Phe Leu Pro
            180   185   190

Leu Gly Phe Ala Leu Lys Ala Leu Val Ser Phe Ser Asp Tyr Gln Ile
        195   200   205

Phe Gln Glu Leu Ile Lys Gly Lys Glu Asp Asp Ser Phe Leu Arg Asn
    210   215   220

Ser Val Ser Gln Met Leu Arg Ile Val Met Glu Glu Gly Cys Ser Thr
225   230   235   240

Gln Lys Gln Val Leu Asn Tyr Leu Gly Glu Cys Phe Arg Val Lys Leu
                245   250   255

Asn Val Pro Asp Trp Tyr Pro Asn Glu Gln Ala Ala Glu Phe Leu Phe
            260   265   270

Asn Gln Cys Ile Cys Ile His Leu Lys Ser Asn Thr Glu Lys Phe Tyr
        275   280   285

Met Leu Cys Leu Met Thr Arg Lys Leu Phe Ala Leu Ala Lys Gly Glu
    290   295   300

Cys Met Glu Asp Asn Pro Asp Ser Leu Val Asn Gln Glu Val Leu Thr
305   310   315   320

Pro Gly Gln Leu Phe Leu Met Phe Leu Lys Glu Lys Leu Glu Gly Trp
                325   330   335

Leu Val Ser Ile Lys Ile Ala Phe Asp Lys Lys Ala Gln Lys Thr Ser
            340   345   350

Val Ser Met Asn Thr Asp Asn Leu Met Arg Ile Phe Thr Met Gly Ile
        355   360   365

Asp Leu Thr Lys Pro Phe Glu Tyr Leu Phe Ala Thr Gly Asn Leu Arg
    370   375   380

Ser Lys Thr Gly Leu Gly Leu Leu Gln Asp Ser Gly Leu Cys Val Val
385   390   395   400

Ala Asp Lys Leu Asn Phe Ile Arg Tyr Leu Ser His Phe Arg Cys Val
                405   410   415

His Arg Gly Ala Asp Phe Ala Lys Met Arg Thr Thr Thr Val Arg Arg
            420   425   430

Leu Leu Pro Glu Ser Trp Gly Phe Leu Cys Pro Val His Thr Pro Asp
        435   440   445

Gly Glu Pro Cys Gly Leu Met Asn His Leu Thr Ala Val Cys Glu Val
    450   455   460

Val Thr Gln Phe Val Tyr Thr Ala Ser Ile Pro Ala Leu Leu Cys Asn
465   470   475   480

Leu Gly Val Thr Pro Ile Asp Gly Ala Pro His Arg Ser Tyr Ser Glu
                485   490   495

Cys Tyr Pro Val Leu Leu Asp Gly Val Met Val Gly Trp Val Asp Lys
            500   505   510

Asp Leu Ala Pro Gly Ile Ala Asp Ser Leu Arg His Phe Lys Val Leu
        515   520   525

Arg Glu Lys Arg Ile Pro Pro Trp Met Glu Val Val Leu Ile Pro Met
    530   535   540

Thr Gly Lys Pro Ser Leu Tyr Pro Gly Leu Phe Leu Phe Thr Thr Pro
545   550   555   560

Cys Arg Leu Val Arg Pro Val Gln Asn Leu Ala Leu Gly Lys Glu Glu
                565   570   575

Leu Ile Gly Thr Met Glu Gln Ile Phe Met Asn Val Ala Ile Phe Glu
            580   585   590

Asp Glu Val Phe Ala Gly Val Thr Thr His Gln Glu Leu Phe Pro His
        595   600   605

Ser Leu Leu Ser Val Ile Ala Asn Phe Ile Pro Phe Ser Asp His Asn
    610   615   620

Gln Ser Pro Arg Asn Met Tyr Gln Cys Gln Met Gly Lys Gln Thr Met
625 630 635 640

Gly Phe Pro Leu Leu Thr Tyr Gln Asp Arg Ser Asp Asn Lys Leu Tyr
645 650 655

Arg Leu Gln Thr Pro Gln Ser Pro Leu Val Arg Pro Ser Met Tyr Asp
660 665 670

Tyr Tyr Asp Met Asp Asn Tyr Pro Ile Gly Thr Asn Ala Ile Val Ala
675 680 685

Val Ile Ser Tyr Thr Gly Tyr Asp Met Glu Asp Ala Met Ile Val Asn
690 695 700

Lys Ala Ser Trp Glu Arg Gly Phe Ala His Gly Ser Val Tyr Lys Ser
705 710 715 720

Glu Phe Ile Asp Leu Ser Glu Lys Ile Lys Gln Gly Asp Ser Ser Leu
725 730 735

Val Phe Gly Ile Lys Pro Gly Asp Pro Arg Val Leu Gln Lys Leu Asp
740 745 750

Asp Asp Gly Leu Pro Phe Ile Gly Ala Lys Leu Gln Tyr Gly Asp Pro
755 760 765

Tyr Tyr Ser Tyr Leu Asn Leu Asn Thr Gly Glu Ser Phe Val Met Tyr
770 775 780

Tyr Lys Ser Lys Glu Asn Cys Val Val Asp Asn Ile Lys Val Cys Ser
785 790 795 800

Asn Asp Thr Gly Ser Gly Lys Phe Lys Cys Val Cys Ile Thr Met Arg
805 810 815

Val Pro Arg Asn Pro Thr Ile Gly Asp Lys Phe Ala Ser Arg His Gly
820 825 830

Gln Lys Gly Ile Leu Ser Arg Leu Trp Pro Ala Glu Asp Met Pro Phe
835 840 845

Thr Glu Ser Gly Met Val Pro Asp Ile Leu Phe Asn Pro His Gly Phe
850 855 860

Pro Ser Arg Met Thr Ile Gly Met Leu Ile Glu Ser Met Ala Gly Lys

865 870 875 880

Ser Ala Ala Leu His Gly Leu Cys His Asp Ala Thr Pro Phe Ile Phe
885 890 895

Ser Glu Glu Asn Ser Ala Leu Glu Tyr Phe Gly Glu Met Leu Lys Ala
900 905 910

Ala Gly Tyr Asn Phe Tyr Gly Thr Glu Arg Leu Tyr Ser Gly Ile Ser
915 920 925

Gly Leu Glu Leu Glu Ala Asp Ile Phe Ile Gly Val Val Tyr Tyr Gln
930 935 940

Arg Leu Arg His Met Val Ser Asp Lys Phe Gln Val Arg Thr Thr Gly
945 950 955 960

Ala Arg Asp Arg Val Thr Asn Gln Pro Ile Gly Gly Arg Asn Val Gln
965 970 975

Gly Gly Ile Arg Phe Gly Glu Met Glu Arg Asp Ala Leu Leu Ala His
980 985 990

Gly Thr Ser Phe Leu Leu His Asp Arg Leu Phe Asn Cys Ser Asp Arg
995 1000 1005

Ser Val Ala His Val Cys Val Lys Cys Gly Ser Leu Leu Ser Pro
1010 1015 1020

Leu Leu Glu Lys Pro Pro Pro Ser Trp Ser Ala Met Arg Asn Arg
1025 1030 1035

Lys Tyr Asn Cys Thr Leu Cys Ser Arg Ser Asp Thr Ile Asp Thr
1040 1045 1050

Val Ser Val Pro Tyr Val Phe Arg Tyr Phe Val Ala Glu Leu Ala
1055 1060 1065

Ala Met Asn Ile Lys Val Lys Leu Asp Val Val
1070 1075

<210> 84
<211> 346
<212> PRT
<213> Homo sapiens

<400> 84

Met Ala Ala Ser Gln Ala Val Glu Glu Met Arg Ser Arg Val Val Leu

1 5 10 15

Gly Glu Phe Gly Val Arg Asn Val His Thr Thr Asp Phe Pro Gly Asn
20 25 30

Tyr Ser Gly Tyr Asp Asp Ala Trp Asp Gln Asp Arg Phe Glu Lys Asn
35 40 45

Phe Arg Val Asp Val Val His Met Asp Glu Asn Ser Leu Glu Phe Asp
50 55 60

Met Val Gly Ile Asp Ala Ala Ile Ala Asn Ala Phe Arg Arg Ile Leu
65 70 75 80

Leu Ala Glu Val Pro Thr Met Ala Val Glu Lys Val Leu Val Tyr Asn
85 90 95

Asn Thr Ser Ile Val Gln Asp Glu Ile Leu Ala His Arg Leu Gly Leu
100 105 110

Ile Pro Ile His Ala Asp Pro Arg Leu Phe Glu Tyr Arg Asn Gln Gly
115 120 125

Asp Glu Glu Gly Thr Glu Ile Asp Thr Leu Gln Phe Arg Leu Gln Val
130 135 140

Arg Cys Thr Arg Asn Pro His Ala Ala Lys Asp Ser Ser Asp Pro Asn
145 150 155 160

Glu Leu Tyr Val Asn His Lys Val Tyr Thr Arg His Met Thr Trp Ile
165 170 175

Pro Leu Gly Asn Gln Ala Asp Leu Phe Pro Glu Gly Thr Ile Arg Pro
180 185 190

Val His Asp Asp Ile Leu Ile Ala Gln Leu Arg Pro Gly Gln Glu Ile
195 200 205

Asp Leu Leu Met His Cys Val Lys Gly Ile Gly Lys Asp His Ala Lys
210 215 220

Phe Ser Pro Val Ala Thr Ala Ser Tyr Arg Leu Leu Pro Asp Ile Thr
225 230 235 240

Leu Leu Glu Pro Val Glu Gly Glu Ala Ala Glu Glu Leu Ser Arg Cys
245 250 255

Phe Ser Pro Gly Val Ile Glu Val Gln Glu Val Gln Gly Lys Lys Val
260 265 270

Ala Arg Val Ala Asn Pro Arg Leu Asp Thr Phe Ser Arg Glu Ile Phe
275 280 285

Arg Asn Glu Lys Leu Lys Lys Val Val Arg Leu Ala Arg Val Arg Asp
290 295 300

His Tyr Ile Phe Ser Val Glu Ser Thr Gly Val Leu Pro Pro Asp Val
305 310 315 320

Leu Val Ser Glu Ala Ile Lys Val Leu Met Gly Lys Cys Arg Arg Phe
325 330 335

Leu Asp Glu Leu Asp Ala Val Gln Met Asp
340 345

<210> 85
<211> 346
<212> PRT
<213> Homo sapiens

<400> 85

Met Ala Ala Ser Gln Ala Val Glu Glu Met Arg Ser Arg Val Val Leu
1 5 10 15

Gly Glu Phe Gly Val Arg Asn Val His Thr Thr Asp Phe Pro Gly Asn
20 25 30

Tyr Ser Gly Tyr Asp Asp Ala Trp Asp Gln Asp Arg Phe Glu Lys Asn
35 40 45

Phe Arg Val Asp Val Val His Met Asp Glu Asn Ser Leu Glu Phe Asp
50 55 60

Met Val Gly Ile Asp Ala Ala Ile Ala Asn Ala Phe Arg Arg Ile Leu
65 70 75 80

Leu Ala Glu Val Pro Thr Met Ala Val Glu Lys Val Leu Val Tyr Asn
85 90 95

Asn Thr Ser Ile Val Gln Asp Glu Ile Leu Ala His Arg Leu Gly Leu
100 105 110

Ile Pro Ile His Ala Asp Pro Arg Leu Phe Glu Tyr Arg Asn Gln Gly
115 120 125

Asp Glu Glu Gly Thr Glu Ile Asp Thr Leu Gln Phe Arg Leu Gln Val
    130   135   140

Arg Cys Thr Arg Asn Pro His Ala Ala Lys Asp Ser Ser Asp Pro Asn
145   150   155   160

Glu Leu Tyr Val Asn His Lys Val Tyr Thr Arg His Met Thr Trp Ile
                165   170   175

Pro Leu Gly Asn Gln Ala Asp Leu Phe Pro Glu Gly Thr Ile Arg Pro
            180   185   190

Val His Asp Asp Ile Leu Ile Ala Gln Leu Arg Pro Gly Gln Glu Ile
        195   200   205

Asp Leu Leu Met His Cys Val Lys Gly Ile Gly Lys Asp His Ala Lys
    210   215   220

Phe Ser Pro Val Ala Thr Ala Ser Tyr Arg Leu Leu Pro Asp Ile Thr
225   230   235   240

Leu Leu Glu Pro Val Glu Gly Glu Ala Ala Glu Glu Leu Ser Arg Cys
                245   250   255

Phe Ser Pro Gly Val Ile Glu Val Gln Glu Val Gln Gly Lys Lys Val
            260   265   270

Ala Arg Val Ala Asn Pro Arg Leu Asp Thr Phe Ser Arg Glu Ile Phe
        275   280   285

Arg Asn Glu Lys Leu Lys Lys Val Val Arg Leu Ala Arg Val Arg Asp
    290   295   300

His Tyr Ile Phe Ser Val Glu Ser Thr Gly Val Leu Pro Pro Asp Val
305   310   315   320

Leu Val Ser Glu Ala Ile Lys Val Leu Met Gly Lys Cys Arg Arg Phe
                325   330   335

Leu Asp Glu Leu Asp Ala Val Gln Met Asp
            340   345

<210> 86
<211> 133
<212> PRT
<213> Homo sapiens

<400> 86

Met Glu Glu Asp Gln Glu Leu Glu Arg Lys Ile Ser Gly Leu Lys Thr
1   5   10   15

Ser Met Ala Glu Gly Glu Arg Lys Thr Ala Leu Glu Met Val Gln Ala
            20   25   30

Ala Gly Thr Asp Arg His Cys Val Thr Phe Val Leu His Glu Glu Asp
        35   40   45

His Thr Leu Gly Asn Ser Leu Arg Tyr Met Ile Met Lys Asn Pro Glu
    50   55   60

Val Glu Phe Cys Gly Tyr Thr Thr Thr His Pro Ser Glu Ser Lys Ile
65   70   75   80

Asn Leu Arg Ile Gln Thr Arg Gly Thr Leu Pro Ala Val Glu Pro Phe
                85   90   95

Gln Arg Gly Leu Asn Glu Leu Met Asn Val Cys Gln His Val Leu Asp
            100   105   110

Lys Phe Glu Ala Ser Ile Lys Asp Tyr Lys Asp Gln Lys Ala Ser Arg
        115   120   125

Asn Glu Ser Thr Phe
    130

<210> 87
<211> 1970
<212> PRT
<213> Homo sapiens

<400> 87

Met His Gly Gly Gly Pro Pro Ser Gly Asp Ser Ala Cys Pro Leu Arg
1   5   10   15

Thr Ile Lys Arg Val Gln Phe Gly Val Leu Ser Pro Asp Glu Leu Lys
            20   25   30

Arg Met Ser Val Thr Glu Gly Gly Ile Lys Tyr Pro Glu Thr Thr Glu
        35   40   45

Gly Gly Arg Pro Lys Leu Gly Gly Leu Met Asp Pro Arg Gln Gly Val
    50   55   60

Ile Glu Arg Thr Gly Arg Cys Gln Thr Cys Ala Gly Asn Met Thr Glu
65   70   75   80

Cys Pro Gly His Phe Gly His Ile Glu Leu Ala Lys Pro Val Phe His
                85   90   95

Val Gly Phe Leu Val Lys Thr Met Lys Val Leu Arg Cys Val Cys Phe
            100   105   110

Phe Cys Ser Lys Leu Leu Val Asp Ser Asn Asn Pro Lys Ile Lys Asp
        115   120   125

Ile Leu Ala Lys Ser Lys Gly Gln Pro Lys Lys Arg Leu Thr His Val
    130   135   140

Tyr Asp Leu Cys Lys Gly Lys Asn Ile Cys Glu Gly Gly Glu Glu Met
145   150   155   160

Asp Asn Lys Phe Gly Val Glu Gln Pro Glu Gly Asp Glu Asp Leu Thr
                165   170   175

Lys Glu Lys Gly His Gly Gly Cys Gly Arg Tyr Gln Pro Arg Ile Arg
            180   185   190

Arg Ser Gly Leu Glu Leu Tyr Ala Glu Trp Lys His Val Asn Glu Asp
        195   200   205

Ser Gln Glu Lys Lys Ile Leu Leu Ser Pro Glu Arg Val His Glu Ile
    210   215   220

Phe Lys Arg Ile Ser Asp Glu Glu Cys Phe Val Leu Gly Met Glu Pro
225   230   235   240

Arg Tyr Ala Arg Pro Glu Trp Met Ile Val Thr Val Leu Pro Val Pro
                245   250   255

Pro Leu Ser Val Arg Pro Ala Val Val Met Gln Gly Ser Ala Arg Asn
            260   265   270

Gln Asp Asp Leu Thr His Lys Leu Ala Asp Ile Val Lys Ile Asn Asn
        275   280   285

Gln Leu Arg Arg Asn Glu Gln Asn Gly Ala Ala Ala His Val Ile Ala
    290   295   300

Glu Asp Val Lys Leu Leu Gln Phe His Val Ala Thr Met Val Asp Asn
305   310   315   320

Glu Leu Pro Gly Leu Pro Arg Ala Met Gln Lys Ser Gly Arg Pro Leu
                325   330   335

Lys Ser Leu Lys Gln Arg Leu Lys Gly Lys Glu Gly Arg Val Arg Gly
340 345 350

Asn Leu Met Gly Lys Arg Val Asp Phe Ser Ala Arg Thr Val Ile Thr
355 360 365

Pro Asp Pro Asn Leu Ser Ile Asp Gln Val Gly Val Pro Arg Ser Ile
370 375 380

Ala Ala Asn Met Thr Phe Ala Glu Ile Val Thr Pro Phe Asn Ile Asp
385 390 395 400

Arg Leu Gln Glu Leu Val Arg Arg Gly Asn Ser Gln Tyr Pro Gly Ala
405 410 415

Lys Tyr Ile Ile Arg Asp Asn Gly Asp Arg Ile Asp Leu Arg Phe His
420 425 430

Pro Lys Pro Ser Asp Leu His Leu Gln Thr Gly Tyr Lys Val Glu Arg
435 440 445

His Met Cys Asp Gly Asp Ile Val Ile Phe Asn Arg Gln Pro Thr Leu
450 455 460

His Lys Met Ser Met Met Gly His Arg Val Arg Ile Leu Pro Trp Ser
465 470 475 480

Thr Phe Arg Leu Asn Leu Ser Val Thr Thr Pro Tyr Asn Ala Asp Phe
485 490 495

Asp Gly Asp Glu Met Asn Leu His Leu Pro Gln Ser Leu Glu Thr Arg
500 505 510

Ala Glu Ile Gln Glu Leu Ala Met Val Pro Arg Met Ile Val Thr Pro
515 520 525

Gln Ser Asn Arg Pro Val Met Gly Ile Val Gln Asp Thr Leu Thr Ala
530 535 540

Val Arg Lys Phe Thr Lys Arg Asp Val Phe Leu Glu Arg Gly Glu Val
545 550 555 560

Met Asn Leu Leu Met Phe Leu Ser Thr Trp Asp Gly Lys Val Pro Gln
565 570 575

Pro Ala Ile Leu Lys Pro Arg Pro Leu Trp Thr Gly Lys Gln Ile Phe
580 585 590

Ser Leu Ile Ile Pro Gly His Ile Asn Cys Ile Arg Thr His Ser Thr
595 600 605

His Pro Asp Asp Glu Asp Ser Gly Pro Tyr Lys His Ile Ser Pro Gly
610 615 620

Asp Thr Lys Val Val Val Glu Asn Gly Glu Leu Ile Met Gly Ile Leu
625 630 635 640

Cys Lys Lys Ser Leu Gly Thr Ser Ala Gly Ser Leu Val His Ile Ser
645 650 655

Tyr Leu Glu Met Gly His Asp Ile Thr Arg Leu Phe Tyr Ser Asn Ile
660 665 670

Gln Thr Val Ile Asn Asn Trp Leu Leu Ile Glu Gly His Thr Ile Gly
675 680 685

Ile Gly Asp Ser Ile Ala Asp Ser Lys Thr Tyr Gln Asp Ile Gln Asn
690 695 700

Thr Ile Lys Lys Ala Lys Gln Asp Val Ile Glu Val Ile Glu Lys Ala
705 710 715 720

His Asn Asn Glu Leu Glu Pro Thr Pro Gly Asn Thr Leu Arg Gln Thr
725 730 735

Phe Glu Asn Gln Val Asn Arg Ile Leu Asn Asp Ala Arg Asp Lys Thr
740 745 750

Gly Ser Ser Ala Gln Lys Ser Leu Ser Glu Tyr Asn Asn Phe Lys Ser
755 760 765

Met Val Val Ser Gly Ala Lys Gly Ser Lys Ile Asn Ile Ser Gln Val
770 775 780

Ile Ala Val Val Gly Gln Gln Asn Val Glu Gly Lys Arg Ile Pro Phe
785 790 795 800

Gly Phe Lys His Arg Thr Leu Pro His Phe Ile Lys Asp Asp Tyr Gly
805 810 815

Pro Glu Ser Arg Gly Phe Val Glu Asn Ser Tyr Leu Ala Gly Leu Thr
820 825 830

Pro Thr Glu Phe Phe Phe His Ala Met Gly Gly Arg Glu Gly Leu Ile

835 840 845

Asp Thr Ala Val Lys Thr Ala Glu Thr Gly Tyr Ile Gln Arg Arg Leu
850 855 860

Ile Lys Ser Met Glu Ser Val Met Val Lys Tyr Asp Ala Thr Val Arg
865 870 875 880

Asn Ser Ile Asn Gln Val Val Gln Leu Arg Tyr Gly Glu Asp Gly Leu
885 890 895

Ala Gly Glu Ser Val Glu Phe Gln Asn Leu Ala Thr Leu Lys Pro Ser
900 905 910

Asn Lys Ala Phe Glu Lys Lys Phe Arg Phe Asp Tyr Thr Asn Glu Arg
915 920 925

Ala Leu Arg Arg Thr Leu Gln Glu Asp Leu Val Lys Asp Val Leu Ser
930 935 940

Asn Ala His Ile Gln Asn Glu Leu Glu Arg Glu Phe Glu Arg Met Arg
945 950 955 960

Glu Asp Arg Glu Val Leu Arg Val Ile Phe Pro Thr Gly Asp Ser Lys
965 970 975

Val Val Leu Pro Cys Asn Leu Leu Arg Met Ile Trp Asn Ala Gln Lys
980 985 990

Ile Phe His Ile Asn Pro Arg Leu Pro Ser Asp Leu His Pro Ile Lys
995 1000 1005

Val Val Glu Gly Val Lys Glu Leu Ser Lys Lys Leu Val Ile Val
1010 1015 1020

Asn Gly Asp Asp Pro Leu Ser Arg Gln Ala Gln Glu Asn Ala Thr
1025 1030 1035

Leu Leu Phe Asn Ile His Leu Arg Ser Thr Leu Cys Ser Arg Arg
1040 1045 1050

Met Ala Glu Glu Phe Arg Leu Ser Gly Glu Ala Phe Asp Trp Leu
1055 1060 1065

Leu Gly Glu Ile Glu Ser Lys Phe Asn Gln Ala Ile Ala His Pro
1070 1075 1080

Gly Glu Met Val Gly Ala Leu Ala Ala Gln Ser Leu Gly Glu Pro
1085 1090 1095

Ala Thr Gln Met Thr Leu Asn Thr Phe His Tyr Ala Gly Val Ser
1100 1105 1110

Ala Lys Asn Val Thr Leu Gly Val Pro Arg Leu Lys Glu Leu Ile
1115 1120 1125

Asn Ile Ser Lys Lys Pro Lys Thr Pro Ser Leu Thr Val Phe Leu
1130 1135 1140

Leu Gly Gln Ser Ala Arg Asp Ala Glu Arg Ala Lys Asp Ile Leu
1145 1150 1155

Cys Arg Leu Glu His Thr Thr Leu Arg Lys Val Thr Ala Asn Thr
1160 1165 1170

Ala Ile Tyr Tyr Asp Pro Asn Pro Gln Ser Thr Val Val Ala Glu
1175 1180 1185

Asp Gln Glu Trp Val Asn Val Tyr Tyr Glu Met Pro Asp Phe Asp
1190 1195 1200

Val Ala Arg Ile Ser Pro Trp Leu Leu Arg Val Glu Leu Asp Arg
1205 1210 1215

Lys His Met Thr Asp Arg Lys Leu Thr Met Glu Gln Ile Ala Glu
1220 1225 1230

Lys Ile Asn Ala Gly Phe Gly Asp Asp Leu Asn Cys Ile Phe Asn
1235 1240 1245

Asp Asp Asn Ala Glu Lys Leu Val Leu Arg Ile Arg Ile Met Asn
1250 1255 1260

Ser Asp Glu Asn Lys Met Gln Glu Glu Glu Glu Val Val Asp Lys
1265 1270 1275

Met Asp Asp Asp Val Phe Leu Arg Cys Ile Glu Ser Asn Met Leu
1280 1285 1290

Thr Asp Met Thr Leu Gln Gly Ile Glu Gln Ile Ser Lys Val Tyr
1295 1300 1305

Met His Leu Pro Gln Thr Asp Asn Lys Lys Lys Ile Ile Ile Thr
1310 1315 1320

Glu Asp Gly Glu Phe Lys Ala Leu Gln Glu Trp Ile Leu Glu Thr
1325 1330 1335

Asp Gly Val Ser Leu Met Arg Val Leu Ser Glu Lys Asp Val Asp
1340 1345 1350

Pro Val Arg Thr Thr Ser Asn Asp Ile Val Glu Ile Phe Thr Val
1355 1360 1365

Leu Gly Ile Glu Ala Val Arg Lys Ala Leu Glu Arg Glu Leu Tyr
1370 1375 1380

His Val Ile Ser Phe Asp Gly Ser Tyr Val Asn Tyr Arg His Leu
1385 1390 1395

Ala Leu Leu Cys Asp Thr Met Thr Cys Arg Gly His Leu Met Ala
1400 1405 1410

Ile Thr Arg His Gly Val Asn Arg Gln Asp Thr Gly Pro Leu Met
1415 1420 1425

Lys Cys Ser Phe Glu Glu Thr Val Asp Val Leu Met Glu Ala Ala
1430 1435 1440

Ala His Gly Glu Ser Asp Pro Met Lys Gly Val Ser Glu Asn Ile
1445 1450 1455

Met Leu Gly Gln Leu Ala Pro Ala Gly Thr Gly Cys Phe Asp Leu
1460 1465 1470

Leu Leu Asp Ala Glu Lys Cys Lys Tyr Gly Met Glu Ile Pro Thr
1475 1480 1485

Asn Ile Pro Gly Leu Gly Ala Ala Gly Pro Thr Gly Met Phe Phe
1490 1495 1500

Gly Ser Ala Pro Ser Pro Met Gly Gly Ile Ser Pro Ala Met Thr
1505 1510 1515

Pro Trp Asn Gln Gly Ala Thr Pro Ala Tyr Gly Ala Trp Ser Pro
1520 1525 1530

Ser Val Gly Ser Gly Met Thr Pro Gly Ala Ala Gly Phe Ser Pro
1535 1540 1545

Ser Ala Ala Ser Asp Ala Ser Gly Phe Ser Pro Gly Tyr Ser Pro
1550 1555 1560

Ala Trp Ser Pro Thr Pro Gly Ser Pro Gly Ser Pro Gly Pro Ser
1565 1570 1575

Ser Pro Tyr Ile Pro Ser Pro Gly Gly Ala Met Ser Pro Ser Tyr
1580 1585 1590

Ser Pro Thr Ser Pro Ala Tyr Glu Pro Arg Ser Pro Gly Gly Tyr
1595 1600 1605

Thr Pro Gln Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser
1610 1615 1620

Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Asn Tyr Ser Pro
1625 1630 1635

Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr
1640 1645 1650

Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser
1655 1660 1665

Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro
1670 1675 1680

Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser
1685 1690 1695

Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr
1700 1705 1710

Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser
1715 1720 1725

Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro
1730 1735 1740

Thr Ser Pro Asn Tyr Ser Pro Thr Ser Pro Asn Tyr Thr Pro Thr
1745 1750 1755

Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser
1760 1765 1770

Pro Asn Tyr Thr Pro Thr Ser Pro Asn Tyr Ser Pro Thr Ser Pro
1775 1780 1785

Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser

1790 1795 1800

Tyr Ser Pro Ser Ser Pro Arg Tyr Thr Pro Gln Ser Pro Thr Tyr
1805 1810 1815

Thr Pro Ser Ser Pro Ser Tyr Ser Pro Ser Ser Pro Ser Tyr Ser
1820 1825 1830

Pro Thr Ser Pro Lys Tyr Thr Pro Thr Ser Pro Ser Tyr Ser Pro
1835 1840 1845

Ser Ser Pro Glu Tyr Thr Pro Thr Ser Pro Lys Tyr Ser Pro Thr
1850 1855 1860

Ser Pro Lys Tyr Ser Pro Thr Ser Pro Lys Tyr Ser Pro Thr Ser
1865 1870 1875

Pro Thr Tyr Ser Pro Thr Thr Pro Lys Tyr Ser Pro Thr Ser Pro
1880 1885 1890

Thr Tyr Ser Pro Thr Ser Pro Val Tyr Thr Pro Thr Ser Pro Lys
1895 1900 1905

Tyr Ser Pro Thr Ser Pro Thr Tyr Ser Pro Thr Ser Pro Lys Tyr
1910 1915 1920

Ser Pro Thr Ser Pro Thr Tyr Ser Pro Thr Ser Pro Lys Gly Ser
1925 1930 1935

Thr Tyr Ser Pro Thr Ser Pro Gly Tyr Ser Pro Thr Ser Pro Thr
1940 1945 1950

Tyr Ser Leu Thr Ser Pro Ala Ile Ser Pro Asp Asp Ser Asp Glu
1955 1960 1965

Glu Asn
1970

<210> 88
<211> 1970
<212> PRT
<213> Homo sapiens

<400> 88

Met His Gly Gly Gly Pro Pro Ser Gly Asp Ser Ala Cys Pro Leu Arg
1 5 10 15

Thr Ile Lys Arg Val Gln Phe Gly Val Leu Ser Pro Asp Glu Leu Lys

            20   25   30

Arg Met Ser Val Thr Glu Gly Gly Ile Lys Tyr Pro Glu Thr Thr Glu
        35   40   45

Gly Gly Arg Pro Lys Leu Gly Gly Leu Met Asp Pro Arg Gln Gly Val
    50   55   60

Ile Glu Arg Thr Gly Arg Cys Gln Thr Cys Ala Gly Asn Met Thr Glu
65   70   75   80

Cys Pro Gly His Phe Gly His Ile Glu Leu Ala Lys Pro Val Phe His
                85   90   95

Val Gly Phe Leu Val Lys Thr Met Lys Val Leu Arg Cys Val Cys Phe
            100   105   110

Phe Cys Ser Lys Leu Leu Val Asp Ser Asn Asn Pro Lys Ile Lys Asp
        115   120   125

Ile Leu Ala Lys Ser Lys Gly Gln Pro Lys Lys Arg Leu Thr His Val
    130   135   140

Tyr Asp Leu Cys Lys Gly Lys Asn Ile Cys Glu Gly Gly Glu Glu Met
145   150   155   160

Asp Asn Lys Phe Gly Val Glu Gln Pro Glu Gly Asp Glu Asp Leu Thr
                165   170   175

Lys Glu Lys Gly His Gly Gly Cys Gly Arg Tyr Gln Pro Arg Ile Arg
            180   185   190

Arg Ser Gly Leu Glu Leu Tyr Ala Glu Trp Lys His Val Asn Glu Asp
        195   200   205

Ser Gln Glu Lys Lys Ile Leu Leu Ser Pro Glu Arg Val His Glu Ile
    210   215   220

Phe Lys Arg Ile Ser Asp Glu Glu Cys Phe Val Leu Gly Met Glu Pro
225   230   235   240

Arg Tyr Ala Arg Pro Glu Trp Met Ile Val Thr Val Leu Pro Val Pro
                245   250   255

Pro Leu Ser Val Arg Pro Ala Val Val Met Gln Gly Ser Ala Arg Asn
            260   265   270

Gln Asp Asp Leu Thr His Lys Leu Ala Asp Ile Val Lys Ile Asn Asn
        275   280   285

Gln Leu Arg Arg Asn Glu Gln Asn Gly Ala Ala Ala His Val Ile Ala
    290   295   300

Glu Asp Val Lys Leu Leu Gln Phe His Val Ala Thr Met Val Asp Asn
305   310   315   320

Glu Leu Pro Gly Leu Pro Arg Ala Met Gln Lys Ser Gly Arg Pro Leu
                325   330   335

Lys Ser Leu Lys Gln Arg Leu Lys Gly Lys Glu Gly Arg Val Arg Gly
            340   345   350

Asn Leu Met Gly Lys Arg Val Asp Phe Ser Ala Arg Thr Val Ile Thr
        355   360   365

Pro Asp Pro Asn Leu Ser Ile Asp Gln Val Gly Val Pro Arg Ser Ile
    370   375   380

Ala Ala Asn Met Thr Phe Ala Glu Ile Val Thr Pro Phe Asn Ile Asp
385   390   395   400

Arg Leu Gln Glu Leu Val Arg Arg Gly Asn Ser Gln Tyr Pro Gly Ala
                405   410   415

Lys Tyr Ile Ile Arg Asp Asn Gly Asp Arg Ile Asp Leu Arg Phe His
            420   425   430

Pro Lys Pro Ser Asp Leu His Leu Gln Thr Gly Tyr Lys Val Glu Arg
        435   440   445

His Met Cys Asp Gly Asp Ile Val Ile Phe Asn Arg Gln Pro Thr Leu
    450   455   460

His Lys Met Ser Met Met Gly His Arg Val Arg Ile Leu Pro Trp Ser
465   470   475   480

Thr Phe Arg Leu Asn Leu Ser Val Thr Thr Pro Tyr Asn Ala Asp Phe
                485   490   495

Asp Gly Asp Glu Met Asn Leu His Leu Pro Gln Ser Leu Glu Thr Arg
            500   505   510

Ala Glu Ile Gln Glu Leu Ala Met Val Pro Arg Met Ile Val Thr Pro
        515   520   525

Gln Ser Asn Arg Pro Val Met Gly Ile Val Gln Asp Thr Leu Thr Ala
    530   535   540

Val Arg Lys Phe Thr Lys Arg Asp Val Phe Leu Glu Arg Gly Glu Val
545   550   555   560

Met Asn Leu Leu Met Phe Leu Ser Thr Trp Asp Gly Lys Val Pro Gln
                565   570   575

Pro Ala Ile Leu Lys Pro Arg Pro Leu Trp Thr Gly Lys Gln Ile Phe
            580   585   590

Ser Leu Ile Ile Pro Gly His Ile Asn Cys Ile Arg Thr His Ser Thr
        595   600   605

His Pro Asp Asp Glu Asp Ser Gly Pro Tyr Lys His Ile Ser Pro Gly
    610   615   620

Asp Thr Lys Val Val Val Glu Asn Gly Glu Leu Ile Met Gly Ile Leu
625   630   635   640

Cys Lys Lys Ser Leu Gly Thr Ser Ala Gly Ser Leu Val His Ile Ser
                645   650   655

Tyr Leu Glu Met Gly His Asp Ile Thr Arg Leu Phe Tyr Ser Asn Ile
            660   665   670

Gln Thr Val Ile Asn Asn Trp Leu Leu Ile Glu Gly His Thr Ile Gly
        675   680   685

Ile Gly Asp Ser Ile Ala Asp Ser Lys Thr Tyr Gln Asp Ile Gln Asn
    690   695   700

Thr Ile Lys Lys Ala Lys Gln Asp Val Ile Glu Val Ile Glu Lys Ala
705   710   715   720

His Asn Asn Glu Leu Glu Pro Thr Pro Gly Asn Thr Leu Arg Gln Thr
                725   730   735

Phe Glu Asn Gln Val Asn Arg Ile Leu Asn Asp Ala Arg Asp Lys Thr
            740   745   750

Gly Ser Ser Ala Gln Lys Ser Leu Ser Glu Tyr Asn Asn Phe Lys Ser
        755   760   765

Met Val Val Ser Gly Ala Lys Gly Ser Lys Ile Asn Ile Ser Gln Val
    770   775   780

Ile Ala Val Val Gly Gln Gln Asn Val Glu Gly Lys Arg Ile Pro Phe
785 790 795 800

Gly Phe Lys His Arg Thr Leu Pro His Phe Ile Lys Asp Asp Tyr Gly
805 810 815

Pro Glu Ser Arg Gly Phe Val Glu Asn Ser Tyr Leu Ala Gly Leu Thr
820 825 830

Pro Thr Glu Phe Phe Phe His Ala Met Gly Gly Arg Glu Gly Leu Ile
835 840 845

Asp Thr Ala Val Lys Thr Ala Glu Thr Gly Tyr Ile Gln Arg Arg Leu
850 855 860

Ile Lys Ser Met Glu Ser Val Met Val Lys Tyr Asp Ala Thr Val Arg
865 870 875 880

Asn Ser Ile Asn Gln Val Val Gln Leu Arg Tyr Gly Glu Asp Gly Leu
885 890 895

Ala Gly Glu Ser Val Glu Phe Gln Asn Leu Ala Thr Leu Lys Pro Ser
900 905 910

Asn Lys Ala Phe Glu Lys Lys Phe Arg Phe Asp Tyr Thr Asn Glu Arg
915 920 925

Ala Leu Arg Arg Thr Leu Gln Glu Asp Leu Val Lys Asp Val Leu Ser
930 935 940

Asn Ala His Ile Gln Asn Glu Leu Glu Arg Glu Phe Glu Arg Met Arg
945 950 955 960

Glu Asp Arg Glu Val Leu Arg Val Ile Phe Pro Thr Gly Asp Ser Lys
965 970 975

Val Val Leu Pro Cys Asn Leu Leu Arg Met Ile Trp Asn Ala Gln Lys
980 985 990

Ile Phe His Ile Asn Pro Arg Leu Pro Ser Asp Leu His Pro Ile Lys
995 1000 1005

Val Val Glu Gly Val Lys Glu Leu Ser Lys Lys Leu Val Ile Val
1010 1015 1020

Asn Gly Asp Asp Pro Leu Ser Arg Gln Ala Gln Glu Asn Ala Thr

1025 1030 1035

Leu Leu Phe Asn Ile His Leu Arg Ser Thr Leu Cys Ser Arg Arg
1040 1045 1050

Met Ala Glu Glu Phe Arg Leu Ser Gly Glu Ala Phe Asp Trp Leu
1055 1060 1065

Leu Gly Glu Ile Glu Ser Lys Phe Asn Gln Ala Ile Ala His Pro
1070 1075 1080

Gly Glu Met Val Gly Ala Leu Ala Ala Gln Ser Leu Gly Glu Pro
1085 1090 1095

Ala Thr Gln Met Thr Leu Asn Thr Phe His Tyr Ala Gly Val Ser
1100 1105 1110

Ala Lys Asn Val Thr Leu Gly Val Pro Arg Leu Lys Glu Leu Ile
1115 1120 1125

Asn Ile Ser Lys Lys Pro Lys Thr Pro Ser Leu Thr Val Phe Leu
1130 1135 1140

Leu Gly Gln Ser Ala Arg Asp Ala Glu Arg Ala Lys Asp Ile Leu
1145 1150 1155

Cys Arg Leu Glu His Thr Thr Leu Arg Lys Val Thr Ala Asn Thr
1160 1165 1170

Ala Ile Tyr Tyr Asp Pro Asn Pro Gln Ser Thr Val Val Ala Glu
1175 1180 1185

Asp Gln Glu Trp Val Asn Val Tyr Tyr Glu Met Pro Asp Phe Asp
1190 1195 1200

Val Ala Arg Ile Ser Pro Trp Leu Leu Arg Val Glu Leu Asp Arg
1205 1210 1215

Lys His Met Thr Asp Arg Lys Leu Thr Met Glu Gln Ile Ala Glu
1220 1225 1230

Lys Ile Asn Ala Gly Phe Gly Asp Asp Leu Asn Cys Ile Phe Asn
1235 1240 1245

Asp Asp Asn Ala Glu Lys Leu Val Leu Arg Ile Arg Ile Met Asn
1250 1255 1260

Ser Asp Glu Asn Lys Met Gln Glu Glu Glu Glu Val Val Asp Lys
1265 1270 1275

Met Asp Asp Asp Val Phe Leu Arg Cys Ile Glu Ser Asn Met Leu
1280 1285 1290

Thr Asp Met Thr Leu Gln Gly Ile Glu Gln Ile Ser Lys Val Tyr
1295 1300 1305

Met His Leu Pro Gln Thr Asp Asn Lys Lys Lys Ile Ile Ile Thr
1310 1315 1320

Glu Asp Gly Glu Phe Lys Ala Leu Gln Glu Trp Ile Leu Glu Thr
1325 1330 1335

Asp Gly Val Ser Leu Met Arg Val Leu Ser Glu Lys Asp Val Asp
1340 1345 1350

Pro Val Arg Thr Thr Ser Asn Asp Ile Val Glu Ile Phe Thr Val
1355 1360 1365

Leu Gly Ile Glu Ala Val Arg Lys Ala Leu Glu Arg Glu Leu Tyr
1370 1375 1380

His Val Ile Ser Phe Asp Gly Ser Tyr Val Asn Tyr Arg His Leu
1385 1390 1395

Ala Leu Leu Cys Asp Thr Met Thr Cys Arg Gly His Leu Met Ala
1400 1405 1410

Ile Thr Arg His Gly Val Asn Arg Gln Asp Thr Gly Pro Leu Met
1415 1420 1425

Lys Cys Ser Phe Glu Glu Thr Val Asp Val Leu Met Glu Ala Ala
1430 1435 1440

Ala His Gly Glu Ser Asp Pro Met Lys Gly Val Ser Glu Asn Ile
1445 1450 1455

Met Leu Gly Gln Leu Ala Pro Ala Gly Thr Gly Cys Phe Asp Leu
1460 1465 1470

Leu Leu Asp Ala Glu Lys Cys Lys Tyr Gly Met Glu Ile Pro Thr
1475 1480 1485

Asn Ile Pro Gly Leu Gly Ala Ala Gly Pro Thr Gly Met Phe Phe
1490 1495 1500

Gly Ser Ala Pro Ser Pro Met Gly Gly Ile Ser Pro Ala Met Thr
1505 1510 1515

Pro Trp Asn Gln Gly Ala Thr Pro Ala Tyr Gly Ala Trp Ser Pro
1520 1525 1530

Ser Val Gly Ser Gly Met Thr Pro Gly Ala Ala Gly Phe Ser Pro
1535 1540 1545

Ser Ala Ala Ser Asp Ala Ser Gly Phe Ser Pro Gly Tyr Ser Pro
1550 1555 1560

Ala Trp Ser Pro Thr Pro Gly Ser Pro Gly Ser Pro Gly Pro Ser
1565 1570 1575

Ser Pro Tyr Ile Pro Ser Pro Gly Gly Ala Met Ser Pro Ser Tyr
1580 1585 1590

Ser Pro Thr Ser Pro Ala Tyr Glu Pro Arg Ser Pro Gly Gly Tyr
1595 1600 1605

Thr Pro Gln Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser
1610 1615 1620

Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Asn Tyr Ser Pro
1625 1630 1635

Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr
1640 1645 1650

Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser
1655 1660 1665

Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro
1670 1675 1680

Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser
1685 1690 1695

Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr
1700 1705 1710

Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser
1715 1720 1725

Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro
1730 1735 1740

Thr Ser Pro Asn Tyr Ser Pro Thr Ser Pro Asn Tyr Thr Pro Thr
1745 1750 1755

Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser
1760 1765 1770

Pro Asn Tyr Thr Pro Thr Ser Pro Asn Tyr Ser Pro Thr Ser Pro
1775 1780 1785

Ser Tyr Ser Pro Thr Ser Pro Ser Tyr Ser Pro Thr Ser Pro Ser
1790 1795 1800

Tyr Ser Pro Ser Ser Pro Arg Tyr Thr Pro Gln Ser Pro Thr Tyr
1805 1810 1815

Thr Pro Ser Ser Pro Ser Tyr Ser Pro Ser Ser Pro Ser Tyr Ser
1820 1825 1830

Pro Thr Ser Pro Lys Tyr Thr Pro Thr Ser Pro Ser Tyr Ser Pro
1835 1840 1845

Ser Ser Pro Glu Tyr Thr Pro Thr Ser Pro Lys Tyr Ser Pro Thr
1850 1855 1860

Ser Pro Lys Tyr Ser Pro Thr Ser Pro Lys Tyr Ser Pro Thr Ser
1865 1870 1875

Pro Thr Tyr Ser Pro Thr Thr Pro Lys Tyr Ser Pro Thr Ser Pro
1880 1885 1890

Thr Tyr Ser Pro Thr Ser Pro Val Tyr Thr Pro Thr Ser Pro Lys
1895 1900 1905

Tyr Ser Pro Thr Ser Pro Thr Tyr Ser Pro Thr Ser Pro Lys Tyr
1910 1915 1920

Ser Pro Thr Ser Pro Thr Tyr Ser Pro Thr Ser Pro Lys Gly Ser
1925 1930 1935

Thr Tyr Ser Pro Thr Ser Pro Gly Tyr Ser Pro Thr Ser Pro Thr
1940 1945 1950

Tyr Ser Leu Thr Ser Pro Ala Ile Ser Pro Asp Asp Ser Asp Glu
1955 1960 1965

Glu Asn

1970

<210> 89
<211> 1174
<212> PRT
<213> Homo sapiens

<400> 89

Met Tyr Asp Ala Asp Glu Asp Met Gln Tyr Asp Glu Asp Asp Asp Glu
1 5 10 15

Ile Thr Pro Asp Leu Trp Gln Glu Ala Cys Trp Ile Val Ile Ser Ser
20 25 30

Tyr Phe Asp Glu Lys Gly Leu Val Arg Gln Gln Leu Asp Ser Phe Asp
35 40 45

Glu Phe Ile Gln Met Ser Val Gln Arg Ile Val Glu Asp Ala Pro Pro
50 55 60

Ile Asp Leu Gln Ala Glu Ala Gln His Ala Ser Gly Glu Val Glu Glu
65 70 75 80

Pro Pro Arg Tyr Leu Leu Lys Phe Glu Gln Ile Tyr Leu Ser Lys Pro
85 90 95

Thr His Trp Glu Arg Asp Gly Ala Pro Ser Pro Met Met Pro Asn Glu
100 105 110

Ala Arg Leu Arg Asn Leu Thr Tyr Ser Ala Pro Leu Tyr Val Asp Ile
115 120 125

Thr Lys Thr Val Ile Lys Glu Gly Glu Glu Gln Leu Gln Thr Gln His
130 135 140

Gln Lys Thr Phe Ile Gly Lys Ile Pro Ile Met Leu Arg Ser Thr Tyr
145 150 155 160

Cys Leu Leu Asn Gly Leu Thr Asp Arg Asp Leu Cys Glu Leu Asn Glu
165 170 175

Cys Pro Leu Asp Pro Gly Gly Tyr Phe Ile Ile Asn Gly Ser Glu Lys
180 185 190

Val Leu Ile Ala Gln Glu Lys Met Ala Thr Asn Thr Val Tyr Val Phe
195 200 205

Ala Lys Lys Asp Ser Lys Tyr Ala Tyr Thr Gly Glu Cys Arg Ser Cys

210 215 220

Leu Glu Asn Ser Ser Arg Pro Thr Ser Thr Ile Trp Val Ser Met Leu
225 230 235 240

Ala Arg Gly Gly Gln Gly Ala Lys Lys Ser Ala Ile Gly Gln Arg Ile
245 250 255

Val Ala Thr Leu Pro Tyr Ile Lys Gln Glu Val Pro Ile Ile Ile Val
260 265 270

Phe Arg Ala Leu Gly Phe Val Ser Asp Arg Asp Ile Leu Glu His Ile
275 280 285

Ile Tyr Asp Phe Glu Asp Pro Glu Met Met Glu Met Val Lys Pro Ser
290 295 300

Leu Asp Glu Ala Phe Val Ile Gln Glu Gln Asn Val Ala Leu Asn Phe
305 310 315 320

Ile Gly Ser Arg Gly Ala Lys Pro Gly Val Thr Lys Glu Lys Arg Ile
325 330 335

Lys Tyr Ala Lys Glu Val Leu Gln Lys Glu Met Leu Pro His Val Gly
340 345 350

Val Ser Asp Phe Cys Glu Thr Lys Lys Ala Tyr Phe Leu Gly Tyr Met
355 360 365

Val His Arg Leu Leu Leu Ala Ala Leu Gly Arg Arg Glu Leu Asp Asp
370 375 380

Arg Asp His Tyr Gly Asn Lys Arg Leu Asp Leu Ala Gly Pro Leu Leu
385 390 395 400

Ala Phe Leu Phe Arg Gly Met Phe Lys Asn Leu Leu Lys Glu Val Arg
405 410 415

Ile Tyr Ala Gln Lys Phe Ile Asp Arg Gly Lys Asp Phe Asn Leu Glu
420 425 430

Leu Ala Ile Lys Thr Arg Ile Ile Ser Asp Gly Leu Lys Tyr Ser Leu
435 440 445

Ala Thr Gly Asn Trp Gly Asp Gln Lys Lys Ala His Gln Ala Arg Ala
450 455 460

Gly Val Ser Gln Val Leu Asn Arg Leu Thr Phe Ala Ser Thr Leu Ser
465   470   475   480

His Leu Arg Arg Leu Asn Ser Pro Ile Gly Arg Asp Gly Lys Leu Ala
                485   490   495

Lys Pro Arg Gln Leu His Asn Thr Leu Trp Gly Met Val Cys Pro Ala
            500   505   510

Glu Thr Pro Glu Gly His Ala Val Gly Leu Val Lys Asn Leu Ala Leu
        515   520   525

Met Ala Tyr Ile Ser Val Gly Ser Gln Pro Ser Pro Ile Leu Glu Phe
    530   535   540

Leu Glu Glu Trp Ser Met Glu Asn Leu Glu Glu Ile Ser Pro Ala Ala
545   550   555   560

Ile Ala Asp Ala Thr Lys Ile Phe Val Asn Gly Cys Trp Val Gly Ile
                565   570   575

His Lys Asp Pro Glu Gln Leu Met Asn Thr Leu Arg Lys Leu Arg Arg
            580   585   590

Gln Met Asp Ile Ile Val Ser Glu Val Ser Met Ile Arg Asp Ile Arg
        595   600   605

Glu Arg Glu Ile Arg Ile Tyr Thr Asp Ala Gly Arg Ile Cys Arg Pro
    610   615   620

Leu Leu Ile Val Glu Lys Gln Lys Leu Leu Leu Lys Lys Arg His Ile
625   630   635   640

Asp Gln Leu Lys Glu Arg Glu Tyr Asn Asn Tyr Ser Trp Gln Asp Leu
                645   650   655

Val Ala Ser Gly Val Val Glu Tyr Ile Asp Thr Leu Glu Glu Glu Thr
            660   665   670

Val Met Leu Ala Met Thr Pro Asp Asp Leu Gln Glu Lys Glu Val Ala
        675   680   685

Tyr Cys Ser Thr Tyr Thr His Cys Glu Ile His Pro Ser Met Ile Leu
    690   695   700

Gly Val Cys Ala Ser Ile Ile Pro Phe Pro Asp His Asn Gln Ser Pro
705   710   715   720

```
Arg Asn Thr Tyr Gln Ser Ala Met Gly Lys Gln Ala Met Gly Val Tyr
                725                 730                 735

Ile Thr Asn Phe His Val Arg Met Asp Thr Leu Ala His Val Leu Tyr
            740                 745                 750

Tyr Pro Gln Lys Pro Leu Val Thr Thr Arg Ser Met Glu Tyr Leu Arg
        755                 760                 765

Phe Arg Glu Leu Pro Ala Gly Ile Asn Ser Ile Val Ala Ile Ala Ser
    770                 775                 780

Tyr Thr Gly Tyr Asn Gln Glu Asp Ser Val Ile Met Asn Arg Ser Ala
785                 790                 795                 800

Val Asp Arg Gly Phe Phe Arg Ser Val Phe Tyr Arg Ser Tyr Lys Glu
                805                 810                 815

Gln Glu Ser Lys Lys Gly Phe Asp Gln Glu Glu Val Phe Glu Lys Pro
            820                 825                 830

Thr Arg Glu Thr Cys Gln Gly Met Arg His Ala Ile Tyr Asp Lys Leu
        835                 840                 845

Asp Asp Asp Gly Leu Ile Ala Pro Gly Val Arg Val Ser Gly Asp Asp
    850                 855                 860

Val Ile Ile Gly Lys Thr Val Thr Leu Pro Glu Asn Glu Asp Glu Leu
865                 870                 875                 880

Glu Ser Thr Asn Arg Arg Tyr Thr Lys Arg Asp Cys Ser Thr Phe Leu
                885                 890                 895

Arg Thr Ser Glu Thr Gly Ile Val Asp Gln Val Met Val Thr Leu Asn
            900                 905                 910

Gln Glu Gly Tyr Lys Phe Cys Lys Ile Arg Val Arg Ser Val Arg Ile
        915                 920                 925

Pro Gln Ile Gly Asp Lys Phe Ala Ser Arg His Gly Gln Lys Gly Thr
    930                 935                 940

Cys Gly Ile Gln Tyr Arg Gln Glu Asp Met Pro Phe Thr Cys Glu Gly
945                 950                 955                 960

Ile Thr Pro Asp Ile Ile Ile Asn Pro His Ala Ile Pro Ser Arg Met
                965                 970                 975
```

Thr Ile Gly His Leu Ile Glu Cys Leu Gln Gly Lys Val Ser Ala Asn
            980   985   990

Lys Gly Glu Ile Gly Asp Ala Thr Pro Phe Asn Asp Ala Val Asn Val
        995   1000   1005

Gln Lys Ile Ser Asn Leu Leu Ser Asp Tyr Gly Tyr His Leu Arg
    1010   1015   1020

Gly Asn Glu Val Leu Tyr Asn Gly Phe Thr Gly Arg Lys Ile Thr
    1025   1030   1035

Ser Gln Ile Phe Ile Gly Pro Thr Tyr Tyr Gln Arg Leu Lys His
    1040   1045   1050

Met Val Asp Asp Lys Ile His Ser Arg Ala Arg Gly Pro Ile Gln
    1055   1060   1065

Ile Leu Asn Arg Gln Pro Met Glu Gly Arg Ser Arg Asp Gly Gly
    1070   1075   1080

Leu Arg Phe Gly Glu Met Glu Arg Asp Cys Gln Ile Ala His Gly
    1085   1090   1095

Ala Ala Gln Phe Leu Arg Glu Arg Leu Phe Glu Ala Ser Asp Pro
    1100   1105   1110

Tyr Gln Val His Val Cys Asn Leu Cys Gly Ile Met Ala Ile Ala
    1115   1120   1125

Asn Thr Arg Thr His Thr Tyr Glu Cys Arg Gly Cys Arg Asn Lys
    1130   1135   1140

Thr Gln Ile Ser Leu Val Arg Met Pro Tyr Ala Cys Lys Leu Leu
    1145   1150   1155

Phe Gln Glu Leu Met Ser Met Ser Ile Ala Pro Arg Met Met Ser
    1160   1165   1170

Val

<210> 90
<211> 275
<212> PRT
<213> Homo sapiens

<400> 90

Met Pro Tyr Ala Asn Gln Pro Thr Val Arg Ile Thr Glu Leu Thr Asp
1   5   10   15

Glu Asn Val Lys Phe Ile Ile Glu Asn Thr Asp Leu Ala Val Ala Asn
            20   25   30

Ser Ile Arg Arg Val Phe Ile Ala Glu Val Pro Ile Ile Ala Ile Asp
        35   40   45

Trp Val Gln Ile Asp Ala Asn Ser Ser Val Leu His Asp Glu Phe Ile
    50   55   60

Ala His Arg Leu Gly Leu Ile Pro Leu Ile Ser Asp Asp Ile Val Asp
65   70   75   80

Lys Leu Gln Tyr Ser Arg Asp Cys Thr Cys Glu Glu Phe Cys Pro Glu
                85   90   95

Cys Ser Val Glu Phe Thr Leu Asp Val Arg Cys Asn Glu Asp Gln Thr
            100   105   110

Arg His Val Thr Ser Arg Asp Leu Ile Ser Asn Ser Pro Arg Val Ile
        115   120   125

Pro Val Thr Ser Arg Asn Arg Asp Asn Asp Pro Asn Asp Tyr Val Glu
    130   135   140

Gln Asp Asp Ile Leu Ile Val Lys Leu Arg Lys Gly Gln Glu Leu Arg
145   150   155   160

Leu Arg Ala Tyr Ala Lys Lys Gly Phe Gly Lys Glu His Ala Lys Trp
                165   170   175

Asn Pro Thr Ala Gly Val Ala Phe Glu Tyr Asp Pro Asp Asn Ala Leu
            180   185   190

Arg His Thr Val Tyr Pro Lys Pro Glu Glu Trp Pro Lys Ser Glu Tyr
        195   200   205

Ser Glu Leu Asp Glu Asp Glu Ser Gln Ala Pro Tyr Asp Pro Asn Gly
    210   215   220

Lys Pro Glu Arg Phe Tyr Tyr Asn Val Glu Ser Cys Gly Ser Leu Arg
225   230   235   240

Pro Glu Thr Ile Val Leu Ser Ala Leu Ser Gly Leu Lys Lys Lys Leu

245 250 255

Ser Asp Leu Gln Thr Gln Leu Ser His Glu Ile Gln Ser Asp Val Leu
260 265 270

Thr Ile Asn
275

<210> 91
<211> 275
<212> PRT
<213> Homo sapiens

<400> 91

Met Pro Tyr Ala Asn Gln Pro Thr Val Arg Ile Thr Glu Leu Thr Asp
1 5 10 15

Glu Asn Val Lys Phe Ile Ile Glu Asn Thr Asp Leu Ala Val Ala Asn
20 25 30

Ser Ile Arg Arg Val Phe Ile Ala Glu Val Pro Ile Ile Ala Ile Asp
35 40 45

Trp Val Gln Ile Asp Ala Asn Ser Ser Val Leu His Asp Glu Phe Ile
50 55 60

Ala His Arg Leu Gly Leu Ile Pro Leu Ile Ser Asp Asp Ile Val Asp
65 70 75 80

Lys Leu Gln Tyr Ser Arg Asp Cys Thr Cys Glu Glu Phe Cys Pro Glu
85 90 95

Cys Ser Val Glu Phe Thr Leu Asp Val Arg Cys Asn Glu Asp Gln Thr
100 105 110

Arg His Val Thr Ser Arg Asp Leu Ile Ser Asn Ser Pro Arg Val Ile
115 120 125

Pro Val Thr Ser Arg Asn Arg Asp Asn Asp Pro Asn Asp Tyr Val Glu
130 135 140

Gln Asp Asp Ile Leu Ile Val Lys Leu Arg Lys Gly Gln Glu Leu Arg
145 150 155 160

Leu Arg Ala Tyr Ala Lys Lys Gly Phe Gly Lys Glu His Ala Lys Trp
165 170 175

Asn Pro Thr Ala Gly Val Ala Phe Glu Tyr Asp Pro Asp Asn Ala Leu

            180   185   190

Arg His Thr Val Tyr Pro Lys Pro Glu Glu Trp Pro Lys Ser Glu Tyr
        195   200   205

Ser Glu Leu Asp Glu Asp Glu Ser Gln Ala Pro Tyr Asp Pro Asn Gly
    210   215   220

Lys Pro Glu Arg Phe Tyr Tyr Asn Val Glu Ser Cys Gly Ser Leu Arg
225   230   235   240

Pro Glu Thr Ile Val Leu Ser Ala Leu Ser Gly Leu Lys Lys Lys Leu
                245   250   255

Ser Asp Leu Gln Thr Gln Leu Ser His Glu Ile Gln Ser Asp Val Leu
            260   265   270

Thr Ile Asn
        275

<210> 92
<211> 275
<212> PRT
<213> Homo sapiens

<400> 92

Met Pro Tyr Ala Asn Gln Pro Thr Val Arg Ile Thr Glu Leu Thr Asp
1   5   10   15

Glu Asn Val Lys Phe Ile Ile Glu Asn Thr Asp Leu Ala Val Ala Asn
            20   25   30

Ser Ile Arg Arg Val Phe Ile Ala Glu Val Pro Ile Ile Ala Ile Asp
        35   40   45

Trp Val Gln Ile Asp Ala Asn Ser Ser Val Leu His Asp Glu Phe Ile
    50   55   60

Ala His Arg Leu Gly Leu Ile Pro Leu Ile Ser Asp Asp Ile Val Asp
65   70   75   80

Lys Leu Gln Tyr Ser Arg Asp Cys Thr Cys Glu Glu Phe Cys Pro Glu
                85   90   95

Cys Ser Val Glu Phe Thr Leu Asp Val Arg Cys Asn Glu Asp Gln Thr
            100   105   110

Arg His Val Thr Ser Arg Asp Leu Ile Ser Asn Ser Pro Arg Val Ile

115 120 125

Pro Val Thr Ser Arg Asn Arg Asp Asn Asp Pro Asn Asp Tyr Val Glu
130 135 140

Gln Asp Asp Ile Leu Ile Val Lys Leu Arg Lys Gly Gln Glu Leu Arg
145 150 155 160

Leu Arg Ala Tyr Ala Lys Lys Gly Phe Gly Lys Glu His Ala Lys Trp
165 170 175

Asn Pro Thr Ala Gly Val Ala Phe Glu Tyr Asp Pro Asp Asn Ala Leu
180 185 190

Arg His Thr Val Tyr Pro Lys Pro Glu Glu Trp Pro Lys Ser Glu Tyr
195 200 205

Ser Glu Leu Asp Glu Asp Glu Ser Gln Ala Pro Tyr Asp Pro Asn Gly
210 215 220

Lys Pro Glu Arg Phe Tyr Tyr Asn Val Glu Ser Cys Gly Ser Leu Arg
225 230 235 240

Pro Glu Thr Ile Val Leu Ser Ala Leu Ser Gly Leu Lys Lys Lys Leu
245 250 255

Ser Asp Leu Gln Thr Gln Leu Ser His Glu Ile Gln Ser Asp Val Leu
260 265 270

Thr Ile Asn
275

<210> 93
<211> 142
<212> PRT
<213> Homo sapiens

<400> 93

Met Ala Ala Gly Gly Ser Asp Pro Arg Ala Gly Asp Val Glu Glu Asp
1 5 10 15

Ala Ser Gln Leu Ile Phe Pro Lys Glu Phe Glu Thr Ala Glu Thr Leu
20 25 30

Leu Asn Ser Glu Val His Met Leu Leu Glu His Arg Lys Gln Gln Asn
35 40 45

Glu Ser Ala Glu Asp Glu Gln Glu Leu Ser Glu Val Phe Met Lys Thr

    50   55   60

Leu Asn Tyr Thr Ala Arg Phe Ser Arg Phe Lys Asn Arg Glu Thr Ile
65   70   75   80

Ala Ser Val Arg Ser Leu Leu Leu Gln Lys Lys Leu His Lys Phe Glu
                85   90   95

Leu Ala Cys Leu Ala Asn Leu Cys Pro Glu Thr Ala Glu Glu Ser Lys
            100   105   110

Ala Leu Ile Pro Ser Leu Glu Gly Arg Phe Glu Asp Glu Glu Leu Gln
        115   120   125

Gln Ile Leu Asp Asp Ile Gln Thr Lys Arg Ser Phe Gln Tyr
    130   135   140

<210> 94
<211> 142
<212> PRT
<213> Homo sapiens

<400> 94

Met Ala Ala Gly Gly Ser Asp Pro Arg Ala Gly Asp Val Glu Glu Asp
1   5   10   15

Ala Ser Gln Leu Ile Phe Pro Lys Glu Phe Glu Thr Ala Glu Thr Leu
            20   25   30

Leu Asn Ser Glu Val His Met Leu Leu Glu His Arg Lys Gln Gln Asn
        35   40   45

Glu Ser Ala Glu Asp Glu Gln Glu Leu Ser Glu Val Phe Met Lys Thr
    50   55   60

Leu Asn Tyr Thr Ala Arg Phe Ser Arg Phe Lys Asn Arg Glu Thr Ile
65   70   75   80

Ala Ser Val Arg Ser Leu Leu Leu Gln Lys Lys Leu His Lys Phe Glu
                85   90   95

Leu Ala Cys Leu Ala Asn Leu Cys Pro Glu Thr Ala Glu Glu Ser Lys
            100   105   110

Ala Leu Ile Pro Ser Leu Glu Gly Arg Phe Glu Asp Glu Glu Leu Gln
        115   120   125

Gln Ile Leu Asp Asp Ile Gln Thr Lys Arg Ser Phe Gln Tyr

130 135 140

<210> 95
<211> 210
<212> PRT
<213> Homo sapiens

<400> 95

Met Asp Asp Glu Glu Glu Thr Tyr Arg Leu Trp Lys Ile Arg Lys Thr
1 5 10 15

Ile Met Gln Leu Cys His Asp Arg Gly Tyr Leu Val Thr Gln Asp Glu
20 25 30

Leu Asp Gln Thr Leu Glu Glu Phe Lys Ala Gln Phe Gly Asp Lys Pro
35 40 45

Ser Glu Gly Arg Pro Arg Arg Thr Asp Leu Thr Val Leu Val Ala His
50 55 60

Asn Asp Asp Pro Thr Asp Gln Met Phe Val Phe Phe Pro Glu Glu Pro
65 70 75 80

Lys Val Gly Ile Lys Thr Ile Lys Val Tyr Cys Gln Arg Met Gln Glu
85 90 95

Glu Asn Ile Thr Arg Ala Leu Ile Val Val Gln Gln Gly Met Thr Pro
100 105 110

Ser Ala Lys Gln Ser Leu Val Asp Met Ala Pro Lys Tyr Ile Leu Glu
115 120 125

Gln Phe Leu Gln Gln Glu Leu Leu Ile Asn Ile Thr Glu His Glu Leu
130 135 140

Val Pro Glu His Val Val Met Thr Lys Glu Glu Val Thr Glu Leu Leu
145 150 155 160

Ala Arg Tyr Lys Leu Arg Glu Asn Gln Leu Pro Arg Ile Gln Ala Gly
165 170 175

Asp Pro Val Ala Arg Tyr Phe Gly Ile Lys Arg Gly Gln Val Val Lys
180 185 190

Ile Ile Arg Pro Ser Glu Thr Ala Gly Arg Tyr Ile Thr Tyr Arg Leu
195 200 205

Val Gln

210

<210> 96
<211> 210
<212> PRT
<213> Homo sapiens

<400> 96

Met Asp Asp Glu Glu Glu Thr Tyr Arg Leu Trp Lys Ile Arg Lys Thr
1 5 10 15

Ile Met Gln Leu Cys His Asp Arg Gly Tyr Leu Val Thr Gln Asp Glu
20 25 30

Leu Asp Gln Thr Leu Glu Glu Phe Lys Ala Gln Phe Gly Asp Lys Pro
35 40 45

Ser Glu Gly Arg Pro Arg Arg Thr Asp Leu Thr Val Leu Val Ala His
50 55 60

Asn Asp Asp Pro Thr Asp Gln Met Phe Val Phe Phe Pro Glu Glu Pro
65 70 75 80

Lys Val Gly Ile Lys Thr Ile Lys Val Tyr Cys Gln Arg Met Gln Glu
85 90 95

Glu Asn Ile Thr Arg Ala Leu Ile Val Val Gln Gln Gly Met Thr Pro
100 105 110

Ser Ala Lys Gln Ser Leu Val Asp Met Ala Pro Lys Tyr Ile Leu Glu
115 120 125

Gln Phe Leu Gln Gln Glu Leu Leu Ile Asn Ile Thr Glu His Glu Leu
130 135 140

Val Pro Glu His Val Val Met Thr Lys Glu Glu Val Thr Glu Leu Leu
145 150 155 160

Ala Arg Tyr Lys Leu Arg Glu Asn Gln Leu Pro Arg Ile Gln Ala Gly
165 170 175

Asp Pro Val Ala Arg Tyr Phe Gly Ile Lys Arg Gly Gln Val Val Lys
180 185 190

Ile Ile Arg Pro Ser Glu Thr Ala Gly Arg Tyr Ile Thr Tyr Arg Leu
195 200 205

Val Gln

210

<210> 97
<211> 127
<212> PRT
<213> Homo sapiens

<400> 97

Met Ser Asp Asn Glu Asp Asn Phe Asp Gly Asp Asp Phe Asp Asp Val
1   5   10   15

Glu Glu Asp Glu Gly Leu Asp Asp Leu Glu Asn Ala Glu Glu Glu Gly
            20   25   30

Gln Glu Asn Val Glu Ile Leu Pro Ser Gly Glu Arg Pro Gln Ala Asn
        35   40   45

Gln Lys Arg Ile Thr Thr Pro Tyr Met Thr Lys Tyr Glu Arg Ala Arg
    50   55   60

Val Leu Gly Thr Arg Ala Leu Gln Ile Ala Met Cys Ala Pro Val Met
65   70   75   80

Val Glu Leu Glu Gly Glu Thr Asp Pro Leu Leu Ile Ala Met Lys Glu
                85   90   95

Leu Lys Ala Arg Lys Ile Pro Ile Ile Ile Arg Arg Tyr Leu Pro Asp
            100   105   110

Gly Ser Tyr Glu Asp Trp Gly Val Asp Glu Leu Ile Ile Thr Asp
        115   120   125

<210> 98
<211> 172
<212> PRT
<213> Homo sapiens

<400> 98

Met Phe Tyr His Ile Ser Leu Glu His Glu Ile Leu Leu His Pro Arg
1   5   10   15

Tyr Phe Gly Pro Asn Leu Leu Asn Thr Val Lys Gln Lys Leu Phe Thr
            20   25   30

Glu Val Glu Gly Thr Cys Thr Gly Lys Tyr Gly Phe Val Ile Ala Val
        35   40   45

Thr Thr Ile Asp Asn Ile Gly Ala Gly Val Ile Gln Pro Gly Arg Gly
    50   55   60

Phe Val Leu Tyr Pro Val Lys Tyr Lys Ala Ile Val Phe Arg Pro Phe
65   70   75   80

Lys Gly Glu Val Val Asp Ala Val Val Thr Gln Val Asn Lys Val Gly
                85   90   95

Leu Phe Thr Glu Ile Gly Pro Met Ser Cys Phe Ile Ser Arg His Ser
            100   105   110

Ile Pro Ser Glu Met Glu Phe Asp Pro Asn Ser Asn Pro Pro Cys Tyr
        115   120   125

Lys Thr Met Asp Glu Asp Ile Val Ile Gln Gln Asp Asp Glu Ile Arg
    130   135   140

Leu Lys Ile Val Gly Thr Arg Val Asp Lys Asn Asp Ile Phe Ala Ile
145   150   155   160

Gly Ser Leu Met Asp Asp Tyr Leu Gly Leu Val Ser
                165   170

<210> 99
<211> 150
<212> PRT
<213> Homo sapiens

<400> 99

Met Ala Gly Ile Leu Phe Glu Asp Ile Phe Asp Val Lys Asp Ile Asp
1   5   10   15

Pro Glu Gly Lys Lys Phe Asp Arg Val Ser Arg Leu His Cys Glu Ser
            20   25   30

Glu Ser Phe Lys Met Asp Leu Ile Leu Asp Val Asn Ile Gln Ile Tyr
        35   40   45

Pro Val Asp Leu Gly Asp Lys Phe Arg Leu Val Ile Ala Ser Thr Leu
    50   55   60

Tyr Glu Asp Gly Thr Leu Asp Asp Gly Glu Tyr Asn Pro Thr Asp Asp
65   70   75   80

Arg Pro Ser Arg Ala Asp Gln Phe Glu Tyr Val Met Tyr Gly Lys Val
                85   90   95

Tyr Arg Ile Glu Gly Asp Glu Thr Ser Thr Glu Ala Ala Thr Arg Leu
            100   105   110

Ser Ala Tyr Val Ser Tyr Gly Gly Leu Leu Met Arg Leu Gln Gly Asp
        115   120   125

Ala Asn Asn Leu His Gly Phe Glu Val Asp Ser Arg Val Tyr Leu Leu
    130   135   140

Met Lys Lys Leu Ala Phe
145   150

<210> 100
<211> 125
<212> PRT
<213> Homo sapiens

<400> 100

Met Glu Pro Asp Gly Thr Tyr Glu Pro Gly Phe Val Gly Ile Arg Phe
1   5   10   15

Cys Gln Glu Cys Asn Asn Met Leu Tyr Pro Lys Glu Asp Lys Glu Asn
            20   25   30

Arg Ile Leu Leu Tyr Ala Cys Arg Asn Cys Asp Tyr Gln Gln Glu Ala
        35   40   45

Asp Asn Ser Cys Ile Tyr Val Asn Lys Ile Thr His Glu Val Asp Glu
    50   55   60

Leu Thr Gln Ile Ile Ala Asp Val Ser Gln Asp Pro Thr Leu Pro Arg
65   70   75   80

Thr Glu Asp His Pro Cys Gln Lys Cys Gly His Lys Glu Ala Val Phe
                85   90   95

Phe Gln Ser His Ser Ala Arg Ala Glu Asp Ala Met Arg Leu Tyr Tyr
            100   105   110

Val Cys Thr Ala Pro His Cys Gly His Arg Trp Thr Glu
        115   120   125

<210> 101
<211> 117
<212> PRT
<213> Homo sapiens

<400> 101

Met Asn Ala Pro Pro Ala Phe Glu Ser Phe Leu Leu Phe Glu Gly Glu
1   5   10   15

Lys Lys Ile Thr Ile Asn Lys Asp Thr Lys Val Pro Asn Ala Cys Leu
20 25 30

Phe Thr Ile Asn Lys Glu Asp His Thr Leu Gly Asn Ile Ile Lys Ser
35 40 45

Gln Leu Leu Lys Asp Pro Gln Val Leu Phe Ala Gly Tyr Lys Val Pro
50 55 60

His Pro Leu Glu His Lys Ile Ile Ile Arg Val Gln Thr Thr Pro Asp
65 70 75 80

Tyr Ser Pro Gln Glu Ala Phe Thr Asn Ala Ile Thr Asp Leu Ile Ser
85 90 95

Glu Leu Ser Leu Leu Glu Glu Arg Phe Arg Val Ala Ile Lys Asp Lys
100 105 110

Gln Glu Gly Ile Glu
115

<210> 102
<211> 115
<212> PRT
<213> Homo sapiens

<400> 102

Met Asn Ala Pro Pro Ala Phe Glu Ser Phe Leu Leu Phe Glu Gly Glu
1 5 10 15

Lys Ile Thr Ile Asn Lys Asp Thr Lys Val Pro Lys Ala Cys Leu Phe
20 25 30

Thr Ile Asn Lys Glu Asp His Thr Leu Gly Asn Ile Ile Lys Ser Gln
35 40 45

Leu Leu Lys Asp Pro Gln Val Leu Phe Ala Gly Tyr Lys Val Pro His
50 55 60

Pro Leu Glu His Lys Ile Ile Ile Arg Val Gln Thr Thr Pro Asp Tyr
65 70 75 80

Ser Pro Gln Glu Ala Phe Thr Asn Ala Ile Thr Asp Leu Ile Ser Glu
85 90 95

Leu Ser Leu Leu Glu Glu Arg Phe Arg Thr Cys Leu Leu Pro Leu Arg
100 105 110

Leu Leu Pro
        115

<210> 103
<211> 115
<212> PRT
<213> Homo sapiens

<400> 103

Met Asn Ala Pro Pro Ala Phe Glu Ser Phe Leu Leu Phe Glu Gly Glu
1   5   10   15

Lys Ile Thr Ile Asn Lys Asp Thr Lys Val Pro Asn Ala Cys Leu Phe
            20   25   30

Thr Ile Asn Lys Glu Asp His Thr Leu Gly Asn Ile Ile Lys Ser Gln
        35   40   45

Leu Leu Lys Asp Pro Gln Val Leu Phe Ala Gly Tyr Lys Val Pro His
    50   55   60

Pro Leu Glu His Lys Ile Ile Ile Arg Val Gln Thr Thr Pro Asp Tyr
65   70   75   80

Ser Pro Gln Glu Ala Phe Thr Asn Ala Ile Thr Asp Leu Ile Ser Glu
                85   90   95

Leu Ser Leu Leu Glu Glu Arg Phe Arg Thr Cys Leu Leu Pro Leu Arg
            100   105   110

Leu Leu Pro
        115

<210> 104
<211> 58
<212> PRT
<213> Homo sapiens

<400> 104

Met Asp Thr Gln Lys Asp Val Gln Pro Pro Lys Gln Gln Pro Met Ile
1   5   10   15

Tyr Ile Cys Gly Glu Cys His Thr Glu Asn Glu Ile Lys Ser Arg Asp
            20   25   30

Pro Ile Arg Cys Arg Glu Cys Gly Tyr Arg Ile Met Tyr Lys Lys Arg
        35   40   45

Thr Lys Arg Leu Val Val Phe Asp Ala Arg
    50   55

<210> 105
<211> 58
<212> PRT
<213> Homo sapiens

<400> 105

Met Asp Thr Gln Lys Asp Val Gln Pro Pro Lys Gln Gln Pro Met Ile
1   5   10   15

Tyr Ile Cys Gly Glu Cys His Thr Glu Asn Glu Ile Lys Ser Arg Asp
            20   25   30

Pro Ile Arg Cys Arg Glu Cys Gly Tyr Arg Ile Met Tyr Lys Lys Arg
        35   40   45

Thr Lys Arg Leu Val Val Phe Asp Ala Arg
    50   55

<210> 106
<211> 67
<212> PRT
<213> Homo sapiens

<400> 106

Met Ile Ile Pro Val Arg Cys Phe Thr Cys Gly Lys Ile Val Gly Asn
1   5   10   15

Lys Trp Glu Ala Tyr Leu Gly Leu Leu Gln Ala Glu Tyr Thr Glu Gly
            20   25   30

Asp Ala Leu Asp Ala Leu Gly Leu Lys Arg Tyr Cys Cys Arg Arg Met
        35   40   45

Leu Leu Ala His Val Asp Leu Ile Glu Lys Leu Leu Asn Tyr Ala Pro
    50   55   60

Leu Glu Lys
65

<210> 107
<211> 67
<212> PRT
<213> Homo sapiens

<400> 107

Met Ile Ile Pro Val Arg Cys Phe Thr Cys Gly Lys Ile Val Gly Asn
1   5   10   15

Lys Trp Glu Ala Tyr Leu Gly Leu Leu Gln Ala Glu Tyr Thr Glu Gly
            20   25   30

Asp Ala Leu Asp Ala Leu Gly Leu Lys Arg Tyr Cys Cys Arg Arg Met
        35   40   45

Leu Leu Ala His Val Asp Leu Ile Glu Lys Leu Leu Asn Tyr Ala Pro
    50   55   60

Leu Glu Lys
65

<210> 108
<211> 1075
<212> PRT
<213> Homo sapiens

<400> 108

Met Lys Ala Asn Glu Lys Val Thr Ser Asp Ala Asp Pro Met Trp Tyr
1   5   10   15

Leu Lys Tyr Leu Asn Ile Tyr Val Gly Leu Pro Asp Val Glu Glu Ser
            20   25   30

Phe Asn Val Thr Arg Pro Val Ser Pro His Glu Cys Arg Leu Arg Asp
        35   40   45

Met Thr Tyr Ser Ala Pro Ile Thr Val Asp Ile Glu Tyr Thr Arg Gly
    50   55   60

Ser Gln Arg Ile Ile Arg Asn Ala Leu Pro Ile Gly Arg Met Pro Ile
65   70   75   80

Met Leu Arg Ser Ser Asn Cys Val Leu Thr Gly Lys Thr Pro Ala Glu
                85   90   95

Phe Ala Lys Leu Asn Glu Cys Pro Leu Asp Pro Gly Gly Tyr Phe Ile
            100   105   110

Val Lys Gly Val Glu Lys Val Ile Leu Ile Gln Glu Gln Leu Ser Lys
        115   120   125

Asn Arg Ile Ile Val Glu Ala Asp Arg Lys Gly Ala Val Gly Ala Ser
    130   135   140

Val Thr Ser Ser Thr His Glu Lys Lys Ser Arg Thr Asn Met Ala Val
145   150   155   160

```
Lys Gln Gly Arg Phe Tyr Leu Arg His Asn Thr Leu Ser Glu Asp Ile
                165                 170                 175

Pro Ile Val Ile Ile Phe Lys Ala Met Gly Val Glu Ser Asp Gln Glu
            180                 185                 190

Ile Val Gln Met Ile Gly Thr Glu Glu His Val Met Ala Ala Phe Gly
        195                 200                 205

Pro Ser Leu Glu Glu Cys Gln Lys Ala Gln Ile Phe Thr Gln Met Gln
    210                 215                 220

Ala Leu Lys Tyr Ile Gly Asn Lys Val Arg Arg Gln Arg Met Trp Gly
225                 230                 235                 240

Gly Gly Pro Lys Lys Thr Lys Ile Glu Glu Ala Arg Glu Leu Leu Ala
                245                 250                 255

Ser Thr Ile Leu Thr His Val Pro Val Lys Glu Phe Asn Phe Arg Ala
            260                 265                 270

Lys Cys Ile Tyr Thr Ala Val Met Val Arg Arg Val Ile Leu Ala Gln
        275                 280                 285

Gly Asp Asn Lys Val Asp Asp Arg Asp Tyr Tyr Gly Asn Lys Arg Leu
    290                 295                 300

Glu Leu Ala Gly Gln Leu Leu Ser Leu Leu Phe Glu Asp Leu Phe Lys
305                 310                 315                 320

Lys Phe Asn Ser Glu Met Lys Lys Ile Ala Asp Gln Val Ile Pro Lys
                325                 330                 335

Gln Arg Ala Ala Gln Phe Asp Val Val Lys His Met Arg Gln Asp Gln
            340                 345                 350

Ile Thr Asn Gly Met Val Asn Ala Ile Ser Thr Gly Asn Trp Ser Leu
        355                 360                 365

Lys Arg Phe Lys Met Asp Arg Gln Gly Val Thr Gln Val Leu Ser Arg
    370                 375                 380

Leu Ser Tyr Ile Ser Ala Leu Gly Met Met Thr Arg Ile Ser Ser Gln
385                 390                 395                 400

Phe Glu Lys Thr Arg Lys Val Ser Gly Pro Arg Ser Leu Gln Pro Ser
```

405 410 415

Gln Trp Gly Met Leu Cys Pro Ser Asp Thr Pro Glu Gly Glu Ala Cys
420 425 430

Gly Leu Val Lys Asn Leu Ala Leu Met Thr His Ile Thr Thr Asp Met
435 440 445

Glu Asp Gly Pro Ile Val Lys Leu Ala Ser Asn Leu Gly Val Glu Asp
450 455 460

Val Asn Leu Leu Cys Gly Glu Glu Leu Ser Tyr Pro Asn Val Phe Leu
465 470 475 480

Val Phe Leu Asn Gly Asn Ile Leu Gly Val Ile Arg Asp His Lys Lys
485 490 495

Leu Val Asn Thr Phe Arg Leu Met Arg Arg Ala Gly Tyr Ile Asn Glu
500 505 510

Phe Val Ser Ile Ser Thr Asn Leu Thr Asp Arg Cys Val Tyr Ile Ser
515 520 525

Ser Asp Gly Gly Arg Leu Cys Arg Pro Tyr Ile Ile Val Lys Lys Gln
530 535 540

Lys Pro Ala Val Thr Asn Lys His Met Glu Glu Leu Ala Gln Gly Tyr
545 550 555 560

Arg Asn Phe Glu Asp Phe Leu His Glu Ser Leu Val Glu Tyr Leu Asp
565 570 575

Val Asn Glu Glu Asn Asp Cys Asn Ile Ala Leu Tyr Glu His Thr Ile
580 585 590

Asn Lys Asp Thr Thr His Leu Glu Ile Glu Pro Phe Thr Leu Leu Gly
595 600 605

Val Cys Ala Gly Leu Ile Pro Tyr Pro His His Asn Gln Ser Pro Arg
610 615 620

Asn Thr Tyr Gln Cys Ala Met Gly Lys Gln Ala Met Gly Thr Ile Gly
625 630 635 640

Tyr Asn Gln Arg Asn Arg Ile Asp Thr Leu Met Tyr Leu Leu Ala Tyr
645 650 655

Pro Gln Lys Pro Met Val Lys Thr Lys Thr Ile Glu Leu Ile Glu Phe
660 665 670

Glu Lys Leu Pro Ala Gly Gln Asn Ala Thr Val Ala Val Met Ser Tyr
675 680 685

Ser Gly Tyr Asp Ile Glu Asp Ala Leu Val Leu Asn Lys Ala Ser Leu
690 695 700

Asp Arg Gly Phe Gly Arg Cys Leu Val Tyr Lys Asn Ala Lys Cys Thr
705 710 715 720

Leu Lys Arg Tyr Thr Asn Gln Thr Phe Asp Lys Val Met Gly Pro Met
725 730 735

Leu Asp Ala Ala Thr Arg Lys Pro Ile Trp Arg His Glu Ile Leu Asp
740 745 750

Ala Asp Gly Ile Cys Ser Pro Gly Glu Lys Val Glu Asn Lys Gln Val
755 760 765

Leu Val Asn Lys Ser Met Pro Thr Val Thr Gln Ile Pro Leu Glu Gly
770 775 780

Ser Asn Val Pro Gln Gln Pro Gln Tyr Lys Asp Val Pro Ile Thr Tyr
785 790 795 800

Lys Gly Ala Thr Asp Ser Tyr Ile Glu Lys Val Met Ile Ser Ser Asn
805 810 815

Ala Glu Asp Ala Phe Leu Ile Lys Met Leu Leu Arg Gln Thr Arg Arg
820 825 830

Pro Glu Ile Gly Asp Lys Phe Ser Ser Arg His Gly Gln Lys Gly Val
835 840 845

Cys Gly Leu Ile Val Pro Gln Glu Asp Met Pro Phe Cys Asp Ser Gly
850 855 860

Ile Cys Pro Asp Ile Ile Met Asn Pro His Gly Phe Pro Ser Arg Met
865 870 875 880

Thr Val Gly Lys Leu Ile Glu Leu Leu Ala Gly Lys Ala Gly Val Leu
885 890 895

Asp Gly Arg Phe His Tyr Gly Thr Ala Phe Gly Gly Ser Lys Val Lys
900 905 910

Asp Val Cys Glu Asp Leu Val Arg His Gly Tyr Asn Tyr Leu Gly Lys
915 920 925

Asp Tyr Val Thr Ser Gly Ile Thr Gly Glu Pro Leu Glu Ala Tyr Ile
930 935 940

Tyr Phe Gly Pro Val Tyr Tyr Gln Lys Leu Lys His Met Val Leu Asp
945 950 955 960

Lys Met His Ala Arg Ala Arg Gly Pro Arg Ala Val Leu Thr Arg Gln
965 970 975

Pro Thr Glu Gly Arg Ser Arg Asp Gly Gly Leu Arg Leu Gly Glu Met
980 985 990

Glu Arg Asp Cys Leu Ile Gly Tyr Gly Ala Ser Met Leu Leu Leu Glu
995 1000 1005

Arg Leu Met Ile Ser Ser Asp Ala Phe Glu Val Asp Val Cys Gly
1010 1015 1020

Gln Cys Gly Leu Leu Gly Tyr Ser Gly Trp Cys His Tyr Cys Lys
1025 1030 1035

Ser Ser Cys His Val Ser Ser Leu Arg Ile Pro Tyr Ala Cys Lys
1040 1045 1050

Leu Leu Phe Gln Glu Leu Gln Ser Met Asn Ile Ile Pro Arg Leu
1055 1060 1065

Lys Leu Ser Lys Tyr Asn Glu
1070 1075

<210> 109
<211> 534
<212> PRT
<213> Homo sapiens

<400> 109

Met Thr Gln Ala Glu Ile Lys Leu Cys Ser Leu Leu Leu Gln Glu His
1 5 10 15

Phe Gly Glu Ile Val Glu Lys Ile Gly Val His Leu Ile Arg Thr Gly
20 25 30

Ser Gln Pro Leu Arg Val Ile Ala His Asp Thr Gly Thr Ser Leu Asp
35 40 45

```
Gln Val Lys Lys Ala Leu Cys Val Leu Val Gln His Asn Leu Val Ser
    50                  55                  60

Tyr Gln Val His Lys Arg Gly Val Val Glu Tyr Glu Ala Gln Cys Ser
65                  70                  75                  80

Arg Val Leu Arg Met Leu Arg Tyr Pro Arg Tyr Ile Tyr Thr Thr Lys
                85                  90                  95

Thr Leu Tyr Ser Asp Thr Gly Glu Leu Ile Val Glu Glu Leu Leu Leu
            100                 105                 110

Asn Gly Lys Leu Thr Met Ser Ala Val Val Lys Lys Val Ala Asp Arg
        115                 120                 125

Leu Thr Glu Thr Met Glu Asp Gly Lys Thr Met Asp Tyr Ala Glu Val
    130                 135                 140

Ser Asn Thr Phe Val Arg Leu Ala Asp Thr His Phe Val Gln Arg Cys
145                 150                 155                 160

Pro Ser Val Pro Thr Thr Glu Asn Ser Asp Pro Gly Pro Pro Pro Pro
                165                 170                 175

Ala Pro Thr Leu Val Ile Asn Glu Lys Asp Met Tyr Leu Val Pro Lys
            180                 185                 190

Leu Ser Leu Ile Gly Lys Gly Lys Arg Arg Arg Ser Ser Asp Glu Asp
        195                 200                 205

Ala Ala Gly Glu Pro Lys Ala Lys Arg Pro Lys Tyr Thr Thr Asp Asn
    210                 215                 220

Lys Glu Pro Ile Pro Asp Asp Gly Ile Tyr Trp Gln Ala Asn Leu Asp
225                 230                 235                 240

Arg Phe His Gln His Phe Arg Asp Gln Ala Ile Val Ser Ala Val Ala
                245                 250                 255

Asn Arg Met Asp Gln Thr Ser Ser Glu Ile Val Arg Thr Met Leu Arg
            260                 265                 270

Met Ser Glu Ile Thr Thr Ser Ser Ser Ala Pro Phe Thr Gln Pro Leu
        275                 280                 285

Ser Ser Asn Glu Ile Phe Arg Ser Leu Pro Val Gly Tyr Asn Ile Ser
    290                 295                 300
```

Lys Gln Val Leu Asp Gln Tyr Leu Thr Leu Leu Ala Asp Asp Pro Leu
305   310   315   320

Glu Phe Val Gly Lys Ser Gly Asp Ser Gly Gly Gly Met Tyr Val Ile
                325   330   335

Asn Leu His Lys Ala Leu Ala Ser Leu Ala Thr Ala Thr Leu Glu Ser
            340   345   350

Val Val Gln Glu Arg Phe Gly Ser Arg Cys Ala Arg Ile Phe Arg Leu
        355   360   365

Val Leu Gln Lys Lys His Ile Glu Gln Lys Gln Val Glu Asp Phe Ala
    370   375   380

Met Ile Pro Ala Lys Glu Ala Lys Asp Met Leu Tyr Lys Met Leu Ser
385   390   395   400

Glu Asn Phe Met Ser Leu Gln Glu Ile Pro Lys Thr Pro Asp His Ala
                405   410   415

Pro Ser Arg Thr Phe Tyr Leu Tyr Thr Val Asn Ile Leu Ser Ala Ala
            420   425   430

Arg Met Leu Leu His Arg Cys Tyr Lys Ser Ile Ala Asn Leu Ile Glu
        435   440   445

Arg Arg Gln Phe Glu Thr Lys Glu Asn Lys Arg Leu Leu Glu Lys Ser
    450   455   460

Gln Arg Val Glu Ala Ile Ile Ala Ser Met Gln Ala Thr Gly Ala Glu
465   470   475   480

Glu Ala Gln Leu Gln Glu Ile Glu Glu Met Ile Thr Ala Pro Glu Arg
                485   490   495

Gln Gln Leu Glu Thr Leu Lys Arg Asn Val Asn Lys Leu Asp Ala Ser
            500   505   510

Glu Ile Gln Val Asp Glu Thr Ile Phe Leu Leu Glu Ser Tyr Ile Glu
        515   520   525

Cys Thr Met Lys Arg Gln
    530

<210> 110

<211> 534
<212> PRT
<213> Homo sapiens

<400> 110

Met Thr Gln Ala Glu Ile Lys Leu Cys Ser Leu Leu Leu Gln Glu His
1   5   10   15

Phe Gly Glu Ile Val Glu Lys Ile Gly Val His Leu Ile Arg Thr Gly
            20   25   30

Ser Gln Pro Leu Arg Val Ile Ala His Asp Thr Gly Thr Ser Leu Asp
        35   40   45

Gln Val Lys Lys Ala Leu Cys Val Leu Val Gln His Asn Leu Val Ser
    50   55   60

Tyr Gln Val His Lys Arg Gly Val Val Glu Tyr Glu Ala Gln Cys Ser
65   70   75   80

Arg Val Leu Arg Met Leu Arg Tyr Pro Arg Tyr Ile Tyr Thr Thr Lys
                85   90   95

Thr Leu Tyr Ser Asp Thr Gly Glu Leu Ile Val Glu Glu Leu Leu Leu
            100   105   110

Asn Gly Lys Leu Thr Met Ser Ala Val Val Lys Lys Val Ala Asp Arg
        115   120   125

Leu Thr Glu Thr Met Glu Asp Gly Lys Thr Met Asp Tyr Ala Glu Val
    130   135   140

Ser Asn Thr Phe Val Arg Leu Ala Asp Thr His Phe Val Gln Arg Cys
145   150   155   160

Pro Ser Val Pro Thr Thr Glu Asn Ser Asp Pro Gly Pro Pro Pro Pro
                165   170   175

Ala Pro Thr Leu Val Ile Asn Glu Lys Asp Met Tyr Leu Val Pro Lys
            180   185   190

Leu Ser Leu Ile Gly Lys Gly Lys Arg Arg Arg Ser Ser Asp Glu Asp
        195   200   205

Ala Ala Gly Glu Pro Lys Ala Lys Arg Pro Lys Tyr Thr Thr Asp Asn
    210   215   220

Lys Glu Pro Ile Pro Asp Asp Gly Ile Tyr Trp Gln Ala Asn Leu Asp

225 230 235 240

Arg Phe His Gln His Phe Arg Asp Gln Ala Ile Val Ser Ala Val Ala
245 250 255

Asn Arg Met Asp Gln Thr Ser Ser Glu Ile Val Arg Thr Met Leu Arg
260 265 270

Met Ser Glu Ile Thr Thr Ser Ser Ser Ala Pro Phe Thr Gln Pro Leu
275 280 285

Ser Ser Asn Glu Ile Phe Arg Ser Leu Pro Val Gly Tyr Asn Ile Ser
290 295 300

Lys Gln Val Leu Asp Gln Tyr Leu Thr Leu Leu Ala Asp Asp Pro Leu
305 310 315 320

Glu Phe Val Gly Lys Ser Gly Asp Ser Gly Gly Gly Met Tyr Val Ile
325 330 335

Asn Leu His Lys Ala Leu Ala Ser Leu Ala Thr Ala Thr Leu Glu Ser
340 345 350

Val Val Gln Glu Arg Phe Gly Ser Arg Cys Ala Arg Ile Phe Arg Leu
355 360 365

Val Leu Gln Lys Lys His Ile Glu Gln Lys Gln Val Glu Asp Phe Ala
370 375 380

Met Ile Pro Ala Lys Glu Ala Lys Asp Met Leu Tyr Lys Met Leu Ser
385 390 395 400

Glu Asn Phe Met Ser Leu Gln Glu Ile Pro Lys Thr Pro Asp His Ala
405 410 415

Pro Ser Arg Thr Phe Tyr Leu Tyr Thr Val Asn Ile Leu Ser Ala Ala
420 425 430

Arg Met Leu Leu His Arg Cys Tyr Lys Ser Ile Ala Asn Leu Ile Glu
435 440 445

Arg Arg Gln Phe Glu Thr Lys Glu Asn Lys Arg Leu Leu Glu Lys Ser
450 455 460

Gln Arg Val Glu Ala Ile Ile Ala Ser Met Gln Ala Thr Gly Ala Glu
465 470 475 480

Glu Ala Gln Leu Gln Glu Ile Glu Glu Met Ile Thr Ala Pro Glu Arg
485 490 495

Gln Gln Leu Glu Thr Leu Lys Arg Asn Val Asn Lys Leu Asp Ala Ser
500 505 510

Glu Ile Gln Val Asp Glu Thr Ile Phe Leu Leu Glu Ser Tyr Ile Glu
515 520 525

Cys Thr Met Lys Arg Gln
530

<210> 111
<211> 398
<212> PRT
<213> Homo sapiens

<400> 111

Met Ser Glu Gly Asn Ala Ala Gly Glu Pro Ser Thr Pro Gly Gly Pro
1 5 10 15

Arg Pro Leu Leu Thr Gly Ala Arg Gly Leu Ile Gly Arg Arg Pro Ala
20 25 30

Pro Pro Leu Thr Pro Gly Arg Leu Pro Ser Ile Arg Ser Arg Asp Leu
35 40 45

Thr Leu Gly Gly Val Lys Lys Lys Thr Phe Thr Pro Asn Ile Ile Ser
50 55 60

Arg Lys Ile Lys Glu Glu Pro Lys Glu Glu Val Thr Val Lys Lys Glu
65 70 75 80

Lys Arg Glu Arg Asp Arg Asp Arg Gln Arg Glu Gly His Gly Arg Gly
85 90 95

Arg Gly Arg Pro Glu Val Ile Gln Ser His Ser Ile Phe Glu Gln Gly
100 105 110

Pro Ala Glu Met Met Lys Lys Lys Gly Asn Trp Asp Lys Thr Val Asp
115 120 125

Val Ser Asp Met Gly Pro Ser His Ile Ile Asn Ile Lys Lys Glu Lys
130 135 140

Arg Glu Thr Asp Glu Glu Thr Lys Gln Ile Leu Arg Met Leu Glu Lys
145 150 155 160

Asp Asp Phe Leu Asp Asp Pro Gly Leu Arg Asn Asp Thr Arg Asn Met
165 170 175

Pro Val Gln Leu Pro Leu Ala His Ser Gly Trp Leu Phe Lys Glu Glu
180 185 190

Asn Asp Glu Pro Asp Val Lys Pro Trp Leu Ala Gly Pro Lys Glu Glu
195 200 205

Asp Met Glu Val Asp Ile Pro Ala Val Lys Val Lys Glu Glu Pro Arg
210 215 220

Asp Glu Glu Glu Glu Ala Lys Met Lys Ala Pro Pro Lys Ala Ala Arg
225 230 235 240

Lys Thr Pro Gly Leu Pro Lys Asp Val Ser Val Ala Glu Leu Leu Arg
245 250 255

Glu Leu Ser Leu Thr Lys Glu Glu Glu Leu Leu Phe Leu Gln Leu Pro
260 265 270

Asp Thr Leu Pro Gly Gln Pro Pro Thr Gln Asp Ile Lys Pro Ile Lys
275 280 285

Thr Glu Val Gln Gly Glu Asp Gly Gln Val Val Leu Ile Lys Gln Glu
290 295 300

Lys Asp Arg Glu Ala Lys Leu Ala Glu Asn Ala Cys Thr Leu Ala Asp
305 310 315 320

Leu Thr Glu Gly Gln Val Gly Lys Leu Leu Ile Arg Lys Ser Gly Arg
325 330 335

Val Gln Leu Leu Leu Gly Lys Val Thr Leu Asp Val Thr Met Gly Thr
340 345 350

Ala Cys Ser Phe Leu Gln Glu Leu Val Ser Val Gly Leu Gly Asp Ser
355 360 365

Arg Thr Gly Glu Met Thr Val Leu Gly His Val Lys His Lys Leu Val
370 375 380

Cys Ser Pro Asp Phe Glu Ser Leu Leu Asp His Lys His Arg
385 390 395

<210> 112
<211> 708
<212> PRT

<213> Homo sapiens

<400> 112

Met Ala Asn Glu Glu Asp Asp Pro Val Val Gln Glu Ile Asp Val Tyr
1   5   10   15

Leu Ala Lys Ser Leu Ala Glu Lys Leu Tyr Leu Phe Gln Tyr Pro Val
            20   25   30

Arg Pro Ala Ser Met Thr Tyr Asp Asp Ile Pro His Leu Ser Ala Lys
        35   40   45

Ile Lys Pro Lys Gln Gln Lys Val Glu Leu Glu Met Ala Ile Asp Thr
    50   55   60

Leu Asn Pro Asn Tyr Cys Arg Ser Lys Gly Glu Gln Ile Ala Leu Asn
65   70   75   80

Val Asp Gly Ala Cys Ala Asp Glu Thr Ser Thr Tyr Ser Ser Lys Leu
                85   90   95

Met Asp Lys Gln Thr Phe Cys Ser Ser Gln Thr Thr Ser Asn Thr Ser
            100   105   110

Arg Tyr Ala Ala Ala Leu Tyr Arg Gln Gly Glu Leu His Leu Thr Pro
        115   120   125

Leu His Gly Ile Leu Gln Leu Arg Pro Ser Phe Ser Tyr Leu Asp Lys
    130   135   140

Ala Asp Ala Lys His Arg Glu Arg Glu Ala Ala Asn Glu Ala Gly Asp
145   150   155   160

Ser Ser Gln Asp Glu Ala Glu Asp Asp Val Lys Gln Ile Thr Val Arg
                165   170   175

Phe Ser Arg Pro Glu Ser Glu Gln Ala Arg Gln Arg Arg Val Gln Ser
            180   185   190

Tyr Glu Phe Leu Gln Lys Lys His Ala Glu Glu Pro Trp Val His Leu
        195   200   205

His Tyr Tyr Gly Leu Arg Asp Ser Arg Ser Glu His Glu Arg Gln Tyr
    210   215   220

Leu Leu Cys Pro Gly Ser Ser Gly Val Glu Asn Thr Glu Leu Val Lys
225   230   235   240

Ser Pro Ser Glu Tyr Leu Met Met Leu Met Pro Pro Ser Gln Glu Glu
245 250 255

Glu Lys Asp Lys Pro Val Ala Pro Ser Asn Val Leu Ser Met Ala Gln
260 265 270

Leu Arg Thr Leu Pro Leu Ala Asp Gln Ile Lys Ile Leu Met Lys Asn
275 280 285

Val Lys Val Met Pro Phe Ala Asn Leu Met Ser Leu Leu Gly Pro Ser
290 295 300

Ile Asp Ser Val Ala Val Leu Arg Gly Ile Gln Lys Val Ala Met Leu
305 310 315 320

Val Gln Gly Asn Trp Val Val Lys Ser Asp Ile Leu Tyr Pro Lys Asp
325 330 335

Ser Ser Ser Pro His Ser Gly Val Pro Ala Glu Val Leu Cys Arg Gly
340 345 350

Arg Asp Phe Val Met Trp Lys Phe Thr Gln Ser Arg Trp Val Val Arg
355 360 365

Lys Glu Val Ala Thr Val Thr Lys Leu Cys Ala Glu Asp Val Lys Asp
370 375 380

Phe Leu Glu His Met Ala Val Val Arg Ile Asn Lys Gly Trp Glu Phe
385 390 395 400

Ile Leu Pro Tyr Asp Gly Glu Phe Ile Lys Lys His Pro Asp Val Val
405 410 415

Gln Arg Gln His Met Leu Trp Thr Gly Ile Gln Ala Lys Leu Glu Lys
420 425 430

Val Tyr Asn Leu Val Lys Glu Thr Met Pro Lys Lys Pro Asp Ala Gln
435 440 445

Ser Gly Pro Ala Gly Leu Val Cys Gly Asp Gln Arg Ile Gln Val Ala
450 455 460

Lys Thr Lys Ala Gln Gln Asn His Ala Leu Leu Glu Arg Glu Leu Gln
465 470 475 480

Arg Arg Lys Glu Gln Leu Arg Val Pro Ala Val Pro Pro Gly Val Arg
485 490 495

```
Ile Lys Glu Glu Pro Val Ser Glu Glu Gly Glu Glu Asp Glu Glu Gln
            500                 505                 510

Glu Ala Glu Glu Glu Pro Met Asp Thr Ser Pro Ser Gly Leu His Ser
        515                 520                 525

Lys Leu Ala Asn Gly Leu Pro Leu Gly Arg Ala Ala Gly Thr Asp Ser
    530                 535                 540

Phe Asn Gly His Pro Pro Gln Gly Cys Ala Ser Thr Pro Val Ala Arg
545                 550                 555                 560

Glu Leu Lys Ala Phe Val Glu Ala Thr Phe Gln Arg Gln Phe Val Leu
                565                 570                 575

Thr Leu Ser Glu Leu Lys Arg Leu Phe Asn Leu His Leu Ala Ser Leu
            580                 585                 590

Pro Pro Gly His Thr Leu Phe Ser Gly Ile Ser Asp Arg Met Leu Gln
        595                 600                 605

Asp Thr Val Leu Ala Ala Gly Cys Lys Gln Ile Leu Val Pro Phe Pro
    610                 615                 620

Pro Gln Thr Ala Ala Ser Pro Asp Glu Gln Lys Val Phe Ala Leu Trp
625                 630                 635                 640

Glu Ser Gly Asp Met Ser Asp Gln His Arg Gln Val Leu Leu Glu Ile
                645                 650                 655

Phe Ser Lys Asn Tyr Arg Val Arg Arg Asn Met Ile Gln Ser Arg Leu
            660                 665                 670

Thr Gln Glu Cys Gly Glu Asp Leu Ser Lys Gln Glu Val Asp Lys Val
        675                 680                 685

Leu Lys Asp Cys Cys Val Ser Tyr Gly Gly Met Trp Tyr Leu Lys Gly
    690                 695                 700

Thr Val Gln Ser
705
```

<210> 113
<211> 316
<212> PRT
<213> Homo sapiens

<400> 113

Met Ala Glu Val Lys Val Lys Val Gln Pro Pro Asp Ala Asp Pro Val
1   5   10   15

Glu Ile Glu Asn Arg Ile Ile Glu Leu Cys His Gln Phe Pro His Gly
            20   25   30

Ile Thr Asp Gln Val Ile Gln Asn Glu Met Pro His Ile Glu Ala Gln
        35   40   45

Gln Arg Ala Val Ala Ile Asn Arg Leu Leu Ser Met Gly Gln Leu Asp
    50   55   60

Leu Leu Arg Ser Asn Thr Gly Leu Leu Tyr Arg Ile Lys Asp Ser Gln
65   70   75   80

Asn Ala Gly Lys Met Lys Gly Ser Asp Asn Gln Glu Lys Leu Val Tyr
                85   90   95

Gln Ile Ile Glu Asp Ala Gly Asn Lys Gly Ile Trp Ser Arg Asp Ile
            100   105   110

Arg Tyr Lys Ser Asn Leu Pro Leu Thr Glu Ile Asn Lys Ile Leu Lys
        115   120   125

Asn Leu Glu Ser Lys Lys Leu Ile Lys Ala Val Lys Ser Val Ala Ala
    130   135   140

Ser Lys Lys Lys Val Tyr Met Leu Tyr Asn Leu Gln Pro Asp Arg Ser
145   150   155   160

Val Thr Gly Gly Ala Trp Tyr Ser Asp Gln Asp Phe Glu Ser Glu Phe
                165   170   175

Val Glu Val Leu Asn Gln Gln Cys Phe Lys Phe Leu Gln Ser Lys Ala
            180   185   190

Glu Thr Ala Arg Glu Ser Lys Gln Asn Pro Met Ile Gln Arg Asn Ser
        195   200   205

Ser Phe Ala Ser Ser His Glu Val Trp Lys Tyr Ile Cys Glu Leu Gly
    210   215   220

Ile Ser Lys Val Glu Leu Ser Met Glu Asp Ile Glu Thr Ile Leu Asn
225   230   235   240

Thr Leu Ile Tyr Asp Gly Lys Val Glu Met Thr Ile Ile Ala Ala Lys

245 250 255

Glu Gly Thr Val Gly Ser Val Asp Gly His Met Lys Leu Tyr Arg Ala
260 265 270

Val Asn Pro Ile Ile Pro Pro Thr Gly Leu Val Arg Ala Pro Cys Gly
275 280 285

Leu Cys Pro Val Phe Asp Asp Cys His Glu Gly Gly Glu Ile Ser Pro
290 295 300

Ser Asn Cys Ile Tyr Met Thr Glu Trp Leu Glu Phe
305 310 315

<210> 114
<211> 223
<212> PRT
<213> Homo sapiens

<400> 114

Met Ala Gly Asn Lys Gly Arg Gly Arg Ala Ala Tyr Thr Phe Asn Ile
1 5 10 15

Glu Ala Val Gly Phe Ser Lys Gly Glu Lys Leu Pro Asp Val Val Leu
20 25 30

Lys Pro Pro Pro Leu Phe Pro Asp Thr Asp Tyr Lys Pro Val Pro Leu
35 40 45

Lys Thr Gly Glu Gly Glu Glu Tyr Met Leu Ala Leu Lys Gln Glu Leu
50 55 60

Arg Glu Thr Met Lys Arg Met Pro Tyr Phe Ile Glu Thr Pro Glu Glu
65 70 75 80

Arg Gln Asp Ile Glu Arg Tyr Ser Lys Arg Tyr Met Lys Val Tyr Lys
85 90 95

Glu Glu Trp Ile Pro Asp Trp Arg Arg Leu Pro Arg Glu Met Met Pro
100 105 110

Arg Asn Lys Cys Lys Lys Ala Gly Pro Lys Pro Lys Lys Ala Lys Asp
115 120 125

Ala Gly Lys Gly Thr Pro Leu Thr Asn Thr Glu Asp Val Leu Lys Lys
130 135 140

Met Glu Glu Leu Glu Lys Arg Gly Asp Gly Glu Lys Ser Asp Glu Glu

145 150 155 160

Asn Glu Glu Lys Glu Gly Ser Lys Glu Lys Ser Lys Glu Gly Asp Asp
165 170 175

Asp Asp Asp Asp Asp Ala Ala Glu Gln Glu Glu Tyr Asp Glu Glu Glu
180 185 190

Gln Glu Glu Glu Asn Asp Tyr Ile Asn Ser Tyr Phe Glu Asp Gly Asp
195 200 205

Asp Phe Gly Ala Asp Ser Asp Asp Asn Met Asp Glu Ala Thr Tyr
210 215 220

<210> 115
<211> 223
<212> PRT
<213> Homo sapiens

<400> 115

Met Ala Gly Asn Lys Gly Arg Gly Arg Ala Ala Tyr Thr Phe Asn Ile
1 5 10 15

Glu Ala Val Gly Phe Ser Lys Gly Glu Lys Leu Pro Asp Val Val Leu
20 25 30

Lys Pro Pro Pro Leu Phe Pro Asp Thr Asp Tyr Lys Pro Val Pro Leu
35 40 45

Lys Thr Gly Glu Gly Glu Glu Tyr Met Leu Ala Leu Lys Gln Glu Leu
50 55 60

Arg Glu Thr Met Lys Arg Met Pro Tyr Phe Ile Glu Thr Pro Glu Glu
65 70 75 80

Arg Gln Asp Ile Glu Arg Tyr Ser Lys Arg Tyr Met Lys Val Tyr Lys
85 90 95

Glu Glu Trp Ile Pro Asp Trp Arg Arg Leu Pro Arg Glu Met Met Pro
100 105 110

Arg Asn Lys Cys Lys Lys Ala Gly Pro Lys Pro Lys Lys Ala Lys Asp
115 120 125

Ala Gly Lys Gly Thr Pro Leu Thr Asn Thr Glu Asp Val Leu Lys Lys
130 135 140

Met Glu Glu Leu Glu Lys Arg Gly Asp Gly Glu Lys Ser Asp Glu Glu

145 150 155 160

Asn Glu Glu Lys Glu Gly Ser Lys Glu Lys Ser Lys Glu Gly Asp Asp
165 170 175

Asp Asp Asp Asp Asp Ala Ala Glu Gln Glu Glu Tyr Asp Glu Glu Glu
180 185 190

Gln Glu Glu Glu Asn Asp Tyr Ile Asn Ser Tyr Phe Glu Asp Gly Asp
195 200 205

Asp Phe Gly Ala Asp Ser Asp Asp Asn Met Asp Glu Ala Thr Tyr
210 215 220

<210> 116
<211> 108
<212> PRT
<213> Homo sapiens

<400> 116

Met Leu Leu Phe Cys Pro Gly Cys Gly Asn Gly Leu Ile Val Glu Glu
1 5 10 15

Gly Gln Arg Cys His Arg Phe Ser Cys Asn Thr Cys Pro Tyr Val His
20 25 30

Asn Ile Thr Arg Lys Val Thr Asn Arg Lys Tyr Pro Lys Leu Lys Glu
35 40 45

Val Asp Asp Val Leu Gly Gly Ala Ala Ala Trp Glu Asn Val Asp Ser
50 55 60

Thr Ala Glu Ser Cys Pro Lys Cys Glu His Pro Arg Ala Tyr Phe Met
65 70 75 80

Gln Leu Gln Thr Arg Ser Ala Asp Glu Pro Met Thr Thr Phe Tyr Lys
85 90 95

Cys Cys Asn Ala Gln Cys Gly His Arg Trp Arg Asp
100 105

<210> 117
<211> 1230
<212> PRT
<213> Homo sapiens

<400> 117

Met Ser Ala Leu Cys Trp Gly Arg Gly Ala Ala Gly Leu Lys Arg Ala
1 5 10 15

```
Leu Arg Pro Cys Gly Arg Pro Gly Leu Pro Gly Lys Glu Gly Thr Ala
            20                  25                  30

Gly Gly Val Cys Gly Pro Arg Arg Ser Ser Ser Ala Ser Pro Gln Glu
        35                  40                  45

Gln Asp Gln Asp Arg Arg Lys Asp Trp Gly His Val Glu Leu Leu Glu
    50                  55                  60

Val Leu Gln Ala Arg Val Arg Gln Leu Gln Ala Glu Ser Val Ser Glu
65                  70                  75                  80

Val Val Val Asn Arg Val Asp Val Ala Arg Leu Pro Glu Cys Gly Ser
                85                  90                  95

Gly Asp Gly Ser Leu Gln Pro Pro Arg Lys Val Gln Met Gly Ala Lys
            100                 105                 110

Asp Ala Thr Pro Val Pro Cys Gly Arg Trp Ala Lys Ile Leu Glu Lys
        115                 120                 125

Asp Lys Arg Thr Gln Gln Met Arg Met Gln Arg Leu Lys Ala Lys Leu
    130                 135                 140

Gln Met Pro Phe Gln Ser Gly Glu Phe Lys Ala Leu Thr Arg Arg Leu
145                 150                 155                 160

Gln Val Glu Pro Arg Leu Leu Ser Lys Gln Met Ala Gly Cys Leu Glu
                165                 170                 175

Asp Cys Thr Arg Gln Ala Pro Glu Ser Pro Trp Glu Glu Gln Leu Ala
            180                 185                 190

Arg Leu Leu Gln Glu Ala Pro Gly Lys Leu Ser Leu Asp Val Glu Gln
        195                 200                 205

Ala Pro Ser Gly Gln His Ser Gln Ala Gln Leu Ser Gly Gln Gln Gln
    210                 215                 220

Arg Leu Leu Ala Phe Phe Lys Cys Cys Leu Leu Thr Asp Gln Leu Pro
225                 230                 235                 240

Leu Ala His His Leu Leu Val Val His His Gly Gln Arg Gln Lys Arg
                245                 250                 255

Lys Leu Leu Thr Leu Asp Met Tyr Asn Ala Val Met Leu Gly Trp Ala
```

260 265 270

Arg Gln Gly Ala Phe Lys Glu Leu Val Tyr Val Leu Phe Met Val Lys
275 280 285

Asp Ala Gly Leu Thr Pro Asp Leu Leu Ser Tyr Ala Ala Ala Leu Gln
290 295 300

Cys Met Gly Arg Gln Asp Gln Asp Ala Gly Thr Ile Glu Arg Cys Leu
305 310 315 320

Glu Gln Met Ser Gln Glu Gly Leu Lys Leu Gln Ala Leu Phe Thr Ala
325 330 335

Val Leu Leu Ser Glu Glu Asp Arg Ala Thr Val Leu Lys Ala Val His
340 345 350

Lys Val Lys Pro Thr Phe Ser Leu Pro Pro Gln Leu Pro Pro Pro Val
355 360 365

Asn Thr Ser Lys Leu Leu Arg Asp Val Tyr Ala Lys Asp Gly Arg Val
370 375 380

Ser Tyr Pro Lys Leu His Leu Pro Leu Lys Thr Leu Gln Cys Leu Phe
385 390 395 400

Glu Lys Gln Leu His Met Glu Leu Ala Ser Arg Val Cys Val Val Ser
405 410 415

Val Glu Lys Pro Thr Leu Pro Ser Lys Glu Val Lys His Ala Arg Lys
420 425 430

Thr Leu Lys Thr Leu Arg Asp Gln Trp Glu Lys Ala Leu Cys Arg Ala
435 440 445

Leu Arg Glu Thr Lys Asn Arg Leu Glu Arg Glu Val Tyr Glu Gly Arg
450 455 460

Phe Ser Leu Tyr Pro Phe Leu Cys Leu Leu Asp Glu Arg Glu Val Val
465 470 475 480

Arg Met Leu Leu Gln Val Leu Gln Ala Leu Pro Ala Gln Gly Glu Ser
485 490 495

Phe Thr Thr Leu Ala Arg Glu Leu Ser Ala Arg Thr Phe Ser Arg His
500 505 510

Val Val Gln Arg Gln Arg Val Ser Gly Gln Val Gln Ala Leu Gln Asn
515 520 525

His Tyr Arg Lys Tyr Leu Cys Leu Leu Ala Ser Asp Ala Glu Val Pro
530 535 540

Glu Pro Cys Leu Pro Arg Gln Tyr Trp Glu Glu Leu Gly Ala Pro Glu
545 550 555 560

Ala Leu Arg Glu Gln Pro Trp Pro Leu Pro Val Gln Met Glu Leu Gly
565 570 575

Lys Leu Leu Ala Glu Met Leu Val Gln Ala Thr Gln Met Pro Cys Ser
580 585 590

Leu Asp Lys Pro His Arg Ser Ser Arg Leu Val Pro Val Leu Tyr His
595 600 605

Val Tyr Ser Phe Arg Asn Val Gln Gln Ile Gly Ile Leu Lys Pro His
610 615 620

Pro Ala Tyr Val Gln Leu Leu Glu Lys Ala Ala Glu Pro Thr Leu Thr
625 630 635 640

Phe Glu Ala Val Asp Val Pro Met Leu Cys Pro Pro Leu Pro Trp Thr
645 650 655

Ser Pro His Ser Gly Ala Phe Leu Leu Ser Pro Thr Lys Leu Met Arg
660 665 670

Thr Val Glu Gly Ala Thr Gln His Gln Glu Leu Leu Glu Thr Cys Pro
675 680 685

Pro Thr Ala Leu His Gly Ala Leu Asp Ala Leu Thr Gln Leu Gly Asn
690 695 700

Cys Ala Trp Arg Val Asn Gly Arg Val Leu Asp Leu Val Leu Gln Leu
705 710 715 720

Phe Gln Ala Lys Gly Cys Pro Gln Leu Gly Val Pro Ala Pro Pro Ser
725 730 735

Glu Ala Pro Gln Pro Pro Glu Ala His Leu Pro His Ser Ala Ala Pro
740 745 750

Ala Arg Lys Ala Glu Leu Arg Arg Glu Leu Ala His Cys Gln Lys Val
755 760 765

Ala Arg Glu Met His Ser Leu Arg Ala Glu Ala Leu Tyr Arg Leu Ser
770 775 780

Leu Ala Gln His Leu Arg Asp Arg Val Phe Trp Leu Pro His Asn Met
785 790 795 800

Asp Phe Arg Gly Arg Thr Tyr Pro Cys Pro Pro His Phe Asn His Leu
805 810 815

Gly Ser Asp Val Ala Arg Ala Leu Leu Glu Phe Ala Gln Gly Arg Pro
820 825 830

Leu Gly Pro His Gly Leu Asp Trp Leu Lys Ile His Leu Val Asn Leu
835 840 845

Thr Gly Leu Lys Lys Arg Glu Pro Leu Arg Lys Arg Leu Ala Phe Ala
850 855 860

Glu Glu Val Met Asp Asp Ile Leu Asp Ser Ala Asp Gln Pro Leu Thr
865 870 875 880

Gly Arg Lys Trp Trp Met Gly Ala Glu Glu Pro Trp Gln Thr Leu Ala
885 890 895

Cys Cys Met Glu Val Ala Asn Ala Val Arg Ala Ser Asp Pro Ala Ala
900 905 910

Tyr Val Ser His Leu Pro Val His Gln Asp Gly Ser Cys Asn Gly Leu
915 920 925

Gln His Tyr Ala Ala Leu Gly Arg Asp Ser Val Gly Ala Ala Ser Val
930 935 940

Asn Leu Glu Pro Ser Asp Val Pro Gln Asp Val Tyr Ser Gly Val Ala
945 950 955 960

Ala Gln Val Glu Val Phe Arg Arg Gln Asp Ala Gln Arg Gly Met Arg
965 970 975

Val Ala Gln Val Leu Glu Gly Phe Ile Thr Arg Lys Val Val Lys Gln
980 985 990

Thr Val Met Thr Val Val Tyr Gly Val Thr Arg Tyr Gly Gly Arg Leu
995 1000 1005

Gln Ile Glu Lys Arg Leu Arg Glu Leu Ser Asp Phe Pro Gln Glu
1010 1015 1020

Phe Val Trp Glu Ala Ser His Tyr Leu Val Arg Gln Val Phe Lys
1025 1030 1035

Ser Leu Gln Glu Met Phe Ser Gly Thr Arg Ala Ile Gln His Trp
1040 1045 1050

Leu Thr Glu Ser Ala Arg Leu Ile Ser His Met Gly Ser Val Val
1055 1060 1065

Glu Trp Val Thr Pro Leu Gly Val Pro Val Ile Gln Pro Tyr Arg
1070 1075 1080

Leu Asp Ser Lys Val Lys Gln Ile Gly Gly Gly Ile Gln Ser Ile
1085 1090 1095

Thr Tyr Thr His Asn Gly Asp Ile Ser Arg Lys Pro Asn Thr Arg
1100 1105 1110

Lys Gln Lys Asn Gly Phe Pro Pro Asn Phe Ile His Ser Leu Asp
1115 1120 1125

Ser Ser His Met Met Leu Thr Ala Leu His Cys Tyr Arg Lys Gly
1130 1135 1140

Leu Thr Phe Val Ser Val His Asp Cys Tyr Trp Thr His Ala Ala
1145 1150 1155

Asp Val Ser Val Met Asn Gln Val Cys Arg Glu Gln Phe Val Arg
1160 1165 1170

Leu His Ser Glu Pro Ile Leu Gln Asp Leu Ser Arg Phe Leu Val
1175 1180 1185

Lys Arg Phe Cys Ser Glu Pro Gln Lys Ile Leu Glu Ala Ser Gln
1190 1195 1200

Leu Lys Glu Thr Leu Gln Ala Val Pro Lys Pro Gly Ala Phe Asp
1205 1210 1215

Leu Glu Gln Val Lys Arg Ser Thr Tyr Phe Phe Ser
1220 1225 1230

<210> 118
<211> 1230
<212> PRT
<213> Homo sapiens

<400> 118

Met Ser Ala Leu Cys Trp Gly Arg Gly Ala Ala Gly Leu Lys Arg Ala
1 5 10 15

Leu Arg Pro Cys Gly Arg Pro Gly Leu Pro Gly Lys Glu Gly Thr Ala
20 25 30

Gly Gly Val Cys Gly Pro Arg Arg Ser Ser Ser Ala Ser Pro Gln Glu
35 40 45

Gln Asp Gln Asp Arg Arg Lys Asp Trp Gly His Val Glu Leu Leu Glu
50 55 60

Val Leu Gln Ala Arg Val Arg Gln Leu Gln Ala Glu Ser Val Ser Glu
65 70 75 80

Val Val Val Asn Arg Val Asp Val Ala Arg Leu Pro Glu Cys Gly Ser
85 90 95

Gly Asp Gly Ser Leu Gln Pro Pro Arg Lys Val Gln Met Gly Ala Lys
100 105 110

Asp Ala Thr Pro Val Pro Cys Gly Arg Trp Ala Lys Ile Leu Glu Lys
115 120 125

Asp Lys Arg Thr Gln Gln Met Arg Met Gln Arg Leu Lys Ala Lys Leu
130 135 140

Gln Met Pro Phe Gln Ser Gly Glu Phe Lys Ala Leu Thr Arg Arg Leu
145 150 155 160

Gln Val Glu Pro Arg Leu Leu Ser Lys Gln Met Ala Gly Cys Leu Glu
165 170 175

Asp Cys Thr Arg Gln Ala Pro Glu Ser Pro Trp Glu Glu Gln Leu Ala
180 185 190

Arg Leu Leu Gln Glu Ala Pro Gly Lys Leu Ser Leu Asp Val Glu Gln
195 200 205

Ala Pro Ser Gly Gln His Ser Gln Ala Gln Leu Ser Gly Gln Gln Gln
210 215 220

Arg Leu Leu Ala Phe Phe Lys Cys Cys Leu Leu Thr Asp Gln Leu Pro
225 230 235 240

Leu Ala His His Leu Leu Val Val His His Gly Gln Arg Gln Lys Arg

245 250 255

Lys Leu Leu Thr Leu Asp Met Tyr Asn Ala Val Met Leu Gly Trp Ala
260 265 270

Arg Gln Gly Ala Phe Lys Glu Leu Val Tyr Val Leu Phe Met Val Lys
275 280 285

Asp Ala Gly Leu Thr Pro Asp Leu Leu Ser Tyr Ala Ala Ala Leu Gln
290 295 300

Cys Met Gly Arg Gln Asp Gln Asp Ala Gly Thr Ile Glu Arg Cys Leu
305 310 315 320

Glu Gln Met Ser Gln Glu Gly Leu Lys Leu Gln Ala Leu Phe Thr Ala
325 330 335

Val Leu Leu Ser Glu Glu Asp Arg Ala Thr Val Leu Lys Ala Val His
340 345 350

Lys Val Lys Pro Thr Phe Ser Leu Pro Pro Gln Leu Pro Pro Pro Val
355 360 365

Asn Thr Ser Lys Leu Leu Arg Asp Val Tyr Ala Lys Asp Gly Arg Val
370 375 380

Ser Tyr Pro Lys Leu His Leu Pro Leu Lys Thr Leu Gln Cys Leu Phe
385 390 395 400

Glu Lys Gln Leu His Met Glu Leu Ala Ser Arg Val Cys Val Val Ser
405 410 415

Val Glu Lys Pro Thr Leu Pro Ser Lys Glu Val Lys His Ala Arg Lys
420 425 430

Thr Leu Lys Thr Leu Arg Asp Gln Trp Glu Lys Ala Leu Cys Arg Ala
435 440 445

Leu Arg Glu Thr Lys Asn Arg Leu Glu Arg Glu Val Tyr Glu Gly Arg
450 455 460

Phe Ser Leu Tyr Pro Phe Leu Cys Leu Leu Asp Glu Arg Glu Val Val
465 470 475 480

Arg Met Leu Leu Gln Val Leu Gln Ala Leu Pro Ala Gln Gly Glu Ser
485 490 495

Phe Thr Thr Leu Ala Arg Glu Leu Ser Ala Arg Thr Phe Ser Arg His
500 505 510

Val Val Gln Arg Gln Arg Val Ser Gly Gln Val Gln Ala Leu Gln Asn
515 520 525

His Tyr Arg Lys Tyr Leu Cys Leu Leu Ala Ser Asp Ala Glu Val Pro
530 535 540

Glu Pro Cys Leu Pro Arg Gln Tyr Trp Glu Glu Leu Gly Ala Pro Glu
545 550 555 560

Ala Leu Arg Glu Gln Pro Trp Pro Leu Pro Val Gln Met Glu Leu Gly
565 570 575

Lys Leu Leu Ala Glu Met Leu Val Gln Ala Thr Gln Met Pro Cys Ser
580 585 590

Leu Asp Lys Pro His Arg Ser Ser Arg Leu Val Pro Val Leu Tyr His
595 600 605

Val Tyr Ser Phe Arg Asn Val Gln Gln Ile Gly Ile Leu Lys Pro His
610 615 620

Pro Ala Tyr Val Gln Leu Leu Glu Lys Ala Ala Glu Pro Thr Leu Thr
625 630 635 640

Phe Glu Ala Val Asp Val Pro Met Leu Cys Pro Pro Leu Pro Trp Thr
645 650 655

Ser Pro His Ser Gly Ala Phe Leu Leu Ser Pro Thr Lys Leu Met Arg
660 665 670

Thr Val Glu Gly Ala Thr Gln His Gln Glu Leu Leu Glu Thr Cys Pro
675 680 685

Pro Thr Ala Leu His Gly Ala Leu Asp Ala Leu Thr Gln Leu Gly Asn
690 695 700

Cys Ala Trp Arg Val Asn Gly Arg Val Leu Asp Leu Val Leu Gln Leu
705 710 715 720

Phe Gln Ala Lys Gly Cys Pro Gln Leu Gly Val Pro Ala Pro Pro Ser
725 730 735

Glu Ala Pro Gln Pro Pro Glu Ala His Leu Pro His Ser Ala Ala Pro
740 745 750

```
Ala Arg Lys Ala Glu Leu Arg Arg Glu Leu Ala His Cys Gln Lys Val
        755                 760                 765

Ala Arg Glu Met His Ser Leu Arg Ala Glu Ala Leu Tyr Arg Leu Ser
    770                 775                 780

Leu Ala Gln His Leu Arg Asp Arg Val Phe Trp Leu Pro His Asn Met
785                 790                 795                 800

Asp Phe Arg Gly Arg Thr Tyr Pro Cys Pro Pro His Phe Asn His Leu
                805                 810                 815

Gly Ser Asp Val Ala Arg Ala Leu Leu Glu Phe Ala Gln Gly Arg Pro
            820                 825                 830

Leu Gly Pro His Gly Leu Asp Trp Leu Lys Ile His Leu Val Asn Leu
        835                 840                 845

Thr Gly Leu Lys Lys Arg Glu Pro Leu Arg Lys Arg Leu Ala Phe Ala
    850                 855                 860

Glu Glu Val Met Asp Asp Ile Leu Asp Ser Ala Asp Gln Pro Leu Thr
865                 870                 875                 880

Gly Arg Lys Trp Trp Met Gly Ala Glu Glu Pro Trp Gln Thr Leu Ala
                885                 890                 895

Cys Cys Met Glu Val Ala Asn Ala Val Arg Ala Ser Asp Pro Ala Ala
            900                 905                 910

Tyr Val Ser His Leu Pro Val His Gln Asp Gly Ser Cys Asn Gly Leu
        915                 920                 925

Gln His Tyr Ala Ala Leu Gly Arg Asp Ser Val Gly Ala Ala Ser Val
    930                 935                 940

Asn Leu Glu Pro Ser Asp Val Pro Gln Asp Val Tyr Ser Gly Val Ala
945                 950                 955                 960

Ala Gln Val Glu Val Phe Arg Arg Gln Asp Ala Gln Arg Gly Met Arg
                965                 970                 975

Val Ala Gln Val Leu Glu Gly Phe Ile Thr Arg Lys Val Val Lys Gln
            980                 985                 990

Thr Val Met Thr Val Val Tyr Gly  Val Thr Arg Tyr Gly  Gly Arg Leu
        995                 1000                 1005
```

Gln Ile Glu Lys Arg Leu Arg Glu Leu Ser Asp Phe Pro Gln Glu
1010 1015 1020

Phe Val Trp Glu Ala Ser His Tyr Leu Val Arg Gln Val Phe Lys
1025 1030 1035

Ser Leu Gln Glu Met Phe Ser Gly Thr Arg Ala Ile Gln His Trp
1040 1045 1050

Leu Thr Glu Ser Ala Arg Leu Ile Ser His Met Gly Ser Val Val
1055 1060 1065

Glu Trp Val Thr Pro Leu Gly Val Pro Val Ile Gln Pro Tyr Arg
1070 1075 1080

Leu Asp Ser Lys Val Lys Gln Ile Gly Gly Gly Ile Gln Ser Ile
1085 1090 1095

Thr Tyr Thr His Asn Gly Asp Ile Ser Arg Lys Pro Asn Thr Arg
1100 1105 1110

Lys Gln Lys Asn Gly Phe Pro Pro Asn Phe Ile His Ser Leu Asp
1115 1120 1125

Ser Ser His Met Met Leu Thr Ala Leu His Cys Tyr Arg Lys Gly
1130 1135 1140

Leu Thr Phe Val Ser Val His Asp Cys Tyr Trp Thr His Ala Ala
1145 1150 1155

Asp Val Ser Val Met Asn Gln Val Cys Arg Glu Gln Phe Val Arg
1160 1165 1170

Leu His Ser Glu Pro Ile Leu Gln Asp Leu Ser Arg Phe Leu Val
1175 1180 1185

Lys Arg Phe Cys Ser Glu Pro Gln Lys Ile Leu Glu Ala Ser Gln
1190 1195 1200

Leu Lys Glu Thr Leu Gln Ala Val Pro Lys Pro Gly Ala Phe Asp
1205 1210 1215

Leu Glu Gln Val Lys Arg Ser Thr Tyr Phe Phe Ser
1220 1225 1230

<210> 119

<211> 541
<212> PRT
<213> Homo sapiens

<400> 119

Met Ser Pro Cys Pro Glu Glu Ala Ala Met Arg Arg Glu Val Val Lys
1   5   10   15

Arg Ile Glu Thr Val Val Lys Asp Leu Trp Pro Thr Ala Asp Val Gln
            20   25   30

Ile Phe Gly Ser Phe Ser Thr Gly Leu Tyr Leu Pro Thr Ser Asp Ile
        35   40   45

Asp Leu Val Val Phe Gly Lys Trp Glu Arg Pro Pro Leu Gln Leu Leu
    50   55   60

Glu Gln Ala Leu Arg Lys His Asn Val Ala Glu Pro Cys Ser Ile Lys
65   70   75   80

Val Leu Asp Lys Ala Thr Val Pro Ile Ile Lys Leu Thr Asp Gln Glu
                85   90   95

Thr Glu Val Lys Val Asp Ile Ser Phe Asn Met Glu Thr Gly Val Arg
            100   105   110

Ala Ala Glu Phe Ile Lys Asn Tyr Met Lys Lys Tyr Ser Leu Leu Pro
        115   120   125

Tyr Leu Ile Leu Val Leu Lys Gln Phe Leu Leu Gln Arg Asp Leu Asn
    130   135   140

Glu Val Phe Thr Gly Gly Ile Ser Ser Tyr Ser Leu Ile Leu Met Ala
145   150   155   160

Ile Ser Phe Leu Gln Leu His Pro Arg Ile Asp Ala Arg Arg Ala Asp
                165   170   175

Glu Asn Leu Gly Met Leu Leu Val Glu Phe Phe Glu Leu Tyr Gly Arg
            180   185   190

Asn Phe Asn Tyr Leu Lys Thr Gly Ile Arg Ile Lys Glu Gly Gly Ala
        195   200   205

Tyr Ile Ala Lys Glu Glu Ile Met Lys Ala Met Thr Ser Gly Tyr Arg
    210   215   220

Pro Ser Met Leu Cys Ile Glu Asp Pro Leu Leu Pro Gly Asn Asp Val

225 230 235 240

Gly Arg Ser Ser Tyr Gly Ala Met Gln Val Lys Gln Val Phe Asp Tyr
245 250 255

Ala Tyr Ile Val Leu Ser His Ala Val Ser Pro Leu Ala Arg Ser Tyr
260 265 270

Pro Asn Arg Asp Ala Glu Ser Thr Leu Gly Arg Ile Ile Lys Val Thr
275 280 285

Gln Glu Val Ile Asp Tyr Arg Arg Trp Ile Lys Glu Lys Trp Gly Ser
290 295 300

Lys Ala His Pro Ser Pro Gly Met Asp Ser Arg Ile Lys Ile Lys Glu
305 310 315 320

Arg Ile Ala Thr Cys Asn Gly Glu Gln Thr Gln Asn Arg Glu Pro Glu
325 330 335

Ser Pro Tyr Gly Gln Arg Leu Thr Leu Ser Leu Ser Ser Pro Gln Leu
340 345 350

Leu Ser Ser Gly Ser Ser Ala Ser Ser Val Ser Ser Leu Ser Gly Ser
355 360 365

Asp Val Asp Ser Asp Thr Pro Pro Cys Thr Thr Pro Ser Val Tyr Gln
370 375 380

Phe Ser Leu Gln Ala Pro Ala Pro Leu Met Ala Gly Leu Pro Thr Ala
385 390 395 400

Leu Pro Met Pro Ser Gly Lys Pro Gln Pro Thr Thr Ser Arg Thr Leu
405 410 415

Ile Met Thr Thr Asn Asn Gln Thr Arg Phe Thr Ile Pro Pro Pro Thr
420 425 430

Leu Gly Val Ala Pro Val Pro Cys Arg Gln Ala Gly Val Glu Gly Thr
435 440 445

Ala Ser Leu Lys Ala Val His His Met Ser Ser Pro Ala Ile Pro Ser
450 455 460

Ala Ser Pro Asn Pro Leu Ser Ser Pro His Leu Tyr His Lys His Asn
465 470 475 480

Gly Met Lys Leu Ser Met Lys Gly Ser His Gly His Thr Gln Gly Gly
485 490 495

Gly Tyr Ser Ser Val Gly Ser Gly Gly Val Arg Pro Pro Val Gly Asn
500 505 510

Arg Gly His His Gln Tyr Asn Arg Thr Gly Trp Arg Arg Lys Lys His
515 520 525

Thr His Thr Arg Asp Ser Leu Pro Val Ser Leu Ser Arg
530 535 540

<210> 120
<211> 1135
<212> PRT
<213> Homo sapiens

<400> 120

Met Asp Pro Gly Ser Arg Trp Arg Asn Leu Pro Ser Gly Pro Ser Leu
1 5 10 15

Lys His Leu Thr Asp Pro Ser Tyr Gly Ile Pro Arg Glu Gln Gln Lys
20 25 30

Ala Ala Leu Gln Glu Leu Thr Arg Ala His Val Glu Ser Phe Asn Tyr
35 40 45

Ala Val His Glu Gly Leu Gly Leu Ala Val Gln Ala Ile Pro Pro Phe
50 55 60

Glu Phe Ala Phe Lys Asp Glu Arg Ile Ser Phe Thr Ile Leu Asp Ala
65 70 75 80

Val Ile Ser Pro Pro Thr Val Pro Lys Gly Thr Ile Cys Lys Glu Ala
85 90 95

Asn Val Tyr Pro Ala Glu Cys Arg Gly Arg Arg Ser Thr Tyr Arg Gly
100 105 110

Lys Leu Thr Ala Asp Ile Asn Trp Ala Val Asn Gly Ile Ser Lys Gly
115 120 125

Ile Ile Lys Gln Phe Leu Gly Tyr Val Pro Ile Met Val Lys Ser Lys
130 135 140

Leu Cys Asn Leu Arg Asn Leu Pro Pro Gln Ala Leu Ile Glu His His
145 150 155 160

Glu Glu Ala Glu Glu Met Gly Gly Tyr Phe Ile Ile Asn Gly Ile Glu
165 170 175

Lys Val Ile Arg Met Leu Ile Met Pro Arg Arg Asn Phe Pro Ile Ala
180 185 190

Met Ile Arg Pro Lys Trp Lys Thr Arg Gly Pro Gly Tyr Thr Gln Tyr
195 200 205

Gly Val Ser Met His Cys Val Arg Glu Glu His Ser Ala Val Asn Met
210 215 220

Asn Leu His Tyr Leu Glu Asn Gly Thr Val Met Leu Asn Phe Ile Tyr
225 230 235 240

Arg Lys Glu Leu Phe Phe Leu Pro Leu Gly Phe Ala Leu Lys Ala Leu
245 250 255

Val Ser Phe Ser Asp Tyr Gln Ile Phe Gln Glu Leu Ile Lys Gly Lys
260 265 270

Glu Asp Asp Ser Phe Leu Arg Asn Ser Val Ser Gln Met Leu Arg Ile
275 280 285

Val Met Glu Glu Gly Cys Ser Thr Gln Lys Gln Val Leu Asn Tyr Leu
290 295 300

Gly Glu Cys Phe Arg Val Lys Leu Asn Val Pro Asp Trp Tyr Pro Asn
305 310 315 320

Glu Gln Ala Ala Glu Phe Leu Phe Asn Gln Cys Ile Cys Ile His Leu
325 330 335

Lys Ser Asn Thr Glu Lys Phe Tyr Met Leu Cys Leu Met Thr Arg Lys
340 345 350

Leu Phe Ala Leu Ala Lys Gly Glu Cys Met Glu Asp Asn Pro Asp Ser
355 360 365

Leu Val Asn Gln Glu Val Leu Thr Pro Gly Gln Leu Phe Leu Met Phe
370 375 380

Leu Lys Glu Lys Leu Glu Gly Trp Leu Val Ser Ile Lys Ile Ala Phe
385 390 395 400

Asp Lys Lys Ala Gln Lys Thr Ser Val Ser Met Asn Thr Asp Asn Leu
405 410 415

Met Arg Ile Phe Thr Met Gly Ile Asp Leu Thr Lys Pro Phe Glu Tyr
420 425 430

Leu Phe Ala Thr Gly Asn Leu Arg Ser Lys Thr Gly Leu Gly Leu Leu
435 440 445

Gln Asp Ser Gly Leu Cys Val Val Ala Asp Lys Leu Asn Phe Ile Arg
450 455 460

Tyr Leu Ser His Phe Arg Cys Val His Arg Gly Ala Asp Phe Ala Lys
465 470 475 480

Met Arg Thr Thr Thr Val Arg Arg Leu Leu Pro Glu Ser Trp Gly Phe
485 490 495

Leu Cys Pro Val His Thr Pro Asp Gly Glu Pro Cys Gly Leu Met Asn
500 505 510

His Leu Thr Ala Val Cys Glu Val Val Thr Gln Phe Val Tyr Thr Ala
515 520 525

Ser Ile Pro Ala Leu Leu Cys Asn Leu Gly Val Thr Pro Ile Asp Gly
530 535 540

Ala Pro His Arg Ser Tyr Ser Glu Cys Tyr Pro Val Leu Leu Asp Gly
545 550 555 560

Val Met Val Gly Trp Val Asp Lys Asp Leu Ala Pro Gly Ile Ala Asp
565 570 575

Ser Leu Arg His Phe Lys Val Leu Arg Glu Lys Arg Ile Pro Pro Trp
580 585 590

Met Glu Val Val Leu Ile Pro Met Thr Gly Lys Pro Ser Leu Tyr Pro
595 600 605

Gly Leu Phe Leu Phe Thr Thr Pro Cys Arg Leu Val Arg Pro Val Gln
610 615 620

Asn Leu Ala Leu Gly Lys Glu Glu Leu Ile Gly Thr Met Glu Gln Ile
625 630 635 640

Phe Met Asn Val Ala Ile Phe Glu Asp Glu Val Phe Ala Gly Val Thr
645 650 655

Thr His Gln Glu Leu Phe Pro His Ser Leu Leu Ser Val Ile Ala Asn
660 665 670

Phe Ile Pro Phe Ser Asp His Asn Gln Ser Pro Arg Asn Met Tyr Gln
675 680 685

Cys Gln Met Gly Lys Gln Thr Met Gly Phe Pro Leu Leu Thr Tyr Gln
690 695 700

Asp Arg Ser Asp Asn Lys Leu Tyr Arg Leu Gln Thr Pro Gln Ser Pro
705 710 715 720

Leu Val Arg Pro Ser Met Tyr Asp Tyr Tyr Asp Met Asp Asn Tyr Pro
725 730 735

Ile Gly Thr Asn Ala Ile Val Ala Val Ile Ser Tyr Thr Gly Tyr Asp
740 745 750

Met Glu Asp Ala Met Ile Val Asn Lys Ala Ser Trp Glu Arg Gly Phe
755 760 765

Ala His Gly Ser Val Tyr Lys Ser Glu Phe Ile Asp Leu Ser Glu Lys
770 775 780

Ile Lys Gln Gly Asp Ser Ser Leu Val Phe Gly Ile Lys Pro Gly Asp
785 790 795 800

Pro Arg Val Leu Gln Lys Leu Asp Asp Asp Gly Leu Pro Phe Ile Gly
805 810 815

Ala Lys Leu Gln Tyr Gly Asp Pro Tyr Tyr Ser Tyr Leu Asn Leu Asn
820 825 830

Thr Gly Glu Ser Phe Val Met Tyr Tyr Lys Ser Lys Glu Asn Cys Val
835 840 845

Val Asp Asn Ile Lys Val Cys Ser Asn Asp Thr Gly Ser Gly Lys Phe
850 855 860

Lys Cys Val Cys Ile Thr Met Arg Val Pro Arg Asn Pro Thr Ile Gly
865 870 875 880

Asp Lys Phe Ala Ser Arg His Gly Gln Lys Gly Ile Leu Ser Arg Leu
885 890 895

Trp Pro Ala Glu Asp Met Pro Phe Thr Glu Ser Gly Met Val Pro Asp
900 905 910

Ile Leu Phe Asn Pro His Gly Phe Pro Ser Arg Met Thr Ile Gly Met

```
        915                 920                 925

Leu Ile Glu Ser Met Ala Gly Lys Ser Ala Ala Leu His Gly Leu Cys
    930                 935                 940

His Asp Ala Thr Pro Phe Ile Phe Ser Glu Glu Asn Ser Ala Leu Glu
945                 950                 955                 960

Tyr Phe Gly Glu Met Leu Lys Ala Ala Gly Tyr Asn Phe Tyr Gly Thr
                965                 970                 975

Glu Arg Leu Tyr Ser Gly Ile Ser Gly Leu Glu Leu Glu Ala Asp Ile
            980                 985                 990

Phe Ile Gly Val Val Tyr Tyr Gln  Arg Leu Arg His Met  Val Ser Asp
        995                 1000                 1005

Lys Phe  Gln Val Arg Thr Thr  Gly Ala Arg Asp Arg  Val Thr Asn
    1010                 1015                 1020

Gln Pro  Ile Gly Gly Arg Asn  Val Gln Gly Gly Ile  Arg Phe Gly
    1025                 1030                 1035

Glu Met  Glu Arg Asp Ala Leu  Leu Ala His Gly Thr  Ser Phe Leu
    1040                 1045                 1050

Leu His  Asp Arg Leu Phe Asn  Cys Ser Asp Arg Ser  Val Ala His
    1055                 1060                 1065

Val Cys  Val Lys Cys Gly Ser  Leu Leu Ser Pro Leu  Leu Glu Lys
    1070                 1075                 1080

Pro Pro  Pro Ser Trp Ser Ala  Met Arg Asn Arg Lys  Tyr Asn Cys
    1085                 1090                 1095

Thr Leu  Cys Ser Arg Ser Asp  Thr Ile Asp Thr Val  Ser Val Pro
    1100                 1105                 1110

Tyr Val  Phe Arg Tyr Phe Val  Ala Glu Leu Ala Ala  Met Asn Ile
    1115                 1120                 1125

Lys Val  Lys Leu Asp Val Val
    1130                 1135
```

<210> 121
<211> 122
<212> PRT
<213> Homo sapiens

<400> 121

Met Glu Glu Asp Gln Glu Leu Glu Arg Lys Ala Ile Glu Glu Leu Leu
1 5 10 15

Lys Glu Ala Lys Arg Gly Lys Thr Arg Ala Glu Thr Met Gly Pro Met
20 25 30

Gly Trp Met Lys Cys Pro Leu Ala Ser Thr Asn Lys Arg Phe Leu Ile
35 40 45

Asn Thr Ile Lys Asn Thr Leu Pro Ser His Lys Glu Gln Asp His Glu
50 55 60

Gln Lys Glu Gly Asp Lys Glu Pro Ala Lys Ser Gln Ala Gln Lys Glu
65 70 75 80

Glu Asn Pro Lys Lys His Arg Ser His Pro Tyr Lys His Ser Phe Arg
85 90 95

Ala Arg Gly Ser Ala Ser Tyr Ser Pro Pro Arg Lys Arg Ser Ser Gln
100 105 110

Asp Lys Tyr Glu Lys Arg Ser Asn Arg Arg
115 120

<210> 122
<211> 117
<212> PRT
<213> Homo sapiens

<400> 122

Met Asn Ala Pro Pro Ala Phe Glu Ser Phe Leu Leu Phe Glu Gly Glu
1 5 10 15

Lys Lys Ile Thr Ile Asn Lys Asp Thr Lys Val Pro Asn Ala Cys Leu
20 25 30

Phe Thr Ile Asn Lys Glu Asp His Thr Leu Gly Asn Ile Ile Lys Ser
35 40 45

Gln Leu Leu Lys Asp Pro Gln Val Leu Phe Ala Gly Tyr Lys Val Pro
50 55 60

His Pro Leu Glu His Lys Ile Ile Ile Arg Val Gln Thr Thr Pro Asp
65 70 75 80

Tyr Ser Pro Gln Glu Ala Phe Thr Asn Ala Ile Thr Asp Leu Ile Ser

85 90 95

Glu Leu Ser Leu Leu Glu Glu Arg Phe Arg Val Ala Ile Lys Asp Lys
100 105 110

Gln Glu Gly Ile Glu
115

<210> 123
<211> 1133
<212> PRT
<213> Homo sapiens

<400> 123

Met Asp Val Leu Ala Glu Glu Phe Gly Asn Leu Thr Pro Glu Gln Leu
1 5 10 15

Ala Ala Pro Ile Pro Thr Val Glu Glu Lys Trp Arg Leu Leu Pro Ala
20 25 30

Phe Leu Lys Val Lys Gly Leu Val Lys Gln His Ile Asp Ser Phe Asn
35 40 45

Tyr Phe Ile Asn Val Glu Ile Lys Lys Ile Met Lys Ala Asn Glu Lys
50 55 60

Val Thr Ser Asp Ala Asp Pro Met Trp Tyr Leu Lys Tyr Leu Asn Ile
65 70 75 80

Tyr Val Gly Leu Pro Asp Val Glu Glu Ser Phe Asn Val Thr Arg Pro
85 90 95

Val Ser Pro His Glu Cys Arg Leu Arg Asp Met Thr Tyr Ser Ala Pro
100 105 110

Ile Thr Val Asp Ile Glu Tyr Thr Arg Gly Ser Gln Arg Ile Ile Arg
115 120 125

Asn Ala Leu Pro Ile Gly Arg Met Pro Ile Met Leu Arg Ser Ser Asn
130 135 140

Cys Val Leu Thr Gly Lys Thr Pro Ala Glu Phe Ala Lys Leu Asn Glu
145 150 155 160

Cys Pro Leu Asp Pro Gly Gly Tyr Phe Ile Val Lys Gly Val Glu Lys
165 170 175

Val Ile Leu Ile Gln Glu Gln Leu Ser Lys Asn Arg Ile Ile Val Glu

180 185 190

Ala Asp Arg Lys Gly Ala Val Gly Ala Ser Val Thr Ser Ser Thr His
195 200 205

Glu Lys Lys Ser Arg Thr Asn Met Ala Val Lys Gln Gly Arg Phe Tyr
210 215 220

Leu Arg His Asn Thr Leu Ser Glu Asp Ile Pro Ile Val Ile Ile Phe
225 230 235 240

Lys Ala Met Gly Val Glu Ser Asp Gln Glu Ile Val Gln Met Ile Gly
245 250 255

Thr Glu Glu His Val Met Ala Ala Phe Gly Pro Ser Leu Glu Glu Cys
260 265 270

Gln Lys Ala Gln Ile Phe Thr Gln Met Gln Ala Leu Lys Tyr Ile Gly
275 280 285

Asn Lys Val Arg Arg Gln Arg Met Trp Gly Gly Gly Pro Lys Lys Thr
290 295 300

Lys Ile Glu Glu Ala Arg Glu Leu Leu Ala Ser Thr Ile Leu Thr His
305 310 315 320

Val Pro Val Lys Glu Phe Asn Phe Arg Ala Lys Cys Ile Tyr Thr Ala
325 330 335

Val Met Val Arg Arg Val Ile Leu Ala Gln Gly Asp Asn Lys Val Asp
340 345 350

Asp Arg Asp Tyr Tyr Gly Asn Lys Arg Leu Glu Leu Ala Gly Gln Leu
355 360 365

Leu Ser Leu Leu Phe Glu Asp Leu Phe Lys Lys Phe Asn Ser Glu Met
370 375 380

Lys Lys Ile Ala Asp Gln Val Ile Pro Lys Gln Arg Ala Ala Gln Phe
385 390 395 400

Asp Val Val Lys His Met Arg Gln Asp Gln Ile Thr Asn Gly Met Val
405 410 415

Asn Ala Ile Ser Thr Gly Asn Trp Ser Leu Lys Arg Phe Lys Met Asp
420 425 430

Arg Gln Gly Val Thr Gln Val Leu Ser Arg Leu Ser Tyr Ile Ser Ala
435 440 445

Leu Gly Met Met Thr Arg Ile Ser Ser Gln Phe Glu Lys Thr Arg Lys
450 455 460

Val Ser Gly Pro Arg Ser Leu Gln Pro Ser Gln Trp Gly Met Leu Cys
465 470 475 480

Pro Ser Asp Thr Pro Glu Gly Glu Ala Cys Gly Leu Val Lys Asn Leu
485 490 495

Ala Leu Met Thr His Ile Thr Thr Asp Met Glu Asp Gly Pro Ile Val
500 505 510

Lys Leu Ala Ser Asn Leu Gly Val Glu Asp Val Asn Leu Leu Cys Gly
515 520 525

Glu Glu Leu Ser Tyr Pro Asn Val Phe Leu Val Phe Leu Asn Gly Asn
530 535 540

Ile Leu Gly Val Ile Arg Asp His Lys Lys Leu Val Asn Thr Phe Arg
545 550 555 560

Leu Met Arg Arg Ala Gly Tyr Ile Asn Glu Phe Val Ser Ile Ser Thr
565 570 575

Asn Leu Thr Asp Arg Cys Val Tyr Ile Ser Ser Asp Gly Gly Arg Leu
580 585 590

Cys Arg Pro Tyr Ile Ile Val Lys Lys Gln Lys Pro Ala Val Thr Asn
595 600 605

Lys His Met Glu Glu Leu Ala Gln Gly Tyr Arg Asn Phe Glu Asp Phe
610 615 620

Leu His Glu Ser Leu Val Glu Tyr Leu Asp Val Asn Glu Glu Asn Asp
625 630 635 640

Cys Asn Ile Ala Leu Tyr Glu His Thr Ile Asn Lys Asp Thr Thr His
645 650 655

Leu Glu Ile Glu Pro Phe Thr Leu Leu Gly Val Cys Ala Gly Leu Ile
660 665 670

Pro Tyr Pro His His Asn Gln Ser Pro Arg Asn Thr Tyr Gln Cys Ala
675 680 685

```
Met Gly Lys Gln Ala Met Gly Thr Ile Gly Tyr Asn Gln Arg Asn Arg
    690                 695                 700

Ile Asp Thr Leu Met Tyr Leu Leu Ala Tyr Pro Gln Lys Pro Met Val
705                 710                 715                 720

Lys Thr Lys Thr Ile Glu Leu Ile Glu Phe Glu Lys Leu Pro Ala Gly
                725                 730                 735

Gln Asn Ala Thr Val Ala Val Met Ser Tyr Ser Gly Tyr Asp Ile Glu
            740                 745                 750

Asp Ala Leu Val Leu Asn Lys Ala Ser Leu Asp Arg Gly Phe Gly Arg
        755                 760                 765

Cys Leu Val Tyr Lys Asn Ala Lys Cys Thr Leu Lys Arg Tyr Thr Asn
    770                 775                 780

Gln Thr Phe Asp Lys Val Met Gly Pro Met Leu Asp Ala Ala Thr Arg
785                 790                 795                 800

Lys Pro Ile Trp Arg His Glu Ile Leu Asp Ala Asp Gly Ile Cys Ser
                805                 810                 815

Pro Gly Glu Lys Val Glu Asn Lys Gln Val Leu Val Asn Lys Ser Met
            820                 825                 830

Pro Thr Val Thr Gln Ile Pro Leu Glu Gly Ser Asn Val Pro Gln Gln
        835                 840                 845

Pro Gln Tyr Lys Asp Val Pro Ile Thr Tyr Lys Gly Ala Thr Asp Ser
    850                 855                 860

Tyr Ile Glu Lys Val Met Ile Ser Ser Asn Ala Glu Asp Ala Phe Leu
865                 870                 875                 880

Ile Lys Met Leu Leu Arg Gln Thr Arg Arg Pro Glu Ile Gly Asp Lys
                885                 890                 895

Phe Ser Ser Arg His Gly Gln Lys Gly Val Cys Gly Leu Ile Val Pro
            900                 905                 910

Gln Glu Asp Met Pro Phe Cys Asp Ser Gly Ile Cys Pro Asp Ile Ile
        915                 920                 925

Met Asn Pro His Gly Phe Pro Ser Arg Met Thr Val Gly Lys Leu Ile
    930                 935                 940
```

Glu Leu Leu Ala Gly Lys Ala Gly Val Leu Asp Gly Arg Phe His Tyr
945 950 955 960

Gly Thr Ala Phe Gly Gly Ser Lys Val Lys Asp Val Cys Glu Asp Leu
965 970 975

Val Arg His Gly Tyr Asn Tyr Leu Gly Lys Asp Tyr Val Thr Ser Gly
980 985 990

Ile Thr Gly Glu Pro Leu Glu Ala Tyr Ile Tyr Phe Gly Pro Val Tyr
995 1000 1005

Tyr Gln Lys Leu Lys His Met Val Leu Asp Lys Met His Ala Arg
1010 1015 1020

Ala Arg Gly Pro Arg Ala Val Leu Thr Arg Gln Pro Thr Glu Gly
1025 1030 1035

Arg Ser Arg Asp Gly Gly Leu Arg Leu Gly Glu Met Glu Arg Asp
1040 1045 1050

Cys Leu Ile Gly Tyr Gly Ala Ser Met Leu Leu Leu Glu Arg Leu
1055 1060 1065

Met Ile Ser Ser Asp Ala Phe Glu Val Asp Val Cys Gly Gln Cys
1070 1075 1080

Gly Leu Leu Gly Tyr Ser Gly Trp Cys His Tyr Cys Lys Ser Ser
1085 1090 1095

Cys His Val Ser Ser Leu Arg Ile Pro Tyr Ala Cys Lys Leu Leu
1100 1105 1110

Phe Gln Glu Leu Gln Ser Met Asn Ile Ile Pro Arg Leu Lys Leu
1115 1120 1125

Ser Lys Tyr Asn Glu
1130

<210> 124
<211> 362
<212> PRT
<213> Homo sapiens

<400> 124

Met Leu Leu Val Glu Phe Phe Glu Leu Tyr Gly Arg Asn Phe Asn Tyr
1 5 10 15

```
Leu Lys Thr Gly Ile Arg Ile Lys Glu Gly Gly Ala Tyr Ile Ala Lys
            20                  25                  30

Glu Glu Ile Met Lys Ala Met Thr Ser Gly Tyr Arg Pro Ser Met Leu
        35                  40                  45

Cys Ile Glu Asp Pro Leu Leu Pro Gly Asn Asp Val Gly Arg Ser Ser
    50                  55                  60

Tyr Gly Ala Met Gln Val Lys Gln Val Phe Asp Tyr Ala Tyr Ile Val
65                  70                  75                  80

Leu Ser His Ala Val Ser Pro Leu Ala Arg Ser Tyr Pro Asn Arg Asp
                85                  90                  95

Ala Glu Ser Thr Leu Gly Arg Ile Ile Lys Val Thr Gln Glu Val Ile
            100                 105                 110

Asp Tyr Arg Arg Trp Ile Lys Glu Lys Trp Gly Ser Lys Ala His Pro
        115                 120                 125

Ser Pro Gly Met Asp Ser Arg Ile Lys Ile Lys Glu Arg Ile Ala Thr
    130                 135                 140

Cys Asn Gly Glu Gln Thr Gln Asn Arg Glu Pro Glu Ser Pro Tyr Gly
145                 150                 155                 160

Gln Arg Leu Thr Leu Ser Leu Ser Ser Pro Gln Leu Leu Ser Ser Gly
                165                 170                 175

Ser Ser Ala Ser Ser Val Ser Ser Leu Ser Gly Ser Asp Val Asp Ser
            180                 185                 190

Asp Thr Pro Pro Cys Thr Thr Pro Ser Val Tyr Gln Phe Ser Leu Gln
        195                 200                 205

Ala Pro Ala Pro Leu Met Ala Gly Leu Pro Thr Ala Leu Pro Met Pro
    210                 215                 220

Ser Gly Lys Pro Gln Pro Thr Thr Ser Arg Thr Leu Ile Met Thr Thr
225                 230                 235                 240

Asn Asn Gln Thr Arg Phe Thr Ile Pro Pro Pro Thr Leu Gly Val Ala
                245                 250                 255

Pro Val Pro Cys Arg Gln Ala Gly Val Glu Gly Thr Ala Ser Leu Lys
```

260 265 270

Ala Val His His Met Ser Ser Pro Ala Ile Pro Ser Ala Ser Pro Asn
275 280 285

Pro Leu Ser Ser Pro His Leu Tyr His Lys Gln His Asn Gly Met Lys
290 295 300

Leu Ser Met Lys Gly Ser His Gly His Thr Gln Gly Gly Gly Tyr Ser
305 310 315 320

Ser Val Gly Ser Gly Gly Val Arg Pro Pro Val Gly Asn Arg Gly His
325 330 335

His Gln Tyr Asn Arg Thr Gly Trp Arg Arg Lys Lys His Thr His Thr
340 345 350

Arg Asp Ser Leu Pro Val Ser Leu Ser Arg
355 360

<210> 125
<211> 115
<212> PRT
<213> Homo sapiens

<400> 125

Met Asn Ala Pro Pro Ala Phe Glu Ser Phe Leu Leu Phe Glu Gly Glu
1 5 10 15

Lys Ile Thr Ile Asn Lys Asp Thr Lys Val Pro Lys Ala Cys Leu Phe
20 25 30

Thr Ile Asn Lys Glu Asp His Thr Leu Gly Asn Ile Ile Lys Ser Gln
35 40 45

Leu Leu Lys Asp Pro Gln Val Leu Phe Ala Gly Tyr Lys Val Pro His
50 55 60

Pro Leu Glu His Lys Ile Ile Ile Arg Val Gln Thr Thr Pro Asp Tyr
65 70 75 80

Ser Pro Gln Glu Ala Phe Thr Asn Ala Ile Thr Asp Leu Ile Ser Glu
85 90 95

Leu Ser Leu Leu Glu Glu Arg Phe Arg Thr Cys Leu Leu Pro Leu Arg
100 105 110

Leu Leu Pro

115

<210> 126
<211> 542
<212> PRT
<213> Homo sapiens

<400> 126

Met Ser Pro Cys Pro Glu Glu Ala Ala Met Arg Arg Glu Val Val Lys
1   5   10   15

Arg Ile Glu Thr Val Val Lys Asp Leu Trp Pro Thr Ala Asp Val Gln
            20   25   30

Ile Phe Gly Ser Phe Ser Thr Gly Leu Tyr Leu Pro Thr Ser Asp Ile
        35   40   45

Asp Leu Val Val Phe Gly Lys Trp Glu Arg Pro Pro Leu Gln Leu Leu
    50   55   60

Glu Gln Ala Leu Arg Lys His Asn Val Ala Glu Pro Cys Ser Ile Lys
65   70   75   80

Val Leu Asp Lys Ala Thr Val Pro Ile Ile Lys Leu Thr Asp Gln Glu
                85   90   95

Thr Glu Val Lys Val Asp Ile Ser Phe Asn Met Glu Thr Gly Val Arg
            100   105   110

Ala Ala Glu Phe Ile Lys Asn Tyr Met Lys Lys Tyr Ser Leu Leu Pro
        115   120   125

Tyr Leu Ile Leu Val Leu Lys Gln Phe Leu Leu Gln Arg Asp Leu Asn
    130   135   140

Glu Val Phe Thr Gly Gly Ile Ser Ser Tyr Ser Leu Ile Leu Met Ala
145   150   155   160

Ile Ser Phe Leu Gln Leu His Pro Arg Ile Asp Ala Arg Arg Ala Asp
                165   170   175

Glu Asn Leu Gly Met Leu Leu Val Glu Phe Phe Glu Leu Tyr Gly Arg
            180   185   190

Asn Phe Asn Tyr Leu Lys Thr Gly Ile Arg Ile Lys Glu Gly Gly Ala
        195   200   205

Tyr Ile Ala Lys Glu Glu Ile Met Lys Ala Met Thr Ser Gly Tyr Arg

    210   215   220

Pro Ser Met Leu Cys Ile Glu Asp Pro Leu Leu Pro Gly Asn Asp Val
225   230   235   240

Gly Arg Ser Ser Tyr Gly Ala Met Gln Val Lys Gln Val Phe Asp Tyr
                245   250   255

Ala Tyr Ile Val Leu Ser His Ala Val Ser Pro Leu Ala Arg Ser Tyr
            260   265   270

Pro Asn Arg Asp Ala Glu Ser Thr Leu Gly Arg Ile Ile Lys Val Thr
        275   280   285

Gln Glu Val Ile Asp Tyr Arg Arg Trp Ile Lys Glu Lys Trp Gly Ser
    290   295   300

Lys Ala His Pro Ser Pro Gly Met Asp Ser Arg Ile Lys Ile Lys Glu
305   310   315   320

Arg Ile Ala Thr Cys Asn Gly Glu Gln Thr Gln Asn Arg Glu Pro Glu
                325   330   335

Ser Pro Tyr Gly Gln Arg Leu Thr Leu Ser Leu Ser Ser Pro Gln Leu
            340   345   350

Leu Ser Ser Gly Ser Ser Ala Ser Ser Val Ser Ser Leu Ser Gly Ser
        355   360   365

Asp Val Asp Ser Asp Thr Pro Pro Cys Thr Thr Pro Ser Val Tyr Gln
    370   375   380

Phe Ser Leu Gln Ala Pro Ala Pro Leu Met Ala Gly Leu Pro Thr Ala
385   390   395   400

Leu Pro Met Pro Ser Gly Lys Pro Gln Pro Thr Thr Ser Arg Thr Leu
                405   410   415

Ile Met Thr Thr Asn Asn Gln Thr Arg Phe Thr Ile Pro Pro Pro Thr
            420   425   430

Leu Gly Val Ala Pro Val Pro Cys Arg Gln Ala Gly Val Glu Gly Thr
        435   440   445

Ala Ser Leu Lys Ala Val His His Met Ser Ser Pro Ala Ile Pro Ser
    450   455   460

```
Ala Ser Pro Asn Pro Leu Ser Ser Pro His Leu Tyr His Lys Gln His
465                 470                 475                 480

Asn Gly Met Lys Leu Ser Met Lys Gly Ser His Gly His Thr Gln Gly
                485                 490                 495

Gly Gly Tyr Ser Ser Val Gly Ser Gly Gly Val Arg Pro Pro Val Gly
            500                 505                 510

Asn Arg Gly His His Gln Tyr Asn Arg Thr Gly Trp Arg Arg Lys Lys
        515                 520                 525

His Thr His Thr Arg Asp Ser Leu Pro Val Ser Leu Ser Arg
    530                 535                 540
```